The Syntax of Likpakpaanl Elliptical Constructions

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Dedication

To my **Beloved Mother**

AGNES NANTOB

who watered seedlings but did not live long enough to eat the fruits.

REST WITH THE LORD!

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If I have been able to see farther than others, it was because I stood on the shoulders of Giants*. (Sir Isaac Newton)*

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We came, we moved, we focused, and we conquered.

Ni ni litulm pam. Uwumbɔr piin timɔmɔk pu.

Abstract

This study investigates Modal Complement Ellipsis (MCE), Sluicing, and Fragment answers in Likpakpaanl, a Mabia language spoken in the Northern, Oti, and North-East regions of Ghana. I propose a movement-plus-PF-deletion approach for the ellipsis in Likpakpaanl and argue that deleted constituents under ellipsis possess the same underlying syntactic structure as their non-elided counterparts. I also adopt a minimalist approach to account for the derivation of these ellipsis constructions. The study shows that only root modals **ŋmàà** 'can', and **bàn** 'want' can license MCE. The E-feature is merged on the head of the modal phrase (ModP) and 'deletes' its vP (in the case of **ŋmàà**) or a TP complement (when the Mod-head is **bàn**). I propose a unified account for sluicing and fragment answers and show that both elliptical phenomena target the FocP for deletion and not a tense phrase (TP) as in English or Dutch. The evidence for this is that the focus particle that is present in non-elliptical sluicing and fragment constructions must be elided under ellipsis in conformity with Merchant (2001)'s Sluicing-Comp Generalisation, which proposes that in sluicing, for instance, on the wh-phrase may escape the ellipsis site and not the non-operator like a focus particle. Drawing on Merchant (2001)'s assumption that ellipsis phenomena are licensed by an ellipsis (E)-feature hosted on a single syntactic element, I propose that Likpakpaanl sluicing and fragments are licensed by an E-feature merged on a Licensing Phrase (LP) above FocP. Fragments and sluicing are derived via an Agree relationship between the wh-sluice or fragment element in the vP periphery and a Foc^0 with an interpretable but unvalued focus feature. After feature valuation, the EPP feature on FocP extracts the constituent to Spec-FocP. Once the LP bearing the E-feature and an EPP feature merge in the derivation, it moves the wh-phrase or fragment element to Spec-LP, with the E-feature licensing the deletion of the complement FocP. Evidence from island effects (Complex Noun Phrase, Coordinate Structure, Adjunct), binding effects, and the inability of wh-phrases like **kinyé** 'how' and **bàŋà** 'why' which cannot undergo ex-situ movement, to license sluicing, were used to support the claim for a movement approach to the derivation of MCE, Sluicing and Fragment answers in Likpakpaanl.

Keywords: *Ellipsis*, *PF-Deletion*, *Likpakpaanl*, *Modal Complement Ellipsis*, *sluicing* , *fragment answers.*

Zusammenfassung

Diese Studie untersucht die Modalkomplementellipse (MCE), Sluicing und Fragmentantworten in Likpakpaanl, einer Mabia-Sprache, die in den Regionen Northern, Oti und North-East in Ghana gesprochen wird. Ich schlage einen Ansatz von Bewegung plus Tilgung für die Ellipse in Likpakpaanl vor und argumentiere, dass getilgte Konstituenten unter Ellipse die gleiche zugrunde liegende syntaktische Struktur besitzen wie ihre nicht-elidierten Gegenstücke. Ich verwende auch einen minimalistischen Ansatz, um die Ableitung dieser Ellipsenkonstruktionen zu erklären. Die Studie zeigt, dass nur die Wurzelmodale **ŋmàà** 'können' und **bàn** 'wollen' MCE lizenzieren können. Das E-Merkmal wird am Kopf der Modalphrase (ModP) zusammengeführt und 'löscht' seine vP (im Fall von **ŋmàà**) oder ein TP-Komplement (wenn der Mod-Kopf **bàn** ist). Ich schlage einen einheitlichen Ansatz für Sluicing und Fragmentantworten vor und zeige, dass beide elliptischen Phänomene die FocP für die Tilgung anvisieren und nicht eine Tempusphrase (TP) wie im Englischen oder Niederländischen. Der Beweis dafür ist, dass die Fokuspartikel, die in nicht-elliptischen Sluicing- und Fragmentkonstruktionen vorhanden ist, unter Ellipse getilgt werden muss, in Übereinstimmung mit Merchants (2001) Sluicing-Comp-Generalisierung, die vorschlägt, dass beim Sluicing beispielsweise nur die Wh-Phrase der Ellipse entkommen kann und nicht der Nicht-Operator wie eine Fokuspartikel. Basierend auf Merchants (2001) Annahme, dass Ellipsenphänomene durch ein Ellipsen (E)-Merkmal lizenziert werden, das auf einem einzigen syntaktischen Element beherbergt ist, schlage ich vor, dass Likpakpaanl-Sluicing und -Fragmente durch ein E-Merkmal lizenziert werden, das auf einer Lizenzierungsphrase (LP) oberhalb der FocP zusammengeführt wird. Fragmente und Sluicing werden durch eine Agree-Beziehung zwischen dem Wh-Sluice- oder Fragmentelement in der vP-Peripherie und einem Foc mit einem interpretierbaren, aber nicht bewerteten Fokusmerkmal abgeleitet. Nach der Merkmalbewertung extrahiert das EPP-Merkmal auf FocP die Konstituente nach Spec-FocP. Sobald die LP mit dem E-Merkmal und einem EPP-Merkmal in der Ableitung zusammengeführt wird, bewegt sie die Wh-Phrase oder das Fragmentelement nach Spec-LP, wobei das E-Merkmal die Tilgung des Komplement-FocP lizenziert.Beweise aus Inseleffekten (komplexe Nominalphrase, Koordinationsstruktur, Adjunkt), Bindungseffekten und der Unfähigkeit von Wh-Phrasen wie **kinyé** 'wie' und **bàŋà** 'warum', die keine Ex-situ-Bewegung durchführen können, um Sluicing zu lizenzieren, wurden verwendet, um den Anspruch auf einen Bewegungsansatz zur Ableitung von MCE, Sluicing und Fragmentantworten in Likpakpaanl zu unterstützen.

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List of Abbreviations

COMP- Complementiser

CONJ- Conjunction

COMPLE-Completive marker

DEF- Definite marker

FOC- Focus particle

FUT- Future Tense

HEST.PST- Hesternal Past

PFV- Perfective Aspect

INF- Infinitive marker

INTER- Interrogative marker

INTENS- Intensifier

LOC- Locative

LF- Logical Form

MCE- Modal Complement Ellipsis

NEG- Negative marker

PST- Past tense marker

PFV- Perfective aspect marker

PL- Plural

IL-Indirect Licensing

PF- Phonological Form

SCG- Sluicing-Comp Generalisation

REL- Relative

RECP -Reciprocal pronoun REL.DET- Relative Determiner

RP- Resumptive Pronoun

TP- Tense Phrase

2PL- Second person plural

2SG- Second person singular

SG- Singular

3SG- Third person singular

Chapter 1

General Overview

1.1 Introduction

Ellipsis, a linguistic phenomenon characterised by the omission of overt linguistic material, has captivated scholars for decades due to its intricate relationship between syntactic structure and silence. Despite the apparent absence of overt forms, the intended meaning is preserved, challenging traditional assumptions about the correspondence between form and meaning in language (Frazier, 2019).

In the intricate tapestry of natural language, we weave our intended messages through carefully crafted utterances, where strings of sounds/words are carefully mapped to their corresponding meanings. This intrinsic relationship between form and meaning enables effective communication, allowing interlocutors to interpret the utterances based on the established patterns and contexts.[1] However, the phenomenon of ellipsis challenges this conventional form-to-meaning mapping, introducing an intriguing mismatch between the overt linguistic form and the underlying meaning (Frazier and Clifton, 2006; Frazier, 2019; Kehler, 2019; Merchant, 2019).

Ellipsis entails 'deleting' elements whose interpretation can be inferred from the context, creating a disconnect between the surface form and the intended meaning. Despite the absence of some linguistic material, the meaning is nevertheless understood. Kehler (2019) notes that this seemingly paradoxical situation, where the meaning transcends the overtly expressed form, is a testament to the remarkable cognitive abilities that underlie language processing. He notes that when a speaker utters an elliptical sentence, the other interlocutor's interpretation is enriched beyond what is explicitly pronounced. This phenomenon is exemplified in (1):

[1] Frazier (2008) discusses the cognitive processes in processing elliptical constructions, highlighting the interplay between the overt linguistic form and the contextual information in deriving the intended meaning. There are also other psycho-linguistics studies like that of Arregui et al. (2006), which examine the cognitive mechanisms involved in recovering the intended meaning from contextual information.

(1) The ducks haven't eaten the corn in the barn, but the chickens have [^{VP} ~~eaten the corn in the barn~~.]

In the second conjunct in (1), the elided clause is semantically understood as *the chickens have eaten the corn in the barn* despite the non-pronunciation of the verb phrase. The ~~strikethrough~~ represents the ellipsis site (E-site) where the missing material would have been present in a non-elliptical utterance.

Different ellipsis phenomena, such as *nominal ellipsis*, *gapping,sluicing*, *fragment answers*, and *verb phrase ellipsis*, among others, have been attested in the literature. The core differences between all these types of ellipsis lie in the kind of constituent that is elided and the syntactic environments in which they occur. Nominal ellipsis occurs when the noun and its accompanying modifiers are omitted from a noun phrase. The example in (2) from Aelbrecht (2015) shows a nominal ellipsis with a nominal phrase. NP ellipsis can also target adjectival modifiers and their complements.

(2) Roy's older brother is taller than Jeff's [$_{DP}$ ~~older brother~~].

(Aelbrecht, 2015, 567)

The other examples in 3 showcase some ellipsis phenomena. In (3-a), we observe sluicing, where a wh-phrase is displaced from a TP, and the remaining TP is elided. Another type of ellipsis, gapping, is exemplified in (3-b). In this case, the verb in the second conjunct is omitted, leaving behind the remaining constituents of the VP. Verb Phrase Ellipsis (VPE), demonstrated in (3-c), involves the elision of the verb and its internal arguments.

(3) a. The President gave the house to someone, but I can't tell **who**$_i$ [$_{TP}$ ~~The President gave the house to~~ t$_i$]. *Sluicing*

b. The carpenter bought his mother a house, and [$_{gap}$ ~~bought~~] his father a car. *Gapping*

c. Agnes fried all her yam, but Bianca didn't [$_{VP}$ ~~fry all her yam~~.] *VPE*

Ellipsis phenomena have been extensively investigated in Romance and Germanic languages, including English, French, Hungarian, Dutch, Italian and Spanish (Ross, 1969; Merchant, 2001; Dagnac, 2010; Lipták, 2019, 2020; Aelbrecht and Harwood, 2015; Depiante, 2001), respectively. These studies have shed light on the intricate properties and licensing conditions of various ellipsis constructions across these language families. For instance, Ross (1969)'s seminal work on sluicing and Merchant (2001)'s analysis of the syntax of silence laid the foundation for understanding the syntactic constraints governing ellipsis. Subsequent research by

Dagnac (2010) on modal ellipsis in French and Lipták (2019, 2020)'s investigations into fragments and sluicing in Hungarian further expanded the cross-linguistic perspective. Additionally, Aelbrecht and Harwood (2015) and Depiante (2001) enriched the empirical landscape of ellipsis by contributing insights into ellipsis phenomena in Dutch and Spanish.

Despite the extensive investigation of ellipsis phenomena across various language families, there is a notable paucity of research regarding African languages. With the exception of notable contributions from Bassong (2014, 2019) on Basàà, (Lipták and Aboh, 2013) on Gungbe,Mendes and Kandybowicz (2023) on Nupe and Biloa and Bassong (2015) on Tuki, among others, the study of ellipsis phenomena in Mabia languages and Likpakpaanl, in particular, remains largely unexplored. This dissertation aims to contribute to the ongoing effort to examine the nature of ellipsis licensing from a cross-linguistic perspective by investigating ellipsis phenomena in Likpakpaanl.

1.2 Objectives of the Study

This dissertation focuses on three types of ellipsis phenomena: Modal Complement Ellipsis (MCE), Sluicing, and Fragment answers. The over-arching goal is to broaden our understanding of the licensing conditions, constraints, and evidence of movement account of ellipsis in Likpakpaanl. The specific objectives that underpin this study include the following:

a. Investigate the syntactic elements that bear the E-feature that license Modal Complement ellipsis, sluicing, and fragment answers in Likpakpaanl.

b. Examine the theoretical model that can adequately account for the derivation of these three elliptical phenomena in Likpakpaanl.

c. Discuss and motivate an analysis to support the claim that sluicing, MCE, and fragment answers in Likpakpaanl are derived via A-bar movement followed by a PF-deletion.

d. Identify the constraints in the licensing of modal complement ellipsis, sluicing, and fragment answers in Likpakpaanl.

e. Offer a unified account of sluicing and fragment answers in Likpakpaanl.

1.3 Theoretical Framework

The theoretical framework assumed in this study is Chomsky (2000, 2001)'s Minimalist Program (MP), where structure-building is seen as a purely derivational system driven by 'Merge' and 'Move'. Merge is a recursive syntactic operation that takes two syntactic objects (α and β) and merges them into an unordered set with a label (either α or β, in this case, α). (Chomsky, 2000). The label serves to identify the properties of the phrase, as in (4-a).

(4) a. Merge (α, β) $\rightarrow$ [α, [α, β]]
 b. Merge (drink, water) $\rightarrow$ drink, drink, water

In (4-b), for instance, the noun phrase *bread* and the verb *eat* are merged to generate a new, larger constituent, *eat bread.* The label *verb phrase* is assigned to *eat bread* since the head of the label is a verb. Thus, for *eat bread*, the merge process unfolds as in (4-b). Operation Move, on the other hand, refers to the operation whereby a constituent is extracted from one position in the syntactic structure to a higher position. According to Chomsky (2000), this movement is motivated by the need to check Formal Features (FFs) associated with functional heads in the structure. For instance, in the English question in (5), the constituent "what" has moved from the vP position to the higher Complementiser (C) position to check the question [+Q]-feature on the C-head. Both Move and Merge operations are driven by Formal Features (FFs).

(5) What did John buy?

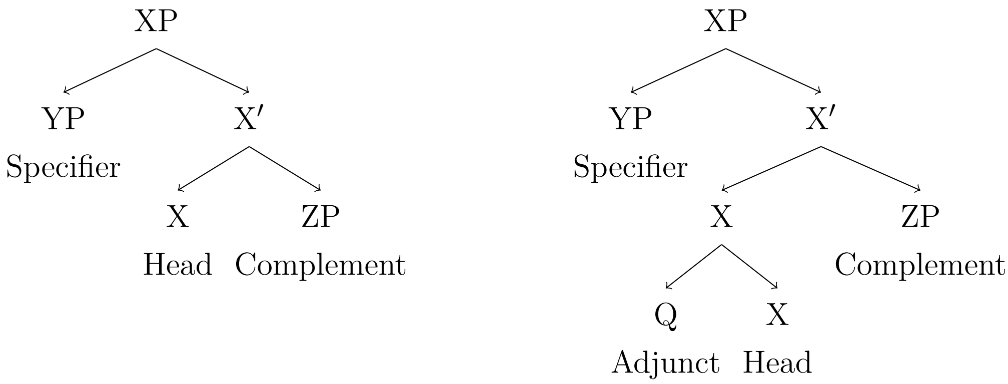
Chomsky (1995) posits that FFs vary in terms of their interpretability, with some being Interpretable [iF] and others being uninterpretable [uF]. [uF] FFs must be checked and deleted at Logical Form (LF) for a derivation to converge, while [iF] FFs do not require checking. Some examples of uninterpretable FFs include the Case feature of DP arguments, the Case-assigning feature of tense head and little verb phrase (VP), as well as phi-features of nouns. Kayne (1994) accounts for the linear order of terminal elements dominated by non-terminal nodes by proposing the *Linear Correspondence Axiom* (LCA). The LCA imposes constraints, dictating that precedence relations must recapitulate asymmetric c-command relations. Asymmetric c-command is defined as follows:

(6) X asymmetrically c-commands Y if and only if X c-commands Y and Y does not c-command X.

(Kayne, 1994, 4)

According to (6), if a head X asymmetrically c-commands another head Y within a sequence, X must precede Y in the linear order. Kayne (1994)'s theory categorically excludes any linear order deviating from the Specifier-Head-Complement (SHC) configuration as a base-generated structure, and the surface order of lexical elements is derived through the displacement of a complement to an asymmetrically c-commanding specifier position. Kayne (1994) claims that the specifier must always precede its head, and the head always precedes its complement (cf. (7-a)), while adjuncts appear to the left of the node to which they adjoin (7-b).

(7) a. *The Spec-Head-Complement* b. *Left Adjoining Adjunct*



Kayne (1994) claims that the linear order of constituents in a sentence directly corresponds to the hierarchical, asymmetric c-command relations in the syntactic structure. According to the LCA, a constituent can only move leftward from a lower position to a higher one in the tree. Downward or rightward movement is not permitted.

The LCA's explanatory power is responsible for accounting for the universal hierarchical structure of phrases and imposing restrictions on the derivation of phrase structures. This has been supported by Cinque (1999)'s hierarchy, which organises functional categories into a hierarchy based on their syntactic positions and roles in sentence structure. Although Kayne (1994)'s LCA has been successfully applied to the analysis of various languages, yielding intriguing results, some studies have identified exceptions to its claims. For example, Ndayiragije (1999) demonstrates that specific constructions in Kirundi, a Bantu language, such as Object Verb Subject (OVS) word orders such as 8, violate the directionality constraints imposed by the LCA. In this Kirundi construction, the LCA is violated due to the rightward A-bar movement of the vP-internal subject to the Spec-Foc position, which affects the positioning of focused constituents and agreement relations.

(8) Ivyo bitabo bi-á-somye Yohani.

those books 3P-PST-read:PERF John
'John (not Peter) read those books.'

(Ndayiragije, 1999, 424)

Given that the central theme of this thesis revolves around ellipsis, I will refrain from delving deeply into the mechanisms of the Minimalist Program in Likpakpaanl Syntax. Instead, my investigation will concentrate on the derivational aspects of ellipsis through the lens of the Minimalist framework.

In current minimalist syntax, the operation Agree is considered to be an operation between a probe carrying uninterpretable and unvalued formal features and a goal with matching interpretable and valued formal features c-commanded by the probe (Chomsky, 2000, 2001). Since uninterpretable features must be deleted at the level of LF and feature valuation is a necessary condition for feature deletion, every uninterpretable feature must be valued through the process of Agree during the derivation. In Chomsky (2000, 2001) Operation Agree is defined as in (9):

(9) Agree: α can agree with β if:
  a. α carries at least one unvalued and uninterpretable feature, and β carries a matching interpretable and valued feature.
  b. α c-commands β.
  c. β is the closest target to α.
  d. β contains an unvalued, uninterpretable feature.

From the Agree definition in (9), a set of specific conditions must be met before *Agree* can take place. Firstly, the probe (α) must carry at least one unvalued and uninterpretable feature, while the goal (β) must possess a matching interpretable and valued feature. The condition in (9-a) means that the probe (α) must have an unvalued and uninterpretable feature, while the goal (β) must have a matching interpretable and valued feature. For example, the [uF] ϕ-features (person, number, gender) on the verb can agree with the [iF] and valued ϕ-features on the subject noun phrase, illustrating why the verb in (10) must agree with the plural subject.

(10) Boys *is/are smart.

The probe α must c-command the goal β, and β must be the closest potential target for α. Thus, the goal β must also contain an unvalued, uninterpretable feature that needs to be valued through the Agree relation. When all these conditions are met, the Agree operation allows the unvalued and uninterpretable features on the probe to be valued and deleted by the interpretable and valued features on the goal, facilitating the convergence of the syntactic derivation and the well-formedness of the resulting sentence.

Two main proposals have been put forth regarding the valuation of uninterpretable features in derivations. Chomsky (2000, 2001) proposes a Biconditional feature approach where an uninterpretable feature is also unvalued and vice versa. Chomsky (2000, 2001) suggests that Agree exists to delete uninterpretable features by establishing a connection between the valuation of unvalued features and the deletion of uninterpretable features. On the contrary, Pesetsky and Torrego (2007) advocate an Agree mechanism that differs from Chomsky (2001)'s bi-conditional feature approach. Pesetsky and Torrego (2007) contend that the lexicon of languages contains items with four sets of features (11) and only unvalued features function as probes.

(11) **Types of features**: (Pesetsky and Torrego, 2007, 269).

a. **Uninterpretable valued** [*uF:Val*] : Interpretable, valued [*iF: Val*]

b. Uninterpretable, unvalued [*uF:Uval*] : **Interpretable, unvalued** [*iF:Uval*]

The main difference between the Agree mechanism proposed by Chomsky (2000, 2001) and that of (11) is that in the former, only an uninterpretable feature can act as a probe, but in the latter, an interpretable feature that is *unvalued* acts as a probe. Pesetsky and Torrego (2007) further suggests that features can iterate by allowing a single feature to be shared between a probe (an unvalued feature) and a goal (a valued feature). Adopting the framework of the Minimalist Program, this thesis aims to offer an account of how elliptical constructions, including sluicing, VP ellipsis, and fragment answers, are derived in Likpakpaanl. The analysis explores the formal feature requirements and structural constraints that govern the syntactic mechanisms underlying these elliptical phenomena.

1.4 Proposal for Ellipsis Licensing in Likpakpaanl

The investigation of Likpakpaanl sluicing and fragment answers construction shows that the focus particle, obligatory in non-elliptical sentences, is deleted in these two elliptical phenomena. The focus head must remain inside the ellipsis site while the focused constituent moves out of Spec-Foc. Island and connectivity effects indicate that the elided constituents display PF-deletion properties. Based on these properties, I adopt Aelbrecht (2010)'s Licensing Phrase (LP) and Merchant (2001)'s idea that the syntactic, semantic, and phonological properties of the Ellipsis [E]-features are on one head to account for how sluicing and fragment answers are derived. The Hypothesis is shown in (12).

(12) **Licensing Phrase with E-feature Hypothesis**:

a. There is a Licensing Phrase (LP) at the left periphery above FocP that bears the E-feature, an Extended Projection Principle (EPP) feature, and an uninterpretable focus feature. LP [E,*u*Foc, EPP].

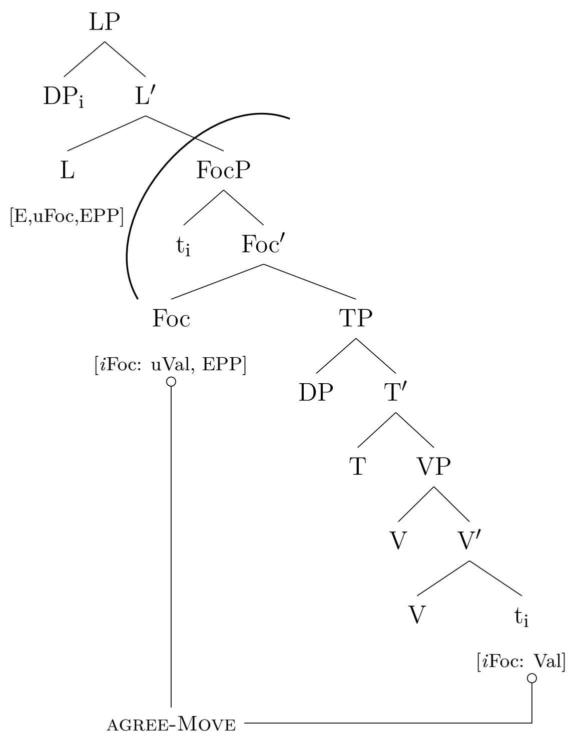
b. The [*u*Foc] on LP is valued via an Agree relation with the [iFoc] on Foc, which occurs after the wh-phrase or the fragment constituent has valued Foc-head.

c. After the valuation, the EPP feature on L triggers A-bar movement of the wh-phrase or fragment element to Spec-LP, and the E-feature now licenses FocP deletion, and everything c-commanded by L is marked for ellipsis by the [E]-feature

d. Ellipsis occurs when LP is merged in the structure, and the [*u*Foc] feature is valued.

Given that the relevant syntactic and semantic requirements are satisfied, the analysis posits that the EPP feature on the Licensing head triggers the movement of the wh-sluice or fragment element to its specifier position. The E-feature associated with the L-head renders all maximal projections within its c-command domain phonetically null, thereby preventing further syntactic operations and phonetic insertion in these elided domains. (13) illustrates the theoretical assumption for sluicing and fragment answers in Likpakpaanl.

(13)



The analyses in Chapters 5 and 6 demonstrate that the LP Hypothesis can account for the grammatical derivations of sluicing and fragment answers in Likpakpaanl. Furthermore, this theoretical proposal successfully rules out the ungrammatical instances of these elliptical constructions in the language.

1.5 Data Sources and Orthography

To investigate the properties of elliptical constructions in Likpakpaanl, the researcher, a native speaker, gathered a selection of sentences based on their intuitions. Additionally, the researcher utilised primary data from various existing works, including folktales, Likpakpaanl Bible (Wycliffe Bible Translators, Inc., 2014), and a Likpakpaanl-English dictionary (Langdon and Steele, 1981), carefully extracting elliptical structures from these materials. Furthermore, the researcher incorporated elliptical sentences from other languages, such as English (Merchant, 2001; Johnson, 2001; Van Craenenbroeck, 2017), Dutch Modal Complement Ellipsis (Aelbrecht, 2010), Bassa (Bassong, 2014, 2019), and Dagbani (Issah, 2020), and translated them into Likpakpaanl.

A total of 90 questions covering Verb Phrase Ellipses (VPE), Sluicing, and fragment answers were administered to 150 participants, comprising seventy-five adults (aged 45 and above) and seventy-five youth (19-44 years old). All participants were speakers of the Lichaabol dialect, one of the five dialects of Likpakpaanl. These sentences were subsequently evaluated for grammaticality by four language consultants (native speakers), two females and two males aged between 45 and 60. The methodology focused on the participants' intuitive linguistic knowledge of Likpakpaanl. The elicitation of the data did not rely on the intuitions of a single native speaker but included different respondents to ensure the accuracy and reliability of the findings, as group-based data can provide a more comprehensive and representative picture of the underlying linguistic system. The data was further cross-checked by independent language consultants to ascertain the naturalness and context appropriateness of the utterances.

The writing system employed in this dissertation largely adheres to the orthographic conventions outlined in the approved document produced by Acheampong et al. (2020). It also considers the conventions used in the *English-Likpakpaanl Dictionary* published in 1981 by Langdon and Steele (1981). Likpakpaanl utilises the Latin alphabet as the basis for its orthography. The current orthography has twenty-three consonants, five digraphs, eighteen monographs, and six vowels *i, e, a, ɔ, o, u/* which are sub-classified into short and long vowels[2]. Although the present orthographic system does not mark tonal distinctions, this dissertation

[2]Long and short vowels are distinguished by the use of single and double letters like **aa**, and **a**, respectively

does mark tone throughout, as tonal information is essential in motivating certain syntactic analyses.

1.6 Structure of the Dissertation

Beyond this broad introduction in this Chapter (Chapter 1), the rest of the dissertation chapters are organised into seven chapters. Chapter 2 reviews the literature on the ellipsis phenomenon and the various approaches adopted in the study of ellipsis by looking at the structural, non-structural, and hybrid approaches. I also examined the theoretical implementation and implication of the PF-deletion approach on elliptical constructions. A typological overview of Merchant (2001)'s E-feature is also discussed in this chapter, focusing on the distribution of the properties of the E-feature and seeking to understand whether they are encoded on a single lexical head or spread across heads in the clausal spine. An overview of verb phrases and modal complement ellipsis and sluicing is also discussed in this chapter. The chapter further touches on the Rizzi (1997)'s Cartography of left peripheral projections, such as the complemetiser Phrase (CP), focus (FOcP) and topic (TopP) phrase.

Chapter 3 addresses in detail some aspects of Likpakpaanl grammar. This is crucial since the language has been relatively under-researched. The chapter provides an overview of Likpakpaanl and its speakers, dialectal variants, occupation, and the population of speakers in Ghana and the Republic of Togo. I also reviewed previous by Tait and Strevens (1954) and Winkelmann (2012) studies on Likpakpaanl Noun Class (NC) system and looked at gaps in the NC inventory. I provided a formal syntactic analysis of the interaction between the NC system and relative, demonstrative, and interrogative pronouns. The chapter further explores properties of grammatical categories such as tense and aspect; I provide empirical data that shows that Likpakpaanl past (remote, hesternal, immediate past) and future tense are morphologically marked by independent particles. In the aspectual domain, I show that while the perfection aspect is the default form, the imperfection and habitual aspects are also marked by particles. The interaction between and aspect indicates that these particles are heads of a TP and AspP projection and that the TP projects above the vP and the AspP in the clausal spine.

Section 3.8 focuses on Likpakpaanl A-bar constructions and demonstrates that the ex-situ strategy of forming wh-questions is characterised by three main processes: (i) An Agree relation between a focus head in the left periphery of the clause with uninterpretable but valued focus feature and the h-phrase which has an interpretable but unvalued Foc feature. ii) An EPP feature on Foc-head, which triggers A-bar movement of the valued wh-phrase from the canonical position to Spec-FocP and (iii) optional presence of the focus head **lé**. I contend that the asymmetry in wh-questions where the moved wh-phrase is optionally focused, but the answer must be obligatorily focused. In addition, in long-distance wh-movement, the extracted wh-phrase can either leave a trace in its base-generated position if it occurs as an internal argument of the verb or leave a Resumptive Pronoun (RP) in the case of subject wh-phrase. Using island effects, I demonstrate that the presence of the RP is a reflex of movement and not base-generation contra to the claim by (McCloskey, 2002). Using island effects and drawing data from The complex NP constraint (CNPC) and coordinate structure constraint, I examined constraints on wh-phrase movement and also showed that the occurrence of the wh-phrase in Spec-Foc is derived by focus movement.

In chapter 4, I discuss Modal Complement Ellipsis (MCE) in Likpakpaanl. I explore how modal verbs in Likpakpaanl express root and epistemic modality and the semantic interpretations arising from their interaction of tense and aspect. Theoretically, I contend that modal verbs project a Modal Phrase ModP above the verb and aspect, following the hierarchical structure proposed by Cinque (1999). I demonstrate that only MCE is licensed by the root modals **ŋmàà** and **bàn** because tense and aspectual markers cannot trigger MCE. Even though both modals can license MCE, the ellipsis site targeted by **ŋmàà** is its vP complement, while **bàn** deletes a CP complement. The empirical facts also indicate that when **ŋmàà** and **bàn** co-occur, it is the higher modal **ŋmàà** that licenses MCE, suggesting it hosts the ellipsis-licensing feature. Evidence from antecedent-contained deletion, focus/topic movement, islands, missing antecedents, and anaphoric binding is used to argue that the ellipsis site has an underlying syntactic structure that becomes phonologically null when the ellipsis feature merges on the modal head.

I examine sluicing constructions in Likpakpaanl and explore the theoretical implications of how they are derived in chapter 5. I propose an analysis of sluicing in Likpakpaanl by adopting Merchant (2001)'s concept that all properties of the E-feature are hosted on a syntactic element. I propose projecting a licensing phrase (LP) based on Aelbrecht (2010)'s approach. However, unlike Aelbrecht (2010), who proposes a split between the semantic and syntactic together phonological features of the E, I propose that the E feature and its feature bundles are all merged on LP in Likpakpaanl allowing L^0. I demonstrate that sluicing in Likpakpaanl is derived through the A-bar wh-movement of the wh-phrase from its base position to Spec-FocP, followed by successive cyclic movement into Spec-LP. This higher functional projection hosts the E-feature and licenses sluicing. I use the analysis proposed by Hatakeyama (1997) for inverse copular sluicing, which suggests that the wh-phrase functioning as the predicate complement is initially merged as a vP complement. This wh-phrase then moves through Spec-FocP and Spec-LP before reaching Spec-PredP.

The data reveals that the overt focus particle accompanying ex-situ wh-movement is obligatorily absent in sluicing. This supports Merchant (2001)'s *Sluicing-COMP Generalisation*, which requires only operators to survive sluicing. I also present various island diagnostics, such as complex NP constraints, adjunct islands, and violations of the coordinate structure constraint. These diagnostics indicate that these island effects are improved under sluicing. The facts indicate that wh-phrases restricted to in-situ occurrences are banned from licensing sluicing in Likpakpaanl. I adopt Merchant (2004)'s PF-theory of islands and argue that sluicing obviates syntactic islands because the PF-uninterpretable feature causing island violations is eliminated at the phonetic form (PF). I show that Likpakpaanl data prove that cross-linguistic sluicing does not always involve TP-ellipsis, as some languages may target larger domains like FocP.

Chapter 6 provides an overview of the fundamental characteristics of Likpakpaanl fragment answers by examining how fragments can respond to matrix, embedded, and polar questions. The data demonstrates that fragments are derived through A-bar movement to a left peripheral position. I contend that fragment responses to matrix and embedded questions are derived through Agree, A- bar movement driven by the EPP feature on FocP and LP, following Merchant (2001, 2004)'s analysis. Fragments are licensed by an E-feature hosted on the L^0, but embedded fragments require the obligatory complementiser **ké**. I follow Kramer and Rawlins (2009)'s analysis for polar question fragments, but I argue that the response particles **Eén**, 'Yes' and **Aáyì**, 'No' are base-generated in the specifier of a polarity phrase, whose head bears the E-feature, rather than sentential adverbs. I present evidence to support the hypothesis that overt A-bar remnant movement to Spec-LP from the base position before ellipsis occurs during fragment derivation. This evidence comes from binding and island effects.

Chapter 7 concludes the dissertation by providing a summary of each of the earlier chapters. It also highlights the dissertation's contributions and relevance to the theory of ellipsis and suggests recommendations for further investigations into elliptical phenomena.

Chapter 2

Approaches to Ellipsis and Typology of the E-Feature

2.1 Introduction

The omission of some aspects from utterances, a phenomenon known as ellipsis, is widespread across the languages of the world. This prevalence is understandable, as ellipsis enables speakers to communicate their intended meaning efficiently by leaving out parts of their speech that can be inferred from the context. Despite this apparent reliance on context for interpretation, there is ongoing debate regarding the precise nature of how elliptical utterances depend on and interact with the surrounding linguistic environment (Van Craenenbroeck and Temmerman, 2019). While there is a consensus that ellipsis involves the omission of material recoverable from the context, the mechanisms underlying this recovery process remain a subject of scholarly contention. Different theoretical approaches have, therefore, been proposed, each offering distinct perspectives and assumptions about the role of syntax, semantics, and pragmatics in resolving ellipsis (Merchant, 2019).

Some analyses prioritise syntactic structure, positing that elided material is structurally present but unpronounced. In contrast, others emphasise semantic and pragmatic interpretive processes over syntactic derivations. Yet other frameworks adopt a hybrid stance, allowing for different types of ellipsis to be analysed through varying combinations of syntactic, semantic, and pragmatic mechanisms. This lack of consensus highlights the complexity of ellipsis and its significance as a window into the intricate workings of grammar and the interfaces between different linguistic domains. Ongoing research continues to shed light on this pervasive phenomenon, advancing our understanding of how speakers expertly navigate the bounds of what must be overtly expressed and what can be pragmatically omitted. Following this introductory overview, the chapter is structured into several sections that delve into various theoretical approaches and frameworks related to ellipsis. Section 2.2 examines the structural, non-structural, and hybrid approaches to analysing ellipsis, exploring the distinct assumptions and mechanisms proposed by each perspective. Section 2.5 section also sheds light on the typology of the E-feature and its cross-linguistic variation. Section 2.7 provides an overview of Rizzi (1997)'s Cartography of syntactic structures, focusing on the information-structural left peripheral projections. An interim summary is given in Section 2.8.

2.2 Approaches to the Study of Ellipsis

Elliptical constructions have been analysed within two major competing frameworks: structural and non-structural and a hybrid approach that combines both structural and non-structural. The following sub-sections briefly outline these approaches and their implication for the theory of ellipsis.

2.2.1 The Structural Approach

The structural approach posits that the ellipsis site contains underlying structures that are null or deleted (Merchant, 2001; Aelbrecht, 2010; van Craenenbroeck, 2010). This approach argues for an unpronounced syntactic structure. It employs theories like LF-Copying and PF-Deletion to support its claims. Most generative syntacticians such as Merchant (2001); Johnson (2001); Merchant (2004); Aelbrecht (2010); Merchant and Simpson (2012); Van Craenenbroeck and Lipták (2013), among others, are advocates of this structural approach. The fundamental position of the structuralists is that the unpronounced structure is either null or deleted. This variation depends on whether one adopts the Logical Form (LF) Copying theory or the Phonological Form (PF) Deletion. Proponents of the LF-Copying approach, like Fiengo and May (1994); Chung et al. (1995), argue that the ellipsis site contains a null category drawn from the lexicon, whose semantic interpretation is filled by copying the semantic component of the correlate of the antecedent clause at Logical Form (LF) level. Thus, they assume that the semantic properties of syntactic objects are copied from the lexicon as an already-built syntactic structure at the ellipsis site at LF and receive semantic interpretation from the discourse context. Chung et al. (1995) adopting the traditional interpretative analysis of ellipsis proposed by Chao (1987) and Wasow (1972) argue that in an elliptical sentence like (14), there exists an elided TP in the complement of the C^0, but the content of the antecedent TP 'They're going to serve the guests' is copied into the unpronounced TP at LF through a 'recycling process' (Chung et al., 1995, 242)[1]

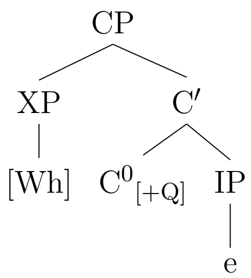
(14) They're going to serve the guests, but it's unclear [CP what [TP e]]. (*Spell-Out*)

[1] Chung et al. (1995) define IP recycling as a broad LF operation involving reusing the content of discourse-available antecedent TP. They argue that recycling best accounts for sluicing licensed at LF.

(Chung et al., 1995, 242)

Thus, the LF-Copying approach rejects a movement plus PF deletion account, contending that sluices such as (14) are base-generated in their surface position with the phonological null TP having no form of internal structure. The null TP [e] in the ellipsis site in (14) is replaced by the operation of structure copying at LF, which takes place before a semantic interpretation is given, as (15) shows (cf. Chao, 1987; Lobeck, 1991).

(15)



(Chung et al., 1995, 242).

Lobeck (1995) and Hardt (1999) further argue that syntactic elements at the ellipsis site are empty pronominals, which receive their interpretation from a co-referred semantic antecedent. For instance, Lobeck (1995), posits that the empty verbal phrase 'buy a skateboard' in (16), comprising the VP and its internal argument, is not derived by a movement operation but occurs in the construction as a null pronominal as depicted in (15). Consequently, the empty pronominal is free in its governing category as stipulated in Principle *B* of the Binding Theory stipulated in (17).

(16) Mary bought a skateboard, and she thinks that Sam should [$_{\text{VP}}$ e] too.

(Lobeck, 1995, 101)

Lobeck (1995) assumes a parallel distribution between pronouns and the null VP by arguing that just as anaphors and pronouns get their semantic interpretation at LF by co-referring with an antecedent, the null VP proform at the E-site, like overt pronouns, has an identical antecedent in the sentence where it is semantically bound.

(17) **Binding Theory**

a. **Principle A**: An anaphor must be bound in its binding domain.
b. **Principle B**: A pronoun must be free in its binding domain.
c. **Principle C**: An R-expression must be free.

(Chomsky, 1981, 190)

A key distinction between Chung et al. (1995) LF-copying and (Lobeck, 1995)'s proform approaches to ellipsis depends on how they view antecedents and the mechanism by which ellipsis sites are semantically interpreted. In the LF-copying approach, antecedents are treated as syntactic objects expressing copies of their syntactic antecedent at LF. In contrast, in the proform approach, an elided structure is treated as a semantic variable assigned the appropriate semantic value from the antecedent they receive their interpretation.

The second account of the structural approach is the movement plus Phonological Form (PF) deletion approach premised on the assumption that A-bar movement feeds ellipsis. The proponents of this approach led by Ross (1969); Merchant (2001); Aelbrecht (2010); van Craenenbroeck (2010), among others, postulate that elided elements at the ellipsis (E)-site have an unpronounced fully-fledged internal structure, which shares semantic and syntactic identity with an antecedent clause. They assume that in (18-b), the phonological content of the elided VP (VP_A) of the second conjunct, though present throughout the derivation, is PF elided and is syntactically and semantically identical to the antecedent VP (VP_A).

(18) a. Lauren can play the guitar, and Mike can, too.

b. [$_{TP}$ Lauren can [$_{VP_A}$ play the guitar], and [$_{TP}$ Mike can [$_{VP_E}$ ~~play the guitar~~], too.

(Merchant, 2019, 2)

It has also been argued by Merchant (2001) that in the sluicing cases such as (19-a), the wh-phrase *what* first undergoes A-bar movement from the complement position of the verb play to Spec-CP in the left periphery. After that, the PF-content of the rest of the TP, out of which the wh-remnant moved, is deleted to derive the sluice (19-b). Although the PF content of the TP [*Lauren can play*] is null at LF, it still exists at PF.

(19) a. Lauren can play something, but I don't know [$_{CP}$ what$_i$ [$_{TP}$ ~~Lauren can play t_i~~ .]]

b.
CP
what$_i$ C'
C ~~TP~~
~~Lauren can play~~ t_i

(Merchant, 2019, 24)

2.2.2 The Non-Structural Approach

The non-structural (N-S) approach argues that there are no lexical items other than what is pronounced in the elliptical structure. In other words, an elliptical clause has no syntactic structure beyond the pronounced remnant visible at LF. Assuming only the uttered structure's categories exist, the N-S approach analyses meaning from the existing surface structure. Proponents of this approach, such as Hankamer and Sag (1976); Ginzburg and Sag (2000), claim that the elliptical site in sluicing, for instance, contains only the wh-remnant, which is considered a complement of its preceding verb. Given that the approach claims no hidden or deleted structures, the interpretation of the entire clause depends on more than the phonetic realisation; i.e., the interface between semantics and syntax helps interpret the elliptical construction Aelbrecht (2010). Using the example in (20), Aelbrecht (2010) opines that the wh-phrase *who* is selected by *know* as its internal argument or complement. There is no underlying syntactic structure in the elided site other than the DP *who*, which we see at LF.

(20) Someone was singing La Marseillaise, but I don't know who.

(Aelbrecht, 2010, 3)

There is also the Simpler Syntax Hypothesis (SSH), proposed by Ginzburg and Sag (2000) and Culicover and Jackendoff (2005), which suggests a *What You See Is What You Get* (WYSIWYG) analysis arguing that syntactic structure at the ellipsis site has no phonological form or any underlying unpronounced structure beyond what is overtly expressed. This SSH rejects the existence of mechanisms like PF deletion or deep anaphora, as stated in (21).

(21) **Simpler Syntax Hypothesis** (Culicover and Jackendoff, 2005, 5)
The most explanatory theory is one that imputes the minimum syntactic structure necessary to mediate between phonology and meaning.

Culicover and Jackendoff (2005) propose an Indirect Licensing (IL) approach and adopt the mechanisms of *matching* and *trace* to account for the interpretation of apparently incomplete or 'orphan' phrases as in (23). In indirect licensing, a phrase is assumed to be syntactically licensed not by the sentence it appears in but via its semantic relationship to an antecedent. One example is topicalisation, such as (22). Instead of assuming the reflexive 'herself' A-bar moves to the specifier of the topic phrase after being licensed by the antecedent 'Francisca', Culicover and Jackendoff (2005) propose indirect licensing without any movement.

(22) Herself$_i$, Francisca loves t$_i$.

The illustration of the indirect licensing in deriving (22) is as follows: First, the antecedent DP 'John' binds the direct object position in the sentence occupied by the trace. Then, the topicalised phrase 'herself' is identified as co-referential with that direct object trace. Establishing this semantic relationship allows the antecedent DP to indirectly bind the reflexive across the trace, even though the reflexive pronoun does not appear in the same clause as its antecedent.

Extending their hypothesis to elliptical constructions, Culicover and Jackendoff (2005) contend that the sluice wh-phrase 'who' in (23) is an orphan phrase bearing features. These features allow it to probe the target indefinite DP 'someone' in the antecedent clause, deriving its semantic and syntactic features through a free variable whose value is determined from the discourse context via indirect licensing (cf. Merchant, 2019, 23).

(23) Someone was singing La Marseillaise, but I don't know [who].

However, the Simpler Syntax Hypothesis is challenged in some ways. First, the indirect licensing approach of Culicover and Jackendoff (2005) theoretically fails to explain how an orphan phrase can target its antecedent, what feature values the free variable contained in the orphan phrase possesses, and how the elided meaning is recovered through indirect licensing.

Furthermore, evidence from Modal complement ellipsis (Chapter 4), slicing (Chapter 5), and fragment answers (Chapter 6) in Likpakpaanl show that the ellipsis sites contain syntactic structure, contrary to the claims by Culicover and Jackendoff (2005). This study adopts the Movement plus PF deletion as the framework since it accommodates the observed ellipsis patterns that will be pursued in this study.

2.2.3 Non-isomorphic Approaches to Ellipsis

Non-isomorphic approaches to ellipsis reject the necessity of complete syntactic structure within ellipsis sites, focusing instead on mechanisms of semantic recovery. This perspective, developed by Dalrymple et al. (1991) and Hardt (1993), prioritise the retrieval of meaning over structural identity. In contrast, isomorphic approaches, initially proposed by Ross (1969) and expanded by Sag (1976) and Williams (1977), posit that ellipsis sites contain a full syntactic structure identical to their antecedents, though unpronounced at the phonological level. These competing views represent distinct pathways for understanding the syntax-semantics interface in ellipsis.

Non-isomorphic approahc to ellipsis represents a significant challenge to traditional approaches to ellipsis, especially the Movement Approach of Merchant

(2001) where the understood interpretation of a sluiced clause cannot be directly derived from a structurally identical antecedent, raises fundamental questions about the nature of ellipsis identity conditions and the architecture of the grammar of a language. Non-isomorphic approaches predict the grammaticality of voice mismatches in VP ellipsis in English, while structural approaches struggle to account for these cases (Merchant, 2013b, 78). Consider example (24):

(24) The janitor must remove the trash whenever it is apparent that it should ~~be removed~~.

The grammaticality of (24) demonstrates that strict structural identity isn't necessary for successful ellipsis resolution since The antecedent clause is in active voice "the janitor must remove", but the sluiced portion occurs in the passive voice "it should be removed", supporting non-isomorphic predictions (Kehler, 2002). Non-structural approaches go further, predicting that such mismatches should be generally available when semantic and discourse conditions are met (Hardt, 1993). Cases of voice mismatches between the antecedent clause and the E-site cannot be adequately accounted for using the Movement approach, which requires strict structural identity between the antecedent and the elided material.

Another adduced to support non-isomorphic involves sprouting where the correlate to the wh-phrase is merely implicit in the antecedent (25):

(25) Joan ate dinner, but I don't know with whom$_i$ ~~Joan ate dinner~~ t_i.

(Chung et al., 1995, 26)

According to Chung et al. (1995) (25) contains only one internally articulated Inflectional Phrase (IP) whose Logical Form (LF) can be recycled into the IP of the sluice: namely, the LF of "Joan ate dinner." Once this LF is copied into the IP of the sluice, there is still no appropriate syntactic position for the displaced constituent to bind until the defect is remedied by sprouting an extra preposition phrase position.

2.2.4 The Hybrid Approach

The hybrid ellipsis approach integrates semantic and syntactic identity requirements but imposes these conditions selectively. Advocates of this approach, including researchers like Kehler (2002); Chung (2006); Chung et al. (2011); Merchant (2013a) among others, propose that different elliptical phenomena involve varying mechanisms. Thus, they note that some elliptical types rely more heavily on syntactic structure, while others draw more from semantic and pragmatic processes. Central to this theory is the notion that the identity condition governing ellipsis is primarily semantic, formalised as Merchant (2001)'s e-GIVENness condition. For instance, the example in (26) shows that an elided pronoun is syntactically different from a wh-phrase in the antecedent clause. Even though Merchant (2001) assumes that such 'Vehicle Change'[2] is covered under e-GIVENness, Rudin (2019) argues that the two aspects are semantically distinct because the antecedent receives an interrogative interpretation, whereas the ellipsis clause is non-interrogative. Rudin (2019) contends that the elided pronoun and *it* in (26) refer to an implicit argument to the question raised by the antecedent. Consequently, since the question is semantically distinct from the answer, the ellipsis site cannot be argued to mutually entail the antecedent clause.

(26) I don't know who$_{\mathrm{i}}$ said what$_{\mathrm{k}}$ or why they$_{\mathrm{i}}$ said it$_{\mathrm{k}}$.

(Rudin, 2019, 15)

The hybrid approach captures the insight that some elliptical constructions are constrained by syntax while semantic and contextual factors heavily influence how other elliptical forms are derived. This flexibility enables proponents of the hybrid approach to comprehensively account for the diverse range of ellipsis data across constructions and languages without being forced to adopt an overly restrictive, purely syntactic or semantic approach. This dissertation adopts the structural and PF-deletion approaches in analysing elliptical constructions in Likpakpaanl. As such, I elaborate on the properties and typology of the Ellipsis [E]-feature, which Merchant (2001) proposes as the feature that cross-linguistically regulates elliptical constructions.

2.3 Ellipsis and the Movement plus PF-Deletion Approach

Island effects have been widely employed in syntactic theory to diagnose movement operations since Ross (1967). The basic assumption is that configurations like relative clauses and wh-islands are islands that block extraction out of them, and this is used as evidence of syntactic movement. Conversely, insensitivity to islands indicates the absence of movement. Consequently, if sluicing derivations involve focus-movement of the wh-remnant from its base position to a left periphery landing site (Spec-FocP or Spec-CP), this movement should obey island constraints like regular focus wh-movement. However, since Ross (1969), ellipsis

[2]Fiengo and May (1994) suggest that the identity condition governing ellipsis should be defined regarding equivalence classes, allowing mismatches between elements in the same equivalence class.

and sluicing constructions have been examined concerning their ability to repair island violations that cause ill-formedness under overt wh-movement dependencies. Islands induce ungrammaticality when a wh-dependency crosses their boundary. In (27-a), sluicing is insensitive to the island violation since it allows the wh-phrase *which (Balkan language)* to be extracted from the relative clause island *someone who speaks a Balkan language.* However, parallel extraction is disallowed in the VP-ellipsis in (27-b), as evidenced by the ungrammaticality.

(27) a. They want to hire someone who speaks a Balkan language, but I don't remember **which**.
b. *ABBY wants to hire someone who speaks a Balkan language, but I don't remember what kind of language BEN does.

(Merchant, 2004, 705)

Sluicing, for example, has been argued to ameliorate island effects based on the acceptability of sentences like (27-a) and the acceptability of such constructions has led to the proposal that island violations are repaired under sluicing (Merchant, 2001; Ross, 1969; Mendes and Kandybowicz, 2023). The fact that sluicing appears to violate islands in many languages, as in Likpakpaanl, where sluicing from islands is grammatical but non-elliptical wh-movement, raises important theoretical questions.

One line of argument is the apparent ability of sluicing to circumvent island constraints, which supports a base-generation account rather than a movement derivation. For instance, Merchant (2004, 2001) assumes that sluicing may involve direct base-generation of the wh-remnant in its surface position without any movement from within the island, as non-structuralists contend. Under such an approach, sluicing would be expected to show immunity to islands since no extraction is involved.

Merchant (2004), however, postulates that island effects arise from properties of pronounced syntactic structure, not movement itself. Thus, illicit movement adds an uninterpretable PF feature [*] that makes non-elliptical sentences ungrammatical. But in sluicing, the uninterpretable PF feature [*], which will otherwise cause the ungrammaticality, is unpronounced or elided under ellipsis, thereby obviating islands. This 'salvation by deletion', as Mendes and Kandybowicz (2023) term it, rescues otherwise illicit derivations. The apparent island insensitivity of sluicing has, thus, been argued to support a PF-deletion analysis since the effects of syntactic islands are claimed to persist in the derivation but go unpronounced (Lasnik, 2001; Merchant, 2001; Mendes and Kandybowicz, 2023). Therefore, proponents of the PF-deletion approach have advanced several other reasons to support their claim. As explained by Ross (1969) and later supported by Merchant (2001, 2004), Lipták and Bánréti (2022) among others, connectivity effects associated with case marking, number agreement, and binding conditions support this view.

2.3.0.1 Case Matching

Evidence from languages with overt case marking has shown that in sluicing constructions, wh-remnants must bear the same morphological case as their antecedent (cf. Ross, 1967; Merchant, 2001; Lipták and Bánréti, 2022). Lipták and Bánréti (2022) have argued that case matching restriction applies to both wh-focus and relative clause sluicing. They show that in Hungarian sluicing like (28), nominative wh-remnants are prohibited when the antecedent correlate bears an accusative case. Thus, the example (28) shows that a nominative wh-remnant **ki** 'who' is infelicitous when the antecedent DP **valakit** 'someone' bears an accusative case.

(28) Júlia meghívott valakit, de nem tudom,
Júlia PRT.invite.PST.3SG someone.ACC but not know.1SG
kit/*ki.
who.ACC/who.NOM
'Júlia invited someone, but I don't know who (she invited).'

(Lipták and Bánréti, 2022, 148).

Similarly, the German example in (29-a) originally cited in Ross (1969) and repeated in Merchant (2019) demonstrates that the case on the wh-remnant in sluicing must match the case on the antecedent DP. The implication is that when the indefinite correlate bears a dative or accusative case, the wh-remnant must bear the same case. We also see that (29-b) has the same underlying semantic interpretation as its elliptical counterpart in (29-a). The example demonstrates that the predicate *schmeicheln* 'flatter' assigns a dative case while a dative case is assigned by the predicate *loben* 'praise' (29-c). Therefore, the dative or accusative case assigned to *jemandem* and *loben* in the antecedent clause must also be assigned to the wh-phrase in the corresponding sluiced construction for the sentence to the licit (see; Ross, 1969, 253-254).

(29) a. Er will jemand**em** schmeicheln, aber sie wissen nicht
he wants someone.DAT flatter but they know not
*wer /*wen /wem.
who.NOM/who.ACC/who.DAT
'He wants to flatter **someone**, but they don't know **who**.'

b. Sie wissen nicht *wer /*wen /wem er schmeicheln
they know not who.NOM/who.ACC/who.DAT he flatter

will.
want
'They don't know who he wants to flatter.'

c. Er will jemand**en** loben, aber sie wissen nicht
he wants someone.ACC praise but they know not
*wer /wen / *wem.
who.NOM/who.ACC/who.DAT
'He wants to flatter **someone**, but they don't know **who**.'

(Merchant, 2019, 29).

More generally, whenever the indefinite DP in the antecedent clause carries a particular case, the wh-fragment in the sluiced clause must bear the identical case. Violations of case matching lead to ungrammaticality, as seen in the Hungarian and German examples in (28) and (29), respectively. This cross-linguistic evidence on case connectivity in sluicing supports derivational PF-deletion approaches rather than base-generation analyses.

2.3.0.2 Number Agreement

Number agreement within the PF-deletion approach predicts that elements undergoing A-bar movement should exhibit agreement morphology, indicating their origin from an underlying base position prior to movement. This is evidenced in Ross (1969)'s example, where he argues that in (30-a), the singular verb *isn't* agrees with the singular remnant *which problems*, which has A-bar moved out of an elided TP containing the plural noun.

(30) a. He's going to give us some old problems for the test, but which problems isn't clear.

b. *He's going to give us some old problems for the test, but which problems aren't clear.

c. He's going to give us some old problems for the test, but which problems but [$_{CP}$ which problems$_i$ [$_{IP}$ ~~he's going to give us t$_i$~~]] isn't clear.

(Ross, 1969, 256).

Ross (1969) contends that *which problems* in (30-a) is in a derived position since it triggers singular agreement on the matrix verb in the canonical position inside the TP before ellipsis as shown in (30-c). By contrast, the ungrammatical structure in (30-b) shows a failed agreement test because the wh-phrase did not move out of the elided TP containing the plural 'problems'. Therefore, the number agreement facts provides evidence that the remnant undergoes overt movement out of the elided constituent before the IP is deleted, confirming the predictions of the PF-deletion approach (see also Merchant, 2019, 32).

2.3.0.3 Binding Effects

Syntactic binding refers to the relationship between an anaphor or pronoun and its antecedent, where the two need to be in a local configuration for the binding to hold (Reinhart, 1976; Chomsky, 1995; Reinhart, 1993). Constituent Command (C-command) is a sisterhood relation between co-referenced syntactic elements.

(31) **C-command**

Anode α *c-commands* a node β if and only if (a) or (b) hold:

a.β is the sister of α.

b.β is dominated by the sister of α.

Reinhart (1976) notes that the binding principles depend on a local relation between the binder (antecedent) and the bindee. A fundamental principle of c-command is that α and β must be co-referential,i.e., refer to the same individual and share ϕ features such as person, number, and gender. This co-referentiality is indicated by co-indexation. In (32), the antecedent “Francis” c-commands and binds the reflexive “himself” as shown by co-indexation.

(32) Francis$_i$ treated *himself*$_i$ very well.

Merchant (2003) following Lasnik (2001), has argued that binding facts provide evidence for overt wh-movement of the remnant before ellipsis, as claimed under PF-deletion approaches. Consider the example in (33):

(33) Every linguist$_i$ criticised some of his$_i$ work, but I’m not sure how much of his$_i$ work [~~every linguist$_i$ criticised~~ t$_i$].

(Merchant, 2003, 5)

In (33), the third person singular pronoun 'his' in the remnant 'his work' co-references the quantified DP 'every linguist', which is the antecedent and c-commands the anaphor. For this binding to hold, 'his work' must originate in a position where it is c-commanded by the quantified DP. Since the object DP cannot be bound to the subject, the sluiced clause suggests that it is base-generated in the elided TP where binding occurred before extraction. We have examined the literature demonstrating that elliptical constructions have an unpronounced underlying structure subject to the same locality constraints as non-elliptical constructions. The next section examines how island effects are obviated under ellipsis.

2.4 Ellipsis, PF Deletion and Island Repair

There have been many studies on the repair power of ellipsis, dating back to the seminal work of Ross (1969). This section focuses on island repair. Mendes and Kandybowicz (2023) following Merchant (2001); Ross (1967) I propose a 'Salvation-by-Deletion' approach to ellipsis by examining the distinction between grammatical violations permitted under ellipsis and those not. They contend that the possibility or impossibility of repair-by-ellipsis plays a role in deriving well-formed inverse copular sluicing in Likpakpaanl. It will also explain why inverse sluicing cannot occur in non-elliptical settings.

According to Mendes and Nevins (2022) and Mendes and Kandybowicz (2023), ellipsis can repair ungrammatical LF representations. However, if the overt realisation of a structure violates semantic interpretability, it is predicted that it cannot be repaired under ellipsis. Mendes and Kandybowicz (2023) show that in Nupe, A-bar extraction of TP adverbs is possible (34-a), but such extraction of vP-internal elements like complements (34-b), low adjuncts, and material inside perfect clausal complements is disallowed. They also show that object extraction is possible only in Nupe tense clauses (34-c).

(34) a. Ányí$_i$ lẽ t_i Musa á nakàn ba karayín o.
long.ago formerly Musa PFV meat cut.PST carefully FOC
'LONG AGO, Musa had cut the meat carefully.'

b. *Ké$_i$ Musa á t_i pa o?
what Musa PFV pound.PST FOC
Intended: 'What has Musa pounded?'

c. Ké$_i$ Musa t_i pa o?
what Musa pound.PST FOC
Intended: 'What did Musa pound?'

(Mendes and Kandybowicz, 2023, 304-305)

Mendes and Kandybowicz (2023) contend that this restriction is rooted in the mapping from syntax to PF and argue the extraction asymmetry is due to the implementation of Cyclic Linearisation, where vP is treated as a Spell-Out domain rather than VP. The extraction restriction arises because perfect vPs do not allow successive cyclic movement, making A-bar extraction of non-edge vP-internal material 'too long'. They note that the apparent violations of the extraction restriction in Nupe perfect clauses are saved by ellipsis, ensuring that otherwise illicit movements are repaired by the non-Spell-Out of their PF content. For instance, ungrammatical constructions resulting from extraction from perfect clauses are permitted if they occur in elliptical constructions.

(35) a. [A:]

Musa	á	ejan	ndoci	pa.
Musa	PFV	thing	certain	pound.PST

'Musa has pounded something.'

b. [B:]

Ké$_i$	~~Musa~~	~~á~~	~~t_i pa~~		o?
what	Musa	PFV	pound.PST		FOC

'What ~~has Musa pounded~~?'

(Mendes and Kandybowicz, 2023, 308)

Mendes and Kandybowicz (2023)'s observation parallels Merchant (2004)'s PF theory of islands, which investigates the asymmetry in island sensitivity across elliptical constructions like sluicing, VP-ellipsis, and fragments. The PF theory posits that island violations stem from properties of the pronounced syntactic structure rather than constraints on derivations or LF representations. Merchant proposes that intermediate traces of XPs extracted from islands are marked with a PF-uninterpretable feature, labelled (*), merged to the XP upon movement across an island. This PF-uninterpretable feature exists in non-elliptical contexts, rendering island-violating extractions uninterpretable at PF. In the sluicing examples in (36), the defective trace (*) is on the TP, which is deleted, and this deletion eliminates all (*) traces in sluicing, avoiding a PF crash. According to Merchant (2004), this accounts for why sluicing is insensitive to islands.

(36) a. They want to hire someone who speaks a Balkan language, but I don't remember which (they do).

b. [CP [DP $_{\text{which}2}$] [TP [*t_2] [TP [they] [do] [vP [*t_2] [vP [to hire [DP [NP someone] CP[who speaks t_2]]]]]]]]

(Merchant, 2004, 706-707)

He notes further that unlike in sluicing where the TP, hosting the defective trace, is deleted, in VP-ellipsis, the highest *-trace remains after VP deletion, triggering a PF crash and island sensitivity. In the case of fragment answers, Merchant (2004) proposes an additional FP layer dominating CP and TP, allowing the fragment element to raise to Spec-CP, leaving a *-trace on C°. As only TP undergoes ellipsis, the highest *-trace in Spec-CP survives, causing a PF crash that we find in VP-ellipsis. This additional CP layer, absent in sluicing, captures the divergent island sensitivity between fragments and sluicing within a uniform account where pronounced *-traces induce PF ungrammaticality.

Drawing on Brazilian Portuguese and Russian data, Mendes and Kandybowicz (2023) investigate the phenomenon of 'salvation by deletion' in the context of lexical gaps/defectiveness[3]. They make a crucial distinction between defectiveness that can be circumvented via PF deletion, signalling the lack of an eligible allomorph in certain environments within a language, and those that cannot be salvaged by deletion. They demonstrate that in Russian, certain defective verbs like *pret-i-t* 'to repulse' and *oš̃čut-i-t* 'to sense', being second conjugation verbs, cannot be inflected in the 1st person singular or plural forms. However, under ellipsis contexts like (37-a) and (37-b), the elided verbs correspond to their overt antecedents, with *to sense* assigning an accusative case and *pret-i-t* assigning a dative case to their respective complements in the ellipsis site:

(37) a. Na veršine étoj gory ty oš̃čutiš̃ radost, a ja ~~*V~~ strakh.
on top this mountain you sense happiness.ACC and I sense fear.ACC
'At the top of this mountain, you will sense happiness, and I fear.'

b. Ty pretiš̃ mne, a ja ~~*V~~ tebe.
you repulse me.DAT and I repulse you.DAT
'You repulse me, and I you.'

(Mendes and Nevins, 2022, 186)

The examples show that ellipsis constitutes a robust repair strategy that rescues derivations that would otherwise crash due to interface violations. Cross-linguistic empirical evidence from elliptical constructions like sluicing and fragments demonstrate that ellipsis can circumvent ungrammatical constructions caused by pronouncing structures containing defective elements or traces. Thus, the Russian example illustrates that defectiveness constraints can be lifted under PF deletion, allowing otherwise illicit formations to survive via ellipsis repair. Ellipsis enables

[3] Mendes and Kandybowicz (2023) refer to defective verbs as those that exhibit gaps in their inflectional paradigms, specifically lacking certain forms that are expected based on the regular morpho-phonological patterns of the language.

the computational system to converge on a legitimate PF representation by eliding such offending constituents, thereby salvaging the derivation from ill-formed syntactic objects that would otherwise induce a crash at the interfaces.

2.5 The Typology of Merchant (2001)'s Ellipsis Feature

This section is divided into two parts; the first part examines the various properties of the E-feature and how they pattern together to license different ellipsis. The second part focuses on the typology of the E-feature by discussing the three different realisations of the feature across different languages using the works of Merchant (2001), Aelbrecht (2010) and Stigliano (2022).

2.5.1 Properties of the E-Feature

In Merchant (2001)'s movement plus PF-deletion framework, ellipsis licensing is governed by an Ellipsis feature (henceforth, E-feature) with three features: semantic, phonological, and syntactic feature bundles which interact in the syntax to derive elliptical constructions. He argues that syntactically, the E-feature imposes licensing conditions on ellipsis to ensure that it involves a 'local relationship', head-complement relationship, between the licensing head and the ellipsis site. Thus, only specific heads in a language are selected to bear the E-feature to license either sluicing, Verb Phrase, nominal ellipsis, etc. Determining the lexical category that bears the E-feature is a language-specific property. This explains the cross-linguistic variation in the kinds of licensing heads for ellipsis. Johnson (2001); Aelbrecht and Harwood (2015) observe that in English VP ellipsis, finite auxiliaries are obligatorily retained and cannot undergo deletion, as exemplified in (38-a). Aelbrecht and Harwood (2015) note that a finite lexical verb is still elided, even when a dummy *do* is inserted (38-b) to license ellipsis, suggesting strongly that lexical verbs cannot bear the E-feature as the ungrammaticality of (38-c) demonstrates.

(38) a. An elephant can't fly, but maybe a rhino *(could) ~~[fly]~~.
b. The chicken didn't put the tuna on the table, but the penguin did ~~[put the tuna on the table]~~.
c. *The chicken didn't put the tuna on the table, but the penguin put ~~[the tuna on the table]~~.

(Aelbrecht and Harwood, 2015, 67).

Depiante (2001) and Dagnac (2010) also demonstrate that in French, Verb Phrase ellipsis, also known as Modal Complement Ellipsis (MCE), is licensed by a modal auxiliary. Thus, while the E-feature in English is merged on the inflectional head, in French, it is merged on the Modal head, as (39) shows.

(39) Léa lit tous les livres qu'elle [$_{\text{ModP}}$ peut ~~[$_{\text{VP}}$ lire les livres]~~].
Lea reads all the books that she can
'Lea reads all the books that she can.'

(Dagnac, 2010, 159)

Semantically, the E-feature requires an identity isomorphism between the elided constituent and its antecedent. Merchant (2001) adopts Schwarzschild (1999)'s e-GIVENness condition, which imposes a semantic condition that the elided elements must have a mutual entailment relationship with the antecedent clause .Merchant (2001) defines e-GIVENness as stated in (40-a). The focus condition and E-GIVENness on ellipsis ensure that only elements with an appropriate antecedent can be a target for deletion.

(40) **Focus Condition on Ellipsis**-(Merchant, 2001, 26)

a. A VP_α can be deleted only if α is e-GIVEN.

b. e-GIVENness Condition on Ellipsis An expression E counts as e-GIVEN iff, E has a salient antecedent A and modulo $\exists$-type shifting,

i.A entails F-clo(E), and

ii.E entails F-clo(A)

According to Merchant (2001), recoverability and mutual entailment are determined by replacing focus-marked elements with variables and existentially closing the result (Focus closure) of the antecedent clause (VP_A) and the elided clause (VP_E). Consider the example in (41-a):

(41) a. Abby called Chunk an idiot after Ben did ~~[call Chunk an idiot]~~.

b. $\neq$ Abby called Chunk an idiot after Ben did ~~[insult Chunk]~~.

(Merchant, 2001, 27)

The F-closures of the antecedent VP, VP_A, and the elliptical VP, VP_E, are shown in (42-a) and (42-c).

(42) a. Abby [VP_A called Chunk an idiot], after Ben did [VP_E call Chunk an idiot].

b. F-clo (VP_A)= $\exists$x (Ben did x)

c. F-clo (VP_E)= ∃x(Ben did x)

The two F-closures (VP_A) and (VP_E) in (42-a) are not only semantically identical but also entail each other. Therefore, the VP at the ellipsis site is e-GIVEN, and under the focus condition on VP-ellipsis in (40-a), it can be deleted. Merchant (2001) notes that (41-b) is infelicitous because the VP at the ellipsis site fails to satisfy the semantic identity requirement and, therefore, can neither be recovered nor deleted.

The last component of the E-Feature is responsible for the phonological deletion of elements that have met the syntactic and semantic identity requirements. Merchant (2001, 2004) argues that the phonological component of the E-feature governs the non-pronunciation of the complement to the lexical head bearing the feature. He argues that the non-pronunciation of the TP or finite auxiliary complement in English sluicing and VP ellipsis is determined by the phonological component of the E-feature, which operates at the level of Phonological Form (PF).

Merchant (2001) analyses various linguistic properties that enable ellipsis licensing by focusing specifically on sluicing in English. He proposes that the E-feature, situated on the complementiser (C)-head in English, is responsible for deleting its TP complement. The syntactic specification of [E] dictates that only constituent questions with an uninterpretable strong wh- and Q-features [wh, Q] can license sluicing in English. These uninterpretable features are checked against matching features in a local configuration, permitting sluicing to occur exclusively on the complementiser head. Merchant (2004) proposes the lexical entry of the ellipsis feature responsible for English sluicing is illustrated in 43:

(43) a. Syntax of E_S: E_S [uwh*, uQ*]
b. Phonology of E: φTP $\rightarrow \varnothing$/E
c. Semantics of E: $[[E]] = \lambda p : \text{e-GIVEN}(p)[p]$

(Merchant, 2004, 670–672)

(43-a) specifies the licensing condition of ellipsis by determining the syntactic heads or structures in a language permit ellipsis of their complements. For example, the E-feature in English sluicing is compatible only with an uninterpretable wh-phrase and question features on the null C head. This phonological component in (43-b) triggers PF-deletion by instructing the phonological interpretative element of the head bearing the E-feature not to parse its complement. The semantics component (43-c) encodes the e-GIVENness and identity requirement and ensures that the E-feature takes a propositional argument p, and ensures that p is previously mentioned (e-GIVEN) and mutually entailed in the antecedent before eliding it.

2.5.2 Where is the Ellipsis Feature?

Merchant (2001)'s groundbreaking theory of ellipsis, which posits that ellipsis is licensed by an E-feature bearing syntactic, semantic, and phonological properties, has been widely applied to data from different languages, including English, (Johnson, 2001), French (Dagnac, 2010), Hungarian (van Craenenbroeck, 2010), Dutch (Aelbrecht, 2010), Bassá (Bassong, 2014), Gungbe, (Lipták and Aboh, 2013) and Nupe (Mendes and Kandybowicz, 2023), among others.

Merchant (2001, 2019) contends that the ellipsis can occur if the Licensing and identity conditions are met. The licensing condition focuses on analysing the syntactic heads or structures that allow for ellipsis in a language, as well as the locality conditions that govern the relationship between these structures and the ellipses. Conversely, the identity condition investigates the semantic and structural isomorphism between the content at the ellipsis site and its antecedent (Merchant, 2001, 2019). The preceding discussion has indicated three distinct components of the E-feature in the lexicon that license ellipsis. However, various analyses have been given regarding the distribution of these bundles of the E-feature and whether they require a strict head-complement relationship.

To begin with, Merchant (2001, 2004, 2019), and subsequent works have argued that all three components of the E-feature mentioned in (43) are encoded within a single head. Merchant (2001) also observes that there must be a head-complement relationship between the head that carries the E-feature and its complement. Therefore, in the case of licensing sluicing in (44), he assumes that the syntactic, semantic, and phonological features are manifested on the null C-head. The complement TP is also a sister to the C-head with the licensing feature (Merchant, 2004, 670). According to Merchant (2004), the E-feature encompasses all relevant properties that distinguish elliptical structures from their non-elliptical counterparts in the provided example of sluicing in (44).

(44) a. Abby was reading something, but I don't know what$_i$ [~~Abby was reading t_i~~].
b. [$_{CP}$ what $^{[wh]}$ C $^{[wh,Q]}{}_{[E]}$ [~~$_{TP}$ Abby was reading~~]].

While there is consensus among scholars regarding the licensing of various types of ellipsis by the E-feature, there have been divergent views concerning the representation of the syntactic, semantic, and phonological properties encoded by this feature. Aelbrecht (2010), for example, argued that the functions of licensing and non-pronunciation are distinct within the clausal structure and do not necessarily rely on a head-complement relationship as proposed by Merchant (2001). To support her argument, Aelbrecht (2010) provides examples of sentences such as (45) where the licensing head, the final auxiliary *should*, is not in a head-complement relation with its complement. This is due to the presence of non-finite verbs *have been* that intervene between the head and the ellipsis site.

(45) I hadn't been thinking about that. well, you should have been [~~thinking about that~~]!

(Aelbrecht, 2010, 91)

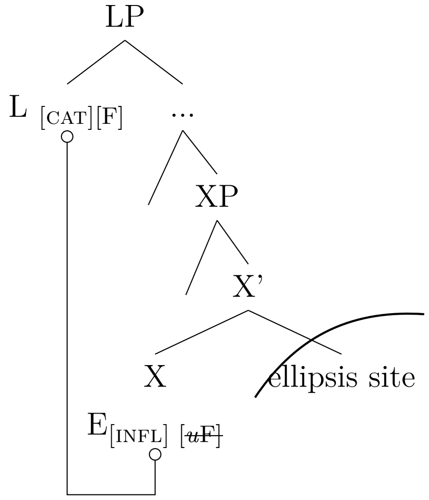
Additionally, Aelbrecht (2010) shows that (*Spading*)(where a demonstrative pronoun follows a wh-sluice), in the Wambeek dialect of Dutch, indicates that the licensor and the ellipsis site are not adjacent. The demonstrative *da* occurs as a complement to the wh-sluice in 46 and thus lies between licensor, *wou*, and ellipsis site.

(46)	Jef	eit	iemand	gezien,	mo	ik	weet	nie	wou	**da**.
	Jef	has	someone	seen	but	I	know	not	who	that

'Jef saw someone, but I don't know who.'

(Aelbrecht, 2010, 93)

These facts challenge the claim made by Merchant (2001, 2004, 2019), among others, that ellipsis always occurs through a head-complement relation between the E-feature and the ellipsis site. Aelbrecht (2010) adopts a similar account to Merchant (2001, 2004)'s implementation of the E-feature but argues that all three properties of the E-feature are encoded on different heads in the syntax. Aelbrecht (2010) notes that each elliptical phenomenon in a language has a specific [E]-feature in the lexicon, with Selection [SEL], Categorial [CAT], and Inflectional [INFL] properties. The SEL(ectional) features allow only certain heads in a language to license ellipsis by deleting the PF content of the the complement to the head. Thus, Unlike Merchant (2001), who assumes that the semantic, syntactic, and phonological properties are situated on one licensing head, Aelbrecht (2010) contends that these features are divided between two distinct heads; the ellipsis licensing head that projects a Licensing Phrase (LP) and the head bearing the E-feature (at least in Dutch). In Aelbrecht (2010)'s E-feature geometry, she assumes an uninterpretable inflectional [*u*INFL] feature that corresponds to the category [CAT] feature of the licensing head [X], which establishes a checking relation with the licensor [L-head]. The [*u*INFL] and E-feature on the licensing head [X] probes upwards for a matching Goal on the LP, and ellipsis is licensed via an Agree relation between the [X]-head (bearing E-feature) and the ellipsis licensing head which projects above XP as illustrated in (47). The licensing is triggered during the derivation as soon as LP is merged in the structure, rendering the ellipsis site inaccessible for further syntactic operations and blocking vocabulary insertion at PF (Chomsky, 1981; Aelbrecht, 2010).

(47)


(Aelbrecht, 2010, 100)

Apart from Aelbrecht (2010) who proposes a split in the properties of the E-feature, Stigliano (2022) also contends that ellipsis in Spanish Wh-Topic-Remnant Elided Questions (wh-TREQs) is governed by two distinct E-features: one that enforces syntactic identity [E^{syn}], located on the Complementiser Head (C), and another that does not impose any specific identity condition, situated on the Topic head (Top) [E]. She observes that in Spanish, CP ellipsis is triggered by the E-feature on Top. In contrast, TP ellipsis is licensed by E^{syn} on C-head. It requires syntactic identity between the elided constituent and the antecedent. Thus, an isomorphic condition is unnecessary for deleted elements between C and Top. Stigliano (2022) emphasises that in examples like (48), there is no syntactic identity requirement before the material between the TP and the CP can be licensed. The wh-phrase is not required to syntactically have a corresponding antecedent since it is assumed to occur between Topic and C-head (see Stigliano, 2022, 149).

(48) a. Sonia sueña con Luciano
Sonia dreams with Luciano
'Sonia dreams about Luciano.'

b. [$_{\text{Top}}$ Y Bruno con [$_{\text{CP}}$ **qué** sueña?]]
and Bruno, with what dreams
'And as for Bruno, what does he dream about?'

She notes that the ellipsis site as in (48-b) must contain **qué** 'what' and not **quién** 'who' (emphasis mine) because a question with **quién** is not possible. After all, when it occurs in a question as in (49-b), it will be infelicitous to answer using (49-c). Crucially, mismatches are allowed between the antecedent clause and the ellipsis site,

(49) a. Sonia sueña con Luciano
Sonia dreams with Luciano
'Sonia dreams about Luciano.'

b. Y Bruno con **quén** sueña?
and Bruno with what dreams
'And as for Bruno, who does he dream about?'

c. ?Con Buenos Aires.
with Buenos Aires
Intended: 'About Buenos Aires.'

(Stigliano, 2022, 149)

Stigliano (2022) contends that no syntactic identity is required for CP ellipsis in Spanish. However, syntactic identity becomes obligatory when the ellipsis targets the TP domain, and this is licensed by the [E^{Syn}] feature. In her analysis, [E^{syn}] can only license ellipsis when E is merged in the structure during the derivation.

(50) $[_{TopP}$ Bruno$_i$ $_{Top}$ [E] $[\ _{CP}$ DP que$_j$ $_C$ E^{syn} $[_{TP}$ t_i $[_{VP}$ sueña t_j]]]]]].

The distinction between Aelbrecht (2010)'s ellipsis licensing and Stigliano (2022)'s lies in the former having the licensing property on the L-head while the ellipsis and identity property are on the X-head. X-head probes upwards to Agree with the L-head. In Spanish, the licensing and ellipsis feature is on the E, while the identity feature [E^{Syn}] is on the C-head. We observed that in Merchant (2001)'s account; the licensing, identity, and ellipsis properties are all on one head. (51) presents the typology of the distribution of the ellipsis feature across ellipsis types and languages.

(51) **Variations of the E-Feature**

a. E-feature's semantic, syntactic, and phonological properties on a single lexical head. (Merchant, 2001, 2004)

b. The categorical (syntactic) property is on a separate head (LP) while the phonological and semantic properties of the E-feature are a different head c-commanded by LP. (Aelbrecht, 2010)

c. The syntactic and phonological properties are on Topic^0 while the semantic property E^{syn} is on the C^0. (Stigliano, 2022)

The observed variations across languages of the E-feature by Merchant (2001), Aelbrecht (2010), and Stigliano (2022) show that the syntactic, semantic, and phonological properties of the E-feature can be encoded across distinct projections within the clausal hierarchy. This distribution manifests as cross-linguistic and cross-constructional variation in the size of the structural domain that hosts the E-feature. This phenomenon resembles the behaviour of other intricate grammatical notions like tense, aspect, and mood, which have been analysed as realised on multiple functional heads in the clause structure. The precise feature specifications associated with each type of ellipsis demonstrate the intricate relationship between the syntactic reflexes and the semantic interpretations of these categories.

The typological overview above is a foundation for analysing sluicing and fragment answers in Likpakpaanl. This investigation aims to substantiate the claim that different types of ellipsis exhibit distinct E-feature configurations, but the realisation of the E-feature itself is subject to cross-linguistic variation. In this vein, the proposed study on Likpakpaanl will examine the E-feature's syntactic, semantic, and phonological reflexes in sluicing and fragment answer constructions within this language. By probing the intricate interplay between the feature bundles and their structural manifestations, this study uncovers language-particular properties that inform the size of the ellipsis domain and the mapping between syntax and interpretation. Consequently, this analysis is expected to contribute to the growing body of evidence that the E-feature, while constituting a universal grammatical primitive, is susceptible to parametric variation in its formal realisation across linguistic systems.

2.6 Overview of Elliptical Constructions

Since the pioneering work of Sag (1976), extensive research has been conducted into different forms of elliptical construction ranging from Verb Phrase Ellipsis (VPE), Sluicing, Fragment answers and gapping, among others. Research on the linguistic phenomenon of ellipsis has been extensively documented across numerous languages worldwide, as evident in the works Merchant (2001, 2013b, 2004) Johnson (2001), Aelbrecht (2010); Aelbrecht and Harwood (2015) Van Craenenbroeck (2017),Dagnac (2010) and Aelbrecht and Harwood (2015, 2019). This section investigates VPE, sluicing, and fragmented answers from a cross-linguistic perspective.

2.6.1 Verb Phrase Ellipsis

Verb Phrase Ellipsis (VPE) refers to the phenomenon where the main predicate of a clause, usually in combination with its internal arguments and adjuncts, is elided. For instance, in (52), the entire verb phrase 'weeding the maize farm' in the second conjunct clause is not pronounced because it is syntactically and semantically recoverable from its antecedent in the first conjunct.

(52) Nicholina is sweeping the room, and Naomi is [$_{vP}$ ~~sweeping the room~~] too.

Unlike in sluicing, which has been observed in many languages to show a universal property where a wh-phrase licenses TP ellipsis (Merchant, 2001, 2004, 2019; Lipták and Aboh, 2013), VPE has been shown to vary cross-linguistically. For example, while VPE is a common phenomenon in English, in other closely related Germanic languages, such as German and Dutch, it has been argued to be non-existent (Lobeck, 1995; Matsuo and Duffield, 2001).

However, recent studies indicate that VPE occurs in various languages, but it manifests in different forms. For instance, unlike the VPE observed in English, where the licensing feature is merged on a finite auxiliary, these alternative VPE constructions involve either the verb moving out of the verb phrase to a higher head prior to ellipsis or a modal licensing ellipsis. Goldberg (2005) refers to the former type of VPE as Verb-Stranding Verb Phrase Ellipsis (V-VPE) while the latter has been termed as Modal Complement Ellipsis (MCE). In V-VPE, the verb is raised out of the verb phrase and escapes the ellipsis, resulting in a distinct VPE pattern. Goldberg (2005) notes that Modern Hebrew, Modern Irish, and Swahili display V-VPE. On the other hand, Dutch and French display MCE, where a root modal licenses the ellipsis (Aelbrecht, 2008, 2012; Dagnac, 2010).

In the following sub-sections, I provide an overview of the English VPE and the MCE. In V-VPE, the verb movement out of the verb phrase escapes the ellipsis, resulting in a distinct VPE pattern. Goldberg (2005) notes that Modern Hebrew, Modern Irish, and Swahili display V-VPE. It has also emerged that Dutch and French display a different ellipsis called *Modal Complement Ellipsis* (MCE), where a root modal licenses the ellipsis (Aelbrecht, 2008, 2012; Dagnac, 2010), among others. The following sub-sections provide an overview of the English VPE and the MCE.

2.6.1.1 English-Type Verb Phrase Ellipsis

Ellipsis is a phenomenon characterised by a lack of overt mapping between the structure and meaning of an utterance, resulting in meaning without a corresponding overt form. In English VP ellipsis, it is not only the verb that is elided but also its internal arguments as illustrated in (53), where the 'deleted' verb phrases in the second conjuncts are recoverable from the discourse due to the presence of an overt antecedent. The elided vPs in (53) have overt antecedents and can be elided. I use ~~strikeout~~ to indicate the ellipsis site.

(53) a. John likes candy, but Bill doesn't [VP ~~like candy~~].
b. Mr Honeypie has never eaten pumpkins, but his wife (surely) has [VP ~~never eaten pumpkins~~].

(Aelbrecht and Harwood, 2019, 504)

Further examples from Aelbrecht and Harwood (2015) note that in English VP ellipsis, finite auxiliaries cannot be omitted, but lexical verbs, whether finite or non-finite, must be deleted, as illustrated in (54-a) and (54-b). A dummy auxiliary *do* is inserted in the structure whenever there is no finite auxiliary to license ellipsis (54-d). Thus, the ill-formedness of (54-c). arises because the ellipsis is incorrectly licensed by the lexical verb 'put' instead of the finite auxiliary verb.

(54) a. An elephant can't fly, but maybe a rhino *(could) [$_{VP}$ fly].
b. I thought the auxiliary hadn't disappeared, but it *(had) [$_{VP}$ ~~disappeared~~].
c. *The chicken didn't put the tuna on the table, but the penguin put [$_{DP}$ ~~the tuna on the table~~].
d. The chicken didn't put the tuna on the table, but the penguin did [$_{VP}$ ~~put the tuna on the table~~].

(Aelbrecht and Harwood, 2015, 63).

According to Merchant (2001), elliptical constructions like those in (53) and (54) must fulfil two conditions: licensing and identification. Licensing refers to the locality constraints on syntactic heads that determine the structures that can be omitted under ellipsis, while syntactic and semantic identification refers to recovering missing information expected to be present in the overt syntax(Merchant, 2019). In (54-b) for instance, the licensing condition is satisfied by the presence of the finite auxiliary verb *had* in the second conjunct that deletes its complement VP. Syntactic and semantic identification, on the other hand, describes the process of recovering the elided information from the antecedent clause. For instance, in (55), the elided verb phrase *bring her farm produce* can be recovered from the parallel overt antecedent in the first conjunct, and it is deleted to reduce what Rooth (1992a) refers to as semantic redundancy. According to Rooth (1992a), the mutual entailment can also be met through inference from the discourse context.

(55) The pineapple farmer$_i$ should bring her$_i$ farm produce and poultry farmer$_j$ should [~~bring her $_j$ farm produce~~] too.

(Lobeck, 1995; Johnson, 2001; Aelbrecht and Harwood, 2019). Merchant (2001) point out that the syntactic identity process involves isomorphism between the ellipsis site and the antecedent structure in the first conjunct such that mutual entailment exists between the antecedent and the ellipsis site.

2.6.1.2 Modal Complement Ellipsis (MCE)

In addition to the well-documented verb phrase ellipsis (VPE) observed in English, another type of predicate ellipsis phenomenon exists in various languages. This alternative form of ellipsis occurs when the infinitival complement of a modal verb is omitted. This type of predicate ellipsis has been referred to as *Modal Complement Ellipsis* (MCE) in the literature. MCE has been attested across various languages, including French (Dagnac, 2010; Authier, 2011) and also in Libyan Arabic (Algryani, 2012). Aelbrecht (2008, 2010, 2012) has also documented MCE in Dutch with McShane (2000) attesting to MCE in Russian, Polish and Czech. Unlike the prevalent English-type VPE, where the entire verb phrase is elided, MCE involves the selective deletion of the infinitival complement under the scope of a modal verb.

Aelbrecht (2012) has noted that Dutch MCE is more restricted than English VPE since it is limited to the deletion of the infinitival complement of root modals and not allowed with epistemic modals or temporal auxiliaries. This is illustrated in (56-a), where the root modal *moeten* 'must' licenses ellipsis. He points out that (56-b) is ungrammatical because *moet* 'must' elicits an epistemic interpretation and can, therefore, not elide the structure in brackets. Aelbrecht (2008, 2012) proposes that ellipsis is licensed through an Agree relation between a Licensing Phrase (LP), the root modal.

(56) a. Jessica wil niet gaan werken morgen, maar ze moet [~~gaan~~
Jessica wants not go work tomorrow but she must go
~~werken~~ ~~morgen~~].
work tomorrow
'Jessica doesn't want to go to work tomorrow, but she has to.'

b. *?Jonas zegt dat hij niet de hele taart heeft opgegeten, maar hij
Jonas says that he not the whole pie has up.eaten but he
moet wel, [de hele taart hebben opgegeten], want er is geen
must PRT the whole pie have up.eaten for there is no
taart meer over.
pie more left
'Jonas says he didn't eat the whole pie, but he must have, for there is no pie left.'

(Aelbrecht, 2012, 3)

Apart from Dutch, similar MCE examples have been found in other languages. Dagnac (2010); Authier (2011) have also demonstrated the prevalence of MCE in French, Spanish, and Italian. They show that root modals license deleting their complements under the identity condition in these languages. For instance, Dagnac (2010) shows in 57 that the French modal *peut* 'can' can license MCE by allowing the ellipsis site to be contained within its own antecedent.(Dagnac, 2010, 159).

(57) Lea lit tous les livres$_k$ qu'elle peut [~~lire~~ ~~t~~$_k$].
Lea reads all the books that'she can read
'Lea reads all the books that she can.'

Furthermore, Dagnac (2010) has observed that Modal Complement Ellipsis (MCE) in French exhibits additional syntactic properties not typically associated with ellipsis in English. For instance, in 58, the verb *embrasser* 'to kiss' is deleted after the modal *peut* 'can'. Crucially, the elided verb phrase allows for the extraction of the wh-word *qui* 'who' out of the ellipsis site in the free relative clause construction (see also Authier, 2011).

(58) Il embrasse qui$_i$ il peut [~~embrasser~~ ~~t~~$_i$].
he kisses who he can kiss
'He kissed whoever he can.'

(Dagnac, 2010, 159)

Despite the intriguing parallels between MCE and VPE, there are also notable differences between the two phenomena, particularly when comparing English and Dutch. According to Aelbrecht (2008), one key distinction is that while English VPE can occur with a wide range of verbs and auxiliaries, Dutch MCE is more restricted to root modal verbs.

Furthermore, Aelbrecht (2008, 2012) has observed that Dutch MCE does not exhibit the same object extractions as English VPE. For instance, extraction of wh-extraction of objects in Dutch MCE is blocked as (59-a) shows but is permissible in English (59-b). Aelbrecht (2008) argues it is, however, possible for subjects to move out of the ellipsis site, suggesting that Dutch MCE supports a movement plus PF deletion account. As

(59) a. *Ik weet niet wie Katrien moet uitnodigen, maar ik weet wie ze
I know not who Katrien must invite but I know who she
niet moet.
not must
Intended: 'I don't know who Katrien should invite, but I know who she shouldn't.'

b. I don't know who Mina should invite, but I know who she shouldn't $[_{vP}$ ~~invite t $_{who}$~~ $]$.

(Aelbrecht, 2008, 11)

These restrictions on object extraction indicate that the syntactic structure and licensing conditions that govern Dutch MCE differ significantly from those of English VPE, which is more permissive. Conclusively, VPE and MCE cross-linguistically provide valuable insights into the complex interplay of similarities and differences in the licensing conditions and structural characteristics of ellipsis phenomena.

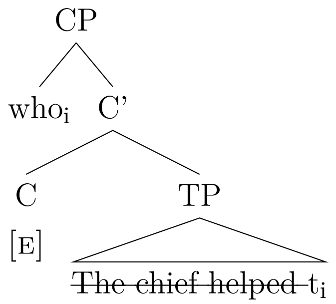
2.6.2 Sluicing

Sluicing refers to an ellipsis phenomenon such as (60), in which an entire clause is deleted, and a wh-phrase survives. The sluice in (60-a) has been argued to express the same meaning as its non-elliptical counterpart in (60-b)(Ross, 1969).

(60) a. The chief helped someone, but I don't know who.

b. The chief helped someone, but I don't know **who**$_i$ [~~the chief helped~~ t_i].

Sluicing has been studied extensively in languages like English, German, Hungarian, Dutch, Arabic, Japanese (Merchant, 2001; Algryani, 2012; Shlonsky, 2022; Aelbrecht, 2010), among others. Research on sluicing in African languages has been undertaken in Gungbe, Nupe and Bassa (cf. Bassong, 2014, 2019; Lipták and Aboh, 2013; Mendes and Kandybowicz, 2023). Despite the plethora of research on sluicing, it still stands to note that the form and derivation depend on every language's internal property. Merchant (2001), following the earlier works of Ross (1969), contends that sluicing is the predictable outcome of two independent processes in a single derivation: wh-movement plus the PF deletion of the TP. He argues that sluicing constructions such as (60-a) are licensed by an E-feature responsible for non-pronunciation of the TP, as represented in (61).

(61)


(adapted from Merchant, 2001, 664)

Merchant (2001) contends that the E-feature is merged on a null complementiser (C) head with interpretable *wh* and question features [+wh, +Q], taking the elided TP as its complement. Thus, the standard assumption in deriving (60-a), under the PF-deletion approach is that the wh-phrase *who* base generated as the internal argument of the verb undergoes A-bar movement within the TP '*the chief helped* t$_i$', to Spec-CP in the left periphery, then followed by the non-pronunciation of the phonological content of the rest of TP, due to a deletion rule, as explained in (40). One of the compelling pieces of evidence, according to Merchant (2001, 2002) for adopting a Movement plus PF deletion analysis of ellipsis, rather than LF-copying or non-structural approaches, comes from the parallel distribution to prepositions in elliptical and non-elliptical constructions. Merchant argues that a language will permit preposition stranding under sluicing if and only if it allows preposition stranding under regular wh-movement. Consequently, in languages like English (62-a) and Frisian (62-c), where wh-phrases can strand a preposition during regular wh-movement, as shown in (62-b) and (62-d), it is also possible to elide the preposition corresponding to the correlate of the wh-phrase in the antecedent clause.

(62)
a. Peter was talking with someone, but I don't know ~~with~~ who.
b. Who was he talking with?
c. Piet hat mei ien prutsen, ma ik wyt net (mei) wa.
   Piet has with someone spoken but I know not with who
d. Wa hat Piet mei prutsen?
   who has piet with spoken

(Merchant, 2002, 291)

On the contrary, it is predicted that languages such as German, Dutch, and Yiddish, in which prepositions are disallowed from being stranded under wh-movement, will obligatorily require the preposition to be retained under sluicing. This prediction is born out, as shown in German (63-a) and Yiddish (63-c) Merchant (2001, 2002) opines that the symmetric relationship in the distribution of the prepositions suggests that the same syntactic constraints regulate them.

(63) a. Anna hat mit jemandem gesprochen, aber ich weiß nicht, *(mit) wem.
Anna has with someone spoken but I know not *(with) who

b. *Wem hat sie mit gesprochen?

c. Zi hot mit emetsn geredt, ober ikh veys nit *(mit) vemen.
she has with someone spoken but I know not *(with) who

d. *Vemen hot zi mit geredt?

Finally, the ‘interpretative approach’ contends that in ellipsis and sluicing, the wh-phrase ‘what’ in the context of (63-a) and Yiddish is base-generated (Ginzburg, 1992; Riemsdijk, 1978; Ginzburg and Sag, 2000). They argue that the transitive predicate *know* Categorially selects the wh-sluice *what*; thus, it is not derived via movement as claimed by proponents of the ‘movement plus deletion’ approach. According to the ‘interpretative analysis’, the wh-phrase *what* and *who* in (62-a) and (62-d), respectively, appear as internal arguments of the verbs ‘know’ and ‘help’ due to an interpretative rule requiring the verbs to C-select an internal argument. In this syntactic approach, the sentence in (62-d) repeated in (64) lacks an underlying structure because the wh-phrase occurs in a canonical position directly selected by the transitive predicate. Thus, rather than proposing that the wh-fragment ‘what’ moves to the left periphery via wh-movement and subsequent TP-deletion, proponents of this viewpoint assert that the wh-phrase emerges as an internal argument (see also: Chao, 1987; Wasow, 1972; Culicover and Jackendoff, 2005).

(64) Peter was talking with someone, but I don’t know $[_{\mathrm{DP}}$ **who**.]

This study adopts the PF-deletion approach to sluicing. As outlined above, proponents of this movement plus deletion analysis, like Merchant (2001, 2004), Ross (1969),Van Craenenbroeck and Lipták (2013), argue that a wh-fragment arrives at the left periphery derivationally via wh-movement, followed by PF-deletion of the TP. They provide evidence against base-generation accounts of sluicing. The following subsection will examine some of the evidence used by proponents of this approach to support a movement plus deletion account for ellipsis and sluicing constructions.

2.7 Cartography and the Split-CP System

Since Chomsky (1986)'s influential work in *Barriers*, where he proposed analysing the Complementiser Phrase (CP) as having an articulated structure with specifier positions like Spec-CP for hosting moved wh-phrases, there have been several studies further mapping out the detailed syntactic structures of natural languages. Rizzi (1997, 2013b), Cinque (1999) and Cinque and Rizzi (2010), among others, explain Cartography as a generative syntactic framework that systematically maps out the functional hierarchies and articulated structures present in syntactic representations. Building on Chomsky's CP analysis, Rizzi (1997)'s *The Fine Structure of the Left Periphery* deconstructs the CP area into fine-grained functional projections and hierarchical structures. He contends that these functional projections cross-linguistically occur in a fixed order, giving rise to complex maps of the left periphery. Analogous to other influential proposals decomposing linguistic units into distinct, more fine-grained functional hierarchies - such as Abney (1987)'s argument for a Determiner Phrase (DP) layer integrating the DP, NP and other nominal modifiers into an articulated hierarchy, Pollock (1989)'s separation of the inflectional domain into Tense, Aspect, Agreement projections, and Cinque (1999)'s mapping of the internal structure and order of adjectival modification inside the DP.

Rizzi (1997) posits that the CP domain can be split into an ordered hierarchy of functional projections and distinguishes between external and internal Merge operations in syntax, noting that External Merge is responsible for building basic argument structures and assigning theta roles to satisfy semantic selectional requirements. In contrast, internal Merge (A-bar movement) creates A-bar chains that establish the proper scope domains, creating functional projections relevant to discourse semantics like topic, focus, etc. This distinction can be illustrated through the examples in (65). In (65-a), the basic argument structure is formed via external Merge - the noun phrase 'John's book' is externally merged as the internal argument of the verb 'like'. However, the examples in (65-b) and (65-c) involve an internal Merge, A-bar movement, of 'John's book' to the left periphery:

(65) a. I very much like [John's book].
b. [John's book]$_i$, I very much like t_i.
c. It is [John's book]$_i$ that I very much like t_i (not Peter's book).

(Rizzi, 2013a, 170).

In (65-b), the object, 'John's book', is internally re-merged in a left-peripheral topic position while, in (65-c), is it is moved to the specifier of the focus projection where the focus features are checked and the required focus interpretation is expressed. Rizzi (1997, 2013a) claims that these movements to the left periphery satisfy argument structure requirements and establish the proper scope domains and discourse-semantic properties associated with operators like topics and foci. The landing sites are specified functional projections in the articulated C-domain. Consequently, Rizzi (1997, 257) and Rizzi (2001) divided the left periphery into a hierarchy of functional heads with discourse semantic features for expressing old (topic), interrogative or new or contrastive (focus) information. The split-CP system is illustrated in (66).

(66) $[_{\text{ForceP}}$ Force $[_{\text{TopP}}$ Top $[_{\text{InterP}}$ Inter $[_{\text{FocP}}$ Foc $[_{\text{*TopP}}$ Top $[_{\text{FinP}}$ Fin $[_{\text{IP}}$ I $]]]]]]]$

Let us provide a brief explanation of each of the functional projections that make up the Split-CP system:

a. **Force Phrase (ForceP**: The ForceP represents the highest projection within a clause, determining whether a sentence is declarative, interrogative, or relative. He asserts that relative operators and preposed exclamative wh-expressions occupy the specifier position of ForceP. Therefore, ForceP is responsible for indicating the clause type in the sense of Cheng (1991) or what Chomsky (1995) refers to as the specification of Force.

b. **Topic Phrase (TopP)**: This refers to a fronted element above TP expressing old, previously-mentioned information in the discourse. The topic is structurally set off from the rest of the clause by 'comma intonation' and splits into a Topic-Comment structure. Rizzi (1997) argues that while the *comment* is the complex predicate introducing new information, the *topic* is the old information moved to the Spec-TopP position. In languages like English and Italian (67-a), it is phonetically null, while in Likpakpaanl and related languages, the Top head is overtly realised as an overt particle **mà** (67-b). He points out that while the topic can be recursive in the causal spine, there is only functional projection for focus.

(67) a. Il tuo libro, lo ho letto.
your book, 1.SG have read
'Your book, I have read it.'

(Rizzi, 1997, 384).

b. Lì-nùù-l gbààn$_i$ **mà**, Neina nàn ŋmɔn t$_i$.
7.SG-yam-7 DEF TOP N. PST eat
'As for the yam, Neina ate (it).'

Rizzi argues that both overt and null Top heads in (67-a) and (67-b) project a Topic Phrase (TopP) as illustrated in the X-bar structure schema in 68. Following, it means that in(67-a), 'the book' and 'the yam' in (67-b) each occupy Spec-ToP, shown as XP in (68).

(68)

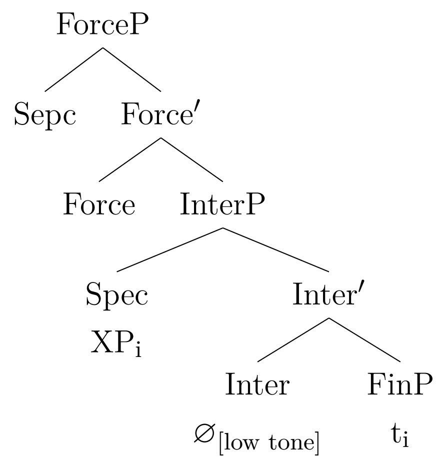
TopP
- XP
  Topic
- Top′
  - Top
    Head
  - YP
    Comment

(Rizzi, 1997, 386)

c. **Interrogative Phrase (InterP)**: Building on previous work by Rizzi (1997), proposing the projection of an interrogative phrase (InterP) in the left periphery, Aboh (2004) shows that the question marker (represented by a floating tone) in Gungbe encodes interrogative force, associated with the head to called InterP. He also notes that the complementiser and the question particle in Gungbe are not in complementary distribution, as shown in the example where the complementiser ɖɔ̃́ 'that', and the interrogative particle can co-occur in the same clause (69-a). He contends that the two elements target different positions in the causal spine. Thus, interrogative constructions such as (69-a) are derived via the leftward movement of the finite phrase to the spec-InterP. As a result, the Gungbe interrogative marker must surface in the sentence-final position, as illustrated in (69-b).

(69) a. Ùn kanbíɔ̃́ ɖɔ̃́ Kòfí ɖù nu͡u.
l.SG ask-PFV that K. eat.PFV thing-INTER
'I asked whether Kofi ate.'

b.



(Aboh, 2004, 319-320)

Rizzi (2001) contends that the interrogative complementiser **se** 'if', which occurs in Italian embedded questions, occupies a position lower and distinct from the position of the declarative complementiser **che** 'that', because while elements that have been topicalised can precede or follow **se**, **che**, on the other hand, can only be followed by a topic. He postulates that **che** is the head of Force, and **se** occupies a distinct Int(errogative) position.

(70) a. Mi domando se questi problemi, potremo mai affrontarli.
'I wonder if these problems, we will ever be able to address them.'
b. Mi domando, questi problemi, se potremo mai affrontarli.
'I wonder, these problems if we will ever be able to address them.'
(Rizzi, 2001, 289)

Furthermore, Aboh and Pfau (2011) show InterP as a distinct functional Projection (FP) pointing out that the projection of InterP in the left periphery offers a unified structure for analysing polar and constituent questions in various languages. They emphasise that the question marker in Gungbe, which occurs as a floating low tone on the verb (**wā**) in (71-a), is the morphological spell-out of the Inter-head, which triggers a Generalised pied-piping of the questioned proposition (FinP) into its spec-InterP. Following Rizzi (2001), Aboh (2004) and Aboh and Pfau (2011) argue that the question marker in Gungbe, Fongbe, has scope over FinP and draws it to Spec-InterP to check the interrogative feature on Inter^0, as illustrated in (71-b).

(71) a. Sέt'ɔ kò wâ?
Seto already come.INTER
'Has Seto arrived yet?'

b. [$_{\text{ForceP}}$ Force$^{\text{o}}$ [... [$_{\text{InterP}}$ [Sέt'ɔ kò]$_{\text{i}}$ [$_{\text{Inter}}$$^{\text{o}}$ wá [$_{\text{FinP}}$ t$_{\text{i}}$]]]]]

(Aboh and Pfau, 2011, 96)

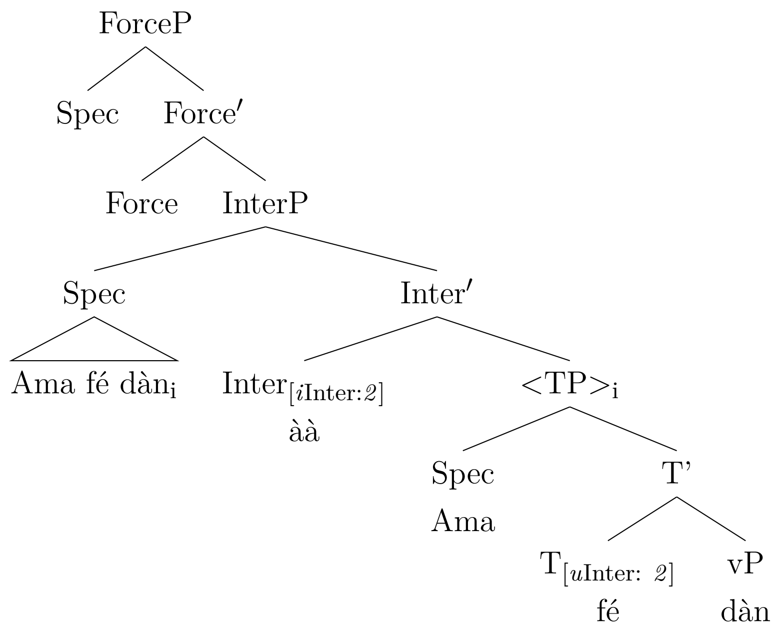
In the Likpakpaanl, an interrogative particle **àà** marking polar questions exists. A declarative statement, such as the one shown in (72-a), can be turned into a polar question like (72-b) when the clause-final interrogative particle is merged into the structure.

(72) a. Ama fé dàn.
A. HEST.PST come
'Ama came (yesterday).'

b. Ama fé dàn àà?
A. HEST.PST come INTER
'Did Ama come (yesterday?'

Adopting the proposals of Rizzi (1997, 2001); Aboh (2004b); Aboh and Pfau (2011), I assume that the interrogative phrase in (72-b) has an interpretable interrogative feature which is unvalued [*i*Inter:_]. In contrast, the questioned proposition has an uninterpretable but valued interrogative feature [*u*Inter: 2]. The unvalued feature on Inter$^{\text{o}}$ probes its c-command domain for a matching valued feature. After valuation, the entire TP proposition is pied-pipped to Spec-InterP, realising the surface order with the question particle occurring clause-finally as (73) illustrates.

(73)



d. **Focus Phrase (FocP):** Rizzi (1997) also assumes a focus projection below the TopP and InterP in the left periphery. He points out that the FocP attracts wh-operators and other focused elements to its specifier based on their focus features. Consequently, the focus head attracts focused elements to its specifier and the background presupposition as its complement.

(74)

```
            FocP
          /      \
        ZP        Foc'
      Focus     /      \
              Foc       WP
                   Presupposition
```

(Rizzi, 1997, 287)

According to Rizzi (1997), the focus-presupposition interpretation in Italian is expressed by assigning focal stress (in this case, in-situ in a low focus position) (cf. Belletti, 2004) as in (75). (75) expresses a contrastive focus interpretation that suggests a set of books belonging to different people, and the speaker picks out the book belonging to the addressee as the one they read, as opposed to the alternative (the book belonging to the other person mentioned).

(75) Ho letto IL TUO LIBRO (, non il suo).
'I read YOUR BOOK, not his.' (Rizzi, 1997, 287)

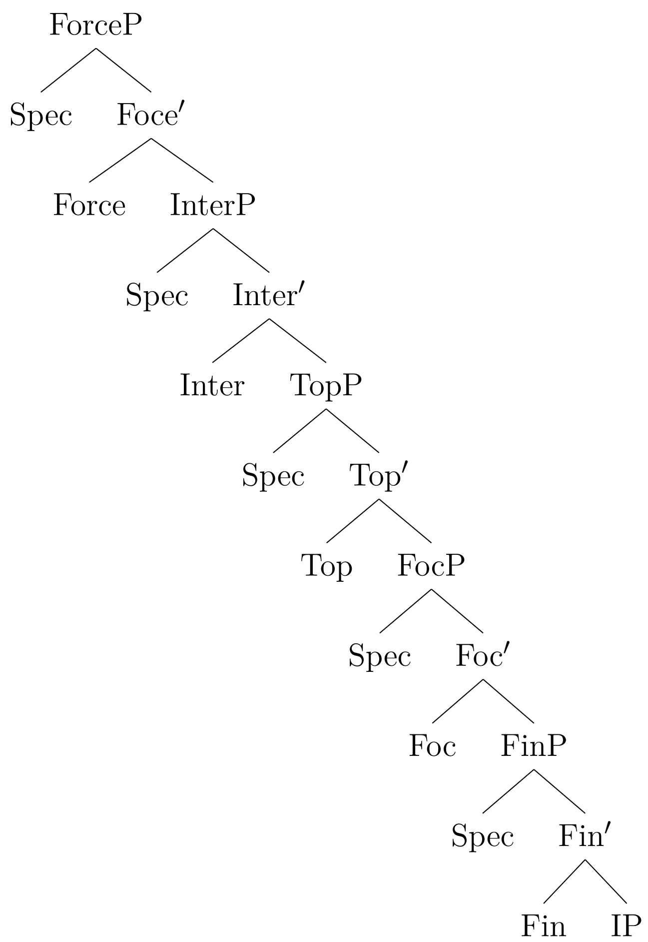
Even though Top^0 and Foc^0 are phonetically null in Italian, English, and related languages, they are morphologically represented by independent particles in Niger-Congo languages like Gungbe, Likpakpaanl, Dagabni, and Akan, among others. For instance, in Likpakpaanl, the particle **lé**[4], the head of Focus projection must be present when a focused element is moved to the clause-initial position as in (76-b).

(76) a. Neina nàn ŋmɔn lì-nùù-l gbààn.
N. PST eat 7.SG-yam-7 DEF
' Neina ate the yam.'

[4] The focus particle is optional in wh-focus movement in Likpakpaanl as we shall see in Section 3.9

b. Lì-nùù-l gbààn$_j$ *(lé) Neina nàn ŋmɔn t$_j$.
7.SG-yam-7 DEF FOC, N. PST eat
Literal: ‘The yam, Neina ate (it).’
Meaning: ‘It is the the yam that Neina ate.’

e. **Finite Phrase (FinP)**: In the hierarchy of projections of Rizzi (1997)’s SPlit-CP system, is one of the functional projections proposed within the expanded left periphery of the clause structure. The Fin-head is assumed to host the finite Inflection (Infl) or the Tense or Agreement features of the clause. FinP is located below higher functional projections like ForceP and connects the Complementiser and Inflectional phrase layers. Thus, in languages like English, where the finite verb raises from v to I, it is further assumed that it moves from I to Fin^0. The cartography of the Left Periphery of the Split-CP system from Rizzi (1997, 2001); Aboh (2004) is illustrated in (77).

(77)


From the discussion, we see that the Split-CP system provides a detailed structural representation of left peripheral projections and shows the interactions between elements associated with clause typing, information structure, and finiteness (Rizzi, 1997, 2001; Aboh, 2004). It has also significantly expanded our understanding of the left periphery and paved the way for subsequent research that has further refined and extended this model. For instance, Benincà (2001) assumes that Italian shows evidence of interrogative, relative, and exclamative pronoun phrases in the CP domain, which occupy the specifier positions of different functional projections.

2.8 Chapter Summary

This chapter discussed the literature on ellipsis theory, specifically focusing on the different approaches and the nature of Merchant (2001)'s ellipsis licensing feature. I began by discussing the notion of ellipsis and the main analytical approaches to ellipsis: structural, non-structural, and hybrid approaches. Proponents of the structuralist approach led by Merchant (2001); Johnson (2001); Aelbrecht (2010); Merchant and Simpson (2012) etc. assert that the ellipsis site possesses the same underlying syntactic configuration as their non-elliptical counterparts except that the phonological content of the former is unpronounced. The fundamental position of structuralists is that the silent structure is either null or deleted at PF. This variation depends on whether one adopts the Logical Form (LF) Copying theory or the Phonological Form (PF) Deletion. Thus, while the PF-Deletion theory assumes that elliptical constructions are derived via A-bar movement followed by deleting the PF-content of the remnant constituent, the LF-Copying theory rejects the movement process but proposes that an empty TP category containing a copy of the element that is base-generated in its base position (Chao, 1987; Lobeck, 1995). and how they cross-linguistically license different ellipses.

The chapter further discussed the various properties of the E-feature from a cross-linguistic perspective and showed that the E-feature can be encoded on different functional heads within the clausal structure. Thus, while Merchant (2001) proposes that in English ellipsis, the semantic, syntactic and phonological features are merged on the complementiser head and occur in a head-complement relationship with the ellipsis site, Aelbrecht (2010), on the other hand, proposes a split between the syntactic, semantic and phonological features values of E-feature. According to her, licensing and PF-deletion functions of the E-feature are distinct and do not rely on a head-complement relationship as proposed by Merchant (2001). She contends these features are spread between the ellipsis licensing head that projects a Licensing Phrase (LP) and a lexical head bearing the E-feature.

Further data from Stigliano (2022) also showed that the licensing of ellipsis in Spanish wh-Topic-Remnant Elided Questions (wh-TREQs) is governed by two distinct E-features: one that enforces syntactic identity, called the [E^{syn}], located on the C^0, and the other that does not impose any specific identity condition, merged on Top^0 [E]. According to Stigliano (2022), in Spanish, the E-feature on Top licenses CP-ellipsis, while TP ellipsis is licensed by E^{syn}. The latter requires strict syntactic identity between the elided constituent and the antecedent, but such an isomorphic condition is not required under CP-ellipsis. Finally, this chapter examines Rizzi (1997)'s proposal for a more intricate structure for the left periphery of the clause. Rizzi (1997), Cinque (1999) and **?**, among others, assert that distinct functional projections serve different informational and discourse functions like expressing topic, focus, and interrogative force. The Split-CP system provides a framework for capturing the derivation of ellipsis phenomena and licensing conditions across languages.

Chapter 3

Aspects of Likpakpaanl Grammar

3.1 Introduction

This Chapter offers a comprehensive grammatical description of Likpakpaanl that goes beyond standard preparatory discussions. While it includes the essential grammatical information necessary for the chapters on ellipsis, it also provides deeper insights into the fundamental aspects of Likpakpaanl grammar. Although the primary focus of the thesis is on ellipsis, this chapter is crucial for two reasons: first, it establishes a solid foundation for understanding the elliptical derivations in Likpakpaanl; second, it contributes to a broader understanding of the syntax of the language, which has received very little attention in terms of research. Section 3.2 gives a background to the language and its speakers, while Section 3.3 outlines the noun class system of Likpakpaanl. A detailed examination of the morphology of the noun class system is provided in section 3.4, while section 3.5 investigates agreement patterns in the DP, demonstrating that Likpakpaanl nominal agreement is a morpho-syntactic phenomenon exhibiting morphological dependencies between noun class prefixes and other nominal modifiers through feature checking. Section **??** establishes that Likpakpaanl has a Subject Verb Object (SVO) constituent order, and section 3.6 discusses the language's clause structure by analysing tense and aspect, evidencing that tense and aspect are morphologically encoded by lexicalised pre-verbal particles marking imperfective and habitual aspects, while independent particles denote present and future tenses. Section **??** explores the syntactic hierarchy of temporal projections in the clause structure by examining the interactions between tense and aspect. Section 3.7 adopts a Minimalist framework to analyse the syntax of tense and aspect while section 3.8 investigates Likpakpaanl A-bar constructions and associated constraints.

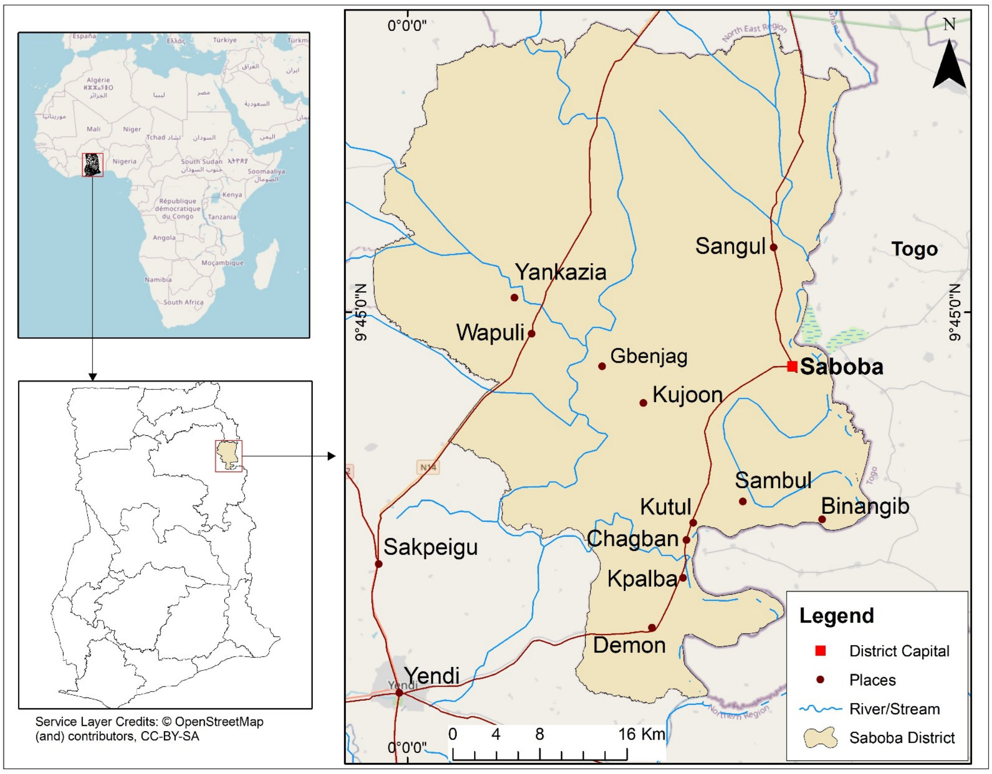
3.2 Background to Likpakpaanl and its speakers

Likpakpaanl is a Mabia (Gur) language (code ISO 639-3: xon) and is genetically classified as belonging to the Gurma subgroup within the Oti-Volta branch of the

North Central Gur languages (Manessy, 1971). Mabia is composed of two lexical items: *ma* which means 'mother', and *bia* (bie or biiga), meaning 'child', is a terminology used to explain the genetic relationship of languages of the Western Oti-Volta (Bodomo, 1994, 1997; Bodomo et al., 2020). The Mabia languages, according to Bodomo et al. (2020), comprise approximately 80 different but mutually intelligible languages spoken in six West African countries: Ghana, Togo, Burkina Faso, Ivory Coast, Mali, and Benin (Mous, 2003). Likpakpaanl, Basaare (Ncham) and Moba, belonging to the Mabia East variety, are collectively classified as Gurma speakers, and these languages have a close linguistic affinity (Acheampong, 2015; Bodomo, 1997). The speakers of Likpakpaanl refer to themselves as *Bikpakpaam* instead of the anglicised name 'Konkomba' that has often been used to refer to both the people and their language. Thus, the language is Likpakpaanl, and the people are *Bikpakpaam*. The homeland of *Bikpakpaam* is known as *Kikpakpaaŋ*. The central geographical territories of the *Bikpakpaam* in Ghana lie within what was formerly referred to as the 'Voltaic' region by French scholars (Tait, 1958). While Bodomo et al. (2020) estimates the Konkomba population to be over 600,000, the 2010 census data indicates that *Bikpakpaam* in Ghana numbered 823,000, and applying the intercensal growth rate places the population at more than one million now. The World Factbook reports in Demographics of Ghana that *Bikpakpaam* are the 8th largest Ethnic group in Ghana, representing 3.5% of the Total population of Ghana (Central Intelligence Agency, 2022). Bikpakpaam are also found in the Republic of Togo, and Schwarz (2009) puts their population at about 50,100. This number has increased following the 2011 census in Togo since data indicates that the total population of *Bikpakpaam* in Dankpen district alone stands at 122,209 (Njindan, 2014). Likpakpaanl is spoken in the eastern part of the northern region of Ghana, specifically in Yendi, Chereponi, Gusheigu, Wulensi, and Bimbila. However, speakers of Likpakpaanl have their major communities in Chaabob (Saboba)[1] and its surrounding villages, such as Waapul (Wapuli), Sambul (Sambuli), Sangul (Sanguli), Kpalb (Kpalba), Kujoon (Kujooni), and Kucha. See Figure 3.1.

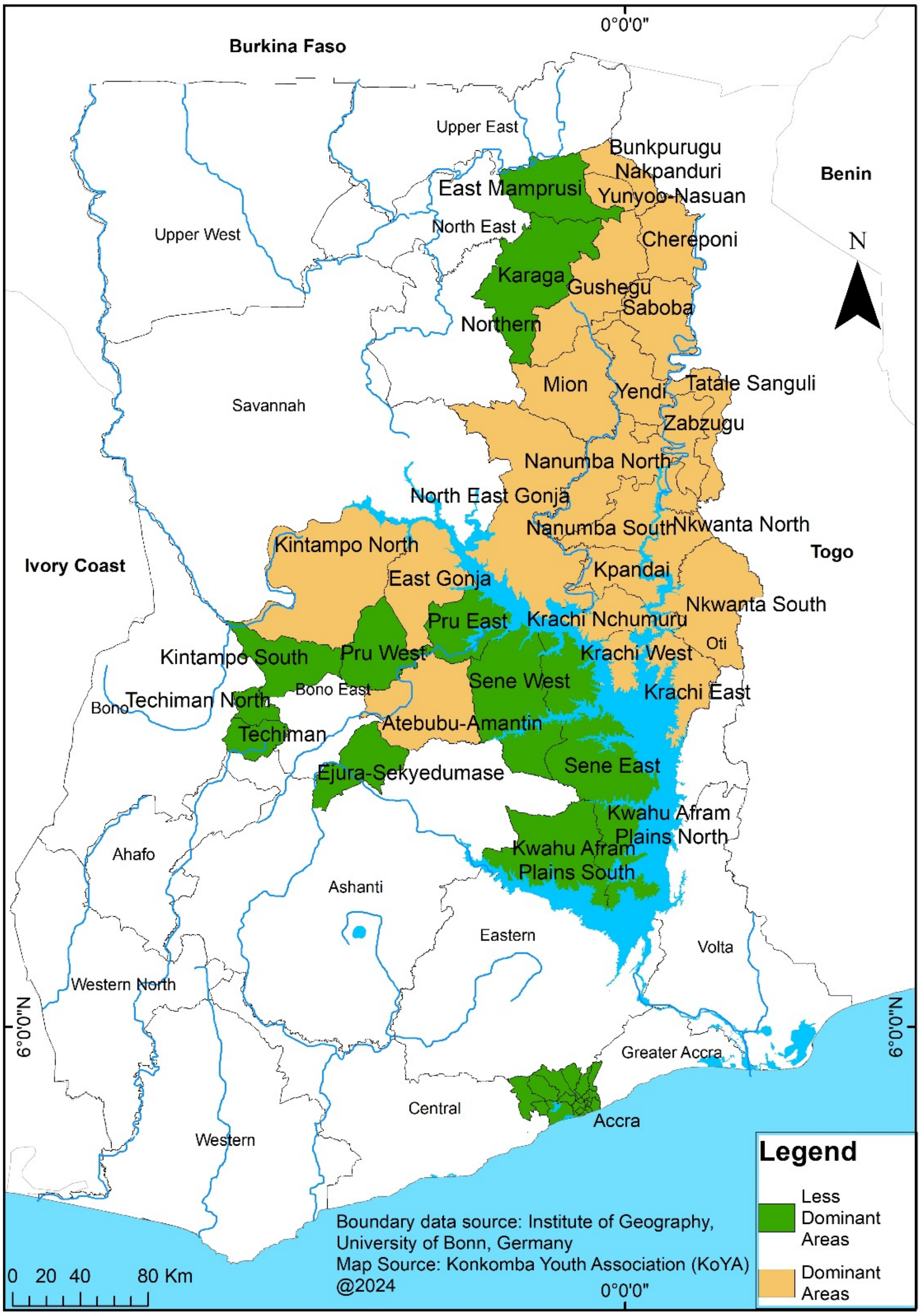
[1]The names in brackets refer to the anglicised versions of these towns

Figure 3.1 The Map of Study Area (Saboba District)



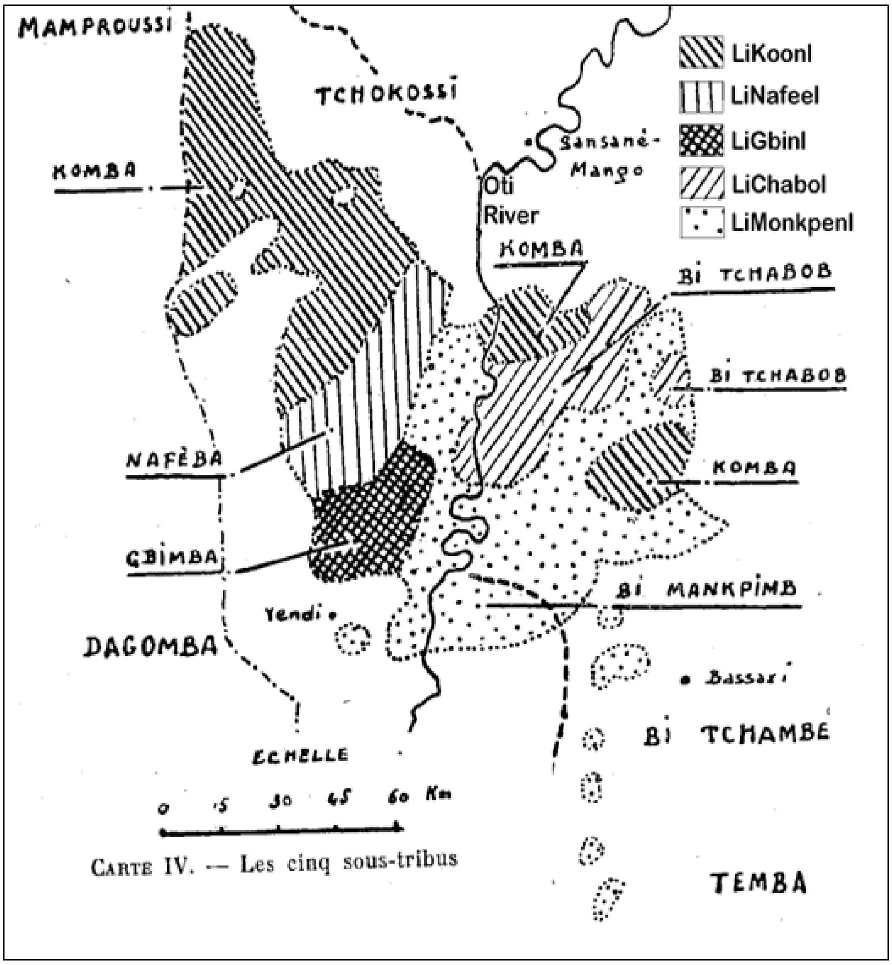
The language is also spoken in and around Tatale and in Chamba, within the Northern Region. Several speakers can also be found in Ghana's Oti, Bono, and Ahafo regions, as illustrated in Figure 3.2.

Figure 3.2 Towns Likpakpaanl is spoken in Ghana



Likpakpaanl has five major dialects with several sub-dialects within each dialectal group. Froelich (1954),Manessy (1971), and Hasselbring (2006) identify five Likpakpaanl dialects: Lichabol, Ligbeln, Likoonl, Limonkpeln and Linafeel, as shown in Figure 3.3. Lichabol is spoken mainly in Saboba, Ligbeln is found mainly in Kujooni, while Likoonli is spoken around Chereponi and Gushegu. Linafeil, on the other hand, is spoken in Wapuli, and Limonkpeln dominates towns such as Kpalba, Sambuli, Kucha and some parts of Northern Volta.

Figure 3.3 Dialectal Map of Bikpakpaam



(Hasselbring, 2006, 25)

3.3 Likpakpaanl Nominal System

This section examines the noun class system of Likpakpaanl. It provides a summary of the earlier works of Tait and Strevens (1954), as well as that of Winkelmann (2012), and I use that as the basis for my discussion of the agreement patterns in Likpakpaanl. I also analyse the structure of the Determiner Phrase (DP) and argue, following Ritter (1993), that number is not only a functional projection within the noun phrase (NumP) but hosts both number and gender features. I use the term 'gender' following Guthrie (1948) to refer to a pair of noun classes that make up the number correlation between singular and plural and also show agreement patterns on other elements such as demonstrative and relative pronouns and 'which' interrogative pronouns (Corbett, 1991). Throughout this dissertation, I use the term 'noun classes' to refer to these 'gender' classes. I also show that Likpakpaanl nominal agreement entails a morpho-syntactic process in which the head noun triggers agreement on demonstratives, relative pronouns and 'which' interrogative pronouns and propose an analysis where gender is a feature that is base-generated on NumP (cf., Ritter, 1993).

Noun classes (NC) have been considered a typological feature of African languages, involving several families of languages. Heine (1982) noted that two-thirds of the African languages he surveyed have noun classes. Nurse and Philippson (2006) have also reported that two out of every three African languages have a noun class system. According to (Dixon, 1968, 109), a language is classified to have a noun class system if there is an agreement between a noun and other parts of speech, "such that specific nouns select one, and certain other nouns the other, independently of person and number considerations". Dixon (1968) notes that noun class markings may be denoted by a prefix to the noun, as is the case in Bantu languages, or maybe be marked by an article, as in French and German, or an inflectional suffix reflecting a portmanteau of case and noun class, as in Latin. According to Hurskainen (2011), Mabia languages possess a noun class system and concord system manifested in three types of noun class marking: prefixes, suffixes, and prefixes plus suffixes (circumfixes). Likpakpaanl noun classes are marked using all three types as stipulated by Hurskainen (2011) (see Bisilki and Akpanglo-Nartey, 2017; Tait and Strevens, 1954).

Dixon (1968) identifies two approaches to setting up nouns into classes cross-linguistically. In the first approach, groups of nouns are set up depending on whether they have singular or plural affixes such that nouns with similar ways of forming their singular forms in the same group may not necessarily occur together when considering the plural forms. In the second approach, groups of nouns are paired based on their singular and plural affixes, and therefore, nouns in the same noun class share similar singular and plural affixes. Tait and Strevens (1954), and later Winkelmann (2012), adopt a morphological approach in classifying the noun classes of Likpakpaanl where a noun class is categorised according to number affixes (i.e. singular and plural markings). Based on this criteria, Tait and Strevens, 1954 concluded that there are '...seven concord classes, nine prefix classes and 59 suffix classes' in Likpakpaanl (p.144). These seven concord classes were subdivided into prefix classes by singular and plural forms. Number correlations are, therefore, the basis for distinguishing genders and noun classes.

Winkelmann (2012), however, identifies fifteen (15)[2] different noun classes in the language based on a bilateral noun class marking system of affixation. She adopts the general noun class frame of Proto-Gurma. She derives what she terms as the 'Proto Konkomba' noun class by selecting only the corresponding noun

[2]Winkelmann (2012) gives a more detailed account of the different sub-classes and affix variants for the noun classes.

classes from the Proto-Gurma and retaining their agreement classes as shown in Table 3.1.

Table 3.1 Noun Class Affixes and Agreement Patterns (Winkelmann, 2012, 473)

NC	Prefix	Pronoun (SG)	Pronoun (PL)	Suffix	NC
1	u...-∅	ù	bi	bi-...b,-m	2
1a	∅ -∅	-	-	∅-...teeb	2a
3	n-...∅	mu	i	i- ∅	4
5	li-...-l	li	ŋì	ŋi-...∅	6
14	bu-...b	bu	i	i-...i	*ci
12,15	ki-...k	ki	mú	n-...m	22
22,23	ki-...∅	ki	tì	ti-...r,l,n	21

The illustration in Table 3.1 shows noun classes 1 to 23 skipping classes that are not found in Likpakpaanl, making it difficult to determine how many noun classes there are on the surface of the Table.Winkelmann (2012) does not capture the affix variant for borrowed words and kinship terms in class 2a of her analysis. There is a NC, which she labels as ***Ci***, and claims that it differs from other genders because it is used in deriving interrogative ‘which’ citing examples like *ilia* and *yilayi*. I will show later in the work that these are a spell-out of the agreement patterns between noun classes and the ‘which’ interrogative pronoun and not a special feature of class *Ci. She further argues that class **Ci* has an allomorph *-g* citing examples such as **bu-sub** (tree, sg) and the plural form **i-sug**. However, such examples do not reflect the current data set because all five dialects of Likpakpaanl use the plural form **i-sui** (tree, pl), but not **i-sug**. It is, thus, plausible that the loss of the NC suffix-**g** results from linguistic changes in the language. In the current analysis, I adopt an approach that abstracts from Winkelmann (2012) and re-categorise the noun classes based on the relationship between their agreement properties and morphological characteristics by treating the singular/plural pair of the same make as distinct classes. This still gives us the 15 classes but with a more sequential labelling system following the pattern proposed by Fiedler (2012) in her analysis of the noun classes of Nawdm, an Oti-Volta language spoken in Northern Togo. She notes that such an account provides an independent characterisation of noun classes in a language while also showing the corresponding anaphoric pronouns. Table 3.2 shows that Likpakpaanl has seven different agreement patterns and one neutral class, resulting in 15 noun classes. Thus, in Likpakpaanl, Classes 1,3,5,7,9,11,13 mark singular nouns, while classes 2,4,6,8,10,12, 14 take plural affixes. Noun class 15 is marked on a neutral pronoun, **nì**, and contains no specific noun.[3]

Table 3.2 Noun Classes

NC	Prefix	Pronoun (SG)	Pronoun (PL)	Suffix	NC
1	ù...-∅	**ù**	bì	**bì**-...b,-m	2
3	∅ -∅	-	-	∅-...tììb, -màm	4
5	n-...∅	**mù**	**ì**	ì-... - ∅	6
7	lì-...l	**lì**	**ŋì**	ŋì-...∅	8
9	bù-...b	**bù**	**ì**	ì-...ì	10
11	kì-...k	**kì**	**mú**	n-...m	12
13	kì-...∅	**kì**	**tì**	tì-...r,l,n	14
15		**nì**			

The existence of multiple variants within the same noun class, as exemplified in noun classes 3 and 4, is not a novel phenomenon. For instance, Fiedler (2012) demonstrated that the noun classes have distinct suffix variants that indicate singular or plural in Nawdm. She points out, for instance, that in noun class 12, the semantic class containing nouns expressing the idea of roundness, animals and plants has the suffix **-re** serves as the primary singular marker in the language; the other two allomorphs,**-de** and **-a** are phonologically conditioned. Fiedler (2012) contend that **-de** usually follows a stem-final alveolar nasal **n**, as well as after a stem-final flap **r** as illustrated in 78.(see Fiedler, 2012, 577).

(78) a. bín-dè
year-NC
'year'

b. híd-ré
name-NC
'name'

The NC pairs of affixes in Likpakpaanl mark ± singular morphology on the nouns such that every noun stem that takes the Class 1 singular prefix also takes the Class 2 prefix in the plural; every noun that takes the Class 9 singular prefix takes the Class 10 prefix in the plural and so on. Only nouns in Class 6 behave differently because they have a singular agreement with Class 1, but in their plurals, they agree with Class 9 nouns as in (79-c).

[3]The noun classes also map onto designated anaphoric pronouns (in **bold**) while the corresponding nominal affixes indicating both number and class are given in regular fonts.

(79) a. ù-ja
1.SG-man
'a man'

b. bì-jà-b gbààn
2.PL-man DEF
'the men'

c. ù-bɔ gbààn
1.SG-dog DEF
'the dog'

d. ì-bɔ gbààn
6.PL-dog DEF
'the dogs'

A further examination of the examples in (80-a) and (79-c) show that 'man' and 'dog' belong to the same singular noun class but differ in their plural class marking as (79-b) and (79-d) demonstrate. The data shows that the classification of nouns into the various classes in Likpakpaanl is not based on core semantic properties such as animacy, sex, and humanness but one based on morphological criteria (Aikhenvald, 2006a). The next subsection looks at the nature of transparency between noun classes and concludes that nouns in the language cannot be grouped based on any specific semantic criteria.

3.4 The Morphology of Likpakpaanl Noun Classes

The relationship between noun classes with lexical meanings in Likpakpaanl is not transparent because no single NC relates exclusively to human, animal, or body parts. Even though Winkelmann (2012) has argued that Class 1 nouns contain humans, this assertion is not entirely valid since some animals also fall into Class 1 but have different plural forms in Class 6. This is illustrated in Table 3.2 and exemplified in (80-b). The fact suggests that Likpakpaanl nouns do not strictly follow a semantic classification where specific nouns are labelled as belonging to a particular class (see Bisilki and Akpanglo-Nartey, 2017).

(80) a. Ì-gbèèr gbààn ŋmɔn ì-dì gbààn.
6.PL-pig DEF eat.PFV 6.PL-millet DEF
'The pigs have eaten the millet.'

b. Ù-gbèèr gbààn ŋmɔn ì-dì gbààn.
1.SG-pig DEF eat.PFV 6.PL-millet DEF
'The pig has eaten the millet.'

Moreover, we notice that nouns in Class 4 contain nouns used for referring to kinship terms, such as *brothers, mothers, fathers, and in-laws.* Many loan words are used in the language, and some animals also fall into this class. Class 4 has no singular prefixes but has the suffix **-tííb** or **-màm**, which Bisilki and Akpanglo-Nartey (2017) contend are used in free variation in the Lichaabol dialect. Contrary to this claim, evidence in sentences such as (81) shows that the suffix **-tííb** can be used when referring to relatives and, thus, **-tííb** and **-màm** can be used in free variation when referring to plural loaned words (81-a). The ungrammaticality of (81-c) indicates that the plural suffix **-tííb** selects nouns denoting kinship relations, while **-màm** is used in free variation.

(81) a. Chòò-tííb gbààn nàn kìr bɔrààdèè-tííb/màm gbààn
in-law-4.PL DEF PST pluck plantain-4.PL DEF
'The in-laws plucked the plantains.'

b. N té-kpèl-tííb nàn tér Sùlà.
1.SG father-old-4.PL PST help S.
'My uncles helped Sula.'

c. Nantob bà kàn nà-*màm/tííb gbààn dìn..
N. PST see mother-4.PL DEF today
'Nantob saw the mothers today.'

Noun classes 5 and 6 are one of the largest noun classes and contain different nouns denoting body parts such as **n-ŋàl**, 'hand', **n-nyùn**, 'water', as well as items 'fire', and 'hearth'. These nouns form plural using the class 6 as shown in (82). In class 6, we also find grains like 'sorghum' and 'rice'.

(82) a. Susan bà gàà n-làn.
S. PST sing 5.SG-song
'Susan sang a song.'

b. Susan bà gàà ì-làn.
S. PST sing 6.PL-song
'Susan sang songs.'

Class 6 also includes animals that morphologically belong to Class 1 in its singular form but form a concord agreement with NC Class 6 when it is plural (83-a). It shows a paradigm where a plural class correlates with different singular classes, and the two are analysed as two different agreement classes (83-b).

(83) a. Ù-bɔ gbààn gà ŋmɔ ì-jàn gbààn .
1.SG-dog DEF FUT eat 6.PL-fish DEF
'The dog will eat the fish.'

b. Ù-kɔlà ù-mìnà kpà tì-kùr jèr ì-nyòòn
1.SG-fowl 1-PROX.DEM have 14.PL-feather than 6.PL-bird
ì-mìnà.
6.PL-PROX.DEM
'This fowl has more feathers than these birds.'

Classes 7 and 8 do not comprise nouns denoting body parts as in (84-a) but also those referring to farm products, (84-b) or landscapes (84-c). This same class also contains fruits **lì-nyóónl** 'lemon', some animals, and seasons of the year.

(84) a. Lì-yìl gbààn ŋàn.
7.SG-head DEF good
'The head is good.'

b. Lì-nùù-l gbààn mɔ.
7.SG-yam-7 DEF sweet
'The yam is sweet.'

c. Ŋ̀ì-jóó gbààn fɔk.
8.PL-mountain DEF tall
'The mountains are high.'

The nouns belonging to NC 9 and 10 are few and contain inanimate objects like 'trees', 'soap', and 'canoe'. According to Winkelmann (2012) identifies the suffixes **-g** and **-ŋ** as part of the plural for this class of morphemes. However, the data shows that the singular NC suffix of class 9 is **bù-** (85-b) while the plural prefix is **i-su-ì** (85-a). The suffixes **-g** and **-ŋ**, which we find in Winkelmann (2012), do not form part of plural formation in the language and plural forms such as (85-c) are ungrammatical in all Likpakpaanl dialects.

(85) a. Bù-sù-b gbààn lò ŋì-sùbìl.
9.SG-tree-9 DEF bear 8.PL-tree.seed
'The tree bears fruits.'

b. Ì-sù-ì gbààn lò ŋì-sùbì-l.
6.PL-tree-6 DEF bear 8.PL-tree.seed
'The trees bear fruits.'

c. *Ì-sìì-g gbààn lò ŋì-sùbìl.
6.PL-tree-6 DEF bear 8.PL-tree.seed
Intended: 'The trees bear fruits.'

Noun classes 11 and 12 consist of smaller body parts, diminutives, sharp cutting objects, lands, and spiritual and abstract nouns, like witchcraft and jealousy. The singular prefix circumfix denoting this NC is **kì-....-k**, and the plural is marked by the circumfix **n-...-m**. The data in (86) shows some nouns in Classes 11 and 12.

(86) a. Kì-jùù-k gbààn gà wɔŋ.
11.SG-knife-11 DEF FUT lost
'The knife will be lost.'

b. N-jùù-m gbààn gà wɔŋ.
12.PL-knife-12 DEF FUT lost
'The knives will be lost.'

Likpakpaanl noun class prefixes can derive diminutives that describe an object's size using the prefixes of Classes 11 and 12 rather than those associated with the gender of the noun being referred to. Thus, Classes 11 and 12 have no particular size characteristics since the morphology associated with them does not indicate size. Furthermore, NC 13 and 14 comprise a group of abstract and mass nouns with the prefix **kì-** and the suffix **tì-**. Noun Class 15 with the pronoun **nì** is ambinumeral and refers to singular and plural objects. Langdon and Steele (1981, 7) describe the function of **nì** as semantically 'non-specific' while Tait and Strevens (1954, 133) refer to it as a "neutral or general category of the pronominal article."

Winkelmann (2012) discards **nì** as a part of Likpakpaanl NC system because its use in the NC system is vague. However, I assume that **nì** applies to nouns with a generic reference and begins with **tìwán** 'thing'. For instance, **nì** is used as a pronoun that refers to inanimate objects with a generic reference, as shown in the nouns that are represented in ((87-b)). Both nouns in (87-a) and (87-b) can be replaced with **nì** as a pronoun, as in (87-c).

(87) a. Tìwán-pèèkààn gbààn nàn wììr.
thing-wearing DEF PST plenty
'The clothing is many.'

b. Tìwán-dòònkààn gbààn dɔ là(chè) ?
thing.SG-sleeping DEF lie where
'Where are the sleeping items/beddings?'

c. Nì dɔ màngù tààb là.
it lie 3.SG.mango under FOC
'It is lying under a mango tree.'

We have so far examined the various noun classes and showed that the noun prefixes in Likpakpaanl carry gender features of ± singular. The examples have also demonstrated that each prefix pair of a noun is determined by the gender Class to which the noun belongs, suggesting that the noun stems bearing singular plural prefixes form one gender pair. It can, therefore, be argued that Likpakpaanl nouns indicate eight stem groups of the 14 noun classes and one neutral one as illustrated in Table 3.3 (see Carstens, 1991).

Table 3.3 Paired Grouping of Likpakpaanl Nouns

Agreement Groups	Classes
Group A	stems of classes 1/2
Group B	stems of classes 3/4
Group C	stems of classes 5/6
Group D	stems of classes 7/8
Group E	stems of classes 9/10
Group F	stems of classes 11/12
Group G	stems of classes 13/14
Group H	stem of class 15

3.5 DP Structure and Agreement patterns in Likpakpaanl

In this sub-section, I examine morphological agreement patterns in the extended nominal projection in Likpakpaanl and show that demonstrative, relative, and 'which' interrogative pronouns agree with the class of a 'controlling' noun. Likpakpaanl nominal 'agreement' is a morpho-syntactic phenomenon that shows morphological dependencies between noun class prefixes and other nominal modifiers through feature checking. I argue gender is a feature that occurs on the functional head, NumP, which selects NP as its complement and not a separate head as Picallo (1991) proposes for Catalan.

3.5.1 Theoretical Assumptions

Agreement patterns within the DP in Likpakpaanl provide a strong argument for the operation Agree as formulated in Chomsky (2001, 2000) and refined in Pesetsky and Torrego (2007). Pesetsky and Torrego (2007) propose feature iteration postulating the ability of a single syntactic feature to participate in multiple agree operations throughout a derivation.

(88) AGREE (Pesetsky and Torrego, 2007, 268)

a. An unvalued feature F (a probe) on a head H at syntactic location α (Fα) scans its c-command domain for another instance of F (a goal) at location β (Fβ) with which to agree.

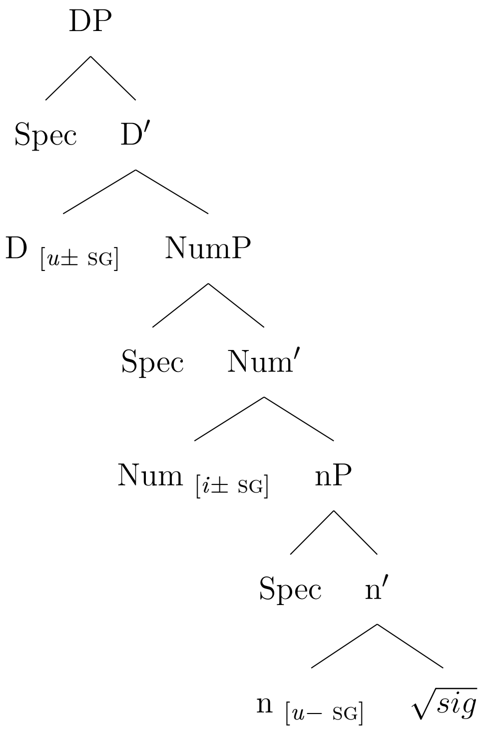
b. Replace Fα with Fβ so that the same feature is present in both locations

I argue in this work that the agreement pattern within the DP in Likpakpaanl, where a single number and gender feature values both the noun and other elements within the DP, can be accounted for by assuming that syntactic features on heads are not necessarily deleted after a single Agree relation but can iterate in successive operations in a single derivation (Pesetsky and Torrego, 2007).

Although Corbett (1991); Guthrie (1948); Picallo (1991); Ritter (1993); Pomino (2017) hold the view that gender influences syntactic derivations by its ability to dictate agreement patterns in the Noun Phrase (DP), they differ on precise nature and syntactic placement of these features. For instance, Kramer (2015); Kouneli (2021) have argued that gender features are hosted on the nominalising head, also referred to as little *n*. Kouneli (2021) thus, notes that in Kipsigis, a Nilo-Saharan language spoken in Kenya, the inherent features of a noun, which are the number features, are merged on *n*. Kouneli (2021)'s analysis, illustrated in **??** indicates that the nominalising head (little *n*) carries uninterpretable number features, which probes upwards for the interpretable binary ± number features on the goal, NumP which determines the interpretation of a noun as singular or plural at LF. (see Ritter, 1991). D also enters the derivation with an unvalued [SG] feature, which is valued through Agree while Num-head bears the [*i*Num]-feature shown in (89-b). Kouneli (2021) proposes a complex left-ward head movement analysis to derive the word order we see at LF in (89-a) (Kouneli, 2021, 1230).

(89) a. sig-iin-ta-it
parent-SG-TH-SEC
'parent' (sìgììndét)

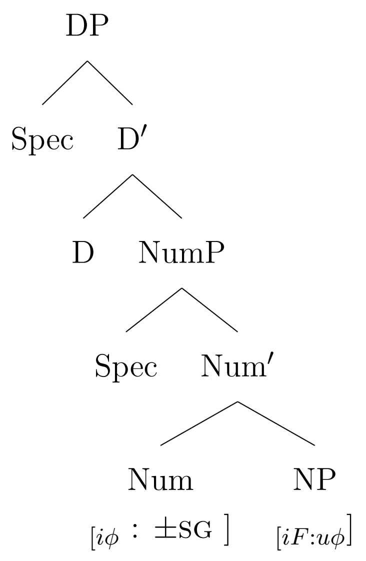
b.



There are also other scholars like Carstens (2010); Ritter (1991, 1993) who gender that gender not only as an inherent property of nouns but that it possesses features that can participate in agreement. In advancing their claims, Ritter (1991), Carstens (1991), among others, contend that number-features are hosted in the functional projection NumP, adding that these features are binary with [± [SG]] values. On the syntactic position of the gender features, Ritter (1993) contends that Num, in addition to its number features, also host gender features. In contrast, Picallo (1991) proposes the existence of a functional projection GenP responsible for gender features. I suggest, following Ritter (1993), that in the case of Likpakpaanl, that number and gender features are conflated on the head of a functional category (NumP) contrary to Picallo (1991)'s proposal of a separate Gender Phrase (GendP) in Catalan.

I claim that the functional Num(ber) phrase has interpretable phi-features [iϕ] while nouns are specified only for unvalued phi-probes [uϕ] along with inherent semantic features. The empirical motivation comes from concord relationships within the nominal phrase, where nouns Agree with targets like demonstratives in number and gender features. Likpakpaanl nouns have unvalued [uϕ] features, which Agree with the valued [iNUM] and [iGen] features on Num^0. Thus, nouns are probes with interpretable nominal but unvalued phi features. Therefore, the nominal morphology within the noun phrase is derived through Agree between unvalued noun probes and the interpretable specifications on Num as shown in 90.

(90)



The assumption that gender is a feature on Num^0 in Likpakpaanl is backed by the fact that noun class affixes are specified for both number and gender as (91) shows. The prefix **ù** in (91) and **ì** in (91-b) does not only indicate the class gender of the noun but also show that the noun is singular or plural. The data give empirical generalisation that Class and Number morphemes combine into a single exponent representing the noun class (Ritter, 1991).

(91) a. **ù**-gbéér gbààn
1.SG-pig
'the Pig'

b. **ì**-gbéér
6.PL-pig DEF
'the Pigs'

In examples like (91), we see an agreement conflation of gender and number triggering agreement between a noun and the demonstrative, relative, and interrogative 'which' pronouns, as I show later in this work. The fact that a single feature can value multiple dependencies in Likpakpaanl, as will be demonstrated in sub-sections 3.5.2, 3.5.3, and 3.5.4, support the iterative property of features put forward by Pesetsky and Torrego (2007). Number and gender markers are realised as pre-fixes on the head noun, and it is proposed that projections in the DP in Likpakpaanl are head-initial. Thus, I assume that the noun **ùgbéér** is the derived word order involving Upward Agree between the noun and the head of the number phrase. I use *Upward Agree* in the sense of Bjorkman and Zeijlstra (2019) in the derivation of nominal agreement as stipulated in (92).

(92) **Upward Agree (Bjorkman and Zeijlstra, 2019, 527)**
α enters an Agree relation with β iff:

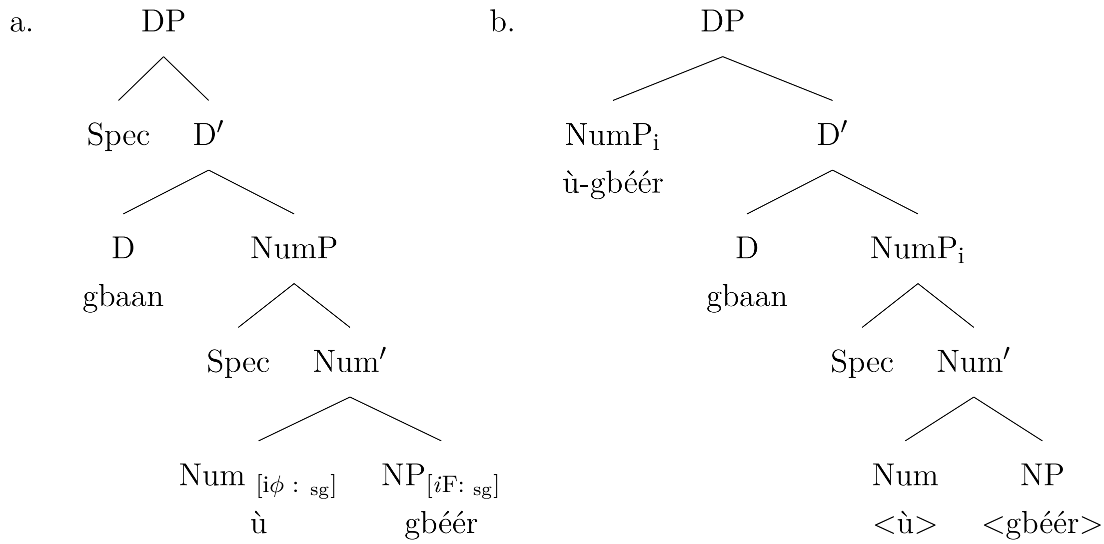
a. α carries at least one uninterpretable feature, and β carries a matching interpretable feature

b. β c-commands α

c. β is the closest goal to α

Thus, the noun like **ù-gbéér** is derived by Agree between the unvalued phi feature on the noun and its valued counterpart on the Num^0 as illustrated in (93-a). After the feature valuation, the EPP feature D attracts the NumP (comprising the noun class prefix and the head noun) to Spec-DP to derive the linear structure in (93-b).

(93)



Bjorkman and Zeijlstra (2019) contend that Upward Agree occurs when a probe with an unvalued [*u*Val] feature searches its c-commanding domain for a goal with matching valued feature [val]. When the probe successfully finds its goal, the unvalued features on the probe are eventually valued by the goal. Extending this analysis to noun classes and in Likpakpaanl, I argue that NumP has an interpretable and valued phi feature, and the various Classes (prefixes) spell out singular or plural on a gender-specific basis. The agreement allows for feature sharing on more than one agreeing element, as I show in the subsequent subsections demonstrating multiple agree with nouns and other elements within the DP.

The preceding analysis has motivated an account of agreement configurations within the determiner phrase in Likpakpaanl whereby the functional Number Phrase (NumP) contains valued, interpretable phi-features [iϕ] serving as a goal for matching probes on the noun. This theoretical framework captures the concord properties between nouns and other elements within the DP. With a motivated feature geometry established for nouns. The rest of the subsections will widen the empirical scope to interrogate further agreement morphology involving other phrasal constituents like demonstratives, interrogatives, and relative pronouns.

3.5.2 Agreement with Demonstrative Pronouns

This sub-section examines agreement patterns within Likpakpaanl nominal projections, focusing on demonstratives, relatives, and 'which' interrogatives. Demonstratives are deictic expressions with dual syntactic and pragmatic functions. Primarily, they focus the addressee's attention on objects in the speech situation through situational reference, often accompanied by pointing. Demonstrative pronouns refer to pronominal forms that stand in place of a noun or noun phrase and, in the case of Likpakpaanl, share the same grammatical features of gender and number morphology with the head noun (Diessel, 1999). Following Diessel (1999), I assume that demonstratives function as nominal pro-forms that share gender and number features with their head nouns. Cross-linguistically, demonstratives typically fall into two distinct categories: proximal, which refers to objects near the speaker, and distal, which indicates objects at a distance.

The structural position of demonstratives has garnered significant attention in syntactic theory. One school of thought, represented by Abney (1987), Brugè (2002, 1996), Giusti (1997), and Carstens (2008), positions demonstratives in specifier positions. An alternative analysis, advocated by Roehrs (2010); Brugè (2002, 1996); Giusti (1997), treats demonstratives as projecting a Demonstrative Phrase (DemP). I adopt this latter framework for analyzing Likpakpaanl demonstratives, treating them as heads of DemP. Following Brugè (2002, 1996) and Giusti (1997), I propose that the post-nominal position of demonstratives in Likpakpaanl results from movement operations. A crucial observation about Likpakpaanl demonstratives is their inability to co-occur with the definite determiner, as exemplified in (94), suggesting that they target the same syntactic position as the D-head.

(94) a. ù-bò ù-mìnà
1.SG-child 1.SG-PROX.DEM
'this child'

b. *ù-bò ù-mìnà gbààn
1.SG-child 1.SG-PROX.DEM DEF
Intended 'this child'

Likpakpaanl has two demonstrative suffix roots, **-mìnà** and **-è** for proximal and distal references, respectively, which are uninflected but occur with the gender and number prefix of the referent noun. The demonstrative pronouns show a strong agreement morphology with the nouns they modify, and the examples in 95 reveal a consistency in the agreement pattern in gender (95-a) and number (95-b) between demonstratives and their referent nouns. The distal demonstrative in (95-a) bears the gender and number prefix (**ù**) of the agreeing noun, similar to the observation in (95-b) and (95-c).[4]

[4]The gender/number prefix will change from (**-ì**) in the plural to (**-ù**) in the plural if the head noun is singular, as shown below:

(i) ŋmà bà dàà **ù**-fíín **ù**-bàà **ù**-mìnà?
who PST buy 1-owl 1-one 1-PROX.DEM
'Who bought this one owl?'.

(95) a. **Ù**-pìì **ù**-è kpà ì-kɔlà.
1.SG-woman 1.SG-DIST.DEM have.PFV 6-fowl
'That woman has fowls.'

b. **Bì**-pìì-b **bì**-è nàn kpà ì-kɔlà.
2PL-woman 2-DIST.DEM PST have 6-fowl
'Those women have a lot of fowls.'

c. Ŋmà (*lé) bà dàà **ì**-fíín **ì**-ŋmù **ì**-mìnà?
who FOC PST buy 6.PL-owl 6.PL-five 6.PL-PROX.DEM
'Who bought these five owls?'

The examples in (95) indicate that the demonstrative pronouns have the same marked morphological features of gender and number. It is also important to point out that in Likpakpaanl, the demonstrative, and the definite article are in complementary distribution: compare (96-a), (96-b) and the ungrammaticality of (96-c).

(96) a. **Kì**-gbàŋ **kì**-mìnà.
13.SG-book 13.SG-PROX.DEM
'This book.'

b. **Kì**-gbàŋ gbààn.
13.-book DEF
'the book'

c. ***Kì**-gbàŋ **kì**-mìnà gbààn.
13.SG-book 13.SG-PROX.DEM DEF
Intended:'this book'

The ungrammaticality in (96-c) is caused by the co-occurrence of the definite article **gbààn** and the demonstrative pronoun. The fact that both the noun and demonstrative pronoun show agreement in phi-features as indicated by the prefix **kì**, provides empirical motivation for Pesetsky and Torrego (2007) conception of Agree and the claim that a single feature can enter into multiple Agree operations in one derivation. Pesetsky and Torrego (2007) propose feature iteration, whereby a valued feature does not become inactive after a single Agree relation but can still participate in further operations. In their analysis of DP-internal structure in Buli, Dagbani, Kabiye, and Gurene, Hiraiwa et al. (2017) propose that the post-nominal position of clausal determiners, including demonstratives, in these languages, as shown in (97), is due to the movement of NP to the specifier of DP. They assume the D-head contains an EPP feature that attracts elements to its specifier position.

(97) a. [$_{DP}$ Naa bu$_i$ la$_{[EPP]}$ [$_{NP}$ t$_i$]]. (Buli)
cow NC DEM
'that cow'

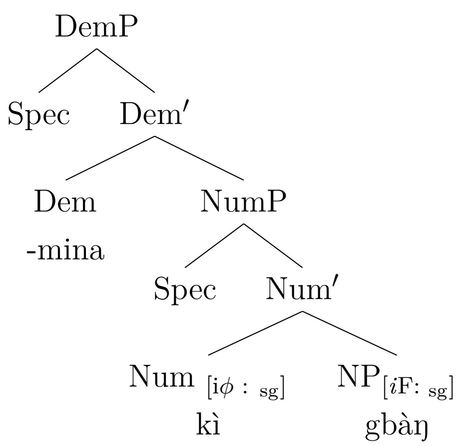
b. [DP Nimdi$_i$ D la$_{[EPP]}$ [NP t$_i$]]. (Dagbani)
meat DEM
'that meat'

(Hiraiwa et al., 2017, 6)

The analysis espoused by Hiraiwa et al. (2017) follows Kayne (1994)'s antisymmetry theory, which postulates that the hierarchical structures of natural languages conform to a universal Specifier-Head-Complement (SHC) branching order. Kayne (1994)'s antisymmetry hypothesis posits that any phrasal configuration exhibiting a surface order, such as in (95) that deviates from the specifier-head-complement sequence has undergone syntactic movement operations that disrupt the underlying order. The antisymmetry theory asserts that all languages are head-initial at the base, and heads are only preceded by their complements through displacement operations.

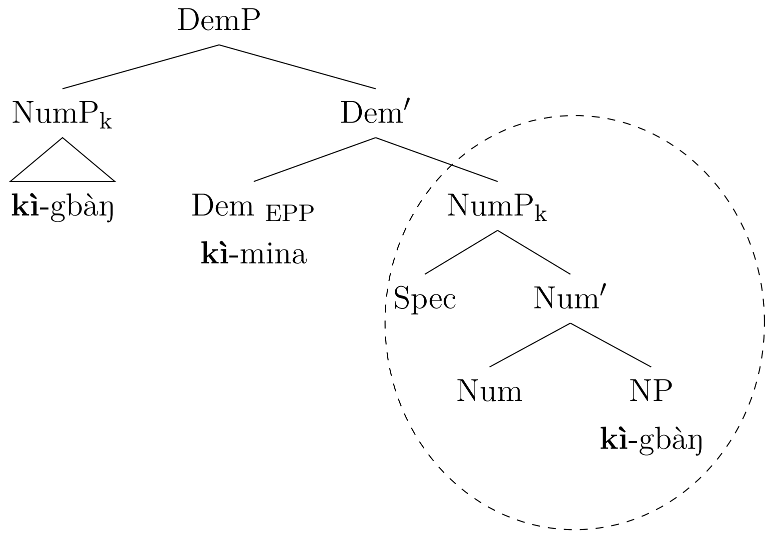
Following Kayne (1994) and Chomsky (2000, 2001), I propose that in the framework of Probe-Goal relations and feature valuation, the surface DP structure where the NP dominates the D-head in Likpakpaanl is derived as follows: The Dem^0 of the Demonstrative Phrase (DemP) is a Probe that bears uninterpretable and unvalued phi features [uϕ:uVal]. The phi feature on Num^0, which contains the N^0 head interpretable but unvalued phi features [iF:uϕ] and the head of NumP bearing both interpretable and valued phi-features. Following Pesetsky and Torrego (2007)'s system that allows a single feature to enter into Agree relations if they match in type (in this case, phi-features) regardless of their interpretability or valuation status, we have two derivational processes in Agree with the demonstrative pronoun. The first phase entails an upward agreement between the NP and NumP, where the interpretable but unvalued phi-features on the noun probe upward and find the NumP with interpretable and valued phi-features. The NP's features are valued after Agree, as already presented in (93-a).

(98) Feature valuation and Agree



To license Agree with the demonstrative pronoun, the DemP with uninterpretable and unvalued phi-features probes downward (in a standard Agree fashion) and finds the NumP with interpretable and valued phi-features. The DemP's features get valued, leading to the morphological spell out of the phi-feature on the demonstrative. Thus, the same feature on NumP is iterated between the elements. Ultimately, all the DemP, NumP, and NP) share the same phi-feature instance, which is now interpretable on NP and NumP. Following the Agree operation, the EPP feature on Dem^0, which requires its specifier to be filled, triggers the movement of the entire NumP to Spec-DemP to derive the surface order where the NP precedes the demonstrative in Likpakpaanl as shown in the derived structure in 99

(99) Derived Structure



Having explored noun class agreement with demonstrative pronouns, the subsequent subsection sheds further light on the agreement morphology between noun classes and interrogative pronouns.

3.5.3 D-link Wh-Interrogatives and NC Agreement

Another interesting observation in noun class agreement occurs with the D(iscourse)-link wh-questions. D-linking refers to wh-phrases for which "the range of felicitous answers is limited by a set both speaker and hearer have in mind" (Pesetsky, 1987, 108). Thus, D-link questions refer to background information in the proposition. The wh-questions in 100 contrast the discourse-linking status of the wh-phrase with consequences for the presupposed answer set.

(100) a. Who won the first prize?
b. Which student won the first prize?

As observed by Pesetsky (1987, 2000), wh-phrases may either introduce a new discourse referent as in (100-a) or draw from a closed set of salient referents in the prior discourse context (100-b). Specifically, while ‘what’ in (100-a) lacks any assumed link to previously established discourse entities, ‘which’ in (100-b) encodes the condition that the answer should be selected from a salient set already introduced. This reflects Pesetsky (1987, 2000)’s theoretical distinction between D-(iscourse)-linked phrases like ‘which’, which enters the derivation indexed to previously parsed discourse entities and non-D-linked ‘what’, which lacks assumed ties to a circumscribed referential domain. Thus, the ‘which’ interrogative functions discourse-pragmatically to constrain the response to a choice among contextually determined options (cf. Karttunen, 1977; Dayal, 2016; Ginzburg and Sag, 2000).

In Likpakpaanl, interrogative constructions demonstrate with the D-linked element ‘which’ shows a morphological process where class-marking circumfixes interact with bound morphemes. This phenomenon is particularly evident in 101, where the interrogative morpheme **-là-** agrees with the noun class of the modified noun. As shown in the (101-a), the class 9 singular marker **bù-** appears as both a prefix and suffix enclosing the interrogative morpheme **-là-**. The obligatory nature of this class agreement is demonstrated by the ungrammaticality of (101-b), where the attempted use of a different class marker renders the construction unacceptable.

In (101-c), the presupposition imposed by ‘which’ that the answer comprises contextually salient food options in a discourse-linked reference. The response cannot include arbitrary food items but is constrained to specify one or more delimited entities from the food set introduced in the preceding context. This contrasts with the wholly open denotation of non-D-linked ‘what’.

(101) a. **Bù**-sùb **bù**-là-**bù** nàn wùn?
9.SG-tree 9.SG-which-9 PST burn
‘Which tree was burnt?’

b. ***Bù**-sùb **ì**-là-**ì** nàn wùn?
9.SG-tree 9.SG-which-9 PST burn
Intended: ‘Which tree was burnt?’

c. Mpòpììn bà jìn **tì**-jìkàà-r **tì**-là-**tì**?
M. PST eat 14-food-14 14-which-14
‘Which food did Mpopiin eat?’

Semantically, ‘which’-interrogative phrase denotes a selection of alternatives and carries the pragmatic implication that the answer will be chosen from that closed set. This contrasts with ‘what’ questions, which place no limitations on the open domain of discourse from which the alternatives may be selected. The morpho-syntactic behaviour of Likpakpaanl interrogative circumfixes can be captured through the Minimalist operations of Feature Checking and Agree (Chomsky, 2000). In these D-linked constructions, the interrogative element enters the derivation with unvalued noun class features [*u*Class] and unvalued phi-features [$u\phi$], probes for a matching goal. The noun head-noun, bearing matching valued class and phi features [class:ϕ], serves as the goal in this operation. After the unvalued NC and phi-features of the D-link interrogative element are valued via the Agree operation, both elements share identical classes and phi-features at LF. This feature valuation then triggers specific morphological spell-out rules:

(102) After feature valuation, the noun prefix, α, appears in the suffix position, while the D-link interrogative element β is spelled out as **-là-**, flanked by α.

The ungrammaticality of forms like (101-b) follows directly from this analysis, as it would involve a mismatch between the valued class features obtained through Agree [class:9] and the morphological realisation of a different class marker [class:5], causing the derivation to crash. This approach not only accounts for the obligatory nature of the class agreement in Likpakpaanl interrogatives but also explains the systematic appearance of circumfixal morphology as a manifestation of syntactic feature checking and agreement processes.

It is worth noting that Likpakpaanl D-linked wh-phrases can appear as a full which-phrase or an elliptical form (shown in brackets) in which the head noun is missing but is implicitly understood from the previous context, and each case shows morphological agreement. The orders in 103 appear in such a way that the D-linked subject must precede the locative wh-adverb in the canonical sentence as in (103-a) or have the object precede the temporal adverb in (103-b). The D-linked direct object must precede the manner wh-adverb **kìnyè** in (103-c). This order could be because adverbs in Likpakpaanl are sentence final. They can A-bar move to the sentence-initial but not to the clause-medial position.

(103) a. **Ù**-bò (**ù**-là-**ù**) nàn bùèn làchè?
1.SG-child 1.SG-which-1 PST go where
'Which child went where?'

b. Ama gà gàà **bù**-sùb (**bù**-là-**bù**) bà-dáál?
A. FUT cut 9.SG-tree 9.SG-which-9 what-day
'When did Ama cut which tree?'

c. Kòònjà nàn mànn (**ì**-jàn) **ì**-là-**ì** **kìnyè**?
K. PST smoke 6.SG-fish 6.SG-which-6 how
'How did Koonja smoke which fish?'

The same observation is made in di-transitive construction like (104-b) where we see the verb **tìì** 'give' with two which-objects, the direct object 'which meat' and the indirect object 'which child'. The direct object and adverb follow the 'which' interrogative phrase. The adverb can, however, be extracted to Spec-Foc as (104-b) indicates.

(104) a. John nàn dì tì-nàn tì-là-tì tìì ù-bò
J. PST take 14.PL-meat 14.PL-which-14 give 1.SG-child
ù-là-ù bá-dààl?
1.SG-which-1 what-day
'When did John give which meat to which child?'

b. Bá-dààl$_i$ (lé) John nàn dì tì-nàn tì-là-tì tìì
what-day FOC J. PST take 14.PL-meat 14.PL-which-14 give
ù-bò ù-là-ù ?
1.SG-child 1.SG-which-1
'When did John give which meat to which child?'

The unanswered question from the foregoing is the position of the 'which' interrogative phrase in the DP in Likpakpaanl. Drawing insight from Wiltschko (1998); Rett (2006), I analyse Likpakpaanl through the lens of German wh-phrase questions, where a crucial distinction exists between pronominal and determiner wh-phrases. Wiltschko (1998); Rett (2006) argue that pronominal wh-phrases such as **er, sie, es** 'he, she, it', respectively, do not take an NP complement. On the contrary, *d*-pronouns such as **der, die, das**, 'the', which she refers to as wh-determiner phrases have an NP as their complements. According to Wiltschko (1998), *d*-pronouns are used in headed relative clauses in German '(105-a), while in a headless relative clause, the relative pronoun appears as a wh-word (105-b). (see Wiltschko, 1998, 146).

(105) a. Nimm das Buch, das/*was du willst.
Take the book d-pron/what you want.
'Take the book you want.'

b. Nimm , was/*das du willst.
Take what/d-pron you want
'Take whatever you want.'

She further posits that 'd'-pronouns are full DPs that select an empty NP, but personal pronouns are "mere spell out of phi-features, i.e., an instantiation of AgrD, rather than an instantiation of D" (see Wiltschko, 1998, 148). Longobardi (1994) also proposed that wh-interrogatives are determiners that function as operators binding a variable whose range is defined by the head noun. Thus, 'which' acts as an operator indicating a set of possible values defined by the nominal references in the real world. I extend this conclusion to the which-interrogative phrase in Likpakpaanl and argue that it is a wh-determiner that takes an NP complement.

Evidence for this assumption comes from the co-occurrence restriction between the definite determiner and the wh-interrogative phrase in Likpakpaanl, as shown by the ungrammatical sentence in (106-a). Thus, I assume that the wh-interrogative head is base-generated in the head of D and enters into Agree with the NumP to have its unvalued phi-features valued.

(106) a. *Bì-jàb bì-là-bì gbààn gà yóór bì-sàpuàm
2.PL-men 2.PL-which-2 DEF FUT take 2.PL-young.girls
bì-là-bì?
2.PL-which-2
Intended: 'Which young girls will which of the men marry?'

b. [DP **Bì**-jàb$_j$ [D **bì-là**-bì [$_{\text{NumPj Num}}$ [$_{\text{NP}}$]]]]

In Likpakpaanl, the interrogative marker **-là-** exhibits properties parallel to German *d*-pronouns in that it obligatorily occurs with an NP complement, as evidenced in the structure **bù-sùb bù-là-bù** 'which tree'. However, Likpakpaanl introduces an additional layer of complexity through its circumfixal agreement pattern. Unlike German, where agreement is marked solely through the determiner form (**der, die, das**), Likpakpaanl realises agreement through a discontinuous morpheme *NC-là-NC*. his analysis aligns with Wiltschko (1998)'s broader typology while acknowledging the unique properties of Likpakpaanl nominal morphosyntax. The parallel between German and Likpakpaanl extends to the semantic domain as well, as both languages employ these constructions in specific contexts requiring definiteness and specificity, though their morphological manifestations significantly differ.

3.5.4 Relative Pronouns and NC Agreement

In Likpakpaanl, the agreement morphology marked on the relative pronoun is determined by the properties of the head noun. Specifically, number agreement between the relative pronoun and its antecedent is regulated by the grammatical number value encoded on the head noun. This dependency is instantiated in relative clause constructions, as illustrated in (107-a) and (107-b).

(107) a. Francis gà kàn **lì**-dìchàn-l **lì** Pious nàn màà nà.
F. FUT see 7.SG-house-7 7.SG-REL P. PST build REL.DET
'Francis will see the house that Pious built.'

b. **Kì**-tìŋ **kì** Jabon gà dàà nà kpà sàlmà.
13.SG-land 13.SG.REL J. FUT buy REL.DET have 3.SG.gold
'The land that Jabon will buy has gold.'

In (107), the singular class markers **lì** and **kì** on the relative pronoun enter into an Agree relation with the singular number of the head nouns **lìdìchànl** (house) and **kìtìŋ** (land), respectively. This reflects the broader generalisation that the specific form of the relative pronoun is contingent on the morpho-syntactic features associated with its head noun referent. Number concord through this dependency applies systematically across relative, interrogative, and demonstrative clause structures in Likpakpaanl. Thus, the language exhibits regular syntactic agreement patterns whereby the grammatical number value of the head noun regulates the morphology carried by the agreeing pronouns. Table 3.4 summarises the agreement patterns between demonstrative, relative, and which-interrogative pronouns in Likpakpaanl.[5]

The syntactic structure of Likpakpaanl relative clauses reveals a complex interaction between the DP and CP domains, as evidenced in two key configurations in (108-b) and (108-c). The derivation proceeds as follows: the C-head, which is initially null at the start of the derivation, carries uninterpretable class and phi features. To value these features, it probes its c-command domain to find the NP **tì-gbàn** as a matching goal. This feature matching occurs through the Agree mechanism. Subsequently, the EPP feature on the C-head triggers A-bar movement of the NP to Spec-CP. This movement results in the overt realisation of the Agree relation, marked by the noun class (NC) exponent associated with the head noun (108-b). After this, the entire CP is pied-piped to Spec-DP with the relative determiner occurring clause finally in the relative clause (108-c).

(108) a. Jonah nàn kàrn [$_{\text{Relative clause}}$ tì-gbàn$_i$ tì Taanan nàn
J. PST read 14.PL-book 14.PL.REL T. PST
dàà t$_i$ nà/*gbààn.]
buy REL.DET/DET
'Jonah read the book that Taanan bought.'

b. [DP D^0 [CP **tì**-gbàn$_k$ **tì**$_{[\text{EPP},u\phi,u\text{class}]}$ [TP Taanan nàn [VP dàà [$_{\text{NP}}$ t$_k$[$i\phi$,$i$class]]]]]]

c. [DP [CP tì-gbàn tì Taanan nàn dàà t$_i$] D^0 nà t$_i$]

The proposed analysis is that Likpakpaanl relative clauses involve a leftward movement of the head noun from an internal argument of the verb to Spec-CP and the complementiser head c-selected by the noun class feature (NumP) on the NP. After that, the entire complex CP is pied-pipped to Spec-DP to check the definite feature on D-head to derive the head-final linear order as illustrated in (108-c).

Unlike Likpakpaanl, where there is a co-occurrence restriction between the relative determiner and the specificity definite article as (108-a) indicates, Aboh

[5] I consider suffixes **-mìnà**, **-è** and **là** and as the primary forms of demonstrative and interrogative pronouns which occurs in nearly all phonological environments.

(2005) shows that relative clauses in Gungbe and Dagaare can either be bare as in (109) or the head noun, and the relative particle can occur with a definite marker to the right edge to allow the noun to receive a specific interpretation. Bodomo and Hiraiwa (2010) makes a similar observation in Dagaare.[6]

(109) Kòfí xɔ́ àgásá [ɖě mí wlé] lɔ́
Kofi buys crab that$_{\text{Rel}}$ 1PL catch DET
'Kofi bought the [aforementioned] crab that we caught'

(Aboh, 2005, 271)

From the discussion, we can also conclude that the noun class system is central in the grammar of Likpakpaanl, and most noun classes share semantic features ranging from human (Kinship relation), animate/inanimate, mass/abstract, and body parts. Wilson (1971)'s examination of the noun class system in the Oti-Volta language family, to which Likpakpaanl belongs, corroborates the robust concord patterns observed between the noun classes and other grammatical elements of these languages. He states that there is "... a very high degree of correspondence between the class membership of cognate nouns in the various languages so that the animals" and "body parts within a given singular/plural class pair are nearly all in these classes throughout the sub-group" (Wilson, 1971, 80). The data indicates that Likpakpaanl noun classes command agreement and whose scope is limited to agreement morphology inside the DP but does not extend to the verb phrase.

[6]Interested reader can check Aboh (2005) for a thorough analysis of Gungbe relative and factive clauses where he proposes a DP-extraction to the specifier of the CP followed by movement of the relative clause to check the number and specificity features on spec-NumP and spec-TopP.

Table 3.4 Agreement patterns in Likpakpaanl

NC	Affixes	Relative		Proximal Dem		Distal Dem		InterP	
		SG	PL	SG	PL	SG	PL	SG	PL
1/2	ù-. . . ø/bì-. . . -b/-m	ù	bì	ù-mìnà	bì-mìnà	ù-è	bì-è	ù-là-ù	bì-là-bì
3/4	ø. . . ø /-tíìb	ù	bì	ù-mìnà	bì-mìnà	ù-è	bì-è	ù-là-ù	bì-là-bì
5/6	n-. . . ø/ ì-. . . ø	mù	ì	mù-mìnà	ì-mìnà	mù-è	ì-è	mù-là-mù	ì-là-ì
7/8	lì-. . . -l/ ŋì-. . . ø	lì	ŋì	lì-mìnà	ŋì-mìnà	lì-è	ŋì-è	lì-là-lì	ŋì-là-ŋì
9/10	bù-. . . -b/ ì-. . . -ì	bù	ì	bù-mìnà	ì-mìnà	bù-è	ì-è	bù-là-bù	ì-là-ì
11/12	kì-. . . -k/ n-. . . -m	kì	mù	kì-mìnà	mù-mìnà	kì-è	mù-è	kì-là-kì	mù-là-mù
13/14	kì-. . . /tì-. . . -r/-l	kì	tì	kì-mìnà	tì-mìnà	kì-è	tì-è	kì-là-kì	tì-là-tì
15	nì	nì	nì	nì-mìnà	nì-mìnà	nì-è	nì-è	nì-là-nì	nì-là-nì

In the next subsection, I outline the clausal structure of Likpakpaanl as well as the tense and aspect system and their syntax. The assumptions underpinning the discussion are (i) the particles and affixes that encode tense and aspect are heads of maximal projections projecting Tense Phrase (TP) and Aspectual Phrase (AspP) in the clausal spine, (ii) Tense and Aspectual interpretations are licensed via an Agree relation between the features on T and Asp heads in the language.

3.6 Clause Structure, Tense and Aspect in Likpakpaanl

Structurally, Likpakpaanl is an **SVO** language where the verb precedes the direct and indirect objects in the canonical sentence (110-a). (110-b) is ungrammatical because the verb does not select rice, which is the internal argument of the transitive verb.

(110) a. Tamanja nàn gbìì lì-núú-l.
T. PST dig 7.SG-yam-7
'Tamanja harvested yam.'

b. *Pònípíír í-mùùl ŋààn dìn.
P. 6.SG-rice cook.PFV today

In di-transitive constructions, the indirect object precedes the direct object, as illustrated in (111).

(111) Namuel gà tìì Simon kì-gbàŋ.
N. FUT give S. 13.SG-book
'Namuel will give Simon a book.'

The direct object can, however, precede the indirect object in Likpakpaanl in serial verb constructions. According to Aikhenvald (2006b), serial verbs act as a single predicate without any overt marker of coordination or subordination. In (112-a), the verb **dì** 'take', like a 'do' insertion in English, has to select the direct object as its complement, else the resulting sentence will be ungrammatical (112-b).

(112) a. Namuel gà dì kì-gbàŋ tìì Simon.
N. FUT take 13.SG-book give S.
'Namuel will give a book to Simon.'

b. *Namuel gà dì Simon tìì kì-gbàŋ.
N. FUT take S. give 13.SG-book
Intended: 'Namuel gave a book to Simon.'

The discussion in subsection 3.5.2 proposed a head-initial order in the DP, arguing that the head of the D-head in the surface is due to the movement of the complement to Spec-DP. Similarly, in Likpakpaanl Verb Phrases (VPs), Complementiser

Phrases (CPs), Tense Phrases (TPs), and Aspectual Phrases (AspPs), the heads linearly precede their phrasal complements. For instance, in (113-a), the head of the phrase **jìn** 'eat' c-commands its complement **sákɔlà** 'fufu'. Switching the linear order, as in the case of the DP, is ungrammatical (113-b).

(113) a. Sanja nàn jìn sákɔlà gbààn .
S. PST eat 4.fufu DEF
'Sanja ate the fufu.'

b. *Sanja nàn sákɔlà gbààn jìn.
S. PST 4.fufu DEF eat

c. [TP Sanja T nàn [VP [V **jìn**] [DP [NP sákɔlà] D gbààn]]]

Likpakpaanl also exhibits head-initial directionality in complementiser phrases, as exemplified in (114-a) where the complementiser head **kè** precedes its complement, the tense phrase (TP).

(114) a. Kè Tamanja gà ŋààn lì-nùù-l.
COMP T. FUT cook 7.SG-yam-7
'That Tamanja will cook yam.'

[CP Kè [TP Tamanja [T gà] [VP [V ŋààn [DP lì-nùù-l]]]]]

Additionally, grammatical markers of tense and aspect typically precede the main verb, suggesting that verb phrases analysed as heads of TP or AspP are also head-initial. In (114-a), the future tense marker **gà** precedes the VP 'cook yam'. Similarly, in (115), the perfect aspect marker appears before the verb phrase, which serves as its complement.

(115) Kunji bì kùùr kì-sàà-k gbààn nì.
K. IPFV weed 11.SG-farm-11 DEF in
'Kunji is wedding the farm.'

3.6.1 Tense Marking in Likpakpaanl

Tense is a deictic category that involves the location of the time of an event relative to the moment of speech Comrie (1985). Cross-linguistically, the tense can be encoded in the morphology of the verb or using particles. Demirdache and Uribe-Etxebarria (2000) note that tense semantically deals with the temporal location of the event(s) relative to an utterance and distinguishes three locations in time: Speaking time (S), which references the time an utterance is made, Event time (E) describes the time an event occurs while Reference time (R) relates to the time interval at which the event takes place (see also: Reichenbach, 1947; Demirdache and Uribe-Etxebarria, 2007).

Tense marking represents a core component of the grammatical system in Likpakpaanl. Likpakpaanl verbs do not inflect for aspect, number, or tense, as we demonstrated under section 3.6.4. Instead, tense distinctions are encoded through independent preverbal particles[7] that grammaticalise temporal deixis, contrasting past and non-past (future) events. Likpakpaanl makes a basic tense distinction between past and non-past events. The following subsections will further elaborate on past and non-past tense expressions.

3.6.2 The past Tense

Past tense in Likpakpaanl distinguishes three sub-categories: the immediate past for describing events that occurred earlier of the same day, the hesternal past for events that happened a day before the speaking time (yesterday), and the remote past for temporally distant events.

A. Immediate Past Tense

The immediate past (or today's past) is marked by the particle **bà**, which occurs preverbally and shows no agreement morphology for either singular (116-a) or plural (116-b) subjects, and therefore, reflecting the lack of subject agreement on tense marking particles in Likpakpaanl. The immediate past tense marker **bà** thus encodes immediate past tense reference irrespective of the subject noun phrase.

(116) a. Mpopiin **bà** ŋààl bù-ŋɔ-b.
M. PST drive 9.SG-canoe-9
'Mpòpíín paddled a canoe.'

b. Waaja nì Amà **bà** nyùn n-nyùn.
W. and A. PST drink 5.SG-water
'Waaja and Ama drank water.'

B. The Remote Past

The remote or distant past refers to events that occurred at significant intervals before the present moment—such as three days, a week, months, or even years ago. This temporal framing allows speakers to contextualise past actions relative to other events in the narrative, emphasising the distinction between actions that are perceived as relatively recent and those that are considered more distant. In (117), the preverbal tense marker **nán** serves a crucial role in encoding this remote past tense interpretation. Its presence indicates that the action described—weeded in this case—has been completed in a time frame that is not only past but also significantly distanced from the present context.

[7] Bodomo (1997); Atintono (2013) refer to these particles as Time Depth Particles (TDPs)

(117) Bilijo nàn kùùr kì-sáá-k gbààn.
B. PST paddle 11.SG-farm-11 DEF
'Bilijo weeded the farm.'

C. The Hesternal past

The hesternal (yesterday's) past tense in Likpakpaanl is marked by the particle **fè**, which encodes events that took place a day before the utterance time of (Demirdache and Uribe-Etxebarria, 2000). Semantically, **fè** carries the feature [+yesterday], reflecting its meaning of 'yesterday' past,(118-b). Therefore, the use of the overt temporal adverb like 'yesterday' would be redundant, and the adverb is typically omitted in Likpakpaanl as in (118-b).

(118) a. David **fè** gbìì bànchí (fènnà).
D. HEST.PST dig 3.SG.cassava (yesterday)
'David dug cassava yesterday.'

b. Ù-tààl **fè** tèèn bù-sù-b Yààjòl.
1-rain HEST.PST shout 9.SG-tree-9 Y.
Literary: 'Rain shouted at a tree in Yaajol (yesterday).'
'Thunder struck a tree in Yaajol (yesterday).'

3.6.3 The Future Tense

In Likpakpaanl, the non-past or future tense encodes situations where the event time (E) is posterior to the speech time (S), as defined by Demirdache and Uribe-Etxebarria (2000). The future tense is formed using the preverbal particle **gà**, as exemplified in (119-a) and (119-b). The use of **gà** in these sentences places the denoted event at a temporally distant point in the future relative to the utterance time.

(119) a. Ntebi **gà** dàà chééchè.
N. FUT buy 3.SG.bicycle
'Ntébí will buy a bicycle.'

b. Dawun nì Taado **gà** pùí ì-dí.
D. and T. FUT thresh 6.PL-millet
'Dawun and Taado will thresh millet.'

3.6.4 Aspect Marking in Likpakpaanl

Aspect has been defined aspect as the various "...ways of viewing the internal temporal constituency of a situation." (Comrie, 1976, 3). Other scholars such as Bhat (1999) also note that aspect "indicates the temporal structure of an event, i.e., the way in which the event occurs in time (on-going or completed, beginning or ended, iterative or semelfactive etc." According to Demirdache and Uribe-Etxebarria (2000), aspectual markings are used to determine whether an event/activity is unitary (perfective) with a definite endpoint in time, continuous and expressing an ongoing event (imperfective) or repetitive (habitual). Likpakpaanl distinguishes between the three aspectual forms; the perfective, the imperfective, and the habitual aspects. Likpakpaanl has lexicalised preverbal particles to mark imperfective, and habitual aspects but the perfection, which is morphologically unmarked and therefore treated as the 'default' aspect.

3.6.4.1 The Perfective Aspect

The perfective aspect is unmarked in Likpakpaanl and verbs receive a default perfective interpretation when they are not preceded by any preverbal aspectual or tense particle. The perfective aspect describes an event in the past that had a definite endpoint. Following Tenny (1987), I assume that aspect is a syntactic category that projects an Aspectual Phrase (AspP) in the clausal structure. The example in (120-a) shows a singular subject with an unmarked verb, and in (120-b), though there is a plural subject, there is no morphological change of the verb. (120-c) shows the verb in its bare form but expresses a completed action. The intransitive verb in (120-c) in its default form indicates that 'Mpopiin' has already slept. All these examples show that verbs in Likpakpaanl expressing completed action do not show overt affixes like what we find in languages such as English. Thus, the perfective head of the AspP in Likpakpaanl is morphologically null.

(120) a. Nàkɔjà chùù ù-ŋùò gbààn.
N. catch.PFV 1.SG-goat DEF
'Nàkɔjà has caught the goat.'

b. Ntébí nì Amà chùù ì-jàn.
N. and A. catch.PFV 6.PL-fish
'Ntébí and Ama have caught fish.'

c. Mpopiin gèèn.
M. sleep.PFV
'Mpopiin has slept.'

Similar examples from Atintono (2013, 103) and Issah (2023, 5) demonstrate that the perfective aspect is unmarked in Gurene (121-a) and Dagbani (121-b). Bodomo (1997), and Dumah (2017), among others have shown similar unmarked perfection patterns in Dagaare and Sisaali.

(121) a. A lobe la kugere. (Gurene)
3SG throw.PFV .FOC stone
'S/he has thrown a stone.'

b. Bí-hí máá dá sábìlí. (Dagbani)
child-PL DEF buy.PFV rat
'The children have bought a rat.'

Cross-linguistically, there is a diversity in how different languages express perfective aspects. For instance, in Slavic languages such as Polish, and Russian, both perfective and imperfective aspects are marked morphologically on verbs. This linguistic asymmetry between Mabia and Slavic languages contrasts with the patterns observed by Comrie (1976) and Dahl (1985) in their studies of aspectual markings.

3.6.4.2 The Imperfective Aspect

In contrast to the unmarked perfective aspect, the imperfective aspect in Likpakpaanl is overtly indicated by the preverbal particle **bì**, as exemplified in (122-a). This imperfective marker does not exhibit agreement morphology reflecting the number of the subject since the same imperfective form **bì** is used with both singular and plural subjects as in (122-b). Both examples indicate an ongoing activity and are expressed by the imperfective marker that precedes the verb. Additionally, the imperfective particle must precede the verb, as demonstrated by the ungrammaticality of (122-c).

(122) a. Sapu bì pù wùlíjù.
S. IPFV roast 3.SG.potato
'Sapu is roasting potato.'

b. Baadak nì Simon bì píí lí-wó-l.
B. and S. PFV blow 5.SG-flute-5
'Baadak and Simon are blowing a flute.'

c. *Nantob wíí bì.
N. cry PFV
Intended: 'Nantob is crying.'

3.6.4.3 Habitual Aspect

The third aspectual form in Likpakpaanl is the habitual aspect employed to express an event as repetitive or indicate the continuous existence of a state of affairs. Bybee et al. (1994) note in their cross-linguistic survey that habitual markers are often associated with imperfective aspects, resulting in potential ambiguity between a habitual interpretation and a progressive one.

Likpakpaanl employs a habitual marker that attaches as an enclitic to the verb stem, as shown in examples (123-b) and (124-b). In these data, the habitual suffixes **-nì** or **-r** indicate the repetitive events denoted by the verb. Thus, without the habitual suffixes, we get a perfective interpretation of the sentences as shown in (123-a) and (124-a).

(123) a. Taadime tàm lì-kèkèn-l.
T. fold.PFV 7.SG-cloth-7
'Taadime has folded a clothe.'

b. Nakɔjà tám-**nì** lì-kèkèn-l
N. fold-HAB 7.SG-cloth-7
'Nakoja folds a cloth.'

(124) a. Alima gàà ì-sù-ì.
A. cut.PFV 6.PL-tree-6
'Alima has cut trees.'

b. Alima gáá-**r** ì-sù-ì.
A. cut-HAB 6.PL-tree-6
'Alima cuts trees.'

As predicted by Bybee et al. (1994), the examples in (124-b) induce ambiguous interpretations because they can be understood as a repetitive action where either 'Alima' cuts trees daily or a single event of cutting trees for a particular purpose. Similarly, example (123-b) could mean that 'Taadime' folds clothes as a habit (or for a living), or it's just a single event of him doing so. Based on the examples in (123-b) and (124-b), the descriptive generalisations that can be made about the habitual aspect in Likpakpaanl is that the form of the verb determines the morphological spell-out of the aspectual marker. Thus, verbs ending with long vowels are marked by the habitual suffix **-r** while those that end with nasal sounds *m,n, ŋ* and other consonant sounds are suffixed by **-nì**. Table 3.5 provides a summary of the habitual markings on verbs depending on their form.

Table 3.5 Some Verbs and their Habitual Markings

Verb	**Gloss**	**Habitual**	**Gloss**
lèè	choose	lèè-**r**	chooses eat
kúú	shave	kùù-**r**	shaves
ŋmàl	mix	ŋmàl-**nì**	mixes
wùŋ	crouch down	wùŋ-**nì**	crouches down
géén	sleep	géén-**nì**	sleeps

I have shown that tense and aspect marking in Likpakpaanl are achieved through the use of dedicated preverbal particles. Specifically, particles such as **gà** (*future*), **bà** (*hodiernal*), and **fè** (*hesternal*) encode tense distinctions. At the same time, aspectual interpretations are marked by **bì** and **nì/-r** for imperfective and habitual aspects, respectively. Across these tense and aspect categories, the preverbal particles do not exhibit agreement with the verb.

An interesting interaction between tense and aspect markers in Likpakpaanl sheds light on the syntactic hierarchy of temporal projections within the clausal structure. Specifically, when tense and aspect co-occur, they surface in the rigid order of tense, (im)perfective, and habitual, as exemplified in (125-a), (125-b) and (125-c), respectively. This indicates that tense projects higher than aspect in the functional sequence of the clause structure.

(125) a. Nàkɔjà nàn bì tíí-r Binlu lì-nùù-l.
N. PST IPFV give.HAB Binlu 7.SG-yam-7
'Nakoja was (habitually) giving Binlu yam.'

b. Kajal nàn tíí-r Mbo ŋ̀ì-tùùn.
K. PST give-HAB M. 8.PL-beans
'Kajal used to (habitually) give Mbo beans.'

c. Tagan fè mánn Nsambaan àsìbìtì nì.
T. PST visit.PFV N. 3.SG.hospital LOC
'Tagan visited Nsambaan in the hospital (yesterday).'

Likpakpaanl data examined in this section provides additional support for the cross-linguistic generalisation that functional heads encoding tense (TP) are projected higher than those encoding aspect in the clausal hierarchy (Cinque, 1999; Julien, 2002). The data shows the base order of Tense-Aspect-Verb in Likpakpaanl, reflecting the theoretical view that tense scopes above AspP in the functional sequence. The proposed syntactic hierarchy projecting tense functional heads above aspect phrases in the clausal spine, with both ordered above the verb phrase, is schematised in (126):

(126) [TP → [$_{\text{AspP}}$ → [VP → [VP]]]]

From the discussion, we have established that tense and aspect marketers in Likpakpaanl constitute independent syntactic elements within the clausal architecture of Likpakpaanl. Furthermore, I also showed the hierarchical relationship between these particles and how they interact to encode various temporal discourse functions. Building upon these discussions, the next section will advance a syntactic derivation of tense and aspect.

3.7 On the Syntax of Tense and Aspect in Likpakpaanl

The previous sections have provided empirical data showing the distribution and interaction between tense and aspect in Likpakpaanl. This section provides a syntactic analysis of the tense and aspect marking particles in Likpakpaanl. The central proposal is that the particles and affixes encoding tense and aspect are Maximal projections in the clausal hierarchy above the verb phrase. Additionally, I argue for a syntactic derivation involving Agree and feature sharing in the verbal domain in Likpakpaanl. I analyse Likpakpaanl tense and aspect particles as heads of maximal projections Tense Phrase (TP) and Aspect Phrase (AspP) within the clausal structure (Demirdache and Uribe-Etxebarria, 2000; Pollock, 1989; Tenny, 1987). I also assume these functional heads have features that enter into feature-checking relationships with the verbal complex to ensure well-formedness and interoperability (Chomsky, 2000, 2001).

Specifically, the tense particles **bà**, **nàn** and **gà** carry interpretable tense features with values as [±past] and [future] (Demirdache and Uribe-Etxebarria, 2000; Zagona, 2013). The heads of the aspectual phrase also bear interpretable aspect features with values [perfective], [imperfective], and [habitual]. Verbs, on the other hand, possess an uninterpretable tense and aspect feature and cannot inherently express tense or aspect distinctions unless via some syntactic operation such as Agree for its features to be valued. Through this feature-valuation mechanism, the tense and aspect projections license the temporal interpretation of the verb.

I adopt the non-standard Upward Agree of Bjorkman and Zeijlstra (2019) discussed in (92) where the probe is c-commanded by the goal, and thus, the probe searches upward for the matching goal. Operating on the assumption of the feature geometry for tense, aspect, and verb in Likpakpaanl shown in 127 T^0 and Asp^0 are goals that carry matching interpretable features that can value the uninterpretable features on the probe, the verb.

(127) a.TP has interpretable and valued features = [iF: Val $_{\pm\text{past,future}}$]

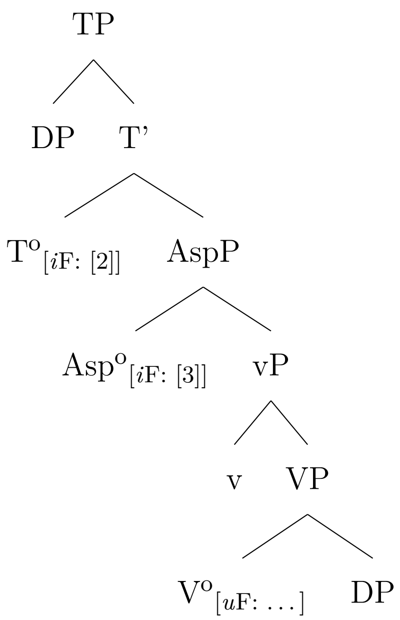
b.AspP has interpretable and valued features = [iF: Val $_{\text{perf, imperf, habitual}}$]

c.VP has uninterpretable and unvalued tense and aspectual features=[uF: uVal]

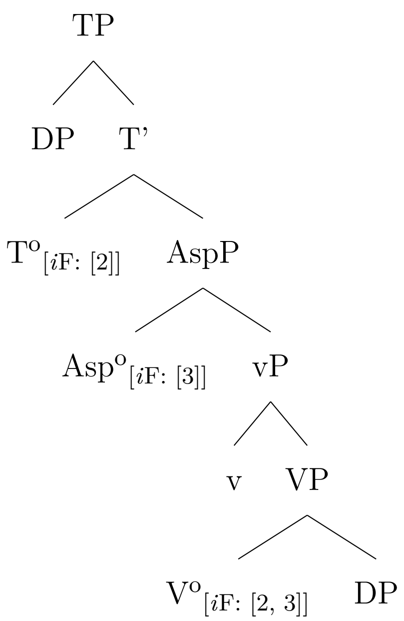
Thus, Agree applies when the probe searches its c-command domain for a matching goal, which it enters into an Agree relation and deletes the uninterpretable features on the probe. Functional heads like TP and AspP act as Goals carrying [iF] features within this Agree mechanism. They enter into Agree with the uninterpretable features [uF] on the verbal Probe. This Agree relationship enables the interpretable features of tense and aspect on TP and AspP to be licensed for tense and aspect interpretation.

In the structure in (128), the tense and aspectual heads have interpretable and valued tense and aspect features [iF:[2]]. The tense value is represented with the number [2]; nothing hinges on this.[8] The verb has interpretable but unvalued tense or aspectual features as illustrated in (128). The unvalued feature on V^0 probes upward in its c-commanded domain for a suitable goal Bjorkman and Zeijlstra (2019). In the second step of the derivation, the value on the tense and aspect heads are shared with the verb (129), and the verb **kìr** receives a tense value and interpretation as indicated by the feature sharing indices.

(128) **Tense and Aspectual Features**



(129) **Agreement and feature valuation**



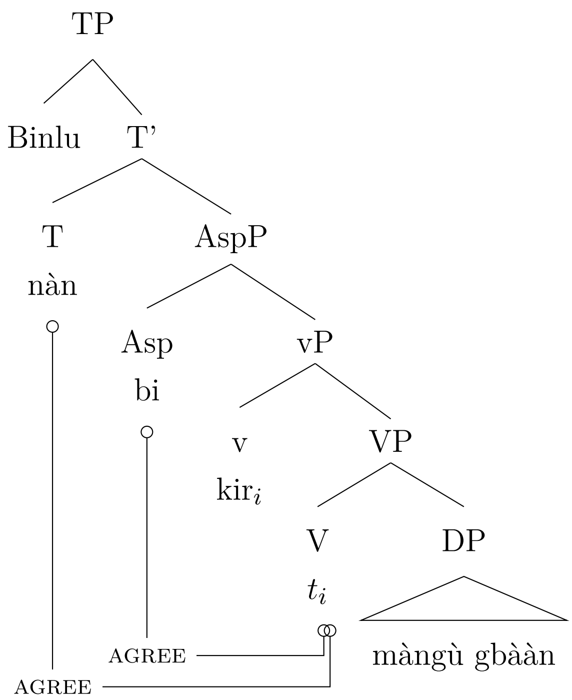
[8]The numerical indices could be any number [n ...]. The number is only a generic representation of the feature.

After the aspectual features of the verb are valued, the unvalued tense features of the verb also go through the similar Agree process and get licensed for tense interpretation. The structural representation in (131) illustrates how Agree relations are established between tense and aspect in Likpakpaanl using the example in (130).

(130) Binlu nàn bì kìr màngù.
B. PST IPFV pluck 4-mango
'Binlu was plucking a mango.'

In 131, the verb enters into multiple Agree with both T^0 and Asp^0 with matching goals (see Hiraiwa, 2001; Carstens, 2001). Both T^0 and Asp^0 only enter Agree relationship with the verb in its base-generated position and license it for tense and aspect interpretation. The verb does not raise to T or AspP.

(131)



Going by the Agree analysis proposed, it is predicted that the habitual head, **nì** precedes the verb since it does not move, but that is not born since the resulting sentence is illicit (132-b).

(132) a. Naakol ŋàŋ-**nì** sìmà.
N. measure-HAB 3.SG-groundnut
'Naakol measures groundnut.'

b. *Naakol **nì** ŋàŋ sìmà.
N. HAB measure 3.SG-groundnut
Intended: 'Naakol measures groundnut.'

The marking of the habitual aspect of Likpakpaanl can be compared to Dagbani (Issah, 2015, 35), and Gurene (Atintono, 2013, 105) which mark the progressive aspect using post-verbal particles. In (133-a) and (133-b), the verbs preceded the imperfective markers. Following the derivational approach to structure building suggests that the verbs **dì** 'eat' and **sige** 'sit' are in derived positions.

(133) a. Chentiwuni dì-rì yìríŋ (Dagbani)
C. eat-IPFV careless
''Chentiwuni eats carelessly.'

b. A sige-ri la kulega bɔbe-ra kinka (Gurene)
3SG go.down-IPFV FOC river tie-IPFV stalks
'S/he has been going to the riverside to tie stalks.' (habitual reading)

The distribution of the habitual marker in Likpakpaanl supports the claims by Krifka (1995) and Boneh and Doron (2013) that habituality expresses as a subtype of genericity. According to Krifka (1995), the sentence in 134 lends itself to two different interpretations. The first implies that every after-dinner Mary usually smokes or that Mary smokes, usually after having her dinner.

(134) Mary smokes after dinner.

In their analysis of Habituality in English, Boneh and Doron (2013) propose the projection of a habitual phrase (HabP), which is selected by the AspP. They contend that HabP is not the realisation of the imperfective aspect as is widely held (cg. Comrie, 1981) but a projection on its own that expresses how events iterate in alternatives to the actual world. AspP, on the other hand, hosts imperfective and perfective aspects. Going by this assumption, I argue that there is a Habitual Phrase in Likpakpaanl with an EPP feature that triggers head movement of the verb to right-adjoin to the Hab^0 to form the complex head. The motivation for assuming HabP is the fact that the habitual marker can combine with the progressive in one sentence, as shown in (135).

(135) [TP Naakol [$_{\text{AspP}}$ bì [HabP ŋàŋ$_i$-**nì** [VP t$_i$ sìmà.]]]]
N. IPFV measure-HAB 3.SG-groundnut
'Naakol is measures groundnut.' (habitual reading)

3.7.1 Interim summary

This section examined the syntactic licensing of tense and aspect in Likpakpaanl and argued that tense and aspect morpho-syntactically marked using particles and, in some cases, enclitics attached to the verb stem. The data showed that tense particles **bà, nàn fè** and **gà** are the heads of maximal projections of TP that occupy a single slot in the clausal spine. They differentiate temporal relations of time, such as immediate, remote, and yesterday's past and future. The discussion further demonstrated that while the perfective aspect is morphologically null, the imperfective and habitual aspects also project an AspP and are marked by particles. I also proposed that tense and aspect are licensed via an Agree relationship established between the unvalued verbal feature that probes upwards in its c-command domain for a matching goal (T-head or aspect-head), with which it Agree to license a past, future, habitual, imperfection or perfective interpretation on the verb.

3.8 Syntax of Likpakpaanl A-bar Constructions

This subsection provides a comprehensive account of the syntax of wh-questions in Likpakpaanl. It investigates the grammatical properties of wh-phrases and lays the foundation for the discussion of sluicing and fragment answers in subsequent discussions. The section characterises the formation of both in-situ and ex-situ wh-questions, showing the asymmetry between subject and non-subject wh-questions and between wh-questions and their answers. I also examined the various strategies employed in deriving ex-situ wh-movement in Likpakpaanl. I adopted a Minimalist analysis where wh-movement to the left periphery is driven by Agree and Move operations.

Cross-linguistic studies have classified wh-phrases into semantic categories and morphological types. Specifically, Heine et al. (1991) identified seven common semantic categories for wh-phrases after surveying 14 typologically diverse languages. These categories relate to person, object, space, time, quantity, cause and purpose. This section provides an overview of and the distribution of Likpakpaanl wh-phrases. Table 3.6 summarises the inventory of Likpakpaanl wh-phrases categorised by semantic class.

Table 3.6 Overview of Likpakpaanl Wh-phrases

Wh-phrase	**Semantic Category**	**Gloss**
ŋmà	person	who
bà	thing	what
bàŋà	reason	why
làchè	place	where
ìŋà	amount	how much
bàdáál	time	when
kínyé	manner	how

(Issah and Acheampong, 2021, 35)

Employing the interrogative pronoun inventory outlined in Table 3.6, the rest of the section examines the syntactic derivation of wh-questions in Likpakpaanl. Three main wh-question formation strategies are in-situ wh-phrases, wh-fronting, and clefted wh-constructions. Data includes examples of each strategy in both matrix and embedded clauses. This analysis elaborates on the variable morpho-syntactic mechanisms for deriving wh-interrogatives from declarative constructions.

3.9 Ex-situ Wh-Questions

Wh-phrases in ex-situ constructions can occur in both matrix and embedded clauses, where a focus particle optionally follows them. I argue that the instance of ex-situ wh-questions in main clauses in Likpakpaanl manifests overtly in at least two distinct ways; they can be extracted to intermediate positions in the clause: partial wh-movement and long-distance ex-situ movement. Ex-situ wh-questions in matrix clauses extract the wh-phrase from its canonical position to the left periphery of the clause, as the examples in (136) illustrate.

(136) a. Ŋmà$_i$ (lé) Sùlà géé t$_i$?
who FOC S. love
'Who does Sùlà love?'

b. Bà$_i$ (lé) Sààjà nàn dì t$_i$ lù n-nyùn?
what FOC S. PST take fetch 5.SG-water
'What did Sààjà use to fetch water?'

c. Ìŋà$_i$ (lé) ù-jà gbààn pàn Wáápù t$_i$?
how.much FOC 1.SG-man DEF pay.PFV W.
'How much did the man pay Wáápù?'

In intermediate focus movement constructions in Likpakpaanl, the wh-phrase moves to an intermediate position and is optionally focus-marked, leaving a trace in its base-generated site. The examples in (137) show the wh-phrase extracted from its canonical position to Spec-FocP in the embedded clause.

(137) a. Bùndì nàn lén kè bà$_i$ (lé) Kòjò nàn bùn t$_i$?
B. PST say COMP what FOC K. PST plant
'What did Bundi say that Kojo planted?'

b. Kumbol dàk kè ŋmà$_i$ (lé) Nàntob bà tér t$_i$?
K. think.PFV COMP who FOC N. PST help
'Who does Kumbol think Nantob will help?'

(138) a. Jàbààb ŋùn kè bà-dààl$_i$ (lé) Nyoja yòòr ù-pìì t$_i$?
J. hear.PFV COMP what-day FOC N. take.PFV 1.SG-woman
'When did Jàbààb hear that Nyoja married.'

b. Neina bì lèn kè làche$_i$ (lé) ì-ŋùò gbààn wɔŋ t$_i$?
N. IPFV say COMP where FOC 6.PL-goat DEF lost
'Where is Neina saying the goats got lost?'

3.9.1 Long-distance wh-movement

In addition to intermediate wh-movement, Likpakpaanl also permits wh-extraction from its base-generated position inside an embedded clause into the left periphery of a matrix clause. Example (139) indicates long-distance extraction where the non-subject wh-element undergoes successive cyclic movement through the Spec-CP position before reaching its final landing site in Spec-FocP of the matrix clause (Chomsky, 1977, 2000). These examples show that in long-distance wh-movement, the base position is left with a trace. The presence of an intermediate trace in the specifier position of the complementiser phrase (CP) indicates the path of the successive cyclic movement of the wh-phrase from the embedded clause before it reaches its final landing site in the specifier of the matrix focus phrase.

(139) a. Ŋmà$_i$ (lé) Waapu lèn t_i ké Kulma gà tér t_i?
who FOC W. say.PFV COMP K. FUT help
'Who does Waapu say Kulma will help?'

b. Bà$_i$ (lé) Jàbààb dàk t_i ké Neina fè kìn t_i?
what FOC J. think COMP N. HEST.PST fry
'What does Jàbààb think Neina fried (yesterday)?

Contrary to what we see in non-subject extraction, subject wh-phrases do not leave traces in their base positions but resumptive pronouns. Let us now consider the case of long-distance movement in (140). The wh-phrase **bà** 'what', originally merged in spec-TP is A-bar moved to Spec-FocP due to EPP features on Foc-head and also the need to satisfy Chomsky (1981)'s 'Extended Projection Principle [EPP]', which requires that Spec-Foc to be filled. A resumptive pronoun is inserted at the base generated position of the moved subject wh-element in the embedded clause rather than a trace, contrasting with the behaviour of non-subject wh-phrases in long-distance dependencies.

(140) a. [$_{FocP}$ Bà$_i$ (lé) [$_{TP}$ Mpopiin lèn [$_{CP}$ t_i kè [$_{TP}$ nì$_i$ nàn
what FOC M. think.PFV COMP RP PST
wìì lì-bùù-l gbààn?]]]]
break 7-pot-7 DEF
'What did Mpopiin say (that it) broke the pot?'

b. [$_{FocP}$ Ŋmà$_i$ (lé) [$_{TP}$ Binlu dàk [$_{CP}$ t_i kè [$_{TP}$ ù$_i$ nàn tìì
what FOC B. say.PFV COMP RP PST give
Kunji bànchì?]]]]
K. 3.SG.cassava
'Who did Binlu think (that he/she) gave Kunji cassava?'

Following Acheampong (2024), I assume that the obligatory presence of resumptive pronouns in embedded subject positions in Likpakpaanl is due to the presence of an EPP feature, which necessitates that the subject position be overtly filled. As a result of this EPP feature, leaving a trace after extracting the embedded subject would violate the structural requirement for an overt subject. However, extracted matrix subject wh-phrases bind traces rather than resumptive pronouns since the presence of resumptive pronouns in such local matrix subject extraction will violate McCloskey (1990)'s Highest Subject Constraint (HSC) as stated in 141 McCloskey (1990, 2002).

(141) **Highest Subject Constraint** (McCloskey, 1990, 77-8)
'The highest subject of a clause cannot be occupied by a resumptive pronoun... however, resumptive pronouns appear freely in the subject position of embedded clauses, both finite and non-finite.'

Thus, the asymmetry between trace and resumptive pronouns can be accounted for by positing that Likpakpaanl requires resumption for extracted embedded subjects to satisfy EPP but traces for extracted matrix subjects to satisfy HSC. Due to these constraints on embedded versus matrix subjects, resumptives and traces are mutually exclusive.

In Likpakpaanl, resumptives are sensitive to island constraints and do not serve to repair island violations. This suggests that resumption in the language patterns with a movement-based analysis and not a base-generation as proposed by Shlonsky (1992); McCloskey (2002). The A-bar dependencies found in complex noun phrase islands, whether involving resumptives (230-b) or gaps (230-c), also exhibit sensitivity to these island constraints.

(142) a. *[$_{\text{FOCP}}$ mà$_i$ (lè) Chàtí tùk [$_{\text{DP}}$ tìbŋùnlkààr [$_{\text{CP}}$ kè [$_{\text{DP}}$ ù$_i$
who FOC C. tell.PFV rumour COMP RP
ŋùn t$_i$]]]?
hear.PFV
Intended: 'Who has Chatí told a rumour that he heard?'

b. *[$_{\text{FOCP}}$ Bà$_i$ (lè) Amà ŋméé [$_{\text{DP}}$ kí-gbààŋ [$_{\text{CP}}$ kè [$_{\text{TP}}$ ù-bò
what foc A. write.PFV 2-book COMP 1-child
kàrn nì$_i$/t$_i$]]]?
read.PFV RP
Intended: 'What has Ama written a book that a child has read (it)?'

(Acheampong, 2024, 38)

3.9.2 Copular Constructions

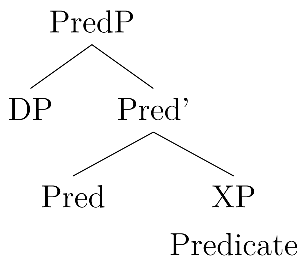
The literature on copular clauses distinguishes between predicational copular clauses (143-a) and equative copular clauses (143-b). There is also the specificational clause (143-c), a sub-type of the equative clause (cf. Mikkelsen, 2005; Bondaruk, 2013, 48).

(143) a. John is fat.
b. She is Mary.
c. The winner is Susan.

Bondaruk (2013) and Mikkelsen (2005) argue that both predicative and specifically copular clauses are subject-initial clauses because they both have one predicative expression and one referential expression. The subject of the predicational clause is referential, comprising the set of individuals or entities. In contrast, the subject of the specifications or inverse copular clause like (143-c) is the canonical predicative complement **the winner**, while the predicate complement is the canonical referential element **Susan**. The analysis by Bondaruk (2013) assumes that despite exhibiting an identical surface structure, equative and predicational clauses elicit different semantic interpretations.

Within the literature, two competing hypotheses have been proposed to account for the syntactic representations of copular clauses. The first is the Specifier Hypothesis (SH), put forth by Stowell (1981) in his dissertation, which posits that the subject in copular clauses originates in the specifier position of a predicative expression XP, which can be realised as a noun, adverbial or prepositional phrase. The second is the Functional Category Hypothesis (FCH) proposed by Bowers (1993, 2001) and supported by scholars such as Mikkelsen (2005) and Bondaruk (2013). In this study, I adopt the latter approach, which stipulates that the copular is the head of a functional category, predicate phrase (PredP), connecting a subject to a predicate and selecting the predicative expression XP as its complement and the subject DP as its specifier, as depicted in (144).

(144)



(Bowers, 1993, 595)

According to Bowers (1993), the subject and predicate of copular constructions form a Predication Phrase (PredP)[9] where the functional Pred head takes the predicate XP as its complement, and the subject is base-generated in Spec-PredP, Adopting the PredP approach is important because it is consistent with the derivation of other functional categories. Bowers (2001, 1993) notes that Pred^o like little *v* mediates between their external arguments (subjects) in their specifiers and predicates in their complements (Chomsky, 2000, 2001). Bowers (2001, 1993), and Chomsky (2000, 2001), among others, have proposed that predicates, both verbal and non-verbal, invariably project functional projections that license external arguments, with subject DPs base-generated in the specifiers of functional heads such as Spec-vP, Spec-VoiceP, Spec-TP, and Spec-PredP.

Moreover, Bowers (2001, 1993)'s FCH offers some advantages over the SH regarding the analysis of Likpakpaanl copular constructions. The PredP configuration structurally encodes the syntactic and semantic predicational relation between a subject and a predicate, such that the PredP provides a structural representation of the predicational relation between the complement of the Pred head and the argument in its specifier. It also captures the semantic notion of predication. Furthermore, the PredP structure has gained acceptance in the literature and aligns with the most recent works on copular clauses within the framework of the Minimalist Program (Bondaruk, 2013; Mikkelsen, 2005; Bowers, 1993, 2001).

In Likpakpaanl, the copular **yè** links the subject of a clause with a subject complement. A predicative copular clause like (145) contains the referential DP-subject, the copular **yè**, and a predicative complement providing additional information about the subject.

(145) Bilisun **yè** ù-kpààl.
B. COP 1.SG-farmer
'Bilisun is a farmer.'

In (145), the referential DP **Bilisun**, which is also the external argument of the copular, is base-generated in Spec-PredP. The Pred^0 **yè** selects the DP **ùkpààl** as its predicative complement. The complement of the copular in Likpakpaanl can be a noun phrase, an adjective phrase or a noun phrase with a focus marker as illustrated in (146-a), (146-b) and (147-b)[10].

[9]Bowers (1993), designates the functional head governing a predicate complement as *Pr.* I used PredP to distinguish the fact that predication in Likpakpaanl does not take a VP complement, but an AP, NP, PP, and even an NP that is focused.

[10]The copular, like lexical verbs in Likpakpaanl, does not inflect for number with the subject as shown in (146-a) where we have a plural subject but the copular morphologically remains the same.

(146) a. Kofi nì Ntekimi **yè** jɔ-tììb.
K. and N. COP friend-4
'Kofi and Ntekimi are friends.'

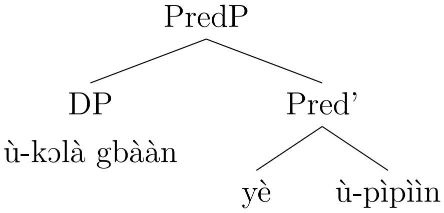
b. Ù-kɔlà gbààn **yè** ù-pìpììn.
1-fowl DEF COP 1-white
'The fowl is white.'

(147) a. [Q:] Kofi nì Ntekimi **yè** bà?
K. and N. COP what
'What are Kofi and Ntekimi?'

b. [A:] Kofi nì Ntekimi **yè** jɔ-tììb là.
K. and N. COP friend-4 FOC
'Kofi and Ntekimi are FRIENDS.'

The representation of (146-b) is illustrated in (148) and shows that the PredP analysis accounts for the syntactic configuration and relationship between subjects and predicates in copular constructions in Likpakpaanl.

(148)



In addition to canonical copular constructions such as (149-a), there is also a distinct construction type referred to as Inverse Copular Constructions (ICC), a terminology introduced by Moro (1997). These constructions are characterised by a reversal of the typical Subject-Copula-Predicative (SPC) order, whereby the predicative DP linearly precedes the canonical subject, as exemplified in (149-b):

(149) a. Mathew is the English teacher.
b. The English teacher is Mathew.

In Likpakpaanl, such ICC constructions are prevalent and are even used in sluicing constructions as I shall demonstrate in 5.3.3. Consider the inverse copular sentence in (150-b) where the predictive DP raises above the subject DP of the predicate clause.

(150) a. Obed nàn yè ù-kpán.
O. PST COP 1.SG-hunter
'Obed was a hunter.'

b. Ù-kpán lé Obed nàn yè .
1.SG-hunter FOC O. PST COP
' OBED was a hunter.'

c. [$_{\text{FocP}}$ Ù-kpán$_i$ lé [TP Obed$_j$ nàn [$_{\text{PredP}}$ t$_j$ yè t$_i$]]]
1.SG-hunter FOC O. PST COP
'OBED was a hunter.'

The derivation of regular copular sentences like (150-a) is straightforward; **Obed** as the subject is base-generated in Spec-PredP and **ù-kpán** 'hunter' occurs as the predicative complement of the Pred head **yè**. However, the inverse copular construction in (150-b) involves more derivational operations. The predicative DP **ù-kpán** 'hunter' originates in the complement position of PredP, with **Obed** as the subject in Spec-PredP. After TP is merged, the EPP feature on T triggers the A-movement of **Obed** from spec-PredP to Spec-TP. In the next stage, a FocP projection is merged above TP, and the Foc head with an interpretable but unvalued focus feature that probes its c-command domain for a matching value, entering into an Agree relation with the predicative DP 'hunter' with uninterpretable but valued focus feature. Following this Agree operation, the unvalued feature on Foc is valued, and the EPP feature triggers the A-bar movement of the predicative DP to Spec-FocP, deriving the observed word order in (150-c).

In his analysis of Inverse Copular Constructions (ICC), Hatakeyama (1997) proposes a series of movement operations to account for sentences like (151-b). He contends that the predicate nominal (NP_{PRED}), 'the culprit' in the canonical sentence (151-a), moves from its base-generated vP-internal position to Spec-CP. At the same time, the copular 'is' raises to C-head, and then the subject nominal (NP_{SUB}) 'John' in the canonical sentence moves to Spec-TP. Hatakeyama (1997) assumes the structural representation in (151-c) to account for the derivation of (151-b) (example (151-c) taken from Hatakeyama, 2004, 204).

(151) a. John is the culprit.
b. The culprit is John.
c. [CP The culprit [$_{\text{C'}}$ is$_j$ [TP John$_k$ [T' t$_j$ [VP t$_k$ [V' t$_j$ t$_i$]]]]]]

(Hatakeyama, 1997, 26)

Hatakeyama (1997) argues that Japanese inverse copular sentences are derived syntactically similar to English matrix wh-questions. He used the examples in (152), where (152-b) represents an ICC derived from the canonical copular sentence in (152-a). A crucial observation is that replacing the particle **wa** attached to the subject nominal in the ICC with **ga** renders the sentence ungrammatical (152-c) (Hatakeyama, 1997, 165-6).

(152) a. Taroo-ga hannin da.
Taro-NOM culprit COP
'Taro is a culprit.'

b. Hannin-wa Taroo da.
Hannin-TOP Taroo COP

c. *Hannin-ga Taroo da.
Hannin-NOM Taroo COP

Hatakeyama (1997) notes that this asymmetry between **wa** and **ga** in Japanese reflects a fundamental difference between subject and predicate nominals in ICS. Adopting the widely accepted view that NP-**ga** is licensed in Spec-IP, while NP-**wa** occupies a position higher than IP (e.g., Spec-CP), and he assumes that NP-**wa** in ICC is licensed in Spec-CP. He, therefore, contends that Japanese ICC and their English counterparts are derived via wh-movement to Spec-CP. With this background, let us consider the third wh-fronting strategy employed in Likpakpaanl, which involves cleft constructions. A cleft construction (or cleft sentence) is a syntactic structure that focuses on a particular sentence constituent. It involves a bi-clausal structure comprising a matrix copular clause and a relative. The matrix clause contains a copular verb with a predicative argument (Lambrecht, 2001).

According to Kiss (1999) and Hartmann and Veenstra (2013), cleft constructions exemplify a focus projection with the cleft constituent in Spec-FocP. Thus, in (153), the cleft phrase 'only chicken wings' expresses the focus of the clause in which it appears. Hartmann and Veenstra (2013) note that the sentence in 153 can either be an answer to an object question (new information focus) or a correction to a proposition where 'Peter' ordered something else for lunch.

(153) It was **only chicken wings** that Peter ordered for lunch.

(Hartmann and Veenstra, 2013, 1).

In Likpakpaanl, cleft and wh-interrogatives can be answered using cleft answers, as shown by the question-answer pairs in (154-a). It is also important to note that in cleft interrogatives like (154-a), the focused constituent must be extracted to the left periphery, or else the resulting sentence will be ungrammatical (154-c).

(154) a. [Q:] Ni yè í-jàn$_i$ lé Mayen bì kìn t$_i$ àà?
it COP 6.PL-fish FOC M. IPFV fry INTER
'Is it fish Mayen is frying?'

b. [A:] Eén, ni yè í-jàn lé Mayen bì kìn.
yes it COP 6.PL-fish FOC M. IPFV fry
'Yes, it is fish that Mayen is frying.'

c. [A1:] *Ni yè Mayen bì kìn í-jàn là.
it COP M. IPFV fry 6.PL-fish FOC
Intended: "Mayen is frying fish."

Apart from cleft interrogatives, in-situ wh-question can be answered using a parallel in-situ response as in (154-b) or with a cleft answer (155-c) where the focused constituent is moved from its base position as the internal argument of the verb **kìn** ‘fry’ to the complement of the copular.

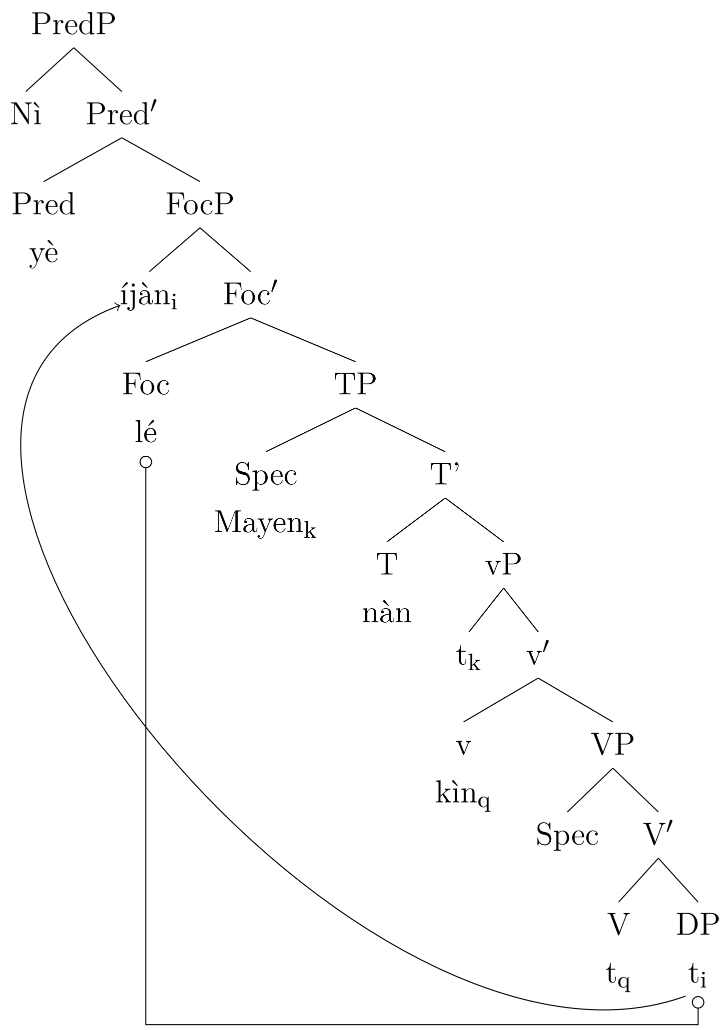
(155) a. [Q:] Mayen bì kìn bà?
M. IPFV fry what
‘What is Mayen frying?’

b. [A:] Mayen bì kìn í-jàn là?
M. IPFV fry 6.PL-fish FOC
‘Mayen is frying fish’

c. [A1:] Ni yè í-jàn$_i$ *(lé) Mayen bì kìn t$_i$.
it COP 6.PL-fish FOC M. IPFV fry
’It FISH that Mayen is frying?’

Adopting a focused analysis for Likpakpaanl cleft constructions predicts that focus movement and clefts are related since the cleft phrase moves to the specifier of a focus projection in the left periphery in the same fashion that ex-situ focus movement is derived. There is, therefore, a syntactic partition of the cleft construction into a focus part (the cleft phrase) and a presupposition (the cleft clause), as argued by Kiss (1998). I propose that Likpakpaanl cleft constructions involve a focus movement where the DP first enters into an Agree relation with Foc^0 in its base-generated position, thereby receiving focus interpretation. The EPP feature on Foc-head then moves the focused constituent to Spec, FocP, where it becomes ‘Criterially frozen’ (since all its features are valued) and no longer eligible for further movement (see Rizzi, 2006; Chomsky, 1977). The representation (156) shows the derived structure for the sentence in (155-c).

(156)



The data has shown that ex-situ wh-questions in Likpakpaanl can occur in long-distance, intermediate movement, or by a cleft strategy. The rest of this chapter examines in-situ wh-questions and the asymmetries in wh-question formation. I also propose a syntactic analysis and provide evidence that fronted wh-phrases undergo focus movement in clefts and long-distance questions.

3.10 In-situ Wh-Questions

This section briefly examines the case of in-situ wh-questions in Likpakpaanl. I discuss in-situ wh-questions in matrix and embedded clauses

3.10.1 Wh-in-situ in Matrix Clauses

Almost all wh-expressions (subject and non-subject) in Likpakpaanl can occur in-situ in main clauses. The examples (157-a), (157-b) illustrate that simple and complex interrogative words (157-c) may appear without any overt morphological focus marking.

(157) a. **Ŋmà** nàn chùù ì-jàn gbààn?
Who PST catch 6.PL-fish DEF
‘Who caught the fish?’

b. **Bà** bì wù kì-sàà-k gbààn nì?
what IPFV burn 11.SG-farm-11 DEF LOC
‘What is burning on the farm?’

c. Bù-sùb **bù-là-bù** nàn kpà mbóón?
9-tree 9-which-9 PST have 5.shade
’Which tree had shade?’

Shifting the focus to object wh-expressions, I show that both subject and non-subject wh-arguments may surface clause-internally in ditransitive and mono-transitive clauses. In (158-a), **ŋmà** ’who’ functions as the direct object of the verb ’give’, following the verb and preceding the indirect object. In (158-b) too, the wh-phrase **bà** ’what’ is the direct object of the ditransitive verb and appears between the indirect object and the clause-final temporal adverb. The positioning of **ŋmà** and **bà** as non-peripheral arguments in these structures illustrates that Likpakpaanl permits clause-internal wh-in-situ for both subject and non-subject interrogative phrases.

(158) a. Nantob fè tìì **ŋmà** kpààchà fènnà?
N. HEST.PST give who cutlass.4 yesterday
‘Who did Nantob give a cutlass yesterday?’

b. Alima gà tìì ù-bò gbààn **bà** dìn?
A. FUT give 1-child DEF what today
‘What will Alima give the child today?’

Similar observations can be made in mono-transitive constructions, like (159-a), with the direct object or the adjunct occurring clause-finally.

(159) a. Wumbei nàn dàà **bà**?
W. PST buy what
’What did Wumbei buy?’

b. Mpopiin gà tér **ŋmà**?
M. FUT help who
’Who will Mpopiin help?’

In Likpakpaanl, not all wh-adjunct phrases may also appear clause internally. The morphologically complex wh-expressions for ‘reason’ **bàŋà** and ‘manner’ **kìnyè** in Likpakpaanl are asymmetrically distributed such that the former obligatorily appears in the left periphery of the clausal (160-a) and cannot be focused (160-c) while the latter must occur clause-internally following the verb phrase at the right edge of the clause (160-d). When the manner adjunct **kìnyè** is displaced to the left edge of the clause, it yields an ungrammatical sentence as shown in (160-e) is ungrammatical.

(160) a. **Bà-ŋà** Kwame dàà lóór gbààn?
what-do K. buy.PFV 3.SG.car DEF
'Why did Kwame sell the car?'

b. *Kwame dàà lóór gbààn **bà-ŋà**?
K. buy.PFV 3.SG.car DEF what-do
Intended: 'Why did Kwame sell the car?'

c. ***Bà-ŋà** lé Kwame dàà lóór gbààn ?
what-do FOC K. buy.PFV 3.SG.car DEF
Intended: 'Why did Kwame sell the car?'

d. Kunji sèè n-kòòn gbààn **kìnyè**?
3PL kill 1-guinea-fowl DEF how?
'How did you kill the

K. burn.PFV 5-charcoal DEF how

'How did Kunji burn the charcoal?'

e. ***Kìnyè** lé Kunji sèè n-kòòn gbààn?
how FOC K. burn.PFV 5-charcoal DEF
Intended 'How did Kunji burn the charcoal?'

The use of interrogative expression **kínyé** in Likpakpaanl, results in semantic ambiguity because it allows for two or more possible interpretations (Cysouw, 2004; Issah and Acheampong, 2021). Thus, the question in (161-a) can be answered by stating the instrument (161-b) that was used for killing the sheep or how it was slaughtered (161-c).

(161) a. Nì kù ù-pìì gbààn **kínyé**?
2PL kill.PFV 1-sheep DEF how
'How did you kill the sheep?'

b. [A:] Tì dì kì-jù-k lé kɔr ù.
3PL take 11-knife-11 FOC kill.PFV 1.it
'We used a knife to slaughter it?'

c. [A1:] Tì kù ù màlá-màlá là.
3PL kill.PFV 1.it quick.quick FOC
'We slaughtered it quickly.'

The occurrence of **bàŋà** 'why' in left clause-peripheral positions in Likpakpaanl (160-a) is topologically not surprising because cross-linguistic data from African and Romance languages have shown that some wh-adjuncts, like 'why', are restricted from appearing clause-internally (cf. Kandybowicz and Torrence, 2017; Torrence and Kandybowicz, 2015; Issah, 2015; Issah and Acheampong, 2021; Adams, 2004; Rizzi, 2001). Issah (2015), for instance, shows that in Dagbani, wh-adjuncts like 'why' and 'how' must obligatorily undergo extraction to the left periphery for the construction to be grammatical (162).

(162) a. Bò-zùγù$_i$ kà bɛ kú kpáŋ máá t$_i$?
what-head FOC 3PL kill.PFV guinea.fowl DEF
'Why did they kill the guinea fowl?'

b. *Bɛ kú kpáŋ máá bò-zùγù?
3PL kill.PFV guinea.fowl DEF what-head
Intended: 'Why did they kill the guinea fowl?'

c. Wúlà$_i$ kà á dí bíndírígú máá t$_i$?
how FOC 2SG eat.PFV food DEF
'How did they eat the food?'

d. *Á dí bíndírígú máá Wúlà?
2SG eat.PFV food DEF how
Intended: 'How did they eat the food?'

(Issah, 2015, 23); (Issah and Acheampong, 2021, 45).

Moreover, Kandybowicz et al. (2021) argue that in Ikpana, the wh-adjunct 'why' must not equally occur clause internally, else the resulting will be ungrammatical as we see in (163-b).

(163) a. **Mɛ** Kofí ɔ-zá a-zaì ɛ-ta?
what Kofi SM-cook.PST CM-beans CM-means/manner
'How did Kofi cook beans?'

b. *Kofí ɔ-zá a-zaì **mɛ** ɛ-ta?
Kofi SM-cook.PST CM-beans what CM-means/manner
'How did Kofi cook beans?

(Kandybowicz et al., 2021, 78).

This restriction is also attested in Italian Rizzi (2001) and Dutch Corver (1990). In his analysis of the Italian wh-adjunct **perché** 'why',Rizzi (2001) proposes a base-generation account noting focus cannot precede the adverbial adjunct even though it can precede focus elements to suggest that it occupies a position distinct from and higher than focus. Consequently, Rizzi (2001) posits that 'why' interrogative phrases in Italian occupy positions higher than the Focus Phrase projection and above positions targeted by all other moving wh-elements.

(164) a. Perché QUESTO avremmo dovuto dirgli, non qualcos'altro?
why this 2PL.would due tell.him not something.else
'THIS we should have said to him, not something else?'

b. *QUESTO Perché avremmo dovuto dirgli, non qualcos'altro?
this why 2PL.would due tell.him not something.else
'THIS why we should have said to him, not something else?'

(Rizzi, 2001, 294)[11]

I have shown that unlike non-subject wh-phrases, which allow optional focus particle marking when extracted ex-situ, subject wh-phrases in Likpakpaanl cannot co-occur with a focus particle, even optionally. This contrast suggests subject wh-elements are uniformly base-generated in Spec-vP and raise to Spec-TP for EPP reasons, while non-subjects can undergo ex-situ displacement. However, an asymmetry arises in that answers to subject questions obligatorily contain the focus particle post-subject. So, while subject wh-phrases remain in-situ, their responses require focus particle marking.

3.10.2 Wh-in-situ in Embedded Clauses

Apart from the main clause, wh-phrases in embedded clauses can occur within an embedded Complementiser phrase (CP). Such embedded wh-questions are introduced by the overt C-head **ké**. All wh-phrases except **baŋà** 'why' may occur in situ in embedded complement clauses, as evidenced by the subject expressions in (165-a), direct objects in (165-b), and adjuncts in (165-c) to (165-e). This clause-internal positioning contrasts with the ungrammatical example in (165-f), which confirms that, like in matrix clauses, **bà-ŋà** is prohibited from surfacing non-peripherally in embedded clauses as well.

(165) a. Liyab bà lèn ké **ŋmà** nàn gbíí bànchì gbààn?
L. PST say COMP who PST dig 3.SGcassava DEF
'Who did Liyab say harvested the cassava?'

b. Liyab bà lèn ké Naakol gà gbíí **bà** dìn?
L. PST say COMP N. FUT dig what today
'What did Liyab say Naakol will harvest today?'

c. Liyab bà lèn ké Naakol gà gbíí bànchì gbààn **bà-dààl**?
L. PST say COMP who PST dig 3.SG.cassava DEF what-day
'When did Liyab say Naakol will harvest the cassava?'

d. Liyab bà lèn ké Naakol gà gbíí bànchì gbààn **kìnyè**?
L. PST say COMP who PST dig 3.SG.cassava DEF how
'How did Liyab say Naakol will harvest the cassava?'

[11]The interlinear glossing is mine.

e. Liyab bà lèn ké Naakol gà gbíí bànchì gbààn **làchè**?
L. PST say COMP who PST dig 3.SG.cassava DEF where
'Where did Liyab say Naakol will harvest the cassava?'

f. *Liyab bà lèn ké Naakol gà gbíí bànchì gbààn **bà-ŋà**?
L. PST say COMP who PST dig 3.SG.cassava DEF why

Likpakpaanl allows wh-in-situ regardless of the depth of clausal embedding. For instance, (166) shows that interrogative expressions may occur even in deeply embedded contexts. (166-b) shows a wh-in-situ within double-clausal embedding suggesting that wh-in-situ is not constrained by structural depth.

(166) a. Liyab bà lèn ké Naakol nyì ké Timor gà gbíí **bà**?
L. PST say COMP N. know.PFV COMP T. FUT dig what
'What did Liyab say Naakol knows that Timor will harvest?'

b. Francis nàn dàk ké Kunji nyì ké Timor bàn **ŋmà**?
F. PST think COMP K. know.PFV COMP T. want who
'Who did Francis think Naakol knew that Timor was looking for?'

The discussion so far has focused on in-situ wh-questions in matrix and embedded clauses and showed that not all wh-phrases appear clause-internally. The data demonstrated that the wh-phrases for 'reason' **bà-ŋà** and 'manner' **kìnyè** are asymmetrically distributed, with the former restricted to the left-periphery of the clause and the latter occurring in the clause-internal position in the vp periphery. In the next section, I examine the asymmetries between wh-question and the answers.

3.10.3 Asymmetries in Wh-questions and the Answers

Likpakpaanl displays asymmetries between wh-questions and their corresponding answers. The data examined thus far has demonstrated that in-situ wh-phrases lack overt marking, while ex-situ wh-phrases may optionally bear focus. However, a crucial difference emerges with respect to the answers to wh-questions, both in-situ and ex-situ. The particle **lé** must obligatorily follow the answers (167-a); otherwise, the sentence is infelicitous (167-b).[12] This contrast reveals an asymmetry between the optional focus marking on fronted wh-phrases in questions and the obligatory presence of **lé** on their answers. The latter appears to be a general requirement when answering wh-questions in Likpakpaanl, independent of the

[12] In the absence of the focus particle, the sentence receives a simple affirmative interpretation rather than conveying new information focus.

position of the wh-phrase.

(167) Q: Ŋmà nàn chùù ì-jàn lì-mùà-l gbààn nì?
Who PST catch 6.PL-fish 7.SG-river-7 DEF LOC
'Who caught fish in the river?'

a. [A:] Nàgbíjà lé nàn chùù ì-jàn lì-mùà-l gbààn nì.
N. FOC PST catch 6.PL-fish 7.SG-river-7 DEF LOC
'NAGBIJA caught fish in the river.'

b. [A:] ?Nàgbíjà nàn chùù ì-jàn lì-mùà-l gbààn nì.
N. PST catch 6.PL-fish 7.SG-river-7 DEF LOC
'NAGBIJA caught fish in the river.'

The in-situ object wh-phrases cannot equally be focused by either **lè** or **là**, but an obligatory focus marker is required in the response. The choice of **lè** or **là**[13] does not depend on whether the verb is transitive or intransitive but instead on whether an overt constituent follows the particle in the same clause, as shown in (168-a) and (168-c).

(168) a. [Q:] Licho bì kàrn **bà**?
L. IPFV read what
'What is Licho reading?'

A: Licho bì kàrn kì-gbàŋ **là**.
L. IPFV read 13.SG-book FOC
'Licho is reading A BOOK today.'

b. [Q:] Taabi gà tér ŋmà dìn?
T. FUT help who today
'Who will Taabi help today?'

[13]This clause-final focus marker is assumed to be similar to the completive affix marker **-à**, which follows the verb as in (i-a) and (i-d).

(i) a. Kanambi nàn kùùr kì-sàà-k gbààn.
K. PST weed 11.SG-farm-11 DEF
'Kanambi weeded the farm.

b. Kanambi nàn kùùr-**à**.
K. PST weed.COMPLE
'Kanambi has weeded (it).'

c. Alima bà lù n-nyùn.
A. PST fetch 5.SG-water
'Alima fetched water.'

d. Alima bà lù-**yà**.
A. PST fetch.COMPLE
'Alima has fetched (it).'

However, when the verb ends with a vowel, the voiced palatal glide [j] is phonologically inserted between the verb and the suffix-**-à** to create a suffixed CV syllable form. Therefore, the occurrence of the particle **là** clause finally in focused constructions is seen as a reflex realisation of the completive marker.

c. [A:]Taabi gà tér Jagir **lè** dìn.
T. FUT help J. FOC today
'Taabi will help Jagir today.'

The asymmetry between wh-phrases and their responses observed in the Lichabol dialect of Likpakpaanl is not an idiosyncrasy of that variety alone but manifests in other Likpakpaanl dialects. For example, in Līnàjùúl, wh-questions likewise lack overt marking, yet answers require the focus particle **ń**. Bisilki (2022) notes that the focus particle **ń** may undergo homorganic assimilation and surface as **ḿ** and **ŋ** depending on the following sound169. Nevertheless, the omission of **ń** from a Līnàjùúl utterance does not result in ungrammaticality but rather leads to a neutral reading.

(169) a. Ŋmà (*ń) bì tárí kīyá?
who FOC IMPFV shout like.that
'Who is shouting like that?'

b. Makinyi *(ḿ) bì tárí kīyá.
M. FOC IMPFV shout like.that
'Makinyi is (the one) shouting like that.'

(Bisilki, 2022, 131)

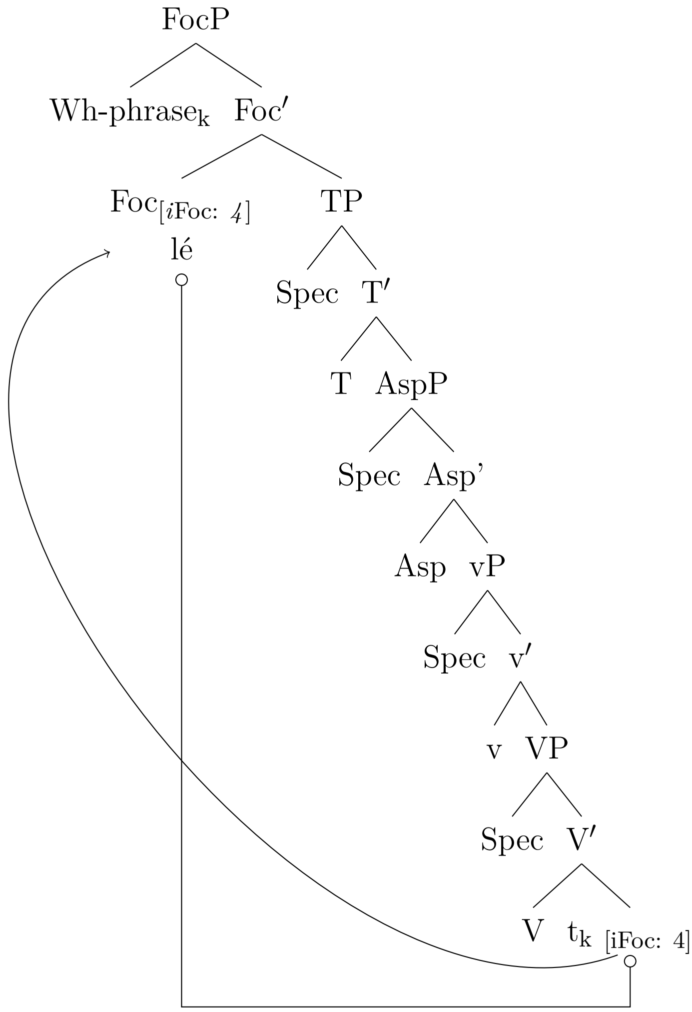
In Kusaal, a Mabia language, Abubakari (2016) also documents an asymmetry in focus-marking between interrogative phrases and their associated responses, while interrogative wh-phrases lack overt focus morphology, their answering counterparts are obligatorily focus-marked. A similar observation is made in Byali, an Oti-Volta language spoken in the Atakora mountains of northwestern Benin, that interrogative wh-phrases are categorically precluded from occurring with the focus marker (Reineke, 2007).

3.11 On the Syntax of Likpakpaanl Constituent Questions

This section presents a syntactic analysis of ex-situ and in-situ wh-questions in Likpakpaanl by systematically discussing the distribution of the particle **lé**, which is absent in in-situ wh-questions but optional in ex-situ constructions. Adopting a probe-goal approach, I show that the focus particle has an interpretable focus feature that contributes to the semantic interpretation of focus. However, it is unvalued (underspecified) and is valued through an Agree operation with a focused constituent with a matching valued feature. FocP also has an EPP feature that is activated in ex-situ focus movement. Under this analysis, the movement of the focused constituent to a Spec-FocP is driven by the EPP feature, and the focus particle's presence is seen as a morphological spell-out of the movement.

The ex-situ data examined in the work suggest that a wh-phrase can undergo extraction to the left periphery (Spec-FocP). I argue that the derivation of ex-situ wh-focus movement is licensed via Agree with a left-peripheral focus head with an interpretable focus but unvalued feature, while the focus constituent has a matching valued feature. Adopting the proposal of Mursell et al. (2022) for focus movement in Likpakpaanl, I posit that after feature valuation between the probe on Foc^0 and the goal on the wh-element, the EPP on the focus head attracts the wh-element to its specifier as represented in (170).

(170)



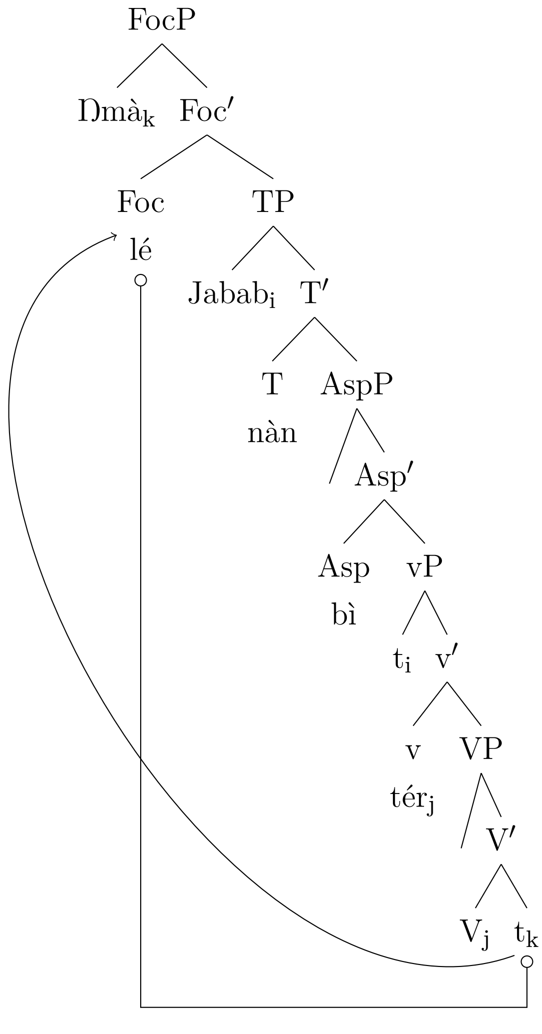
Applying the proposed analysis in (170) to the example in (171-a) derives the underlying structure in (171-b). The structure begins with an internal merge of the wh-phrase as the complement of the verb **tér** 'help'. The uninterpretable [uFoc] feature on Foc^0 searches its c-command domain for a goal and finds the wh-phrase. Following Pesetsky and Torrego (2007)[14], In the second phase of the

[14]According to Pesetsky and Torrego (2007) unvalued features [uVal] on a syntactic element always probe its c-command domain a goal with a matching valued feature [val]. When the probe successfully finds its goal, the unvalued features on the probe are valued by the goal.

derivation, the [EPP] feature on Foc^0 triggers the movement of the wh-phrase to Spec-FocP resulting in the spell out of the wh-phrase in Spec-FocP as shown in ((171-b)).

(171) a. Ŋmà$_i$ (lé) Jabab nàn bì tér t$_i$?
who FOC J. PST IPFV help
'Who was Jabab helping?'

b.



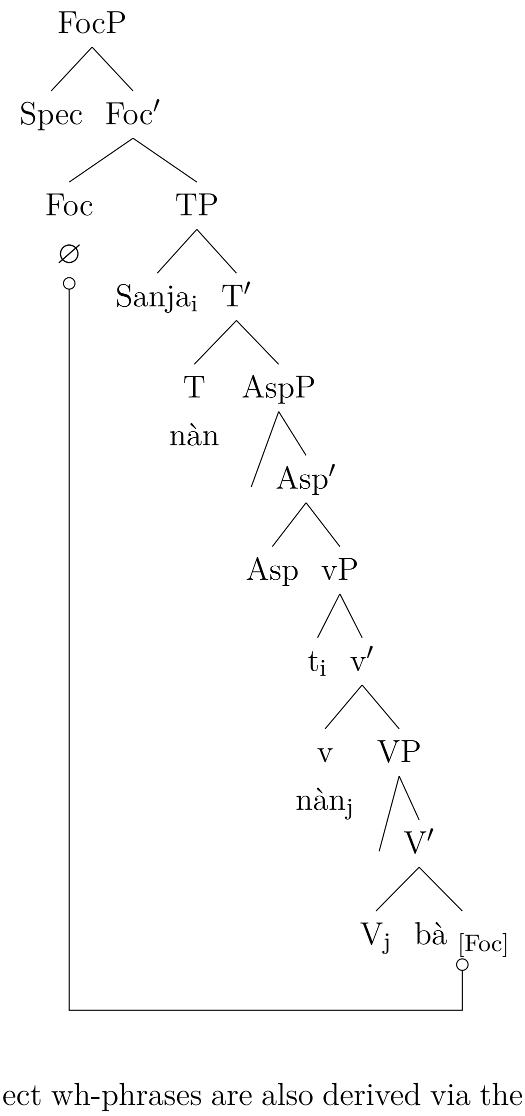
On the analysis of in-situ wh-questions, the data discussed demonstrated that in-situ wh-phrases in Likpakpaanl are not morphologically focus-marked, unlike their ex-situ counterparts, which allow optional focus marking. The in-situ wh-question in (157), repeated here in (172), exemplifies the fact.

(172) a. **Ŋmà** (*lé) nàn chùù ì-jàn gbààn?
Who FOC PST catch 6.PL-fish DEF
'Who caught fish in the river?'

b. Sanja nàn kìr **bà** (*lé/là)?
S PST pluck what FOC
'What did Sanja pluck?'

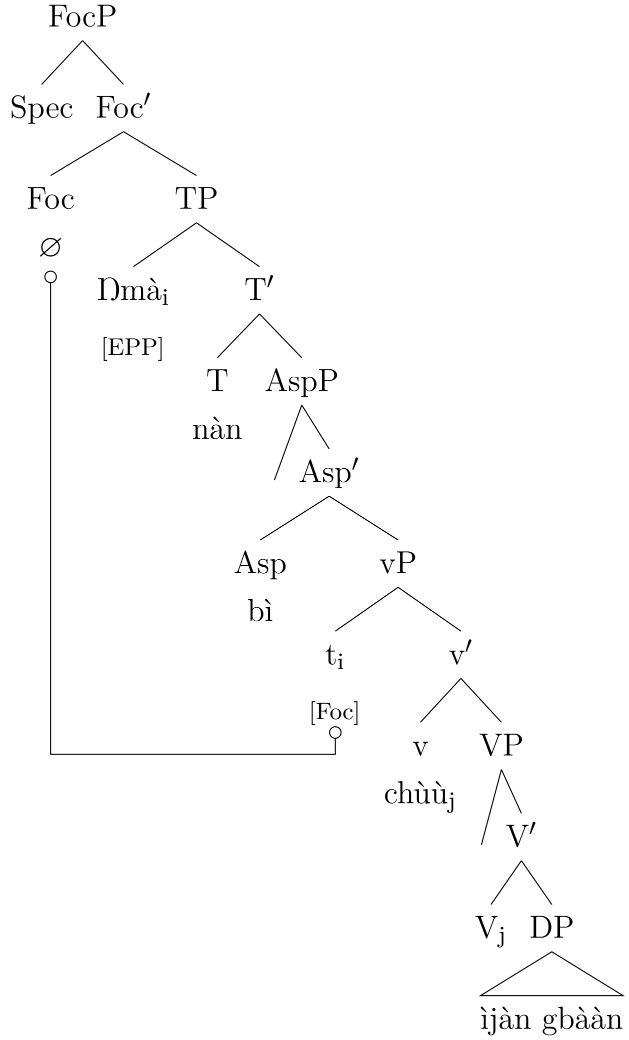
Thus, the absence of the focus marker in wh-questions (even optionally) requires investigating how a focus interpretation of the wh-phrase is derived. I argue that all in-situ wh-phrases, whether subject (172-a) or non-subject (172-b) in Likpakpaanl, are base-generated in their canonical positions and are licensed through an Agree relationship with the focus phrase (FocP), which projects above TP in the left periphery. Since there is no EPP feature to trigger the A-bar movement of the wh-phrase, the focus head is null. I propose that in-situ non-subject wh-phrases in Likpakpaanl are licensed via Agree with a left-peripheral focus head, as illustrated in (173).

(173)



Subject wh-phrases are also derived via the same Agree mechanism with FocP. Thus, the local subject wh-question in (172-a) is derived as shown in (174) as follows: The wh-phrase originating in Spec-vP undergoes movement from its base position to Spec-TP due to an obligatory EPP feature on T in Likpakpaanl. In the second step, Foc^0 probes its c-command domain and finds the matching goal, the wh-phrase **ŋmà**. Foc^0 then establishes an Agree relation with the wh-phrase in its base position in Spec-vP, satisfying Foc^0's uninterpretable focus feature and licensing a focus interpretation on the wh-phrase.

(174)



Empirical evidence examined so far indicates that it is impossible to attach the focus particle **lé** (even optionally) to the subject if it is 'focused' within a minimal clause (see (175-b)). Let me, however, state that this restriction of in-situ focus for subjects applies only to clause-bound A- bar-dependencies. Long-distance wh-subject movement is possible, and in such cases, the focused subject can also be optionally followed by the focus marker **lé** with a resumptive pronoun filling the base position of the moved subject (175-c).[15]

(175) a. Ŋmà bì kìn sìmà dìn?
who IPFV fry 3.SG.groundnut today
'Who is frying groundnuts today?'

[15]The use of RP in the base position does not indicate straight away if this can be taken as evidence of movement or base generation. McCloskey (1990, 2002) and Koopman (1984) have shown that while resumption and gaps can be used interchangeably in Palestinian Arabic, in Vata resumptives occupy syntactic positions after elements are moved. I do not pursue this further but leave it for future research.

b. *Ŋmà lé bì kìn sìmà dìn?
who FOC IPFV fry 3.SG.groundnut today
Intended: 'Who is frying groundnuts today?'

c. Ŋmà (lé) Jato dàk ké **ù** bì kìn sìmà dìn?
who FOC J. think COMP RP IPFV fry 3.SG.groundnut today
'Who does Jato think is frying groundnuts today?'

Additional evidence for the base generation of subject wh-phrases in Spec-TP is provided by constructions containing multiple wh-phrases. This aligns with Rizzi (1997, 2001)'s articulated left periphery, in which a higher Focus Phrase is projected above TopP/IP. I assume, for instance, that in (176-c) and (176-d), the higher wh-phrases **bà-dààl** and **bà** undergo focus movement to Spec-FocP while the subject moves from Spec-vP to Spec-TP for EPP reasons. Sentence (176-e) is ungrammatical even when the subject is optionally focused, indicating one designated Focus position in the left periphery.

(176) a. Ŋmà nàn bì kìr màngù bà-dààl?
who PST IPFV pluck 3.mango what-day
'When was Who plucking a mango?'

b. *Ŋmà$_i$ (lé) bà-dààl kìr màngù t$_i$?
who FOC what-day pluck.PFV 3.mango

c. Bà-dààl$_i$ (lé) ŋmà kìr màngù t$_i$?
what-day FOC who pluck.PFV 3.mango
'Who plucked a mango When?'

d. Bà$_i$ (lé) ŋmà nán dàà t$_i$?
what FOC who PST buy
'Who bought what?'

e. *Bà$_i$ (lé) ŋmà (lé) nán dàà t$_i$?
what FOC where FOC PST buy
Intended: 'What did who buy?'

A similar analysis of the restriction on multiple wh-questions is proposed by Sabel and Zeller (2006) for Duala and Kikuyu. Sabel (2000), for instance, shows, using examples like (177-b), that in multiple wh-questions, only one wh-constituent can be fronted because there is a single focus position in the left periphery to host the wh-element. This supports Rizzi (1997, 2001)'s framework of a single Focus Phrase projection in the left periphery.

(177) a. Neni Kuo a bodi no nje?
how K. he do PTCL what
'What how did Kuo?'

b. *Nje neni Kuo a bodi no?
what how K. he do PTCL
'What, how did Kuo?'

(Sabel, 2000, 426)

The notion that wh-phrases are focused stems from an analysis, which proposes that in a wh-question, the wh-phrase represents the focus or new information being requested, while the rest of the sentence expresses the presupposition or background information that is assumed. Thus, wh-phrases are deemed to be focused because when they question a particular constituent, they highlight that constituent as the one for which the predicate will hold (cf. Jackendoff, 1972; Kiss, 1998; Sabel and Zeller, 2006; Muriungi, 2005).

I have shown the various strategies employed for ex-situ wh-movement in Likpakpaanl in this subsection and shown that Likpakpaanl allows the focus movement of a wh-phrase to an intermediate or clause-initial left peripheral position. I proposed that focus movement is driven by an Agree relation and EPP-triggered movement in the case of ex-situ wh-focus movement and cleft constructions. A base-generation account is proposed for an in-situ subject and a non-subject wh-movement. The analysis lends credence to Rizzi (1997); Aboh (1998)'s analysis regarding the prevalence of distinct functional projections in the left periphery for encoding information structural notions like topic and focus. Further research is needed to account for the optionality of the focus particle in the ex-situ movement of the wh-phrase and the full range of syntactic environments that allow or restrict overt wh-displacement in Likpakpaanl.

3.12 Constraints on wh-movement

Having provided empirical evidence demonstrating that ex-situ wh-constituents in Likpakpaanl undergo extraction to the left periphery of the clause, this section aims to offer syntactic diagnostics for movement using island sensitivity tests Ross (1967). The assumption is that some wh-questions in Likpakpaanl are derived via movement rather than base generation. This analysis carries theoretical implications; if movement derives wh-questions, they are expected to demonstrate sensitivity to syntactic island constraints that restrict displacement operations Ross (1967).

3.12.1 Island Effects and Movement

According to Ross (1967), syntactic islands refer to specific structural configurations within a sentence that create barriers or constraints on the movement of syntactic elements. These configurations result in limitations or restrictions on the grammaticality of certain movements, particularly involving wh-phrases. Ross identified several types of syntactic islands, each characterised by its structural properties that inhibit the movement of wh-phrases out of or across these configurations. However, it is well-established that island effects may be ameliorated by leaving the wh-expression in-situ rather than moving it. Likpakpaanl, similar to Krachi (Torrence and Kandybowicz, 2015), Dagbani (Issah, 2020; Issah, 2018), and Kusaal (Abubakari, 2018) inter alia, allows both wh-movement to the left periphery and wh-in-situ. This section examines whether Likpakpaanl wh-phrases exhibit asymmetric island sensitivity based on their surface position. The prediction is that moved wh-phrases will show island effects while in-situ wh-phrases will not. Testing for this asymmetry will elucidate the interaction between islands and wh-positioning in Likpakpaanl.

3.12.2 Complex Noun Phrase Constraint (CNPC)

The Complex NP Constraint (CNPC) prohibits extraction out of a complex noun phrase. The CNPC relates to the impossibility of extracting an element from a clause that complements a DP. The CNP constraint manifests in two forms; the first blocks movement from the clausal complement of a noun, and the second bans extraction from a relative clause modifying a noun. According to Ross (1967), the CNP constraint ensures that 'elements dominated by a sentence which is dominated by a noun phrase cannot be questioned or relativised' or moved out of that 'noun phrase by a transformation" (Ross, 1967, p. 119). The grammaticality of (178-a) and the illicitness of (178-b) show that extracting an element from a DP island is impossible because DP occupies Spec-CP, the escape hatch for the moved element.

(178) a. [$_{\text{TP}}$ Chàtí lèn [$_{\text{DP island}}$ tìbɔŋùnlkààr [$_{\text{CP}}$ ké Naajo kàn ŋmà]]]?
C. say.PFV rumour COMP N. see.PFV who
'Who did Chàtí spread rumours that Naajo saw?

b. *[$_{\text{FocP}}$ Ŋmà$_{\text{i}}$ (lé) Chàtí lèn [$_{\text{DP Island}}$ tìbɔŋùnlkààr [$_{\text{CP}}$ ké Naajo kàn t$_{\text{i}}$]]]?
who FOC C. say.PFV rumour COMP N see.PFV
Intended: 'Who did Chàtí spread rumours that Naajo saw?'

The data in (179) illustrates that relative clauses constitute island domains for wh-movement, consistent with the CNPC. Example (179-a) shows a relative clause in Likpakpaanl.[16] The sentence in (179-b) demonstrates that extracting the wh-

[16]The noun class system determines the selection of relative operators essential in forming relative clauses because they modify the antecedent of the relative construction. Thus, a noun from class 11 with prefix **kì** will have the same **kì** as its relative operator, as shown below:

phrase **bà** from the relative clause yields ungrammaticality, suggesting that extraction out of a relative clause island blocks overt wh-movement. However, (179-c) shows that an in-situ wh-phrase inside the relative clause is acceptable.

(179) a. [TP Jabab nyì [CP ù-chá ù mùà
J. know.PFV 1.SG-blacksmith 1.SG.REL forge.PFV
ŋì-kùù nà.]]
6.PL-hoe REL.DET
'Jabab knows the blacksmith who forges hoes.'

b. [FocP *Bà$_i$ (lé) [TP Jabab nyì [CP Relative Island
what FOC J. know.PFV
ù-chá ù mùà t$_i$ nà?]]]
1.SG-blacksmith 1.SG.REL forge.PFV REL.DET

c. [TP Jabab nyì [CP ù-chá ù mùà
J. know.PFV 1.SG-blacksmith 1.SG.REL forge.PFV
bà nà.]]
what REL.DET
'Jabab knows the blacksmith who forges what?'

Even though it has been argued that resumptive pronouns can ameliorate island violations like in (179-b), repairing extractions from relative clauses when used in place of the moved element Andrews (2007), this does not appear to be the case in Likpakpaanl. As evidenced by (180-b), using a resumptive pronoun to replace the extracted non-subject argument still results in ungrammaticality despite being inside a relative clause island. Thus, resumptive pronouns do not seem to repair island effects in relative clause islands.

(180) a. Wààjà tér bì-kpààb bì gbìì n-nyún-bùn
W. help.PFV 2.PL-farmer 2.PL.REL dig.PFV 5.SG-water-hole
nà.
REL.DET
'Wààjà helped the farmers who dug wells.'

b. [FocP *N-nyún-bùn mù-là-mù$_i$ (lé) [TP Wààjà tér
5.SG-water-hole 5.SG-which-5.SG FOC W. help.PFV
[CP Relative Island bì-kpààb bì gbìì mù$_i$ nà?]]]
2.PL-farmer 2.PL.REL dig.PFV 5.SG.RP REL.DET

(i) [TP Bakar dàà [CP Kì-bùàkù-k kì$_i$ [TP Ntébi lùù t$_i$ nà]]].
B. buy.PFV 11.SG-basket-11 11.SG.REL N. weave.PFV REL.DET
'Bààkàr bought the basket (that) Ntébi wove.' Refer to Section 3.3 for the discussion on Likpakpaanl noun classes.

3.12.3 Coordinate Structure Constraint (CSC)

Coordinated structures represent another strong island for extraction. According to Ross (1967), the Coordinate Structure Constraint (CSC) prohibits extracting a single conjunct from a coordinate structure, and Likpakpaanl demonstrates island effects, blocking the extraction of individual conjuncts. Extracting a single conjunct from a coordinate structure results in ungrammaticality in Likpakpaanl, as demonstrated by (181-c) and (181-d). The only grammatical extraction possible is when both conjuncts are extracted simultaneously (181-b).[17].

(181) a. Mpopiin nàn ŋààl chééché nì mòtò.
M. PST drive 3.SG.bicycle and 3.SG.motorbike
'Mpopiin can ride a bicycle and a motorbike.'

b. [Chééché nì mòtò]$_i$ lé Mpópíín nàn ŋààl t_i.
3.SG.bicycle and 3.SG.motorbike FOC M. PST drive
'Mpopiin rode a bicycle and a motorbike.'

c. *Bà$_i$ (lé) Mpopiin nàn ŋààl t_i nì mòtò?
what FOC M. PST drive and 3.SG.motorbike

d. *Bà$_i$ (lé) Mpopiin nàn ŋààl chééché nì t_i?
what FOC M. PST drive 3.SG..bicycle and

3.12.4 Connectivity Effects

Evidence supporting a movement analysis for Likpakpaanl clefts emanates from connectivity effects such as anaphor binding and island effects. For instance, (182-a) shows a binding relationship between the extracted cleft phrase and the antecedent. The sentence is grammatical, although the reflexive pronoun c-commands its antecedent, violating Principle A of the Binding theory. This shows that the anaphor at some point in the derivation was in the binding domain of the antecedent and licensed before extraction to the derived position in Spec-FocP. The ungrammaticality of (182-b) results from the lack of ϕ-agreement (number and person) between the anaphor, which is plural, and its singular antecedent.

[17] Nì is homophonous in Likpakpaanl. Apart from functioning as a conjunction, it serves multiple syntactic functions. It can also be used as a 3rd person singular inanimate pronoun (i-a), 2nd-person plural pronoun (i-c) or a locative (181-a):

(i) a. N dàà **nì**.
1SG.NOM buy.PFV 3SG.INAN
'I bought it.'

b. **Nì** nàn chà làché?
2SG.PL PST go where
'Where were you going?'

c. Ù nàn chà sikuu **nì**
3SG.NOM PST go school LOC
'she/he is going to the school.

(182) a. Ni yè bì-bà$_i$ *(lé) bì$_i$ nàn tèr t$_i$.
it COP 3PL-self FOC 3PL PST help
'They helped THEMSELVES.'

b. *Ni yè bì-bà$_i$ *(lé) ù$_i$ nàn tèr t$_i$.
it COP 3PL-self FOC 1SG PST help
Intended: 'They helped themselves'

Likpakpaanl ex-situ constructions exhibit island sensitivity, providing further evidence for a focus movement analysis (Ross, 1967). For example, ex-situ cannot be extracted out of an adjunct island, as in (183-b), neither can a single conjunct be moved out of a coordinate structure island (184-b):

(183) a. [$_{TP}$ Ù-jà gbààn nàn kpò [$_{AP\ Island}$ wààr lé Dànàà
1.SG-man DEF PST die before CONJ D.
màà lì-dìchàn-l gbààn.]]
build.PFV 7.SG-house-7 DEF
'The man died before Danaa built the house.'

b. [$_{FocP}$ *Bà$_i$ (lé) ù-jà gbààn nàn kpò [$_{AP\ Island}$ wààr lè
what FOC 1.SG-man DEF PST die before CONJ
Dànàà màà t$_i$?]]
D. build.PFV
'What did the man build before Danaa died?'

(184) a. [$_{TP}$ Kando gà tér [$_{Coordinate\ Island}$ Amà nì Wàssà]].
K. FUT help A. and W.
'Kando will help Ama and Wassa.'

b. [$_{FocP}$ *Ŋmà$_i$ (lé) [TP] Kando gà tér [$_{Island}$ Amà nì t$_i$]]?
who FOC K. FUT help A. and

3.13 Chapter Summary

This chapter has provided a comprehensive overview of Likpakpaanl grammar, beginning with clausal structure, noun class system, and agreement patterns within the DP. Investigation of tense and aspect marking revealed that temporal and aspectual particles constitute functional projections in the clausal spine, with features that participate in Agree operations to derive attested structures. Furthermore, the chapter explored wh-questions, illuminating an asymmetry between subject and non-subject wh-questions and between questions and answers. Ex-situ wh-movement in Likpakpaanl was analysed under a Minimalist approach, involving feature-driven movement of the wh-phrase to the left periphery coupled with Agree to derive the morpho-syntactic properties. Constraints on wh-movement in Likpakpaanl were also examined, concluding that island effects prohibit the extraction of wh-phrases from embedded clauses.

Chapter 4

Modal Complement Ellipsis

4.1 Introduction

This chapter is dedicated to the syntactic derivation of Modal Complement Ellipsis (MCE), a type of Verb Phrase Ellipsis (VPE) in Likpakpaanl. Likpakpaanl illustrates an elliptical phenomenon reminiscent of Modal Complement Ellipsis (MCE) discussed in Chapter 2, subsection2.6.1.2. The data demonstrates that Likpakpaanl VP ellipsis is licensed by the root modals **ŋmàà** 'can' and **bàn** 'want,' which project separate Modal Phrases (ModP) in the clausal structure Cinque (1999). Thus, Likpakpaanl MCE cannot be licensed by main verbs, aspectual markers, or tense markers. Furthermore, the analysis reveals that within the hierarchical modal system of Likpakpaanl, there is only a single syntactic position where the [E]-feature that licenses MC ellipsis can be hosted - on the higher ModP position occupied by the modal **ŋmàà**. This is evidenced when two modal elements co-occur, the higher ModP, i.e., licenses MC ellipsis. Therefore, in Likpakpaanl, MCE occurs when the licensing ModP head is merged into the syntactic structure, and the vP Phonological Form (PF) content is elided before spell-out. The research questions underpinning this study are:

i. What are the specific strategies or mechanisms that license MCE in Likpakpaanl?

ii. What licensing conditions govern the availability of MCE in Likpakpaanl?

iii. What language-internal evidence supports a movement-plus-PF-deletion analysis of the MCE phenomenon in Likpakpaanl?

Beyond this introduction, the rest of the chapter is structured as follows: Section 4.2 introduces the notion of modality and the semantic differences between epistemic and root modality. It also examines the proposal that deontic modals are lower in structure than epistemic ones. Section 4.3 provides the background on modal constructions in Likpakpaanl

4.2 The Notion of Modals

The concept of modality, which expresses notions such as possibility, necessity, probability, obligation, permission, ability, and volition, has been the subject of extensive linguistic analysis. Kratzer (1981, 2012) identifies two broad classes of modals: epistemic modals, derived from the Greek word *episteme* meaning 'knowledge,' and root modals. Epistemic modality is concerned with what is possible or necessary given the knowledge of an epistemic agent, the speaker. It indicates the degree of assurance of what a speaker says with different dimensions of modal judgment, including the degree of certainty, source of information, and time reference (Cormack and Smith, 2002; Barbiers, 2006).

In contrast, root modality refers to modal verbs that express necessity, *obligation, permission, volition*, or *ability*. These modals interact with tense and aspect to convey different interpretations (Brennan, 1993; Hacquard, 2009). For example, the modal verb *will* in (185-a) conveys a stronger prediction, demonstrating confidence in the speaker's knowledge based on contextual evidence. In contrast, *would* in (185-b), considered a tentative epistemic marker, is used to tone down the speaker's confidence level despite the certainty of the proposition (Picallo, 1990; Barbiers, 2006; Hacquard, 2016).

(185) a. Francis will visit his mother today.
b. Damial would come by evening.

Similarly, the sentence in (186-a) suggests epistemic modality indicating the speaker's assessment and judgment that the writing of the exams by the students is possible, while the second interpretation also suggests that *the students* have been granted permission, by the school authority, to sit the examination. In (186-b), a root interpretation reflects the speaker's judgment of the necessity for the students to write the exams and also implies that the students are obliged to write the exams.

(186) a. The students may the examinations.
b. The students must the examinations.

Despite the categorical distinction between epistemic and root modals, Kratzer (1981, 2012) has shown that the same modal can be used cross-linguistically to express different forms of modality suggesting that the mapping between modal semantics and syntactic representation is not straightforward one (cf: Hacquard, 2016) For instance, in English, the modal *must* can express an epistemic modality (187-a) or various kinds of root modality like deontic (187-b) or teleological (187-c).

(187) a. John is not in his office. He *must* be home.
b. John parked illegally. He *must* pay a fine.
c. John wants to get a PhD. He *must* write a thesis.

(Hacquard, 2016, 46)

Even though the same *must* modal is involved in (187-a)- (187-c), it yields different semantic interpretations when combined with different conversational backgrounds. For instance, *must* in (187-a) indicates that based on what the speaker knows, it means since John is not in his office, he should be at home and no place else. Example (187-b) also means that given the circumstances of John's illegal parking, he must pay a fine, while (187-c) refers to a requirement needed for John to achieve his goal of getting a PhD.

The relative positioning of epistemic and root modals has been a subject of extensive scholarly debate among syntacticians. Numerous researchers, including Brennan (1993), Cinque (1999), and Hacquard (2016), among others, have argued that epistemic modals are merged in a higher structural position than their root counterparts. Picallo (1990) and Cinque (1999)'s analysis of the Catalan and Italian, respectively, provide evidence demonstrating that epistemic modals occupy a higher Inflectional Phrase (IP) position while root modals are generated in the little Verb Phrase (vP) periphery. Picallo (1990) contends that Catalan has two different base positions for epistemic and root modals are derived from the co-occurrence interpretation of adjacent one. According to Picallo (1990), in the Catalan example in (188-a), only the higher modal *due* can have an epistemic interpretation. The only available interpretation that (188-a) can elicit is the epistemic interpretation in (188-b) and not (188-c)

(188) a. En Pere due poder tocar el piano.
the Pere must can play the piano
b. It must be the case that Pere is able/allowed to play the piano.
c. *It must be the case that it is possible that Pere would play the piano.

(Picallo, 1990, 294)

Further evidence for the syntactic distinction between epistemic and root modals comes from the interaction of modals with perfective auxiliaries. In Catalan and Dutch, perfective auxiliaries can precede root modals but not epistemic ones, while root and epistemic modals can precede perfective auxiliaries (Picallo, 1990; Barbiers, 2002). This asymmetry supports the idea that root modals are structurally lower than epistemic modals.

(189) a. En Joan ha pogut anar a Banyoles.
‘Joan has been allowed to go to.Banyoles.’

b. *En Pere havia degut venir.
the Pere had must come

(Picallo, 1990, 293)

Building on these observations, Cinque (1999) proposes a detailed syntactic hierarchy that captures the relative positions of different modal interpretations illustrated in (190=. This hierarchy reflects that necessity modals are merged higher than possibility modals, and obligation modals project higher than ability modals. Cinque (1999) also incorporates positions for volition modals and a distinction between epistemic and alethic modality.

(190) $\mathrm{Mod}_{\mathrm{Epistemic}} ... \rightarrow \mathrm{Mod}_{\mathrm{Necessity}} \rightarrow \mathrm{Mod}_{\mathrm{Possibility}} \rightarrow \mathrm{Mod}_{\mathrm{Volition}} \rightarrow \mathrm{Mod}_{\mathrm{Obligation}} \rightarrow \mathrm{Mod}_{\mathrm{Ability/Permission}}$

(Cinque, 1999, 79).

Further evidence used by Cinque (1999) for arguing that the syntactic position of volition modals comes from Italian sentences such as (191-a) (Cinque, 1999, 80), where the modal *volere* ‘want’ suggesting *possibility* scopes over *ability* and *permission* (191-b). He notes that an inverse interpretation where *ability* scopes over *possibility* are not possible, suggesting a relatively rigid ordering of modals.

(191) a. Gianni potrebbe voler uscire.
‘John could want to go out.’

b. Gianni vorrebbe poter cantare.
‘John would want to be allowed/be able to sing.’

Given this background to the notion of modals and their syntactic position within the clausal spine, the next section focuses on Likpakpaanl modal auxiliaries and their distributional interaction with tense and aspect in the language

4.3 Likpakpaanl Modal Auxiliaries

This subsection provides the background on modal constructions in Likpakpaanl, establishing the foundation for the subsequent discussion of VP ellipsis. Modal verbs, akin to tense, aspectual, and focus markers, are denoted by particles that convey both root and epistemic modality. Modality in this language can be understood as a grammatical domain that primarily encodes expressions of possibility and necessity (Kratzer, 2012; Hacquard, 2006). This section will focus on modal expressions of *possibility*, *ability* and expressions of *necessity*, *need*, *permission* and *obligation*. This delimitation aligns with the general understanding of modality in functional-typological studies of modality (Bybee et al., 1994; Nuyts, 2006) which focuses on the semantic and pragmatic functions of modal expressions within the broader context of language use and communication.

4.3.1 Modal Verbs in Likpakpaanl

Modality in Likpakpaanl is expressed using different modal auxiliaries, which take various elements as their complements. For instance, the root modal **ŋmàà** 'can' takes a vP complement, the modal **bàn** 'want' has a CP as its complement, while the epistemic modal **nìbààkán** 'maybe' takes a whole clause as its complement. Let us consider some detailed examples. The root modal **ŋmàà** is used to express ability and logical possibility. In 192, it indicates **Sula**'s physical or mental capability *to peel the yam*.

(192) Sula nàn ŋmàà yèì lì-nùù-l gbààn.
S. PST can peel 5.SG-yam-5 DEF
'Sula was able to peel the yam.'

The modal **ŋmàà** can also be used to express *permission*, as in (193-a), where the speaker grants approval for Dennis to blow his flute. It can also be used to seek permission or ask for an addressee's consent to provide help as in (196-a).

(193) a. Dennis gà ŋmàà pìì m àà-wòl gbààn.
D. FUT can blow 1SG POSS-flute DEF
'Dennis can blow my flute.'

b. Tì gà ŋmàà tér sì-ì?
3PL FUT can help you-INTER
'Can we help you?'

The modal **ŋmàà** also frequently expresses epistemic modality, conveying notions of 'possibility' or 'uncertainty'. Thus, **ŋmàà** indicates that based on the speaker's knowledge and the available evidence, the proposition expressed in the scope of the modal is likely to be valid. For instance, in a context where the weather is cloudy, the utterance in (194) can be employed to express the epistemic possibility of rain falling:

(194) Ù-táál gà ŋmàà nù.
1SG-rain FUT can fall
'Rain can fall/it can rain.'

In (194), the speaker does not claim the inherent ability or capacity for the rain to fall but instead assesses the probability of the rain falling based on the prevailing atmospheric conditions. The use of **ŋmàà** signals the speaker's judgment that rain may likely fall given the cloudy weather.

Furthermore, **ŋmàà** can be employed to draw inferences as in (195) where the modal verb indicates the speaker's assessment of the implausibility of Joana finishing the work given that it is tedious and requires more time to complete.

(195) Joana ààn kpààn ŋmàà dóó lì-tùn-l gbààn.
J. NEG.FUT already can finish 7.SG-work-7 DEF
'Joan could not have already finished the work.'

These examples show that **ŋmàà** can convey different shades of meaning pragmatically. It expresses the ability to indicate possibility and draw conclusions, depending on the specific context in which it is used. Also, the epistemic modal usage of **ŋmàà** contrasts with its root modal functions, where it may convey dynamic modality (ability/capacity) or deontic modality (permission).

The next modal to be considered is the modal **bàn** 'want', which can also be used to express both root and epistemic modalities. As a root modal, **bàn** 'want' conveys *desire* or *volition*, indicating the subject's internal motivation to act on their own needs, and preferences (Palmer, 2001; Huddleston and Pullum, 2002). For example, in (196-a) **bàn** 'want' demonstrates the intentional willingness of **Mulibi** to help **Sanja** while (196-b) illustrates the subject's desire to build a house.

(196) a. Mulibi$_i$ bàn (kè) ù$_i$ tér Sanja.
M. want COMP RP help S.
'Mulibi wants to help Sanja.'

b. Tì bàn (kè) tì$_i$ màà lì-dìchàn-l.
3.PL want COMP RP build 7.SG-house-7
'We want to build new houses.'

Furthermore, **bàn** can also be used to express epistemic modality, in which case the speaker makes an inference about the likelihood of a proposition being valid based on their knowledge and reasoning. For instance, given the context that 'Mbaye was setting fire in the hearth' will suggest the sentence in (197). This implies the speaker's assessment that cooking is probable, given that the woman has set fire.

(197) Mbaye nàn bàn (kè) ù$_i$ mɔn bùsá.
M. PST want COMP RP cook 9.SG.food
'Mbaye wanted to cook food.'

The versatility of 'want' in conveying both root and epistemic modality allows speakers to shift between expressing their own desires and making inferences about the state of affairs (Palmer, 2001). The sentences exemplified by (196) exhibit a fundamental ambiguity stemming from the referential flexibility of the pronominal element. Specifically, the pronoun contained within these constructions may co-refer with the 'antecedent DP' (the subject of the matrix clause), or it may alternatively refer to some other discourse-salient entity. The only means of disambiguating the intended referent is through appropriate co-indexation conventions that clarify the pronominal reference.

Unlike what we saw in the distribution of **ŋmàà**, where it selects a vP/VP complement, **bàn** does not. The distribution of the modal **bàn** shows that it takes a CP complement, indicating a bi-clausal construction. The complementiser is optional and can thus be either overt or covert. A possible way of capturing the distribution of **ban** would be to argue that it takes larger constituents like CP as its complement. This is evidenced by the ability to extract the DP out of the complement (198-b), which would not be possible if it were an adjunct clause, as observed in (198-c).

(198) a. Kunji nàn bàn (ké) ù kìn ì-jàn gbààn.
K. PST want COMP 3SG fry 6.PL-fish DEF
'Kunji wanted to fry the fish.'

b. Ì-jàn$_i$ ì-là-ì lé Kunji nàn bàn (ké) ù kìn t_i?
6.PL-fish 6.PL-which-6 FOC K. PST want COMP 3SG fry
'Which fishes did Kunji want to fry'

c. *Kì-gbàŋ$_i$ kì-là-kì lé Natug nàn dàà wààr lé
11.SG-book 11.SG-which-11 FOC N. PST buy before CONJ
Thomas kàrn t_i?
T. read.PFV
Intended: 'Which book did Natug buy before Thomas read?'

Similarly, Rowlett (2007) has suggested that the French modal verb **devoir** 'must' selects an infinitival CP complement and not a vP as shown in (199-a). The argument for this analysis comes from the fact that **devoir** can occur under negation, and either the clause containing the modal or the infinitival clause can be independently negated. Jansen et al. (2012) supports this assertion and argues that the grammaticality of both (199-b) and (199-c) indicates that the infinitive with the modal verb constitutes separate clauses.

(199) a. Je dois lire un livre.
I must read a book
'I must read a book.'

b. Marie ne doit pas fumer dans ce restaurant.
Mary ne allowed NEG to.smoke in this restaurant.
'Mary is not allowed to smoke in this restaurant.'

c. Marie doit ne pas fumer dans ce restaurant.
Mary is.obliged ne NEG to.smoke in this restaurant
'Mary is obliged not to smoke in this restaurant.'

(Jansen et al., 2012, 341)

The ability of the negation to target either the modal verb *devoir* or the infinitival clause suggests that the modal and its complement form distinct clausal structures rather than a single, unified clause. This evidence supports the analysis that *devoir* selects an infinitival CP complement.

In Likpakpaanl, interpretations of 'obligation' and 'necessity' arise when the copular **yè** 'be' co-occurs with the complementiser **kè** 'that'. This bi-partite modal construction **yè kè** expresses the necessity or obligation to undertake an activity. For example, in (200), the **yè kè** construction expresses the obligation for a sick person to take their medication in order to get well.

(200) Ù-bún$_i$ yè kè ù$_i$ nyù n-nyɔk.
1SG-sick COP COMP RP drink 5.SG-medicine
'A sick person must take medicine.'

Moreover,**yè kè** can signal different obligatory interpretations depending on the context. In a context where *Bìnlù* is expected to bring fried meat for a ceremony, (201) expresses the fact that *Bìnlù* has to fulfill an obligation and ensure that is done.

(201) Nì yè kè Bìnlù kìn tì-nàn gbààn.
it COP COMP fry.PFV 14.PL-meat DEF
'Binlu must fry the meat.'

The two sentences in (200) and (201) differ in their obligatory interpretations due to their syntactic structures and the information they emphasise. (200) conveys a straightforward obligation requiring **Binlu** to fry the meat with no additional layers of interpretation. (201) employs a cleft sentence, which serves to emphasise **Binlu** as the sole agent responsible for frying the meat and also implies a contrast with others who are not obligated to fry the meat. For example,(201) suggests that while others might be expected to fry the meat, the specific obligation falls on **Binlu**.

The use of bi-partite markers to express modality is not new in the literature of African languages. Many Niger-Congo languages, including Lingala Heine (1991), Ewe (Ameka, 2008), and Igbo (Uchechukwu, 2008), among others, have been shown to employ bi-partite markers comprising two distinct elements to convey modal interpretations. Ameka (2008), for instance, demonstrates that in Ewe, an obligation is expressed when the third person singular pronoun **é** is combined with the expletive verb and followed by the complement clause. In (202), the third person pronoun, which is the subject, is followed by a verb and the complementiser **bé**, which introduces the complement clause to express the obligation in Ewe.

(202) É-dze bé nà-vá kpɔ-m.
3SG-fit that 2SG:SUBJV-VENT see-1SG
'You must come to see me.'

(Ameka, 2008, 147)

Similarly, Uchechukwu (2008) also shows that in Igbo, the deontic meaning of 'compulsion' and 'strong obligation' is expressed through the combination of auxiliary **gà** followed by the verb, a particle and a complementiser as 203 shows.

(203) ó̥ gà-írī-rí.rí. ñrí áhù.
he AUX-EAT-PART-COMP food DET
'He must eat that food.'

(Uchechukwu, 2008, 254)

The modal **nìbààkàn** meaning 'probably/maybe' can be used to express epistemic modality, conveying the speaker's assessment of the likelihood or possibility of a proposition (Palmer, 2001). The examples in (204) present data that show that apart from expressing that the scope proposition is plausible but not certain, **nìbààkàn** can also occur in different positions within the causal spine. The examples also show that **nìbààkàn** can occur in clause-initial, medial or final positions as illustrated in (204).

(204) a. Nìbààkàn, Neina gà lùù n-nyùn gbààn.
maybe N. FUT fetch 4.SG-water DEF
'Maybe/probably, Neina will fetch the water.'

b. Neina gà lùù n-nyùn gbààn, nìbààkàn.
N. FUT fetch 4.SG-water DEF maybe
'Neina will probably fetch the water.'

Modals such as **nìbààkàn** belong to a class of elements that have been referred to in the literature as *modal adverbs* (Cinque, 1999; Nuyts, 1993, 2006), which express epistemic modal meanings while also demonstrating syntactic flexibility in their clausal positioning. Adopting Nuyts (2006)'s criteria for modal adverbs, I analyse **nìbààkàn** as a modal adverb based on its syntactic patterning. Examples

(205-a)-(205-c) demonstrate **nìbààkàn** can appear in a clause's initial, medial, or final position without any change in its meaning.

(205) a. Nìbààkàn Jàbàb gà dàn dìn.
maybe J. FUT come today
'Maybe/Probably, Jabab will come today.'

b. Jàbàb gà dàn dìn, nìbààkàn.
J. FUT come today maybe
'Jabab will come today, maybe/probably.'

c. Jàbàb gà dàn, nìbààkàn dìn.
J. FUT come maybe today
'Jabab will maybe/probably come today.'

This positional flexibility indicates that it syntactically takes sentential scope and modifies the truth value of the entire proposition rather than a single constituent. Semantically, it expresses epistemic modality by qualifying the likelihood of the proposition being true. This modal shares some similar syntactic properties with its Italian counterpart **probabilmente** 'probably'. According to Giorgi (2013), **probabilmente** is able to semantically scope over an entire sentence whether it precedes or follows the subject without resulting in a change of meaning as in ((206-a)). Giorgi (2013) proposes that modal adverbs like **probabilmente** can either occur in the clause-initial position, above the Tense Phrase (TP), or in their base-generated position as the complement of the verb.

(206) a. Probabilmente Gianni vincerà la gara.
Probably G win the race
'Probably Gianni will win the race.'

b. Gianni Probabilmente vincerà la gara.
G Probably win the race
'Probably Gianni will win the race.'

(Giorgi, 2013, p. 99)

The parallels between **nìbààkàn** in Likpakpaanl and **probabilmente** in Italian suggest that modal adverbs may share certain cross-linguistic properties in terms of their syntactic distribution and semantic scope. This aligns with Giorgi (2013)'s predictions about the dual positioning options for modal adverbs within the clausal structure.

Finally, the modal **lì**, 'should,' can convey epistemic or root modal meanings depending on the context. In example (207-a), the use of **lì** expresses a sense of necessity or obligation, indicating that according to rules or moral standards, people must be patient with themselves and their neighbours. The context in (207-b) that an NGO has said that all those who cultivated groundnuts will be given tractors, and Wongol begins to smile with happiness, then the statement is used to suggest the possibility that he farmed groundnuts and therefore heäs smiling because of the gift he would get.

(207) a. Ù-nìì mɔmɔk lì kpà sùlá.
1.SG-person every should have 3.SG.patience
'Everyone should be patient.'

b. Wongol gà lì kùn sìmà.
W. FUT should farm 32G-groundnut
'Wongol might have cultivated groundnuts.'

These examples illustrate that the surrounding context determines the interpretation of **lì**, whether as a root modal expressing necessity or an epistemic modal conveying possibility.

Having explored the forms and distribution of modal verbs in Likpakpaanl, the next step is to examine how these modal elements interact with the tense and aspectual system. The interplay between modality and aspect further illustrates how the language encodes tempo-modal meanings.

4.3.2 The interaction between Modals, Tense and Aspect

The previous section provided an overview of modal verbs in Likpakpaanl. This section focuses on the interaction between modals and the aspectual system in Likpakpaanl. The interaction between tense, aspect, and modals in Likpakpaanl elicits different semantic interpretations. In (208-a), we have a default perfective verb co-occurring with the modal **ŋmàà** to express the present perfective modality, which indicates an action that has been completed. The interaction between the perfective verb and **ŋmàà** suggests that **Bonjei** has acquired the ability or skill to sharpen the cutlass, and this ability is still present in the current time. In (208-b), we have the future tense occurring with the modal to express an action that will occur in the future where **Bonjei** will have the ability or skill to sharpen the cutlass at some point in the future, but this ability is not necessarily present in the current time.

(208) a. Bonjei ŋmàà lùùn kpàchá gbààn.
B. can sharpen.PFV 3SG.cutlass DEF
'Bonjei has been able to sharpen the cutlass.'

b. Bonjei gà ŋmàà lùùn kpàchá gbààn.
B FUT can sharpen.PFV 3SG-cutlass DEF
'Bonjei will be able to to sharpen the cutlass.'

Furthermore, two sentences in (209-a) and (209-b) show the interaction between tense and the modal **bàn** to express different levels of certainty and futurity. In (209-a), the modal is used with the past tense verb to indicate Naajo's past-oriented desire to act. In (209-b), the combination of the future tense and the modal conveys a high degree of certainty that Naajo will develop the desire to peel the yam at some point in the future.

(209) a. Naajo nàn bàn (ké) ù yèì lì-nùù-l.
N. PST want COMP RP peel 7.SG-yam-7
'Naajo wanted to peel yam.'

b. Naajo gà bàn (ké) ù yèì lì-nùù-l.
N. FUT want COMP RP peel 7.SG-yam-7
'Naajo will want to peel yam.'

Moreover, using distinct tense and aspectual elements in the presence of the same modal verb can lead to divergent modal readings. For instance, the particle **lì** may convey a sense of past necessity as in (210-a), where 'Adwoa' was expected to fry some meat. The modal expresses an expectation that was not fulfilled. Conversely, the employment of **lì** together with the future tense and the imperfection aspect in (210-b) indicate a future possibility of the 'Adwoa' frying meat rather than a requirement. The modal 'should' again conveys a strong sense of obligation or ongoing necessity, but the present continuous tense suggests that the necessary action is expected to be in progress.

(210) a. Adwoa bà lì kìn tì-nàn gbààn.
A. PST should fry 14.SG-meat DEF
'Adwoa should have fried the meat.'

b. Adwoa gà lì bì kìn tì-nànn gbààn.
A. FUT should IPFV fry 14.SG-meat DEF
'Adwoa should be frying the meat.'

The bi-partite epistemic modal of necessity and obligation, **yé ké** 'must' can also express different forms of modality when they combine with tense or aspect. In (211), the perfective aspect combines with the modal to convey an immediate necessity for Francis to come without any specific temporal reference.

(211) Nambe$_i$ yè kè ù$_i$ báè tì-gbàn.
N. COP COMP RP learn 14.PL-book
'Nambey must learn (books).'

Example (212) shifts the time frame to the past, suggesting a past obligation required of **Nambey** to learn was unfulfilled. The past tense reduces the strong obligation expressed by 'must' and shifts the focus from a current obligation to a past assumption.

(212) Nambe$_i$ nàn yè kè ù$_i$ báè tì-gbàn.
N. PST COP COMP RP learn 14.PL-book
'Nambey was supposed to learn (books).'

The modal 'must' in Likpakpaanl cannot co-occur with the future tense marker. However, future obligation can be expressed using a temporal adverb as in (213-a) to indicate the future time frame. A more complex construction is illustrated in (213-b) where the main clause expresses a necessity for **Namuel** to dig the yam, and the second clause introduces a progressive aspect, indicating that the digging is in progress at the time of speaking. This bi-clausal structure allows for the simultaneous expression of obligation and the progressive nature of the event.

(213) a. Namuel (*gà) yè kè ù gbìì lì-nùù-l gbààn álìsìmádààl.
N. FUT COP COMP RP dig 7.SG-yam-7 DEF friday
'Namuel must dig the yam on Friday.'

b. Namuel yè kè ù lì bì gbìì lì-nùù-l gbààn.
N. PST COP COMP RP should IPFV dig 7.SG-yam-7 DEF
'Namuel must be digging the yam.'

The discussion so far has shown the form and distribution of various modal verbs in Likpakpaanl. The next question is to determine the syntactic status of the modals. To do that, I follow Cinque (1999)'s influential work on the syntax of functional projections, where he contends that modals are heads of independent functional projections, namely Modal Phrases (ModP). Cinque (1999) proposes a fine-grained hierarchy of functional projections, with different classes of modals occupying distinct positions based on their semantics. I adopt a split-ModP analysis, where each modal particle heads its dedicated projection to explain the contrasting distributions and interpretations of these modals and their interaction with other functional categories like Aspect within the clausal spine in Likpakpaanl.

Likpakpaanl modal particles **ŋmàà** and **bàn** and **lì** in (214) are, therefore, analysed as distinct sub-classes of modality, and the relative ordering of these particles in the clausal structure is argued to be strict and reflect their positions within the hierarchy of modal projections.

(214) a. Dimbu gà ŋmàà lì bàn (kè) ù dàà lòòr gbààn.
D. FUT can should want COMP RP buy 3.SG.car DEF
'Dimbu might want to buy the car.'

b. *Dimbu gà bàn (kè) ù ŋmàà lì dàà lòòr gbààn.
D. FUT want COMP RP can should buy 3.SG.car DEF

c. *Dimbu gà lì ŋmàà bàn (kè) ù dàà lòòr gbààn.
D. FUT should can want COMP RP buy 3.SG.car DEF

The ungrammaticality of sentences (214-b) and (214-c) suggests that the hierarchy of modals in Likpakpaanl follows a strict order, which can be represented as 215:

(215) $\text{Mod}_{\text{possibility/ability}} \rightarrow \text{Mod}_{\text{obligation}} \rightarrow \text{Mod}_{\text{need/desire}}$

In the broader interaction with tense and aspect, the example in (216) shows that the modal **ŋmàà** and **lì** project below TP but above the aspectual phrase and the vP. The ungrammaticality of (216-b) occurs because the aspectual phrase precedes the modal. The data further indicates root modals can co-occur, and when they do, **ŋmàà** encoding 'ability' must precede, **lì** as captured in the ungrammaticality of (216-c).

(216) a. $[_{\text{TP}}$ Binlu gà $[_{\text{ModP1}}$ ŋmàà $[_{\text{ModP2}}$ lì $[_{\text{IPFV}}$ bì $[_{\text{vP}}$ kìr màngù.]]]]]
B. FUT can should IPFV pluck 3.SG.mango
'Binlu could be plucking a mango.'

b. *$[_{\text{TP}}$ Binlu gà $[_{\text{AspP}}$ bì $[_{\text{ModP1}}$ ŋmàà $[_{\text{ModP2}}$ lì $[_{\text{vP}}$ kìr màngù.]]]]]
B. FUT IPFV can should pluck 3.SG.mango

c. *$[_{\text{TP}}$ Binlu gà $[_{\text{ModP2}}$ lì $[_{\text{ModP1}}$ ŋmàà $[_{\text{AspP}}$ bì $[_{\text{vP}}$ kìr màngù.]]]]]
B. FUT should can IPFV pluck 3.SG.mango

I propose the structure in (217) as the relative order of modals as functional categories within Likpakpaanl causal architecture;

(217) $\text{TP} \rightarrow \text{Mod}_{\text{possibility/ability}} \rightarrow \text{Mod}_{\text{obligation}} \rightarrow \text{Mod}_{\text{need/desire}} \rightarrow \text{AspP} \rightarrow \text{vP} \rightarrow \text{VP} \rightarrow \text{DP}$

The preceding section examined the modal verbs in Likpakpaanl and showed that most modals in this language can express both root and epistemic modality. The analysis also discusses how these modal expressions interact with the tense and aspectual heads within the clausal structure to trigger different semantic interpretations. For instance, the modal **yè kè** 'must' was found to convey a strong necessity when occurring in the present tense, but its semantic force is reduced to a mere supposition when combined with the past tense. The strict hierarchy observed among the modal categories suggests systematic syntactic constraints on the co-occurrence and distribution of these modal elements within the clause.

The next section will explore the derivation of Modal Complement Ellipsis (MCE) in Likpakpaanl and show that not all modals in the language can license MCE. Evidence from Island effects will be provided to support the analysis of MCE as involving movement plus PF-deletion.

4.4 The Derivation of Likpakpaanl MCE

This section focuses on the syntax of the phenomenon of Modal Complement Ellipsis (MCE) in Likpakpaanl, and the research questions that underpin the section are:

I. Which syntactic element(s) bear the ellipsis feature that licenses ellipsis in Likpakpaanl, and where is its position in the clausal spine?

ii. Do the elided elements at the ellipsis site (E-site) possess any internal syntactic structure, or are they simply null pro-forms without underlying syntax?

The analysis presented in this section argues that the ellipsis feature that licenses MCE is merged on a Modal head and bears the E-feature that deletes the PF content of the modal complement. Secondly, I contend that the elided elements at the ellipsis site possess internal syntactic structures and are not merely null pro-forms. Evidence for this comes from the fact that the elided material can be reconstructed for interpretation, and it also displays island sensitivity, suggesting the presence of an underlying syntactic constituent. In the previous section, we examined the distribution of modal auxiliaries, and this section proceeds to discuss how modals license the derivation of MCE in Likpakpaanl. Even though Section 4.3) showed the presence of different modals, I propose that it is only the **ŋmàà** and **bàn** that license MC ellipsis in the language. I also argue that when these two modals co-occur, the functionally higher modal **ŋmàà** bearing the E-feature licenses the deletion.

4.4.0.1 The Modal **ŋmàà** and Likpakpaanl MCE

The first modal auxiliary I consider is the root modal **ŋmàà**. In (218-a), the modal is followed by a missing verbal phrase whose interpretation is derived from the antecedent clause. (218-a), has the same interpretation as its non-elliptical counterpart in (218-b). The morphology of the modal in the second conjunct clause varies under ellipsis from when the entire clause is pronounced (218-c).

(218) a. Maadai nàn kìr màngù, àmàà Sulà àà nàn **ŋmàn**
M. PST pluck 3.SG.mango CONJ S. NEG PST can
[~~kìr màngù~~].

'Madai plucked a mongo, but Sula couldn't.'

b. Maadai nàn kìr màngù, àmàà Sulà àà nàn ŋmàà kìr
M. PST pluck 3.SG.mango CONJ S. NEG PST can pluck
màngù.
3.SG.mango
'Madai plucked a mongo, but Sula couldn't pluck a mango .'

c. *Maadai nàn kìr màngù, àmàà Sulà àà nàn **ŋmàà**.
M. PST pluck 3.SG.mango CONJ S. NEG PST can
Intended: 'Madai plucked a mongo, but Sula couldn't.'

Even though the same modal can be present in both the first and the second conjunct clauses as demonstrated in (219), there are observed morphological alterations in the verb form when it licenses ellipsis.

(219) [TP Taabuimi gà [$_{\text{ModP}}$ **ŋmàà** [$_{\text{vP}}$ kpàà [DP ì-nàà àmàà [TP Pònjà
T. FUT can herd 6.PL-cow CONJ P.
áán [$_{\text{ModP}}$ **ŋmàn**/*ŋmàà [$_{\text{vP}}$ ~~kpàà ì-nàà~~.]
NEG.FUT can
'Tabuimi can herd cattle but Ponja can't.'

The ellipsis site can encompass a more extensive clausal domain beyond the verbal element. This is illustrated in (220-a), where the elided vP domain contains not only the verb 'wear' but also its internal argument 'shirts', as well as an adjective modifying the head noun 'new' and a temporal adverb 'today'. As expected, (220-b) is ungrammatical because the vP in the ellipsis site violates the focus condition of ellipsis rule, which requires only an e-GIVEN vPs to be deleted (cf. 40.

(220) a. [TP Bileŋan [$_{\text{vP}}$ lèn [CP kè [TP Ntaadi$_{\text{i}}$ gà [$_{\text{ModP}}$ ŋmàà [$_{\text{vP}}$
B. say COMP N. FUT can
pèèn [DP lì-bùàkú-l pùàn [$_{\text{AdvP}}$ dìn, àmàà [TP tí [$_{\text{vP}}$ nyì
wear 7.SG-shirt-7 new today, CONJ 2PL know
[CP kè [TP ù$_{\text{i}}$ [$_{\text{NegP}}$ ààn [$_{\text{ModP}}$ ŋmàn [$_{\text{vP}}$ ~~pèèn~~
COMP RP NEG.FUT can wear
~~lì-bùàkú-l pùàn dìn~~.]]]]]]]]]]]]]]
7.SG-shirt-7 new today
'Bileŋàn said Ntaadi can wear a new shirt today, but we know (that) he can't.'

b. [TP *Jato bà [$_{\text{ModP}}$ **ŋmàà** [$_{\text{vP}}$ kàrn [DP kì-gbàŋ gbààn àmàà
J. PST can read 11.SG-book DEF but
[TP Kùnjí àà bà [$_{\text{ModP}}$ **ŋmàn** [$_{\text{vP}}$ kìn ì-jàn gbààn]]]]]].
K. NEG PST can fry 6.PL-fish DEF

The ellipsis examples are possible because the elided vP and its internal arguments have corresponding antecedents in the first conjunct, allowing for the semantic recoverability of the elements at the E-site. As predicted by the identify and e-GIVENness condition (Schwarzschild, 1999; Merchant, 2001), (220-b) is disallowed under ellipsis since the deleted VP (the struck-out element in the ellipsis site) is new in the discourse does not have a mutually entailing relationship with the antecedent. Thus, only previously mentioned elements in a discourse can undergo ellipsis

Apart from occurring in matrix clauses, **ŋmàà** can also license MC ellipsis in embedded clauses such as (221-a) where it elides its vP complement in the embedded clause. The unacceptability of (221-b) is due to a violation of the e-GIVENess and identity condition on ellipsis. A new element ‘yam’ has been introduced into the discourse, and since it lacks a salient antecedent, it cannot be elided. Thus, in (221-b), the elided vP at the E-site is neither syntactically nor semantically isomorphic with the verb in the antecedent clause.

(221) a. [TP Chapinn [$_{\text{AspP}}$ bì [$_{\text{vP}}$ yɔk [DP ŋì-nánpùù lé [TP
C. IPFV raise 8.PL-yam.mound and
Olàmbà$_i$ [$_{\text{vP}}$ dàk [CP ké [TP ù$_i$ gà [$_{\text{ModP}}$ **ŋmàn** [$_{\text{vP}}$ ~~yɔk~~
O. think that RP FUT can raise
~~ŋì-nánpùù~~]]]]]]]]].
8.PL-yam.mound
‘Neina is raising yam mounds, and Olamba thinks he will be able to.’

b. [TP ?Chapinn [$_{\text{AspP}}$ bì [$_{\text{vP}}$ yɔk [DP ŋì-nánpùù lé [TP
C. IPFV raise 8.PL-yam.mound and
Olàmbà$_i$ [$_{\text{vP}}$ dàk [CP ké [TP ù$_i$ gà [$_{\text{ModP}}$ **ŋmàn** [$_{\text{vP}}$ ~~yèí~~
O. think that RP FUT can peel
~~lì-nùù-l~~]]]]]]]]].
7.SG-yam
Intended: ‘Neina is pounding fufu, and Olamba thinks he can.’

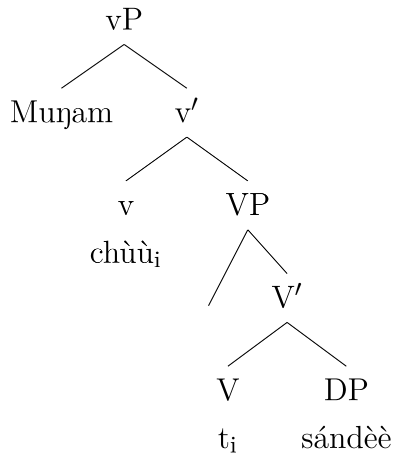
Even though Merchant (2013b) notes that there are instances where VP ellipsis permits voice mismatches between elements at the ellipsis site and the antecedent as illustrated in (222), Likpakpaanl ungrammatical sentences like (220-b) and (221-b) demonstrate that syntactic identity is essential in licensing MCE because only constituents that satisfy the syntactic identity condition can be elided. (222) violates the syntactic identity condition on ellipsis since the passive phrase is elided based on the active antecedent; nevertheless, the sentence is grammatical.

(222) The janitor must remove the trash whenever it is apparent that it should be removed.

(Merchant, 2013b, 78).

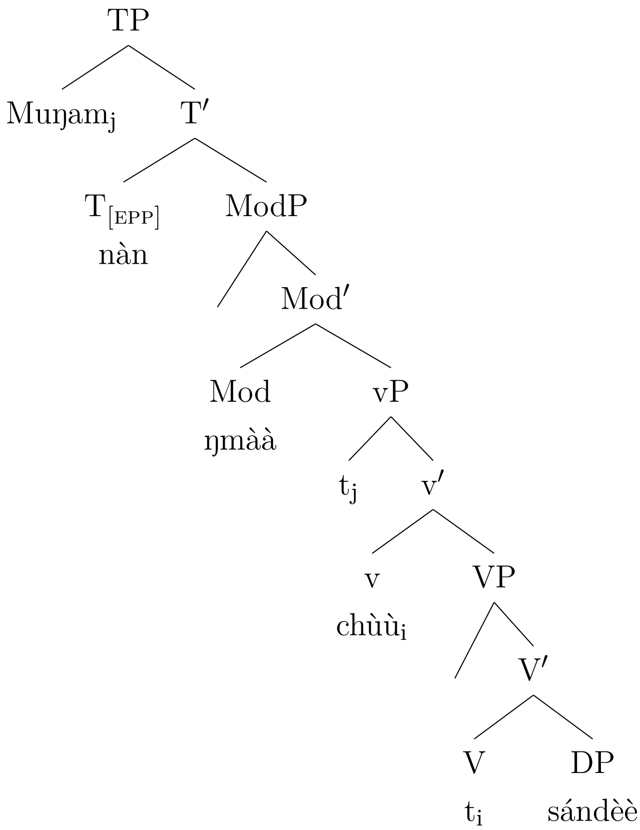
I adopt the standard assumption that Likpakpaanl modal complement ellipsis is licensed by Merchant (2001)'s an E-feature which is syntactically merged on the modal head of ModP. The syntactic component of the E-feature C-selects the root modal **ŋmàà**, while the semantic category ensures that only elements with an antecedent (e-GIVEN) can be elided while the phonological component 'deletes' ModP's vP complement and makes it PF content unpronounced at LF. The example in (223-a) demonstrates how MCE is derived in Likpakpaanl. At the start of the derivation, the DP **sándèè** 'rabbit' is merged as the internal argument of the verb **chùù** 'catch'. The VP is merged with little v, resulting in the movement of the verb from V^0 to v^0.

(223) a. Kusimi àà nàn ŋmàà chùù sándèè, àmàà Muŋam nàn ŋmán [$_{vP}$ ~~chùù~~ ~~sándèè~~.]
K. NEG PST able catch 3.SG.rabbit, CONJ M. PST able catch 3.SG.rabbit
'Kusimi couldn't catch a rabbit, but Muŋam could.'

b.


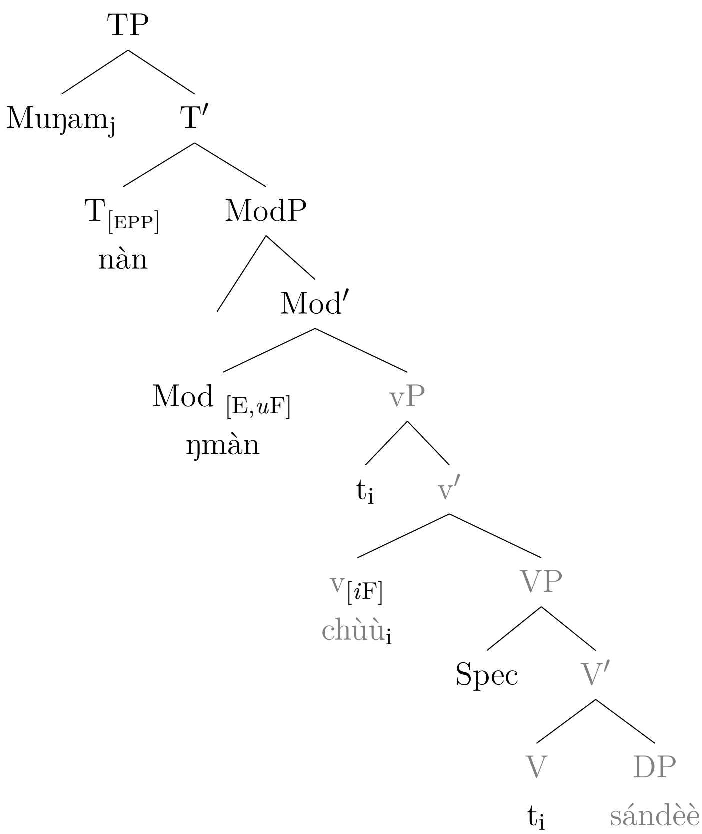
In the next derivation step, ModP with an interpretable root modal feature and an uninterpretable Categorial Feature [*u*F] is merged above vP. ModP probes and checks the interpretable Categorial feature [iF] on the goal, verb head, which it c-commands. Next, overt T-head **nàn**, with en EPP feature, is merged projecting TP. Since the EPP feature on T requires spec-TP to be filled, **Muŋam** base-generated in Spec-vP moves to Spec-TP as illustrated in (224).

(224)



Finally, the E-feature merged on the modal **ŋmàà** as illustrated in (225) licenses the deletion of vP complement to the modal. After the ellipsis, Mod^o, which acquired the E-feature during the derivation, post-syntactically undergoes a morphological change and is Spelt-Out as **ŋmàn**, rather than the default **ŋmàà**. This morphological alteration is necessary to satisfy well-formedness conditions at the Phonological Form (PF) interface, as stipulated in Halle and Marantz (1993)‘s Distributed Morphology framework.

(225)



A major observation from the examples discussed shows a morphological change of the modal auxiliary when it licenses ellipsis. I account for the observation by drawing on Halle and Marantz (1993)'s Distributed Morphology (DM), which conceptualises syntax as an abstract component, with phonological information added in the post-syntactic morphological component through Vocabulary Insertion Rules. They contend that DM also allows morphological operations to alter syntactic nodes to satisfy well-formedness conditions at the Phonological Form (PF) interface (Halle and Marantz, 1993). The implication is that the modal head **ŋmàà** must morphologically change to **ŋmàn** after the ellipsis takes place. Thus, the ellipsis operation at the syntactic level interacts with the morphological component, leading to observable changes in the realisation of the verbal form of the modal.

In the Distributed Morphology (DM) framework adopted for the analysis of **ŋmàà** as a licensor for Likpakpaanl MCE, the observed morphological change of the modal to **ŋmàn** under ellipsis is accounted for as follows; syntax generates the Mod^0 with an E-feature, which licenses the ellipsis of the complement of ModP. At the post-syntactic morphological component, the Mod^0 bearing the E-feature undergoes Vocabulary Insertion, mapping it to the phonological form **ŋmàn** instead of the default **ŋmàà**. This morphological reflex on the modal is required to satisfy well-formedness conditions at the Phonological Form (PF) interface.

4.4.0.2 Likpakpaanl MCE and the Modal **bàn**

This sub-section examines the root modal **bàn** as a licensor for MCE in Likpakpaanl. Although **bàn** is a root modal, its syntactic behaviour is different from its counterpart **ŋmàà**. While **bàn** selects a CP as its complement, its counterpart takes a vP complement as illustrated in the examples under 4.4.0.1. Consider the example in (226-a), where the CP complement 'that he dug a well' in the second conjunct is elided in the presence of **bàn**. The elided vP 'dig a well' is mutually entailed in the antecedent clause, and therefore, it can be elided. This is shown in the non-elliptical construction in (226-b), which expresses the same underlying meaning as its elided version in (226-a). Ellipsis licensed by the complementiser **ké** is illicit, as shown in (226-c) affirming the argument that only modals can license MCE in Likpakpaanl.

(226) a. [TP Ntiyaa nàn [$_{\text{Modp}}$ bàn [CP] (ké) [TP ù [$_{\text{vP}}$ gbìì [DP
N. PST want that RP dig
í-nyùnbùn lé [TP Bingo$_i$ mù nàn [$_{\text{Modp}}$ bàn [CP ~~(ké)~~ [TP
5.SG-well CONJ B. also PST want COMP
~~ù$_i$~~ [vP ~~gbìì í-nyùnbùn~~.]]]]]]]]]
RP dig 5.SG-well
'Ntiya wanted to dig a well, and Bingo also wanted to.'

b. [TP Ntiyaa$_k$ nàn [$_{\text{Modp}}$ bàn [CP] (ké) [TP ù$_k$ [$_{\text{vP}}$ gbìì [DP
N. PST want that RP dig
í-nyùnbùn lé [TP Bingo$_i$ mù nàn [$_{\text{Modp}}$ bàn [CP ~~(ké)~~ [TP
5.SG-well CONJ B. also PST want COMP
~~ù$_i$~~ [vP gbìì í-nyùnbùn.]]]]]]]]]
RP dig 5.SG-well
'Ntiya wanted to dig a well, and Bingo too wanted to dig a well.'

c. *Ntiyaa nàn bàn (ké) ù gbìì í-nyùnbùn lé Bingo$_i$ mù nàn
N. PST want that RP dig 5.SG-well CONJ B. also PST
bàn [CP ké [ù$_i$ ~~gbìì í-nyùnbùn~~]].
want COMP RP dig 5.SG-well
Intended: 'Ntiya wanted to dig a well, and Bingo too wanted to dig a well.'

The foregoing shows that ellipsis is licensed by the root modal **bàn** and the ungrammaticality of example (226-c) demonstrates that MCE in Likpakpaanl is blocked if the licensing head is a complementiser, supporting Merchant (2001)'s claim about selectional restrictions on lexical items in languages to license ellipsis.

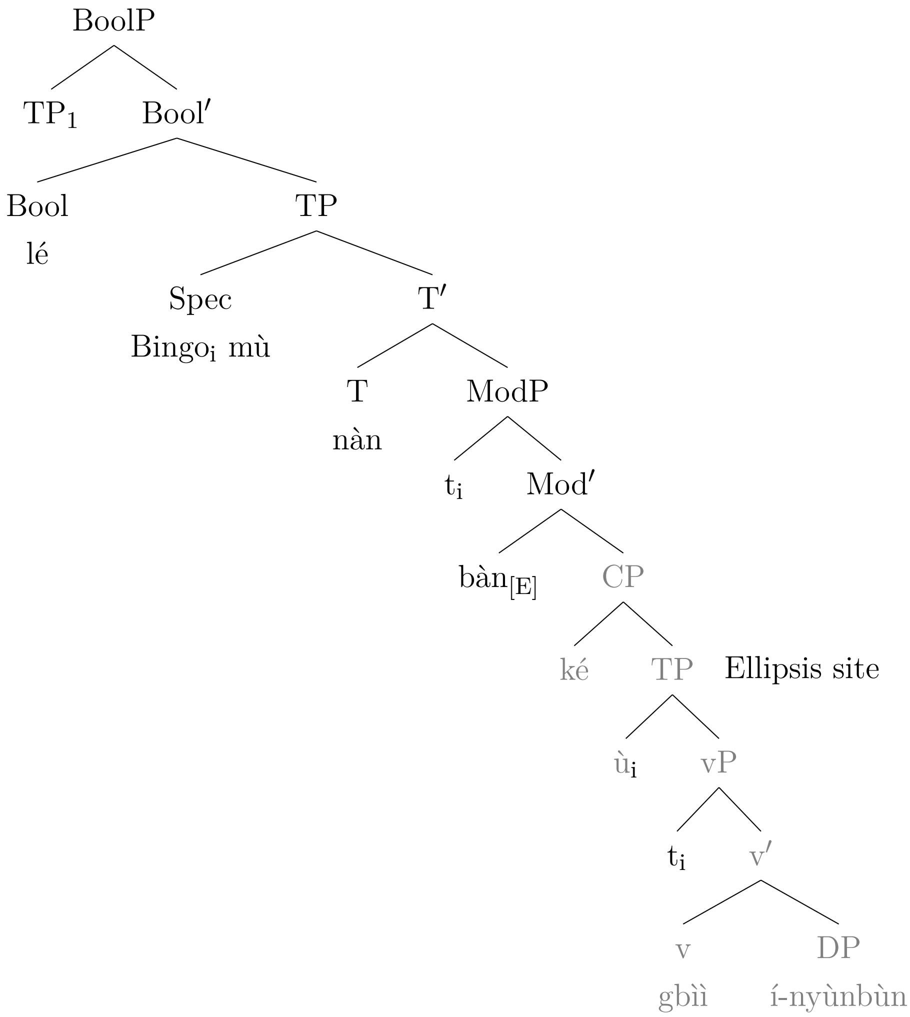
Moreover, in alternative questions with an option to make an 'unbiased' choice between two alternatives, the TP still complements the modal and undergoes deletion, as (227-b) shows. The semantic interpretation of the elided constituent is mutually entailed in the antecedent clause, and, therefore, the elided sentence is understood from the context provided by the antecedent clause (227-a).

(227) a. [TP Mpopiin nì Kunlamni$_k$ bà [$_{ModP}$ bàn [TP bì$_k$ [$_{vP}$ tér [DP
M. and K. PST want 3PL help
Nímòmì bèè [TP bì$_k$ àà bà [$_{ModP}$ bàn [TP ~~bì~~ [$_{vP}$ ~~tér~~
N. or 3PL NEG PST want 3SG help
~~Nímòmì~~]]]]]]]]]
N.
'Did Mpopiin and Kunlamni want to help Nimomi, or they didn't want to?'

b. [TP Mpopiin nì Kunlamni$_k$ bà [$_{ModP}$ bàn [TP bì$_k$ [$_{vP}$ tér [DP
M. and K. PST want 3PL help
Nímòmì bèè [TP bì$_k$ àà bà [$_{ModP}$ bàn [TP ~~bì~~ [$_{vP}$ tér
N. or 3PL NEG PST want 3SG help
Nímòmì.]]]]]]]]]
N.
'Did Mpopiin and Kunlamni want to help Nimomi, or they didn't want to help him?'

I propose the structure in (228) to account for MC ellipsis licensed by the modal **ban**. I assume the same underlying process that occurred in **ŋmàà**-licensed MCE indicated in (225). The only difference is that in the present case, **ban** targets TP for deletion, and the E-feature is merged on the modal head.

(228)



The observation that the modal **bàn** targets a TP for deletion is not novel since French, Italian, and Spanish modals have been found by Dagnac (2010); Depiante (2001) to behave similarly. In Likpakpaanl, the modal **bàn** licenses the ellipsis of an optional TP complement, deviating from the typical VP ellipsis phenomenon where the ellipsis site contains only the vP and its adjuncts. It has been argued that the French modal **pouvoir** 'can' in and the Italian **potere** 'can' license the ellipsis of a TP clausal complement rather than a VP. Dagnac (2010) argues that the French modal *pouvoir* is a subject-raising verb that selects a TP complement, as illustrated in (229-a).

(229) a.

Marie	lit	tous	les	livres	qu'elle	peut.	(French)
Mary	reads	all	the	books	that.she	can:3SG	

'Mary reads all the books that she can.'

b.

Maria	legge	tutti	i	libri	che	può.	(Italian)
Mary	reads	all	the	books	that.she	can:3SG	

'Mary reads all the books that she can.'

c. $[_{\text{VPk}}$ modal $[_{\text{TP}}$ t_{sujet} $[\ldots$ T $\ldots$ $[_{\text{AspP}}$ $\cdots$ $[_{\text{vP}}$ t_{sujet} $[_{\text{VP}}$ $\cdots.$ $]]]]]]$ (French)

(Dagnac, 2010, 161-162).

In this analysis, Dagnac (2010) contends that in (229-a), the subject, **elle**, raises from the embedded vP to the specifier of the embedded TP. She assumes an EPP feature on ModP in French, Italian, and Spanish, which extracts the subject from Spec-vP to the specifier of the conjugated modal verb **pouvoir** and the Italian **potere**. This is similar to the Likpakpaanl data, where the modal **bàn** licenses the ellipsis of the TP complement and, in some cases, extends the ellipsis site to contain the optional CP complement. Data such as (226-a) and (229-a) highlight the language-specific nature of MCE, where the modal verbs' selectional properties and the syntactic structures involved in the ellipsis process vary. Thus, while MCE in Likpakpaanl, French, and Italian share the core property of clausal ellipsis beyond VP ellipsis, the specific targeted constituents and derivations, including the landing site of the raised subject, are subject to cross-linguistic variation.

The syntax of **bàn** constructions raises the question regarding the presence of resumptive pronouns (RPs). An examination of the distribution of **bàn** shows RPs in spec-TP of the embedded clause. The distribution of resumptive pronouns has received different cross-linguistic analyses. (McCloskey, 1990, 2002) notes that in Palestinian Arabic, resumptives are used interchangeably with gaps created by wh-movement, while in Irish and Vata, as (Koopman, 1984) points out, resumptive pronouns can occupy syntactic positions where gaps are expected after an element has undergone A-bar movement to derived position. Scholars like (McCloskey, 2002) opine that resumptive pronouns are deployed as a 'Last-resort' mechanism in derivations where movement is blocked. However, the contrasting views hold that resumptives indicate traces of movement.

In Likpakpaanl, resumptives are island-sensitive, do not repair islands, and indicate a movement approach to resumption. The A-bar dependency in the complex noun phrase island in (230-b) is grammatical when the wh-phrase remains in its canonical position. However, (230-c) shows that presumptive are island-sensitive since the attempt to extract the wh-phrase to the left peripheral position is ungrammatical. The sentence is illicit in both resumptive and gap, indicating the impossibility of extracting the wh-phrase from a clause that complements the DP, 'rumour'.

(230) a. Kwayaja nàn ŋùn tìbŋùnlkààr ké Chàtí nàn mùò màngù.
K. PST hear rumour COMP C. PST eat 3.SG.mango
'Kwayaja heard the rumour that Chati ate a mango'

b. Kwayaja nàn ŋùn [DP tìbŋùnlkàår [CP ké Chàtí nàn mùò bà?]]
K. PST hear rumour COMP C. PST eat what
'Kwayaja heard the rumour that Chati ate what?'

c. *Bà lé Kwayaja nàn ŋùn [DP tìbŋùnlkàår [CP ké Chàtí
what FOC K. PST hear rumour COMP C. PST eat
nàn mùò ù$_i$/t$_i$.]]
RP

The discussion and data examined have revealed that unlike **ŋmàà**, which targets a vP for deletion, the ellipsis site for **bàn** is larger since it selects a TP as its complement. The data also shows that the distribution of **bàn** shares some similarities with French and Italian; it differs in allowing an optional complementiser to intervene between the licensing head and its complement.

4.5 Constraints on Likpakpaanl MCE

Cross-linguistic studies on ellipsis have demonstrated that the E-feature is syntactically constrained, resulting in not all lexical items being able to license ellipsis (Merchant, 2001; Aelbrecht, 2010; Van Craenenbroeck, 2017). The data shows there are constraints on the licensing of modal complement ellipsis (MCE) in Likpakpaanl since modal verbs like **yè kè** 'must', **nìbààkàn** 'maybe' and **lì** 'should', as well as tense, aspect and negation particles cannot license MC ellipsis in the language. The ungrammatical sentence in (231-a) involving the modal **yè kè** lends credence to the claim.

(231) a. *Bì-jàb gbààn$_i$ yè kè bì kùùr kí-sàà-k gbààn àmàà
2.PL-man DEF COP that RP weed 11SG-farm-11 DEF but
Mpopiin nì Nambey àà yè kè [~~bì$_i$~~ ~~kùùr~~ ~~kí-sàà-k~~
M. and N. NEG COP that RP weed 11.SG-farm-11
~~gbààn~~.]
DEF
Intended: 'The men must weed the farm, but Mpopiin and Nambey mustn't.'

In English, the modal 'must' can license VP ellipsis as 232 shows where in the pragmatically controlled context, the negated modal *mustn't* licenses the VP ellipsis. Van Craenenbroeck (2017) gives a context where someone is about to reach for some cookies that just came out of the oven. One speaker points first to a batch of cookies on a different plate and then to the hot batch, saying:

(232) a. These, you may _____. Those you mustn't _____, at least not until they cool down.

(Van Craenenbroeck, 2017, 16)

The examples in (233) further demonstrate that neither the deontic modal **lì** 'should' (233-a) nor the negative particle (233-b) can license MC ellipsis. The ungrammaticality of (233-a) and (233-b) illustrates this fact. It is important to state that these sentences are only unacceptable when the verb phrase is elided but are perfectly correct in non-elliptical contexts, as (233-c) and (233-d) indicate.

(233) a. *Kojo lì dàà chééchè àmàà Fred ààn lì [$_{vP}$ ~~dàà chééchè~~].
K. should buy.PFV 3.SG.bicycle but F. NEG.FUT should buy.PFV 3.SG.bicycle
Intended: 'Kojo should have bought a bicycle, but Fred shouldn't have.'

b. *Nìbààkàn Yelson gà gbìì bànchè gbààn àmàà Philip ààn [$_{vP}$ ~~gbìì bànchè DEF~~].
maybe Y. FUT dig 3.SG.cassava DEF but P. NEG.FUT dig 3.SG.cassava DEF
Intended: 'Maybe Yelson will dig the cassava, but Philip might/will not.'

c. Yelson gà gbìì bànchè gbààn àmàà nìbààkàn Philip ààn gbìì bànchè gbààn.
Y. FUT dig 3.SG.cassava DEF but maybe P. NEG.FUT dig 3.SG.cassava DEF
'Maybe Yelson will dig the cassava, but Philip might/will not dig the cassava.'

d. Kojo lì dàà chééchè àmàà Fred ààn lì dàà chééchè.
K. should buy.PFV 3.SG.bicycle but F. NEG.FUT should buy.PFV 3.SG.bicycle
'Kojo should have bought a bicycle, but Fred shouldn't have bought a bicycle.'

Furthermore, functional categories such as tense (234-a) and aspect heads (234-b) are blocked from licensing MCE in Likpakpaanl, confirming Merchant (2001) and Aelbrecht (2010, 100)'s assertion that there is an ellipsis feature in the lexicon for each type of elliptical construction in a language. The E-feature in Likpakpaanl MCE is, therefore, merged on root modal heads **ŋmàà** and **bàn**.

(234) a. *Bààyì ààn kpììn ì-ŋùò gbààn, àmàà [$_{TP}$ Nímbé **gà** [$_{vP}$ ~~kpììn ì-ŋùò gbààn~~.]]
B. NEG.FUT feed 6.PL-goat DEF but N. FUT feed 6.PL-goat DEF
Intended: 'Bààyì will not feed the goats, but Nimbe will.'

b. *Neina àà nàn bì màà kì-dìì-k gbààn, àmàà [TP Waaki
N. NEG PST IPFV build 11.SG-room-11 DEF but W.
nàn [AspP **bì** [vP ~~màà kì-dìì-k gbààn~~.]]]
PST IPFV build 11.SG-room-11 DEF

Dagnac (2010) notes that aspectual and voice head projections in French and Italian do not also license MC ellipsis. Dagnac (2010) contends that in (235), **avoir** and **avere** meaning 'to have', heads of AspP intervene between the licensing modal head 'can' and the vP, and yet the sentence is ungrammatical. Dagnac takes this as evidence that MCE cannot be licensed by an aspectual head in French and Italian and, secondly, that MCE is TP and not vP ellipsis. In (236), she argues that the voice heads projecting above vP cannot also elide the vP complement because they lack the E-feature to do so, resulting in the ungrammatical sentences (see: Dagnac, 2010, 163).

(235) a. *Tom peut avoir fini en juin, et Lea peut aussi avoir.
Tom can have finished in June and/also Lea can also have
Intended: 'Tom can have finished in June, and Lea can have, too.'

b. *Tom può aver finito in giugno e anche Lea può avere.
Tom can have finished in June and also Lea can have
Intended: 'Tom can have finished in June, and Lea can have, too.'

(236) a. *Paul peut être muté, et Kiko aussi peut être.
Paul can be transferred, and/(also) Kiko (also) can be
Intended: 'Paul can be transferred, and Kiko can be, too.'

b. *Paolo può essere trasferito, e anche Kiko può essere.
Paul can be transferred, and (also) Kiko (also) can be
Intended: 'Paul can be transferred, and Kiko can be, too.'

Despite these observations in Likpakpaanl, Italian, and French, available data in English has shown that it is possible for an aspectual and a voice head, as illustrated in (237-a) and (237-b), respectively, to intervene between the finite auxiliary bearing the E-feature and the elided verb Merchant (2013a, 78).

(237) a. Paul managed to repair this stuff, Luke *couldn't* have [~~repared this stuff~~.]

b. The janitor should remove the trash whenever it is apparent that it needs to *be* [~~removed~~.]

The data presented demonstrate that cross-linguistic variation in selectional constraints suggests that the ellipsis feature is subject to syntactic restrictions regarding which elements can serve as licensors for generating MC or VP ellipsis. I have also demonstrated that in Likpakpaanl, Italian, and French, only root modals can elide the PF content of their complements. We noted that MCE licensing in Likpakpaanl requires a rigid head-complement relationship between the licensor and the elided constituent. This accounts for why functional categories such as tense, aspect, and negation cannot license MCE.

4.6 Motivating a Movement plus PF Deletion Approach

The analysis thus far has contended that Likpakpaanl exhibits a type of ellipsis phenomenon where a root modal elides its complement. Evidence has indicated that a constraint exists, preventing all modals from bearing the E-feature to license MCE. It has also been argued that an underlying syntactic structure is present at the ellipsis site. This section employs extraction, island, and missing antecedent tests to support a Movement plus PF-deletion account of Likpakpaanl MCE. The fundamental assumption by proponents of the PF-deletion approach such Ross (1969); Merchant (2001); Johnson (2001); van Craenenbroeck (2010); Aelbrecht (2010); Dagnac (2010), among others, is that MCE or VPE will permit the extraction of elements from the ellipsis site if and only if the E-site contains sufficient structure to host a movement trace or copy.

4.6.1 MCE and Antecedent Contained Deletion (ACD)

The first evidence supporting the presence of underlying syntactic structure at the E-site comes from Antecedent-Contained Deletion (ACD) constructions. ACD is a syntactic phenomenon whereby an elided verb phrase is contained within its antecedent (Fox, 2002). A crucial diagnostic for distinguishing null proforms from PF-deletion ellipses lies in the (im)possibility of extraction. The extraction facts in Likpakpaanl, where modal ellipsis permits ACD, support the PF-deletion analysis. The examples in (238-a) and (238-b) demonstrate that Likpakpaanl modal ellipsis can have elements in the elided VP at the ellipsis site contained within its antecedent.

(238) a. Kùnjàm nàn tàŋ ŋì-nààtàk ŋì-mɔmɔk$_i$ ù nàn ŋmàn [$_{vP}$ ~~tàŋ~~ t$_i$ nà.]
K. PST wear 8.PL-sandal 8.PL-all 3SG 8.PST can REL.DET
'Kunjam wore every sandal that he could.'

b. Suisɔŋ nì Bìmà gà tèr ù-nì ù-mɔmɔk$_i$ bì gà bàn [TP ~~bì~~ ~~tèr~~ t$_i$ nà.]
S. and B. FUT help 1.SG-person 1.SG-all 3SG FUT want 3SG help REL.DET
'Suisɔŋ and Bima would help everybody that they want to.'

c. *Kùnjàm nàn tàŋ ŋì-nààtàk ŋì-mɔmɔk$_i$ Dimbu nàn tùk [DP
K. PST wear 8.PL-sandal 8.PL-all D. PST say
tìbŋùnlkààr [CP kè [TP ù ŋmàn [$_{vP}$ tàŋ t_i nà.]]]
rumour COMP 3SG can wear REL.DET
Intended: 'Kunjam was able to wear every sandal that Dimbu told a rumour that he could.'

In (238-a), and (238-b), for instance, the quantificational DPs 'every sandal' and 'everybody' are extracted from their canonical position, the complement to the verbs **táŋ** 'wear' and **tèr** 'help', respectively. After the movement, the E-feature on the licensor, **ŋmàà** or **bàn**, is merged and licenses the deletion of the PF content of the licensing head's complement. Such ACD constructions in Likpakpaanl, in which a phrase is moved from a base position inside the gap prior to ellipsis, are also island sensitive and, thus, confirm that movement has indeed taken place. (238-c) is ungrammatical because the complex noun phrase island blocks the extraction from the island. In their analysis of French (239-a) and Spanish (239-b) modal ellipsis, Dagnac (2010) and Fernández-Sánchez (2023) adduce similar ACD tests as proof of PF-deletion (see Aelbrecht, 2010; Alsaud, 2023) for similar evidence for PF-deletion in Dtuch and Najdi Arabic.

(239) a. Léa lit tous [DP les livres$_i$ [CP t_i qu'elle peut [~~lire~~ t_i]]].
L. read all the books that'she can
'Lea reads all the books that she can.'

(Dagnac, 2010, 159)

b. Ella siempre ve todas las películas que puede.
she always watch.PRS.3SG all the films that can.PRS.3SG
'She always watches all the movies she can.'

(Fernández-Sánchez, 2023, 907)

4.6.2 A-bar Movement in Likpakpaanl MCE

Further evidence for a movement plus PF-deletion analysis comes from the A-bar movement of constituents to left peripheral focus and topic positions. Cross-linguistically, the ability to extract elements from an ellipsis site has been taken as a diagnostic for the presence of internal syntactic structure at that site (Merchant, 2001; van Craenenbroeck, 2010; Aelbrecht and Harwood, 2019). The Likpakpaanl data demonstrate that constituents can undergo A-bar movement out of the modal complement ellipsis site to focused and topicalised positions in the left periphery. This extraction from the ellipsis site supports an underlying syntactic representation being present at PF but pronounced. Consider the focused movement of the DP in (240-a) and the topicalised object in (240-b) (both elements in bold here for emphasis).

(240) a. Wààjà bà bàn (ké) ù jì sàkɔlà àmàà tì nyì ké [
W. PST want COMP 3SG eat 3.SG.fufu but 2PL know COMP
$_{\text{FocP}}$ **sàkɔlà** lé [TP Jàlùlà mù àà [$_{\text{ModP}}$ bàn [CP ~~(ké)~~ [TP ~~ù~~
3.SG.fufu FOC J. too NEG want.
[$_{\text{vP}}$ jì t_i]]]]]].

'Waaja wants to eat fufu, but we know that it is fufu that Jalula, too, doesn't want.'

b. Kàndò dàk kè Torbi$_i$ gà ŋmàà kùùr kì-sàà-k gbààn,
K. think COMP T. FUT can weed 12.SG-farm-12 DEF
àmàà [TP tì nyì [CP kè [TOPP **kì-sàà-k gbààn**$_j$
but 3PLL know COMP 12.SG-farm-12 DEF
mà-kàn [TP ù$_i$ ààn [$_{\text{ModP}}$ ŋmàn [VP ~~kùùr~~ t_j]]]]]].
TOP RP NEG.FUT can
'Kando thinks (that) Torbi can weed the farm, but we know that as for the farm, he can't.'

In (240-a), the object DP **sàkɔlà** 'fufu' undergoes focus movement from its base position as the complement of the verb 'eat' to Spec-FocP in the left periphery of the embedded CP. Crucially, the DP is A-bar moved from the ellipsis site, as evidenced by the trace. In (240-b), the DP **kì-sàà-k gbààn** 'the farm' is extracted out of the ellipsis site to the specifier of the topic head before ellipsis occurs. Extracting constituents from within the ellipsis site to focus on the topic provides strong evidence of a movement account.

The ungrammaticality of the sentence (241-a) can be explained by the Adjunct Island Constraint, which states that constituents cannot be extracted from an adjunct clause, such as a temporal or causal adjunct (Ross, 1967). Since the topicalised DP in Spec-DP in (241-a) is base-generated inside the adjunct clause as the trace indicates, and therefore, cannot be extracted from the island without. This supports the view that the MCE examples in 240 involve movement. However, if a resumptive pronoun is inserted in the position of the gap, as shown in (241-b), the resulting sentence becomes grammatical.

(241) a. *Lì-bùù-l gbààn$_i$ má-kán, Naajo nán lùù n-nyùn dóó [
7.SG-pot-7 DEF TOP N. PST fetch 5.SG-water finish
$_{\text{adjunt island}}$ wáár lé Sanja kán t_i.]
before CONJ S. see.PFV

b. Lì-bùù-l gbààn má-kán, Naajo nán lùù n-nyùn dóó
7.SG-pot-7 DEF TOP N. PST fetch 5.SG-water finish
wáárlé Sanja kán lì.
before CONJ S. see.PFV RP
Intended: 'As for the pot, Naajo finished fetching the water before Simon saw it.'

Using the resumptive pronoun ameliorates the island violation by ensuring that the topic DP is base-generated in Spec-DP and binds the pronoun inside the island. McCloskey (2002) has proposed using resumptive pronouns to circumvent island violations as a last resort strategy.

4.6.3 Missing Antecedents and Pronominal Agreement

Within the PF deletion approach to ellipsis, it has been contended that the availability of anaphoric dependencies with missing antecedents in verb phrase ellipsis constructions indicates the presence of syntactically unpronounced structures underlying these ellipsis sites. The example in (242-a) shows that the pronoun **lì**[1] must morphologically agreement with the antecedent **lìnyóónl** 'lemon', which is contained the elided vP complement to **ŋmàà**. (242-a) demonstrates that the pronoun **lì** needs to have its antecedent in the elided verb phrase 'pluck lemon' within the local clausal domain. The sentence is ungrammatical in (242-b) because the pronoun lacks an explicit antecedent DP to which it can bind. However, for the pronoun **lì** to properly refer back to the object antecedent, DP must c-command the pronoun within the local domain of the second clause. Since the antecedent cannot c-command the pronoun due to its occurrence in a different clause, this violates the binding conditions.

(242) a. Tinyoor àà nàn ŋmàà kìr lì-nyóón-l$_j$, àmàà Philemon nàn
T. NEG PST can pluck 7.SG-lemon-7 CONJ P. PST
ŋmàn [$_{vP}$ ~~kìr lì-nyóón-l~~$_i$], lé lì$_i$ nàn gálg pám.
can pluck 7.SG-lemon-7 and 7.SG.it PST sour INTENS
'Tinyoor couldn't pluck a lemon, but Philemon could, and it was very sour.'

[1]Pronouns in Likpakpaanl are the same as the noun class prefix of the antecedent noun. They share the same number features as shown below:

a. Saaja nán chóón bù-sùb àmáá bù kpó.
S. PST plant 9.SG-tree but 9.SG.it die
'Saaja planted a tree but it died.'.

b. *Philemon àà nàn ŋmàà kìr lì-nyóón-l$_i$, lé lì$_i$ nàn gálg
P. NEG PST can pluck 7.SG-lemon-7 CONJ it PST sour
pám.
INTENS

Fernández-Sánchez (2023) also suggested that the ability to anaphorically link a pronoun to an indefinite expression within an elided VP is evidence that the ellipsis site contains an internal structure.

Fernández-Sánchez (2023) also presents data from Spanish indicating that when an indefinite DP is under the scope of negation, it cannot anaphorically refer to a weak pronoun as shown in the ungrammatical sentence in (243-a). However, in MCE, where the complement to the modal verb is elided, as in (243-b), the pronoun can bind the indefinite expression to the PF-deleted structure. Consequently, establishing these anaphoric dependencies from missing antecedents in MC and VP ellipsis supports syntactic structure at the E-site.

(243) a. Javier no pudo comprar una casa en la montaña, pero
Javier not can.PST.3SG buy.INF a house in the mountain but
Pedro sí pudoo [~~comprar una casa en la montaña~~]$_i$, y
P. yes can.PST.3SG buy.INF a house in the mountain and
la$_i$ decoró fatal.
it decorated terribly
'Javier couldn't buy a house in the mountain, but Pedro could, and he decorated it terribly.'

b. *Javier no pudo comprar [una casa en la montaña]$_i$, y
Javier not can.PST.3SG buy.INF a house in the mountain but
la$_i$ decoró fatal.
yes can.PST.3SG terribly
Intended: 'Javier couldn't buy a house$_i$ in the mountain, and he decorated it$_i$ terribly.'

(Fernández-Sánchez, 2023, 894)

4.7 Chapter Summary

This chapter examined a type of Verb Phrase ellipsis referred to in the literature as Modal Complement Ellipsis (MCE). The chapter discussed the syntactic distribution of modal verbs in Likpakpaanl and how they express root and epistemic modality. It also explored the semantic interpretations that arise from the interaction between tense, aspect, and modals. For example, it demonstrates that the bi-partite modal for 'must' can convey 'immediate' or 'past necessity' by combining with the perfective aspect or past tense, respectively. The chapter further presented data to show that modal verbs project a ModP above the verb and aspect, following the hierarchical structure proposed by Cinque (1999). I also showed that when **ŋmàà** and **bàn** co-occur, it is the higher modal **ŋmàà** licenses MCE, suggesting that there is the only place in the lease spine that the E-feature can be hosted. Consequently, **bàn** moves to the higher modal position occupied by **ŋmàà** when it is the only modal to license Likpakpaanl MCE. I also showed that a restricted group of modals must license Likpakpaanl MCE since only **ŋmàà** and **bàn** license MCE, suggesting that the E-feature is merged on **ŋmàà** and licenses ellipsis through an Agree relation between the verb and the modal. This restriction supports the cross-linguistic claim that there was a restriction on the heads that could license ellipsis Aelbrecht (2008, 2010); Dagnac (2010); Lipták and Aboh (2013). In terms of the size of the ellipsis site, the modal **ŋmàà** targets the vP complement for deletion, while **bàn** deletes a CP complement. Using evidence from Antecedent-Contained Deletion, focus and topic movement, island constraints, missing antecedents, and anaphoric binding, I argued that the E-site in Likpakpaanl MCE has an underlying syntactic structure which becomes phonologically null when the E-feature was merged on the modal head.

Chapter 5

On Likpakpaanl Sluicing

5.1 Introduction

Although sluicing constructions have been extensively studied in a variety of languages such as English, Hungarian, Korean, and Arabic (Van Craenenbroeck and Lipták, 2013; Merchant, 2001; Park, 2014; Algryani, 2012), there has been comparatively little syntactic investigation of sluicing in African languages, with analyses of sluicing in Gungbe Lipták and Aboh (2013), Nupe Mendes and Kandybowicz (2023), and Bassa Bassong (2014) as notable exceptions in the literature. This chapter, therefore, provides the first analysis of sluicing in Likpakpaanl, examining its core properties and implications for syntactic theory. The investigation demonstrates that sluicing occurs in Likpakpaanl and can be derived from regular wh-questions via wh-movement followed by TP deletion. I argue that sluicing constructions derive from canonical wh-questions and align with focus movement, though with some variation in how the focus head is spelt out under sluicing. This chapter, thus, provides novel evidence and theoretical analysis of sluicing in an understudied African language, advancing our understanding of sluicing typology and the theoretical mechanisms involved in deriving sluicing cross-linguistically. The overarching goal of this chapter is to investigate the derivation of sluicing in Likpakpaanl, provide a syntactic analysis of sluicing and the nature of Merchant's E-feature in the language, and adduce syntactic evidence to support a PF-Deletion account of sluicing in Likpakpaanl.

Beyond this introduction, the remainder of the chapter is structured as follows: Section 5.2 provides a recap of the notion of sluicing. Section 5.3 discusses the properties of sluicing in Likpakpaanl and the constraints on sluicing in the language. Section 5.6 examines the syntax of Likpakpaanl sluicing, while Section 5.7 provides empirical evidence supporting a movement plus PF deletion approach to Likpakpaanl sluicing. Section 5.8 examines Merchant (2001)'s Sluicing-COMP Generalisation and the Baltin (2010) by analysing data from Likpakpaanl and related languages, along with the implications for the syntax of sluicing cross-linguistically. Finally, Section 5.9 gives a chapter summary of the main ideas illustrated in the discussion.

5.2 The Notion of Sluicing

Sluicing as a linguistic phenomenon was first identified by Ross (1969) and refers to an elliptical construction in which only the wh-phrase remains from an underlying constituent question. The examples from Merchant (2004) and Ross (1969) show that the sluice structure contains a wh-remnant but omits the rest of the clause but has the same semantic interpretation as the corresponding non-elliptical counterpart (strikeout) in (244-b) and (245-b) whereby the TP is elided after the wh-remant has A-bar moved to Spec-CP.

(244) a. Somebody just left—guess **who**.
b. Somebody just left—guess **who** ~~just left.~~

(Ross, 1969, 252).

(245) a. Jack bought something, but I don't know **what**.
b. Jack bought something, but I don't know **what**$_i$ ~~he bought t_i~~.

(Merchant, 2001, 664).

Although Ross (1969) originally defined sluicing as applying solely to embedded clauses such as (244), contemporary syntacticians analyse sluicing more broadly as an ellipsis of the TP complement of a constituent question, leaving only a wh-phrase remnant as argued by Lipták and Bánréti (2022). Under this definition, sluicing can occur in both embedded and root clauses. The Slovenian examples in (246-b) and (246-c) demonstrate matrix and embedded sluicing, respectively.

(246) a. [A:] Valaki úszott a tóban.
someone swim.$_{pst}$.3SG the lake.Ine
'Someone was swimming in the lake.'
b. [B:] Igen? Ki?
yes who
'Was that the case? Who?'
c. Valaki úszott a tóban, de nem tudom, ki.
someone swim.PST.3SG the lake.Ine but not know.1SG who
*Someone was swimming in the lake, but I don't know who.

(Lipták and Bánréti, 2022, 140)

While the non-isomorphic approach to ellipsis discussed in 2.2.3 presents compelling evidence against strict structural identity approaches in many languages, in Likpakpaanl, the evidence typically adduced for non-isomorphic approaches –

voice mismatches and sprouting – is notably absent from the grammatical system. Thus, the prediction of flexibility in ellipsis identity conditions in non-isomorphic approaches isn't attested in Likpakpaanl. Consider that in Likpakpaanl where passive constructions are not possible as shown in **??** but also the voice system does not permit the alternations necessary for voice mismatches in sluicing (247-b).

(247) a. *Kì-gbàŋ gbààn nàn kárn (nì Naajo).
7.SG-book DEF PST read with N.
Intended:' The book was read by Naajo.'

b. *Kì-gbàŋ gbààn nàn kárn ámàà m àà bé ké nì yé ŋmà.
7.SG-book DEF PST read CONJ 1.SG NEG know COMP it COP what
Intended:' The book was read but I don't know by whom.'

This language-specific constraint suggests that while non-isomorphic approaches offer elegant solutions for languages that permit such structures, they may not be universally applicable across all language types. As such, for the analysis of Likpakpaanl sluicing, I adopt the Movement Approach, which can be clearly demonstrated and empirically tested within the language's grammatical system.

5.3 Likpakpaanl Sluicing Constructions

This section discusses sluicing in Likpakpaanl, including the sluicing sites, eligible wh-phrases, potential survival of remnants, and island sensitivity. Sluicing has been argued to be parasitic on focus-movement, and many languages exhibiting overt focus-movement permit sluicing (Merchant, 2001; Vicente, 2009). Even though in Section 3.8 I discussed the focus movement of Likpakpaanl wh-phrases, I will briefly recapitulate wh-questions in Likpakpaanl to establish the foundation for discussing sluicing. Likpakpaanl wh-phrases can occur both in-situ and ex-situ. While in-situ wh-questions (both subject and non-subject) are morphologically unmarked, as shown in 248 and 249, the answers to such in-situ wh-questions must be marked by either **lè** or **là**.

(248) a. [Q:] Waaja gà pèèn bà?
W. FUT wear what
'What will Waaja wear?'

b. [A:] Waaja gà pèèn lì-bùákú-l là.
W. FUT wear 7.SG-shirt-7 FOC
'Waaja will wear a SHIRT.'

(249) a. [Q:] Wumbei nàn tìì ŋmà sìmá?
W. PST give who 3.SG.groundnut
'Who did Wumbei give groundnut?'

b. [A:] Wumbei nàn tìì Naajo lè sìmá.
W. PST give who 3.SG.groundnut
'Wumbei gave NAAJO groundnuts/ It is NAAJo that Wumbei gave groundnuts.'

Ex-situ wh-phrases, on the other hand, can be optionally focus-marked after they are extracted to the left periphery of the clause in Spec-FocP. The response to the wh-question must, however, be obligatorily marked as (250) shows.

(250) a. [Q:] Bà$_i$ (lé) Mpòpíín gà ŋààn t$_i$?
what FOC M. FUT cook
'What will Mpopiin cook?'

b. [A:] Lì-nùù-l *(lé) Mpòpíín gà ŋààn t$_i$
7.SG-yam-7 FOC M. FUT cook
'Mpopiin will cook YAM?'

The optionality of focus marking in Likpakpaanl wh-questions suggests the interface between syntax and pragmatics (Rizzi, 1997; Hartmann and Zimmermann, 2008). There is no truth-conditional semantic difference in contexts where the focus marker is present versus absent, suggesting that the variation serves discourse-pragmatic functions rather than affecting core meaning (Kiss, 1998). Following Hartmann and Zimmermann (2008)'s information structural approach suggests that when the focus marker appears with the wh-phrase, it adds a layer of pragmatic prominence to the discourse without altering its fundamental semantic interpretation (Beaver and Clark, 2008). This follows the assumption that grammaticalised focus marking can serve redundant pragmatic functions even when the semantic focus is independently established through wh-movement (Aboh, 2004a).

5.3.1 Sluicing with an Explicit Argument

In Likpakpaanl, sluicing is restricted to interrogatives with constituent wh-questions, as has been established since Ross (1969), Lobeck (1995), and Merchant (2001). It is characterised by FocP ellipsis after extracting the wh-phrase, as shown in the matrix, and embedded sluicing in (251-b) and (252-a), respectively.

(251) a. [Q:] Naajo bì yìn ù-nì ù-bàà.
N. IPFV call 1.SG-person 1.SG-INDEF
'Naajo is calling someone.'

b. [A:] Mbàmɔn àà? **Ŋmà**$_i$ [$_{FocP}$ ~~(lé)~~ [TP ~~Naajo bì yìn~~ t$_i$?]]
true INTER who FOC N. IPFV call

'Is that so? Who?'

c. [A1:] *Mbàmɔn àà? **Ŋmà**$_i$ **(lé)** [TP ~~Naajo bì yìn~~ t$_i$?]
true INTER who FOC N. IPFV call
Intended: 'Is that so? Who is Naajo calling?'

In the embedded sluicing in (252-a), the sluiced wh-phrase **bà** 'what' corresponds to the overt indefinite correlate **tì-wàn nì-bàà** 'something' that serves as the direct object of the verb **nyù** 'drink.' We notice that the wh-phrase **bà** is introduced by the complementiser **kè**, a common feature in Likpakpaanl sluicing constructions. The complementiser appears obligatory in embedded sluicing contexts, suggesting rich left peripheral projections above FocP, where the sluice moves before the TP ellipsis takes place. It is also important to note that the overt indefinite correlate in the antecedent clause is crucial for the sluicing construction to be felicitous. The presence of this correlate sets up the context for the sluiced wh-phrase to determine the 'identity' of the indefinite entity. The ungrammaticality of (251-c) and (252-c) is due to the overt focus particle in the wh-sluice.

(252) a. [TP Chati bì nyù tì-wàn nì-bàà, àmàà m àà bè [CP
C. IPFV drink 15-thing 15-INDEF CONJ 1SG NEG know
kè [PredP nì yè [FocP **bà**$_i$]]]].
COMP it COP what
'Chati is drinking something, but I don't know what.'

b. [TP Chati bì nyù tì-wàn nì-bàà, àmàà m àà bè [CP
C. IPFV drink 15-thing 15-INDEF CONJ 1SG NEG know
kè [PredP nì yè [FocP **bà**$_i$ ~~**lé**~~ [TP ~~Chati bì nyù~~ t$_i$.]]]]
COMP it COP what FOC C. IPFV drink
'Chati is drinking something, but I don't know what ~~Chati is drinking~~.'

c. *Chati bì nyù tì-wàn nì-bàà, àmàà m àà bè kè
C. IPFV drink 14-thing 14-INDEF CONJ 1SG NEG know COMP
nì yè **bà**$_i$ **lé** ~~TP t$_i$~~
it COP what FOC

Apart from argument wh-phrase, Likpakpaanl sluicing can also occur with an implicit argument like an adjunct wh-phrases as shown in (253-a) and (253-c). The non-elliptical counterparts in (253-b) and (253-d) have the same semantic interpretation as the sluices in (253-a) and (253-c).

(253) a. Mpopiin lèn kè ù gà dàn, lè Tàmànjà bàé kè nì
M. say COMP 3SG FUT come CONJ T. ask COMP it
yè **bà-dààl** [~~(lè) Mpopiin gà dàn~~ t$_i$?]?
COP what-day FOC M. FUT come
'Mpopiin said she will come, and Tamanja wants to find out when.'

b. Mpopiin lèn kè ù gà dàn, lè Tàmànjà bàé kè nì
M. say COMP 3SG FUT come CONJ T. ask COMP it

yè **bà-dààl**$_i$ (lè) Mpopiin gà dàn t$_i$?
COP what-day FOC M. FUT come t$_i$
'Mpopiin said she will come, and Tamanja wants to find out when.'

c. Wongol dàk kè Julia nàn sùn ŋì-nààtàk, lè tì bàn
W. think COMP J. PST steal 8.PL-sandal CONJ 3PL want
tì bé kè nì yè **làchè**$_i$ [~~(lè) Julia nàn sùn ŋì-nààtàk~~ t$_i$.]
3PL know COMP it COP where FOC J. PST steal 8-sandal
'Wongol thinks (that) Julia stole a pair of sandals, and we want to know where.'

d. Wongol dàk kè Julia nàn sùn ŋì-nààtàk, lè tì bàn
W. think COMP J. PST steal 8.PL-sandal CONJ 3PL want
tì bé kè nì yè **làchè**$_i$ (lé) Julia nàn sùn ŋì-nààtàk t$_i$.
3PL know COMP it COP where FOC J. PST steal 8-sandal
'Wongol thinks (that) Julia stole a pair of sandals, and we want to know where Julia stole a pair of sandals.'

Another characteristic of Likpakpaanl sluicing is that the wh-phrase can comprise d-linked wh-phrases as in (254).

(254) Ì-gbéér gbààn pónnì ì-bàà gà kpó, ámàà m àà ŋmàà
6.PL-pig DEF among 6.PL-INDEF FUT die, CONJ 1SG NEG can
bé ké nì yè ì-pónnì ì-làì
know COMP it COP 6.PL-among 6.PL-which
'Some of the pigs will die, but I don't know which ones.'

In (254), the wh-remnant is a complex wh-phrase consisting of the wh-phrase **ì-pónnì ì-làì** 'which one' referring to the indefinite nominal *some of the pigs.* The presence of this complex wh-phrase in the sluiced clause suggests that Likpakpaanl allows for more intricate wh-remnants in sluicing constructions beyond simple wh-words like *what, who, when,* or *where.*

5.3.2 Sluicing with an Implicit Argument

Moreover, Likpakpaanl also exhibits instances of implicit sluicing, as illustrated in (255-a), where the fronted wh-remnant corresponds to an argument that is not overtly realised in the antecedent clause but is licensed by the argument structure of the predicate and the context.

Context:
Sanja is sick and sitting under a tree while his son and his friends are playing. Then Emmanuel hears Sanja calling. Emmanuel tells his friend that,

(255) a. Sanja nàn bì yìn, àmàà m àà nyì ké nì yè ŋmà$_i$ [~~(lé)~~
S. PST IPFV call CONJ 1.SG NEG know COMP it COP who FOC
~~Sanja nàn bì yìn~~ t$_i$.]
S. PST IPFV call
Lit: 'Sanja was calling, but I don't know who.'

In (255-a), the verb **yìn** 'call' takes an object argument, though it is unpronounced in the antecedent clause. The sluiced wh-phrase **ŋmà** 'who' could refer to the implicit argument 'his son', making this an instance of implicit sluicing. Merchant (2001) uses the availability of implicit sluicing as evidence that sluicing targets full clausal structure, with implicit arguments present in the syntax prior to ellipsis. Consequently, licensing an implicit wh-remnant would be unexpected if the antecedent lacked syntactically represented arguments. Thus, (255-a) supports syntactic approaches to sluicing in which the derivation involves the wh-movement of a structurally present element within a full clausal syntax that gets deleted under identity with the antecedent clause.

5.3.3 Inverse Copular Sluicing

The phenomenon of inverse copular sluicing, as exemplified in the provided example, can be linked to the analysis and properties of inverse copular constructions discussed earlier in Section 3.9.2. Thus, it is possible to have sluicing where the wh-phrase appears above the copular verb **yè** without the expletive marker **nì**. The example in (256) demonstrates the point.

(256) a. John tér ù-nì ù-bàà lé m bàn bè **kè** **ŋmà** **yè**.
J. help.PFV 1-person 1.INDEF CONJ 1SG want know COMP who COP
'John helped someone, and I want to know who.'

b. John dáá tì-wàn nì-bàà lé m bàn bè **kè** **bà** **yè**.
J. buy.PFV 14-thing 14.INDEF CONJ 1SG want know COMP what COP
'John bought something, and I want to know what.'

In (256), the wh-sluice **ŋmà** 'who' precedes the copular verb **yè**, suggesting that the wh-sluice has undergone movement to a higher structural position in the clausal architecture. This positioning of the wh-remnant is reminiscent of the word order observed in inverse copular constructions, where the predicative expression, the wh-phrase, appears to have inverted to a pre-verbal position. The availability of inverse copular sluicing in Likpakpaanl provides further evidence for the articulated structure of the left periphery and the potential for wh-movement to target higher positions within this domain.

5.4 Mù, 'else' modification in Likpakpaanl Sluicing

In his seminal work on sluicing, Merchant (2001) presents the distribution of 'else' modification as an argument against analysing sluicing in English as involving a cleft structure. He observes that while 'else' can modify wh-phrases in regular interrogatives, as in (257-c), it is disallowed in clefts, as shown in (257-b). Based on this asymmetry, Merchant contends that the elided constituents in English sluicing constructions are not clefts but regular wh-interrogative clauses.

(257) a. Harry was there, but I don't know who else.'
b. . . . but I don't know who (*else) it was that was there.
c. . . . but I don't know who else was there.

(Merchant, 2001, 122)

Example (258) presents a similar pattern as in the English sentence in (257-a) where a wh-phrase is followed by the particle **mù**. It can also occur in certain contexts where the wh- phrase can occur with the sluice. Given the context in (259) and the list of the people who received the presents is not exhaustive, it implies that (259) introduces a non-exhaustive interpretation since the sentence is grammatical.

(258) A: Kofi dàà lòòr pùàn.
K. buy.PFV 4.car new
'Kofi has bought a new car.'

B: Kì-nà-à? Nì bà **mù**
11-so-Q and what else
'Is that so? And what else?'

The presence of non-exhaustive **mù** 'else' in Likpakpaanl sluicing constructions provides evidence for a copular source analysis, following theoretical frameworks developed by Merchant (2001) and van Craenenbroeck (2010). The underlying structure in (259-b)can be analysed as a specificational copular clause rather than a standard focus movement construction. The first observation supporting this analysis is that the presence of **mù** 'else' indicates that the wh-phrase operates over a presupposed set of alternatives, a characteristic property of copular constructions (Merchant, 2001). The non-exhaustive reading triggered by **mù** aligns with the semantics of specificational copular clauses, where the wh-phrase functions as a variable ranging over a partially identified set. This contrasts with direct questioning of the event structure, which typically yields exhaustive readings (van Craenenbroeck, 2010).

(259) **Context**: *A rich man throws a party and gives presents to some guests. But there are some people to whom he still also gave the presents.* The statement in ((259-a)) is uttered, and the sluicing question in (259-a) is posed.

a. Ù-kpà-dààn gbààn nàn tìì ù-nì ù-bàà mù
1.SG-have-owner DEF PST give 1.SG-person 1.SG-INDEF else
ì-pììn.
6.PL-present
'The rich man gave someone else presents.'

b. Kì-nà-à? **ŋmà mù**$_i$ ~~[(lé) ù-kpà-dààn gbààn nàn tìì~~ t$_i$
11-so-Q who else FOC 1-have-owner DEF PST give
~~ì-pììn]~~?.
6-presents.
'Is that so? Who else.'

It is also important to point out that the particle **mù**, which I identify with the English word 'else', forms a constituent with the wh-element as indicated in (260-b), where it is extracted together with the wh-phrase to the left-periphery. The ungrammaticality of (260-c) occurs because **mù** is stranded in its base-position and separated from its wh-constituent. Therefore, it is not surprising that **mù** survives sluicing with the wh-phrase while the focus head is 'deleted', as shown in (259-b). The fact that **mù** is not elided under sluicing suggests that it is not a head attracting wh-movement to its Spec but an element locally adjoined to the wh-phrase. This observation probably indicates the **mù** forms a constituent of the wh-phrase, and both undergo focus movement.

(260) a. Ù-bò gbààn nàn kìr bà mù?
1-child DEF PST pluck what else
'What else did the child pluck?'

b. Bà mù$_i$ lé ù-bò gbààn nàn kìr t_i ?
what else FOC 1-child DEF PST pluck
'What else did the child pluck?'

c. *Bà$_i$ lé ù-bò gbààn nàn kìr t_i mù?
what FOC 1-child DEF PST pluck else
Intended: 'What else did the child pluck?'

While the evidence explored so far demonstrates that wh-phrases play a crucial role in forming sluicing constructions in Likpakpaanl, certain restrictions prevent some wh-phrases from participating in the sluicing process. The subsequent section examines some constraints on wh-phrases in Likpakpaanl sluicing.

5.5 Constraints in Likpakpaanl Sluicing

In Section 3.10.1, it was demonstrated that adjunct wh-phrases such as **bàŋà** 'why' and **kìnyè** 'how' are restricted to occurring in-situ in clause-initial and final positions, respectively. The examples (160-b) and (160-c), reproduced in 261, illustrate an ungrammatical construction when **bàŋà** undergoes extraction to Spec-Foc and (262-b) is also illicit because **kìnyè** has been moved from its base-generated position to Spec-FocP in the clausal left periphery.

(261) a. **Bà-ŋà** Kwame dàà lóór gbààn?
what-do K. buy.PFV 3.car DEF
'Why did Kwame buy the car?'

b. *Kwame dàà lóór gbààn **bàŋà**?
K. buy.PFV 3.car DEF what-do
Intended: 'Why did Kwame buy the car?'

c. ***Bàŋà** lé Kwame dàà lóór gbààn?
what-do FOC K. buy.PFV 3.car DEF

Intended: 'Why did Kwame buy the car?'

(262) a. Kunji sèè n-kòòn gbààn **kìnyè**?
K. burn 5-charcoal DEF how?
'How did Kunji burn the charcoal?'

b. *__Kìnyè__ lé Kunji sèè n-kòòn gbààn?
how FOC K. burn.PFV 5-charcoal DEF
Intended 'How did Kunji burn the charcoal?'

As is expected, these two adjunct wh-phrases cannot occur in sluicing constructions in the language, as shown in the data in (263-a) and (264-a). The ungrammaticality of the sluice (263-a) and its non-elliptical counterpart in (263-b), as well as (264-a) and (264-b), indicate that sluicing in Likpakpaanl supports a movement plus deletion approach since wh-phrases that cannot be extracted to the left periphery are blocked from participating in sluicing constructions.[1]

(263) a. *Tamanja lén kè ù gà gbàà ù-bò gbààn lé m báè ù ké nì yè **bàŋà**.
T. say.PFV COMP RP FUT beat 1.SG-child DEF CONJ 1SG ask RP COMP it COP why
Intended: 'Tamanja said he would beat the child, and I wanted to know why.'[2]

b. *Tamanja lén kè ù gà gbàà ù-bò gbààn lé m báè ù ké nì yè **bàŋà** lé Tamanja gà gbàà ù-bò gbààn.
T. say.PFV COMP RP FUT beat 1.SG-child DEF CONJ 1SG ask RP COMP it COMP why FOC T. FUT beat 1-child DEF
Intended: 'Tamanja said he will beat the child, and I wanted to know why Tamanja said he will beat the child.'

[1]Adams and Tomioka (2012) have also shown that in Chinese, the wh-adjunct *zenme(yang)* 'how' is barred from occurring in sluicing constructions as we see below;

(i) Dawu xiuhao le che; women dou zai cai (shi) zenme(yang).
Dawu fix-good ASP car 1PL all ASP guess COP how
Intended: '*Dawu fixed the cat, and we all wonder how.'

(Adams and Tomioka, 2012, 224)

[2]The sentence is still ungrammatical even if we remove the expletive and the copular leaving only the C-head and the wh-phrase. The same applies to (264-a)

(264) a. *Anifa lén kè ù nàn tér Akor lé Beso bàn bè ké nì yè **kìnyè**
A. say.PFV COMP RP PST help A. CONJ B. want know COMP it COP how
Intended: 'Anifa said she helped Akor and Beso wants to know how.'

b. *Anifa lén kè ù nàn tér Akor lé Beso bàn bè ké nì yè **kìnyè** lé Anifa nàn tér Akor.
A. say.PFV COMP RP PST help A. CONJ B. want know COMP it COMP how FOC A. PST help A.
Intended: 'Anifa said she helped Akor, and Beso wants to know how Anifa helped Akor.'

The data on sluicing in Likpakpaanl demonstrate that sluicing can occur in matrix and embedded clauses. Additionally, the data indicates constraints on wh-phrases in sluicing constructions. The data from Likpakpaanl sluicing yield the following descriptive observations. Firstly, the overt optional focus particle found in non-elliptical wh-questions does not survive ellipsis in sluicing. In terms of linearisation, sluicing appears to target FocP for 'deletion', as the licensing feature for ellipsis elides complements the XP bearing the E-feature proposed by Merchant (2001). This suggests the presence of a left periphery functional projection above FocP. It is, however, possible that structurally, a lot takes place than we see at Spell-out. Lastly, sluicing occurs in the presence of the complementiser head, and the copular is required to answer embedded sluicing. Section 5.6 provides a syntactic analysis of sluicing in Likpakpaanl and adopts a movement approach.

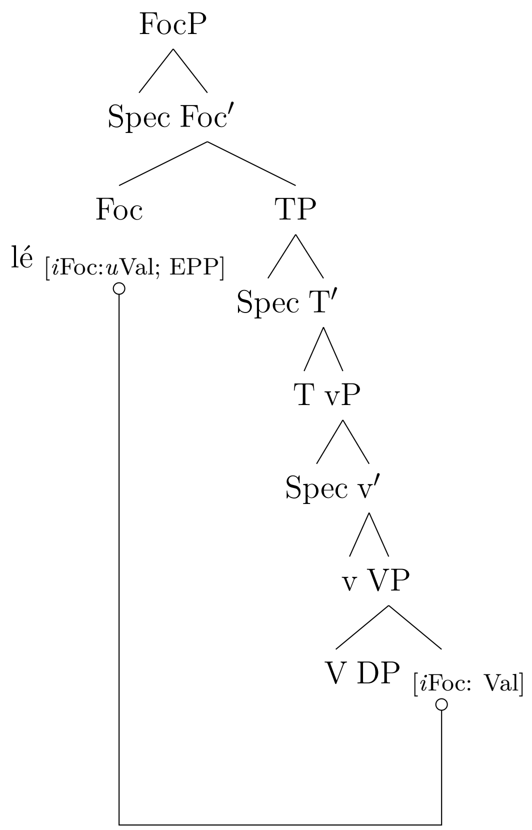
5.6 On the Syntax of Sluicing in Likpakpaanl

The preceding sections illustrated the operation of sluicing in Likpakpaanl. This section aims to account for the derivation of Likpakpaanl sluicing by adopting Merchant (2001)'s idea that sluicing is licensed by an ellipsis E-feature that projects on a lexical head in the left periphery syntax. The examined data has shown that wh-phrases in Likpakpaanl sluicing occur without the overt focus particle found in non-elliptical wh-ex-situ movement. I show that while both regular and inverse copular sluicing are derived through the same mechanism, they differ in the landing site of the wh-phrase. Thus, while Spec-LP attracts the wh-element in regular sluicing, it is Spec-PredP that hosts the wh-remnant in inverse copular sluicing.

5.6.1 Regular Sluicing

Given that the focus particle in ex-situ focus movement is absent in sluicing, it might suggest a symmetric relationship between in-situ wh-questions and sluicing in Likpakpaanl.However, the analysis pursued in this study is a movement plus PF-deletion approach, as there is evidence in the language to support this claim (see 5.7), leading to the second analysis for the derivation of sluicing. I adopt the movement plus PF-deletion approach and argue that Likpakpaanl sluicing arguing that it involves A-bar wh-movement of the sluiced wh-phrase into Spec-FocP followed by deletion of the FocP. (Merchant, 2001, 2003). The derivation of sluicing proceeds in two phases: the first step of the first phase involves an Agree between the interpretable and *unvalued* focus feature [*i*Foc: *u*Val] on Foc^0 and the interpretable but *valued*[*i*Foc: Val] on the wh-phrase constituent. The value is represented with the number [3]; nothing hinges on this[3] (Pesetsky and Torrego, 2007). The unvalued feature on Foc^0 probes for a suitable goal in its c-command domain. It finds the focus constituent,*wh-phrase*, which itself has a matching *valued* feature. The unvalued feature on the focus head is valued by the wh-phrase. Then, the EPP-feature on the Foc^0 A-bar moves the wh-phrase to Spec-FocP as (265) shows.

(265) **Agree and and ex-situ wh-movement**

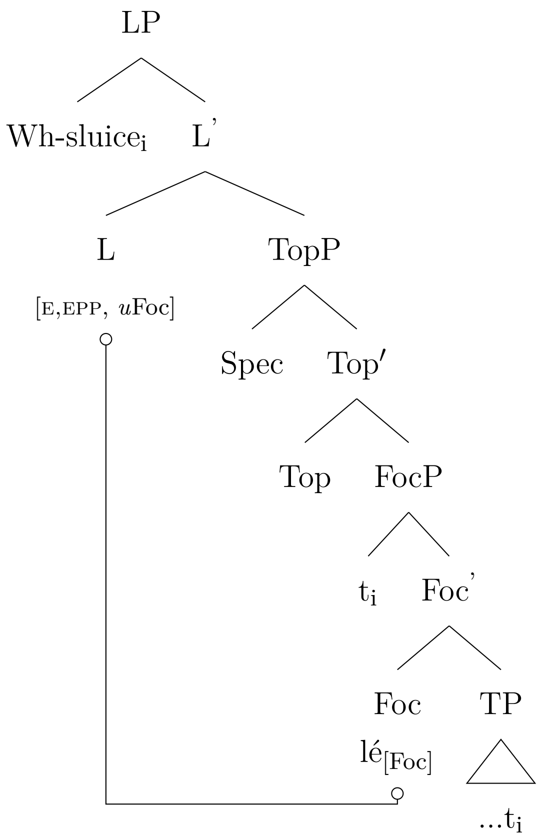


[3]In Pesetsky and Torrego (2007)'s feature valuation numeration, the number is used solely as a generic representation and could be any number [n ...].

For sluicing to occur, the second phase of the derivation is triggered when the Licensing Phrase (LP) is merged into the structure. As previously discussed in Chapter 2 section 2.5, I assume the projection of a Licensing Phrase (LP) following Aelbrecht (2010). However, unlike Aelbrecht (2010), who does not consider a head-complement relationship between the LP and the ellipsis site (since elements can intervene between the LP and the E-site), the analysis adopted in this work follows Merchant (2001), where a single E-feature defines the licensor and the ellipsis site. While the E-feature is merged on the C-head in English, it occurs on the L-head in Likpakpaanl. The licensor and the ellipsis site in Likpakpaanl lie in a head-complement relation with no intervening elements.

Under sluicing, the LP with the ellipsis and EPP features is merged above FocP. I assume the three feature bundles encoded on E-feature (syntactic, semantic, phonological) are all on the head of the LP. Since the uninterpretable focus feature (*u*Foc) on L must be valued, it enters into an Agree relation with the Foc head **lé**, whose uninterpretable feature has already been checked by the wh-phrase (see 265). Consequently, L probes for a matching goal and locates the FocP with a matching interpretable focus [*i*Foc] feature, establishing an Agree relation between the heads of LP and FocP. After that, the EPP feature on the Licensing head attracts the focused constituent to Spec-LP. Finally, the E-feature licenses the deletion of the complement to LP, i.e., FocP, rendering the ellipsis site inaccessible for further syntactic operations and blocking vocabulary insertion at PF. The illustration in (266) shows that the LP projection has the semantic property of selecting a focus phrase.

(266) **Agree between LP and FocP**

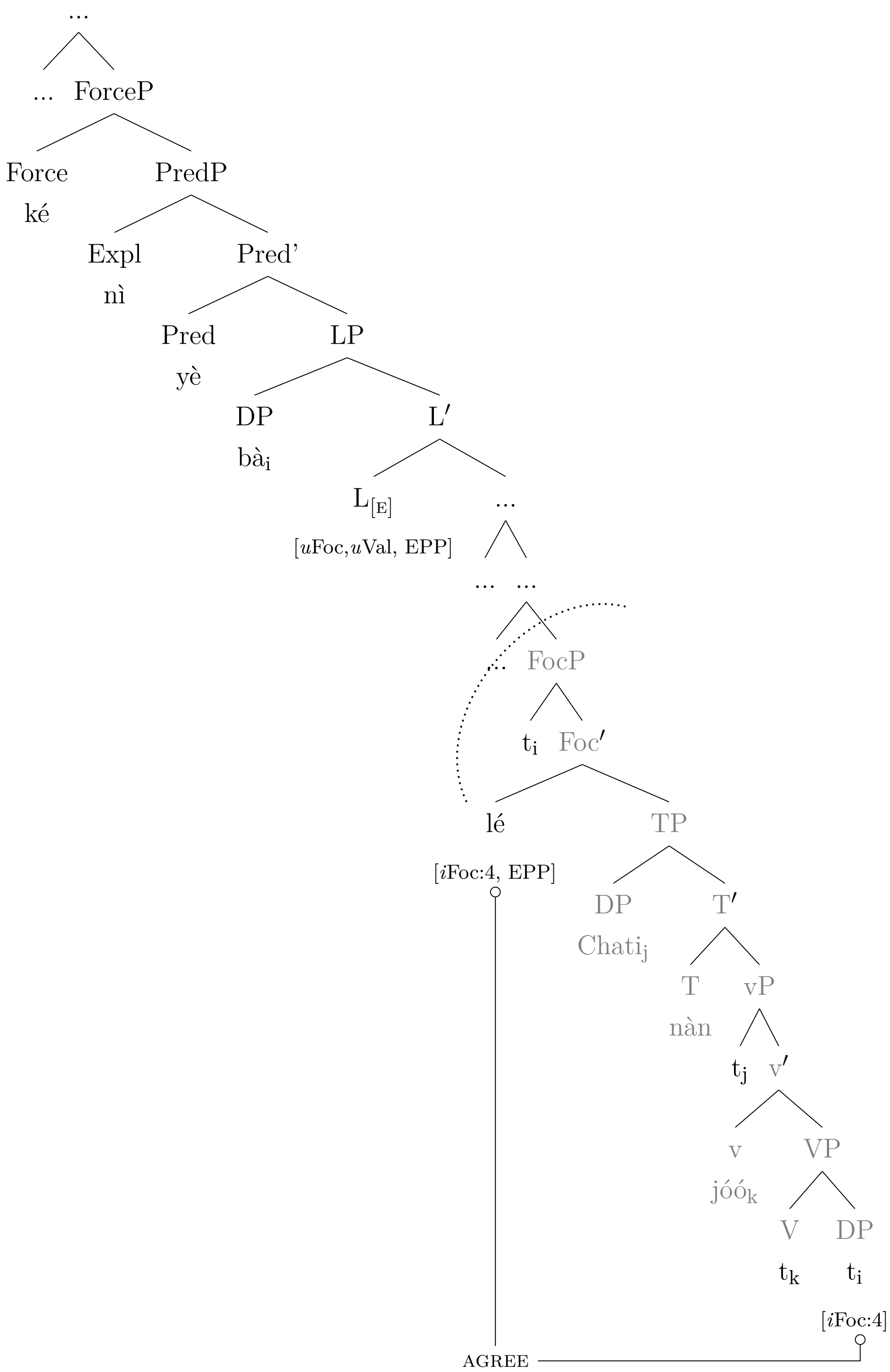


The analysis adopted in accounting for Likpakpaanl sluicing is Movement and PF-deletion, where the wh-phrase is initially merged in the vP periphery. The wh-phrase with an interpretable and valued feature serves as a goal to the probe on FocP with unvalued features. The Agree operation licenses a focus interpretation after the [uVal]-feature on Foc is checked. After valuation, the EPP feature on Foc^0 triggers A-bar movement of the wh-phrase/focused constituent to Spec-Foc as I earlier illustrated in (265).

In the second stage of the derivation, the phonologically null L^0, which bears the E-feature, is then merged above FocP licensing the deletion of the PF-content of its complement (FocP). Therefore, the sluiced in (267-a) is derived systematically as explained 266 where after feature valuation takes place, the EPP feature on Foc^0 then extracts the focus constituent (wh-phrase) to Spec-FocP. The E-feature, which is phonologically null on L^0, then licenses the ellipsis of the FocP. Thus, applying the analysis to the example in (267-a) yields the derived structure in (267-b).

(267) a. Chati nàn jóó tì-wàn nì-bàà, ámàà m àà bé ké nì
C. PST hold 14-thing 14-INDEF CONJ 1SG NEG know COMP it
yè [$_{LP}$ bà$_i$ [$_{FocP}$ t$_i$ ~~lé~~ [TP ~~Chati nàn~~ [$_{vP}$ ~~jóó~~ t$_i$.]]]]
COP what FOC C. PST hold
'Chati held something, but I don't know what.'

b.



In his analysis of sluicing in Slovenian, Shlonsky (2022) argues that the discourse particle **že** is merged above FocusP and followed by movement of FinP, which is the complement of Foc^o, to Spec-XP. Deletion then targets FinP, the complement of Focus^o. Thus, the XP above FocP is projected to accommodate ellipsis phenomena. He notes that the sluice in (268-a) is derived as shown in (268-c) (see: Shlonsky, 2022, 2).

(268) a. Vid je srečal nekoga. Koga že?
Vid AUX met someone. Who PTCL
'Vid met someone. Remind me, who ~~did he meet~~?'

b. $[_{XP}$ $[_{FinP}$ $[_{TP} \ldots t_{wh} \ldots]]$ X^o $[_{FocP}$ wh Foc^o $t_{FinP}]]$.

c. $[_{FocP}$ wh $_{Foc}{}^o$ t_{FinP}]...**že** $[_{XP}$ $[_{FinP}$ $[_{TP} \ldots t_{wh} \ldots]]$ $_X{}^o$ t_{FocP}].

The derivation of sluicing in Slovenian, as proposed by Shlonsky (2022), involves a series of intricate syntactic operations. The process commences with the merger of the Foc-head and the projection of the FocusP, followed by the movement of the wh-phrase to Spec-Foc. Next, a new head X is externally merged, projecting XP above FinP. Notably, the FinP, which is the complement of Foc^o, undergoes movement to the specifier position of XP, yielding the structure in (268-b). The final step involves the merger of **že** and the movement of the FocP above it, resulting in the configuration in (268-c). According to Shlonsky (2022), the movement operations and the projection of functional heads such as FocP and XP capture the word order and the structural properties of sluicing constructions in Slovenian and contribute to our understanding of the cross-linguistic variation in elliptical constructions.

The proposed analysis supports Merchant (2001)'s claim that all the properties of the E-feature are hosted on a single syntactic element in Likpakpaanl, the LP. I also argue that sluicing is derived through the A-bar wh-movement of the wh-phrase from its base position to Spec-FocP, followed by a further successive cyclic movement into Spec-LP, a higher functional projection hosting the ellipsis E-feature that licenses sluicing (Merchant, 2001). The data further indicates that the left periphery of clauses hosts a hierarchy of functional projections, including Force phrase (ForceP), Topic Phrase (TopP), Focus Phrase (FocP), Tense Phrase (TP), and Mdoal Phrase (ModP), among others, serving as domains for various types of A-bar movements (Rizzi, 1997, 2004; Cinque, 1999; Aboh, 1998, 2004a). Likpakpaanl sluicing, therefore, gives rise to articulated left-peripheral projections, such as the Force, Predicate, Licensing, and Focus Phrases, associated with different scope-discourse interpretations. Having looked at regular sluicing, I now shift focus to examine how sluicing with inverse copular sluicing is derived.

5.6.2 Inverse Copular Sluicing

I showed in sub-section 5.3.3 where the wh-sluice occupies the Specifier of a higher projection above FocP as in (269-a), and I refer to this type of case as 'Inverse copular sluicing' (ICS). The ungrammaticality of (269-b) and (269-c) demonstrate that the focus marker cannot be present as in regular sluicing. This subsection attempts a syntactic analysis of such constructions in Likpakpaanl. I start by examining some syntactic analysis of inverse copular structures as proposed by Hatakeyama (1997).

(269) a. Mbigmin nàn bì mùó tì-wàn nì-bàà lé Adelina nàn
M. PST IPFV suck 14-thing 14.INDEF CONJ A. PST
bàn ù bè kè bà yè.
want RP know COMP what COP
'Mbigmin was sucking something, and Adelina wanted to know what.'

b. *Mbigmin nàn bì mùó tì-wàn nì-bàà lé Adelina nàn
M. PST IPFV suck 14-thing 14.INDEF CONJ A. PST
bàn ù bè kè bà lé yè.
want RP know COMP what FOC COP

c. *Mbigmin nàn bì mùó tì-wàn nì-bàà lé Adelina nàn
M. PST IPFV suck 14-thing 14.INDEF CONJ A. PST
bàn ù bè kè bà yè lé.
want RP know COMP what COP FOC

In analysing Likpakpaanl inverse copular sluicing, I adopt Hatakeyama (1997)'s proposal and argue that the wh-phrase selected by the main verb inside the TP (see 5.6.2 for details of Hatakeyama (1997)'s analysis). From this base position, the wh-phrase undergoes successive cyclic movement through the specifier positions of the functional projections FocP and LP to its landing site in Spec-PredP. I don't, however, follow Hatakeyama (1997)'s proposal that the predicate nominal moves to Spec-CP since, in Likpakpaanl, the CP headed by **ké** projects higher above the LP. The proposed structure involves the wh-phrase occupying Spec-PredP, with the copular verb as the head of PredP (270). The PredP replaces the CP and accounts for how inverse copular sluicing constructions in Likpakpaanl are derived—the CP headed by the c-head **ké** projects above the PredP in Likpakpaanl.

(270) $[_{\text{PreP}}$ wh-phrase$_i$ ye $[_{\text{LP}}$ t_i $[_{\text{FocP}}$ t_i $[_{\text{TP}}$ DP$_k$ [t_k vP t_i]]]]]]

Inverse Copular Sluices are derived following the same syntactic process as proposed in (267-b) with a difference in the landing site of the extracted wh-phrase prior to ellipsis. It is important to state that ungrammatical sentences like (271-b) resulting from extraction from focus phrase to Spec-PredP in non-elliptical constructions in Likpakpaanl are repaired by ellipsis, thus demonstrating Mendes and

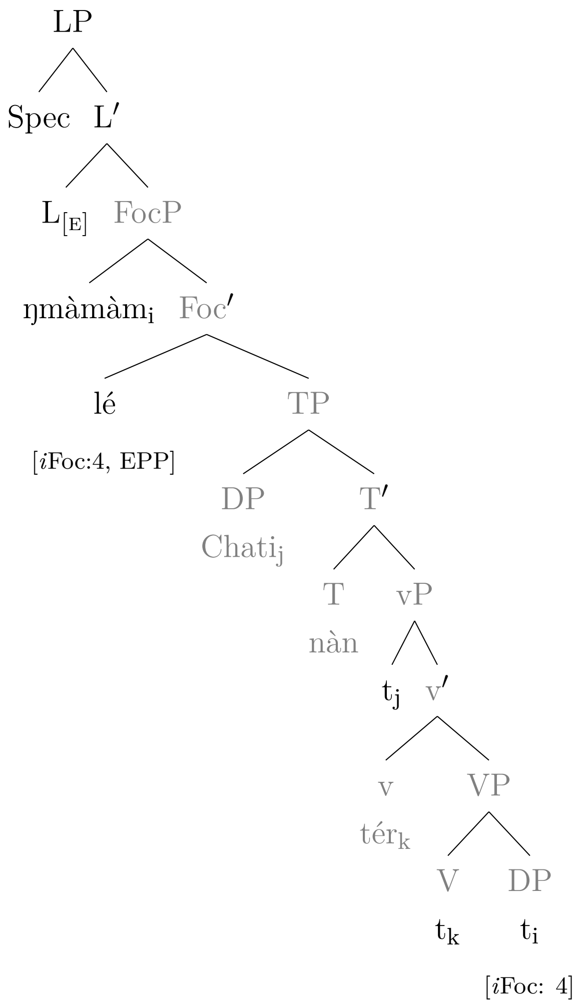
Kandybowicz (2023)'s phenomenon of 'Salvation by deletion'.

(271) a. John nàn tér bì-nìb bì-bàà lé Amos bàn ù bè
J. PST help 2.PL-person 2.PL.INDEF CONJ A. want RP know
kè ŋmà-màm yè.
COMP who-4.PL COP
'John wanted to help some people, and Amos wants to know who.'

b. *John nàn tér bì-nìb bì-bàà lé Amos bàn ù bè
J. PST help 2.PL-person 2.PL.INDEF CONJ A. want RP know
kè ŋmà-màm$_i$ yè t_i lé John nàn tér t_i.
COMP who-4.PL COP FOC J. PST help
Intended: 'John wanted to help some people, and Amos wants to know who John wanted to help.'

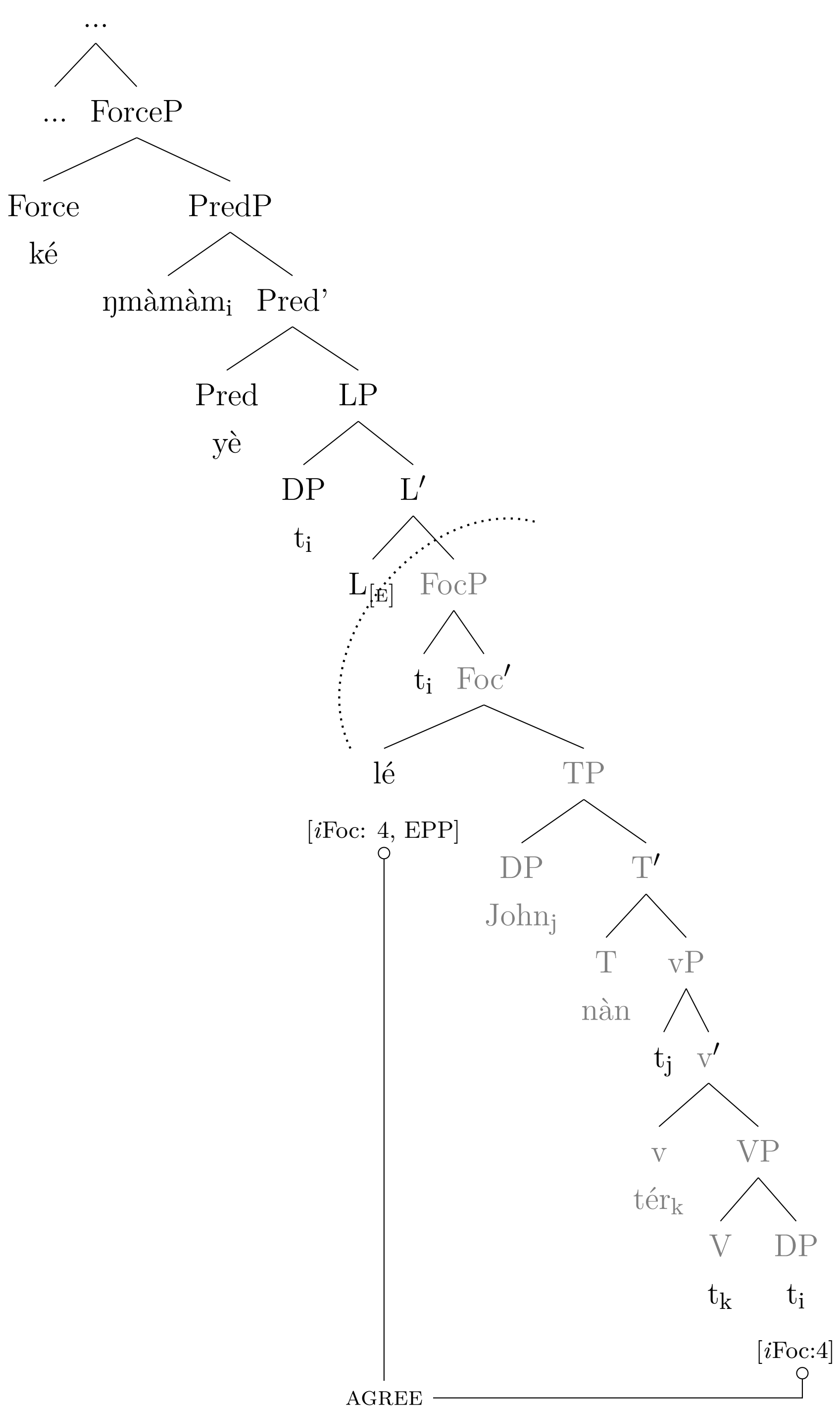
To derive the inverse sluice in (271-a), the wh-phrase **ŋmàmàm** is generated in the vP periphery as the internal argument of the verb enters into Agree relation with FocP merged above TP and the *u*Foc-features on Foc are checked and deleted. The EPP feature on Foc° A-bar moves the wh-phrase to Spec-FocP. Next, the morphologically null head of the licensing phrase, with the ellipsis feature, is projected above FocP as illustrated in (272).

(272)



After that, the predicate phrase is merged above LP because Spec-PredP is empty but has an EPP feature that requires Spec to be filled, **ŋmààm** undergoes successive cyclic movement from Spec-FocP through Spec-LP to Spec-PredP. After that, the E-feature on LP licenses the ellipsis of the FocP, making it inaccessible to any further syntactic operation as illustrated in (273).

(273)



I assume that the derivational timing for inverse copular ellipsis is delayed since deletion does not take place immediately after LP is merged but after the wh-phrase has been extracted to Spec-PredP. In their analysis of Verb phrase ellipsis in English licensed my 'do' Aelbrecht (2010) and Baltin (2012) claimed that the ellipsis of VP complement is not licensed at the same moment in derivational time when v is externally merged. They claim that in VP, an ellipsis licensed by 'do' only *v* can license the ellipsis without requiring the finite auxiliary in T as the adapted example in (274). Thus, they posit *v* licenses the ellipsis of its VP complement by first extracting the non-object wh-phrase from VP to a higher FP, Spec-VoiceP, which selects vP at the edge of the clause. Baltin (2012) argues that VoiceP serves as the clause-internal phase and that is why it can select vP as its complement.

(274) The horses might graze on the field today, and the cattle$_i$ might do ~~graze on the field today~~ t$_i$, too.

5.6.3 Interim Summary

So far, this chapter has aimed to account for the derivation of Likpakpaanl sluicing by adopting Merchant (2001)'s idea that sluicing is licensed by an ellipsis E-feature that projects on a lexical head in a language. I also adopted Aelbrecht (2010)'s Licensing phrase (LP), but unlike her who assumes that the LP projects to license agreement with the head bearing the E-feature, I contend that the LP in Likpakpaanl sluicing hosts the E-feature. The analysis up to this point has demonstrated that while both regular and inverse copular sluicing are derived through the similar mechanism of Agree and successive cyclic movement, targeting the FocP for deletion, they differ in the landing site of the wh-phrase. While Spec-LP attracts the wh-element in regular sluicing, Spec-PredP hosts the wh-remnant in the inverse copular case. The data also shows that wh-phrases in Likpakpaanl sluicing occur without the overt focus particle that we find in non-elliptical wh-ex-situ movement. The next subsection will provide evidence to support the claim that Likpakpaanl sluicing is derived from a movement plus PF-deletion operation, building on the analysis presented so far.

5.7 Evidence for a Movement plus PF Deletion Approach to Likpakpaanl Sluicing

A key prediction of the movement plus PF-deletion analysis for deriving Likpakpaanl sluicing is that it should obey locality constraints on long-distance A-bar dependencies, such as island constraints. The proposed derivation involves overt focus movement of the wh-remnant from its base-generated position to a higher clause in the left periphery. If this analysis is on the right track, we expect Likpakpaanl sluicing to exhibit island sensitivity effects, a core diagnostic for movement dependencies since Ross (1967)'s seminal work. Testing whether Likpakpaanl sluicing structures respect island constraints can provide crucial evidence for or against the movement plus deletion approach advocated here.

However, the cross-linguistic facts have shown a pattern where island violations are absent in sluicing constructions (Merchant, 2001, 2004; Van Craenenbroeck and Lipták, 2013; Mendes and Kandybowicz, 2023), among others. For instance, Merchant (2004) and Mendes and Kandybowicz (2023) contend that the deletion of the PF of otherwise ungrammatical constructions in islands 'saves' the structure since the 'defective' PF-uninterpretable features (represented by astericks [*] by Merchant (2004) that would otherwise cause ungrammaticality are deleted under sluicing (see also; Merchant, 2001; Lasnik, 2001; Lipták and Aboh, 2013). Likpakpaanl exhibits island sensitivity in non-elliptical wh-extraction, but sluicing repairs these violations, which is consistent with the cross-linguistic observation that sluicing repairs islands. The data from Likpakpaanl sluicing show that they are insensitive to islands, as shown with examples from the Complex Noun Phrase Constraint (CNPC), Coordinate Structure Constraint (CSC) and the absence of in-situ wh-phrases in sluicing construction.

5.7.1 Complex Noun Phrase Constraint (CNPC)

Complex noun phrases are constructions in which a DP is contained a CP. The CP can be a complement of the noun, where the noun takes a CP internal argument or an adjunct to the DP, a relative clause. The CNPC is a syntactic constraint that makes it impossible to extract an element from complex DP. CNPC forms part of Ross's Islands, domains from which extraction of wh-phrases is disallowed or leads to degraded acceptability. Let us examine the CNPC and how it interacts with sluicing in Likpakpaanl.

5.7.1.1 Relative Clause Island

The first fact that needs to be established is that Likpakpaanl relative clauses are islands and block extraction out of the relative clause island, as the ungrammatical sentence in (275-c) shows. Contrary to (275-c), where the wh-phrase A-bar moved to the left periphery, the wh-phrase remains in-situ, as in (275-b), and the sentence is licit because no island violation occurs.

(275) a. Bondaan nyì [DP ù-pìì [CP ù màà ŋì-bùù] nà.]
B. know 1.SG-woman REL build 8.PL-pot REL.DET
'Bondaan knows the woman who makes pots.'

b. Bondaan nyì [DP ù-pìì [CP ù màà bà] nà.]
B. know 1.SG-woman REL build 8.PL-pot REL.DET
'Bondaan knows the woman who makes what?''
('What is the x, such that Bondaan knows the woman who builds x?')

c. *Bà$_i$ (lé) Bondaan nyì [DP ù-pìì [CP ù màà
what FOC B. know 1.SG-woman REL build
t$_i$] nà.]
8.PL-pot REL.DET?

The island violation in relative clauses, however, is ameliorated under sluicing as exemplified in (276-b).

(276) a. Bì bàn tì yòòr ù-pì ù ŋmàà chààr n-jìkàbìm
3PL want 1PL take 1-woman REL can winnow 7-grain
nà, àmàà bì àà chà tì bè kè nì yè lì-jìkàbìl
REL.DET CONJ 3PL NEG let 1.PL know COMP it COP 7-grain
lì-là-lì.
7-which-7
'They want us to take a woman who can winnow grains, but they did not tell us which (type of) grain.'

b. *..., àmàà bì àà chà tì bè kè nì yè lì-jìkàbìl lì-là-lì
CONJ 3PL NEG let 1PL know COMP it COP 7-grain 7-which-7
lé bì bàn tì yòòr ù-pì ù ŋmàà chààr nà.
FOC 3PL want 1.PL take 1-woman REL can winnow REL.DET

The observation follows the cross-linguistic generalisation that ellipsis possess the power to repair or save otherwise ungrammatical constructions because the PF-uninterpretable or defective traces are 'deleted' at the interface before Spell-Out (See Section 2.4 on how the *Salvation by deletion* mechanism operates).

5.7.1.2 Coordinate Structure Constraint

The coordinate structure constraint (CSC) is a syntactic island constraint that prohibits extraction out of a conjunct in a coordinate structure Ross (1967). For example, in (277), extracting either conjunct tout of the coordinate conjunct leads to ungrammaticality. In both (277-a) and (277-b), the island effect arises because the conjuncts in a coordinate structure constitute strong islands for extraction.

(277) a. *Bà$_i$ (lé) Bowan nàn jóó kpààchá nì t$_i$?
what FOC B. PST hold 3.SG.cutlass and
'What did Bowan hold a cutlass and '

b. *Kpààchá$_i$ (lé) Bowan nàn jóó t$_i$ nì bà?
3.SG.cutlass FOC B. PST hold and what
'A cultass did Bowan hold and what?'

Interestingly, however, the CSC island effect is systematically absent under sluicing. The FocP and the TP constituent containing the gap are elided in this elliptical construction, leaving only a wh-remnant. In (278-a), the wh-phrase 'who' has been extracted from the coordinate clause and to Spec-FocP, and this would typically induce a CSC violation, yet the sentence is perfectly grammatical under sluicing. Compare with the ungrammatical sentence in (278-b).

(278) a. Evelyn nàn tùk [DP Kòyàjà nì ù-jà ù-bàà] kè bì
E. PST tell K. CONJ 1.SG-man 1.SG-INDEF COMP 3PL
bù ŋì-tùùn àmàà m àà ŋmàà bè kè nì yè ŋmà$_i$.
plant 8PL-beans CONJ 1SG NEG can know COMP it COP who
'Evelyn told Koyaja and someone to plant beans, but I don't know who.'

b. *Evelyn nàn tùk [DP Kòyàjà nì ù-jà ù-bàà] kè bì
E. PST tell K. CONJ 1.SG-man 1.SG-INDEF COMP 3PL
bù ŋì-tùùn , àmàà m àà ŋmàà bè kè nì yè ŋmà
plant 8PL-beans CONJ 1SG NEG can know COMP it COP who
(lé) Evelyn nàn tùk Kòyàjà nì t$_i$ kè bì bù ŋì-tùùn
FOC E. PST tell K. CONJ COMP 3PL plant 8.PL-beans

Merchant (2001, 2004) has proposed a PF theory of islands to explain why sluicing repairs. He argues that island violations are not derived from constraints on movement or LF representations but rather from pronouncing defective elements like traces at PF. Specifically, traces of movement out of islands are marked with a PF-uninterpretable *-feature. In non-elliptical contexts like (278-b), these *-marked traces cannot be deleted, resulting in a crash of the derivation at PF. However, in (278-a), the entire FocP and TP constituent containing the *-traces undergo deletion, rescuing the derivation from a PF crash.

5.7.2 Adjunct Island

The adjunct island constraint also has implications for the analysis of sluicing. Regular focus wh-movements such as in (279-a) are sensitive to adjunct islands, making it impossible to extract a wh-phrase from it as shown in the ungrammatical sentence in (279-b).

(279) a. Nalarb nàn gbìì lì-nùù-l [$_{\text{Adjunt island}}$ wààr lé Kojo kàn
N. PST dig 7.SG-yam-7 before CONJ K. see.PFV
ŋmà?]
who
'Who did Kojo see before Nalarb dug the yam?'

b. *ŋmà$_i$ (lé) Nalarb nàn gbìì lì-nùù-l [$_{\text{Adjunt island}}$ wààr lé
who FOC N. PST dig 7.SG-yam-7 before CONJ
Kojo kàn t$_i$?]
K. see.PFV

However, in Likpakpaanl, sluicing repairs such adjunct island violations as (280-a) shows. In (280-a), the sluiced wh-phrase **bà** 'what' is extracted out of an adjunct island, yet the sentence is grammatical, but its non-elliptical counterpart in (280-b) is illicit.

(280) a. Nimomi nàn dàà lóór wààr lé Bilegnan ŋáán
N. PST buy 3.SG.car before CONJ B. cook.PFV
tìwàn nì-bàà, àmàà m àà bè kè nì yè bà$_i$.
14.thing 14.INDEF but 1SG NEG know COMP it is what
'Nimomi bought a car before Bilegnan bought something, but I don't know what.'

b. *Nimomi nàn dàà lóór wààr lé Bilegnan ŋáán
N. PST buy 3.SG.car before CONJ B. cook.PFV
tìwàn nì-bàà, àmàà m àà bè kè nì yè bà$_i$ (lé)
14.thing 14.INDEF but 1SG NEG know COMP it is what FOC
Nimomi nàn dàà lóór wààr lé Bilegnan ŋáán t$_i$
N. PST buy 3.SG.car before CONJ B. cook.PFV

5.7.3 Further Evidence for Movement in Sluicing

In addition to the standard syntactic tests, language-internal evidence from Likpakpaanl further supports a PF-deletion analysis of sluicing. As shown earlier in subsection 3.10.1, the wh-adjunct **kìnyè** 'how' can only occur in situ, as demonstrated by the ungrammaticality of ex-situ **kìnyè** in (281-b).

(281) a. Lì-kpùù-l gbààn nàn tóór kìnyè?
7.SG-funeral-7 DEF PST perform how
'How was the funeral performed?'

b. ***Kìnyè**$_i$ **lé** lì-kpùù-l gbààn nàn tóór t$_i$?
how FOC 7.SG-funeral-7 DEF PST perform
Intended: 'How was the funeral performed?'

c. *Naadi nàn lén ké ù gà dàà mótò, àmàà m àà
N. PST say COMP RP FUT buy 3.SG.motor CONJ 1SG NEG
bè kè nì yè **kìnyè**
know COMP it COP how.

Since **kìnyè** is restricted to in situ positions, the movement + deletion approach predicts it should be impossible as a sluicing remnant. This prediction is borne out - it cannot occur in sluicing, as in (281-c). This contrasts with other wh-elements that can undergo leftward dislocation and occur in sluicing. The inability of **kìnyè** to license sluicing follows directly from the fact that sluicing involves movement.

This section of sluicing in Likpakpaanl draws on various island diagnostics, including complex NP constraints, adjunct islands, and coordinate structure constraint violations, and demonstrates that these island effects are ameliorated under sluicing. A crucial empirical observation is that wh-phrases restricted to occurring in-situ cannot license sluicing in the language. This premise and the island insensitivity motivate an analysis where sluicing is derived by overt focus movement of the wh-phrase from its base-generated position to the through Spec-FocP, to the specifier of a Licensing Phrase (LP), followed by deletion targeting the FocP constituent containing the trace of the displaced wh-phrase. The Likpakpaanl data, therefore, provide novel evidence that cross-linguistic sluicing does not uniformly equate to TP-ellipsis since some languages may target larger site domains like FocP.

5.8 Sluicing-COMP Generalisation and Sluicing in Likpakpaanl

The data on sluicing that we have analyzed demonstrates that the optional focus particle in focus wh-movement is absent in sluicing and fragment answers (see Chapter 6 on fragment answers). From a theoretical standpoint, the absence of the focus head under sluicing can be accounted for by employing Merchant (2001)'s Sluicing-COMP Generalisation (henceforth, SCG) outlined in (282), which necessitates only the wh-phrase to escape the ellipsis site.

(282) **Sluicing-COMP Generalisation**
... in sluicing, no non-operator material may appear in COMP

(Merchant, 2001, 62)

In Merchant (2001)'s explanation, an 'operator' describes a 'syntactic wh-XP', while a 'material' refers to any pronounced or overt element following the operator. The complementiser also describes 'material dominated by CP but external to IP'. The SCG predicts that only the wh-operator should survive the ellipsis under sluicing. I adopt Merchant (2001)'s SCG in accounting for the absence of the focus particle, **lé**, in Likpakpaanl sluicing because it provides explanatory grounds for why only the wh-operator survives ellipsis and not the focus particle. I have argued that **lé** is realised as a focus head, and this is based on the fact that (i) though optional, the appearance of **lé** is dependent on wh or focus fronting; (ii) **lé** also introduces focus phrases, and (iii) under sluicing, **lé** is missing. Thus, the absence of the focus marker in Likpakpaanl sluicing provides concrete support to the Merchant's SCG.

In English, modal auxiliaries, like 'will', are argued to move to the C-head in matrix wh-questions as shown in (283-a). However, the modal auxiliary verb 'will' is elided in English sluicing constructions, as illustrated in (283-d). As predicted by the SCG, it is only the wh-phrase that moves to Spec-CP that must survive sluicing; the auxiliary verb, which is a non-operator, must be elided.

(283) a. [CP What will [TP the students [$_{vP}$ eat for dinner?]]]
b. The students will eat something for dinner.
c. [A:] The students will eat something for dinner.
d. [B:] [ii: What (*will)?

A parallel phenomenon can be observed in (284-c) where the wh-phrase in ex-situ focus movement is followed by an optional **lé** as shown in (284-b). In sluicing (284-c), the focus particle must obligatorily delete for the sentence to be accepted.

(284) a. Balado gà tér ù-nì ù-bàà.
B. FUT help 1-person 1-INDEF
'Balado will help someone.'
b. [FOCP Ŋmà$_j$ (lé) [TP Balado gà [$_{vP}$ tér t$_j$]]].
who FOC B. FUT help
'Who will Balado will help?'
c. Ŋmà (*lé)?
who FOC
'Who?'

The observation in Likpakpaanl and English is analogous to the Brazilian Portuguese (BrP) pattern. As noted by Mendes and Kandybowicz (2022), in BrP, the particle **que** can intervene between the fronted wh-phrase and the remainder of the clause. In sluicing, **que** is obligatorily omitted, as evidenced by the data in (285-c).

(285) a. Quem (que) saiu? (Wh-question)
who COMP left
'Who left?'
b. Alguém saiu.
Someone left
'Someone left.'
c. Quem (*que)? (Sluicing)
Who COMP
'Who?'

(Mendes and Kandybowicz, 2022, 45).

Merchant (2001) has demonstrated the cross-linguistic consistency of the Sluicing-COMP generalisation phenomenon in languages such as Norwegian, Danish, and Yiddish, among others. Despite this widespread pattern, there are also languages such as Hungarian, Slovenian, Gunbge, and Nupe (Shlonsky, 2022; Mendes and Kandybowicz, 2023; Lipták and Aboh, 2013; Merchant, 2001) with counterexamples to SCG. In these languages, the obligatory focus marker on fronted wh-phrases is retained under sluicing contra the SCG. For instance, in Gungbe, Lipták and Aboh (2013) demonstrates that wh-question formation involves fronting of the wh-phrase and the insertion of a focus particle immediately following the fronted wh-phrase (286-a). Contrary to the prediction of SCG, the focus particle together with the wh-phrase survives ellipsis, and only the TP complement is targeted for deletion (286-b) shows.

(286) a. Mɛ́nù ***(wɛ́)** wá?
who FOC come
'Who came / Who arrived?'

b. Kòfí ná yrɔ́ mɛ́ ɖé àmɔ́n má ny'ɔn m'ɛɖě
K. FUT call person INDEF but 1SG.NEG know person.REL
*(wɛ́)
FOC
Lit.: 'Kofi will call someone, but I don't know the person who.'

(Aboh and Pfau, 2011, 93),(Lipták and Aboh, 2013, 105).

Also, Mendes and Kandybowicz (2023) have shown that the obligatory sentence-final focus particle in Nupe wh-questions (287-a) survives sluicing (287-c), indicating a counter-example Merchant (2001)'s SCG.

(287) a. Ké Musa pa ***(o)**? (Wh-question)
what Musa pound.PST 'FOC
'What did Musa pound?'

b. Musa pa ejan ndoci.
Musa pound.PST thing certain
'Musa pounded something.'

c. Ké ***(o)**? (Sluicing)
what FOC
'What ~~did Musa pound~~?'

(Mendes and Kandybowicz, 2023, 301).

Further evidence that violates (Merchant, 2001, 62)'s Sluicing-COMP Generalisation comes from Slovenian. Marušič et al. (2016) note that in Slovenian, the discourse particle **pa**[4], which appears in ex-situ wh-questions (288-a), is also retained in sluicing constructions as (288-b) shows (cf: Shlonsky, 2022).

[4](Marušič et al., 2016, 202) argue that even though **pa** is a discourse particle it can function as a topic marker or a contrastive focus marker in declarative sentences in Slovenian.

(i) Petra pa še nisem videl plesati.
Peter.GEN PA yet NEG.AUX see dance.INF
'As for Peter, I have not seen him dance yet.'

(288) a. Koga **pa** je Peter videl?
who.ACC PA AUX Peter.NOM saw
'Who did Peter see?'

b. Vid je srećal nekoga. Koga **pa**?
Vid AUX met someone. Who PTCL
'Vid met someone. Remind me, who ~~did he meet~~?'

The data from all these languages constitute genuine counterexamples to the Sluicing-COMP Generalisation because the focus particle, 'a non-operator' is retained with the wh-operator (Merchant, 2001, 62). Having discussed Merchant (2001)'s Sluicing-COMP Generalisation, let us turn our focus to another generation in the literature, which is known as the Baltin (2010)'s Generalisation.

Apart from SCG, which has detailed elements that must be deleted or retained under sluicing, another influential proposal is Baltin (2010)'s Generalisation (henceforth, BG) on sluicing, which proposes that if a language has an overt focus head, that focus head should not be deleted in sluicing. Thus, BG does not only predict that sluicing should target Foc^{o} complement in wh-focus movement languages but also that the Foc^{o} must survive sluicing(cf: Baltin, 2010, 331-332).

While languages like Likpakpaanl, Dutch, Norwegian, Danish, and Yiddish strictly adhere to Merchant (2001)'s SCG, they, on the other hand, violate the Baltin (2010)'s Generalisation since their focus particles are deleted. One might be tempted to assume that this dichotomy between languages that obey and those that violate the SCG and BG, respectively, is based on language families like Mabia and Kwa. While this is a plausible assumption, let us consider some examples from two Kwa languages, Akan and Yoruba.

Data from Akan, a Kwa language spoken in Ghana, shows that the focus marker **na**, which is obligatory for focus movement, deletes sluicing. The example in (289-a) shows a wh-question with **na** while (289-b) illustrates sluicing with the absence of the focus particle. In the non-elliptical sentence in (289-c), just like the ex-situ question (289-a), the focus particle is obligatory.

(289) a. **Déén** *(na) Kofi pɛ́?
what FOC K. like
'What does Kofi like?'

b. Kofi re-hwehwɛ biribi nanso min nim sɛ ɛ yɛ
K. IPFV-look thing.INDEF but 1SG.NEG know COMP it COP
déén **(*na)**.
what FOC
'Kofi is looking for something, but I don't know what.'[5]

[5]All the Akan data that are not cited are constructed by me as a native speaker of the language.

c. Kofi re-hwehwɛ biribi nanso min nim sɛ ɛ yɛ déén
K. IPFV-look thing but 1SG.NEG know COMP it COP what
(*na) Kofi re-hwehwɛ.
FOC K. IPFV-look
'Kofi is looking for something, but I don't know what Kofi is looking for.'

While languages like Likpakpaanl and Akan violate Baltin's generalisation by eliding the focus particle under sluicing (though adheres to Merchant (2001)'s generalisation), other languages such as Gungbe, Nupe, Slovenian (examples (286-b), (287-c), (288-b), respectively and Yoruba (290-b) disobey the SCG but satisfy the BG.

(290) a. **Ki** ***(ni)** Bọ́lá rí ní àná?
what FOC B. see LOC yesterday
'What did Bọ́lá see yesterday?'

b. Bọ́lá n ra nkan, ṣùgbón mi ò mọ **kí** [**ni** ~~Bọ́lá~~
B. PROG buy something but 1SG NEG know what FOC B.
~~n~~ ~~ra~~ t_i.].
PROG buy
'Bọ́lá is buying something, but I don't know what.'

(Acheampong and Aremu, 2024)

Despite being genetically related Kwa languages that morphologically mark focus, Akan and Gungbe exhibit divergent patterns under sluicing constructions. While the focus marker **na** obligatorily undergoes deletion in Akan sluicing, its counterpart **wɛ́** survives ellipsis in Gungbe. This micro-parametric variation within the same language family raises questions about the factors that condition the variation in the survival of the focus marker across these two closely related languages in the context of sluicing.

A preliminary cross-linguistic assessment of the divergence in the distribution of the focus marker under sluicing suggests a link between the choice of overt focus marker and clausal conjunctive particles in a language. The hypothesis is that:

(291) If a language has an overt focus head that does not also function as a clausal conjunction, it should survive ellipsis under sluicing and vice versa.

The realisation of clausal coordination in Likpakpaanl (292-a), Yoruba (292-b), Akan (292-c), and Gungbe (292-d) throws some light on the hypothesis.

(292) a. Francis nàn jìn bùsà **lé** Jonas nàn kùùr kì-sàà-k
F. PST eat 5.SG.food CONJ J. PST weed 11.SG-farm-11

gbààn. (Likpakpaanl)
DEF
'Francis ate food, and Jonas weeded the farm.'

b. Bọ́lá jẹ ẹ̀wa, Adé **sì** mu omi. (Yoruba)
B. eat beans A. CONJ drink water
'Bọ́lá ate beans, and Adé drank water.'

(Acheampong and Aremu, 2024)

c. Bayire **na** Kofi tɔɔ **na** Ama noayɛ́ (Akan)
yam FOC K. buy.PS CONJ A. cook.PFV
'Kofi bought yam, and Ama cooked it.'

d. Sɛsínú ɖà lɛ́sì bɔ́/***wɛ́** Sùrù' ɖù' nus'ɛnù. (Gungbe)
S. cook rice CONJ S. eat soup
'Sesinou cooked rice, and Suru ate soup.'

(Aboh, 2009, 4)

A comparison of the particles used for focus marking (FM) and clausal coordination (CC) in 292 suggests that languages that obey the BG employ different particles for FM and CC. However, languages like Likpakpaanl, Dagbani, Akan, etc., which use the same particle for both FM and CC, violate the BG but satisfy the Sluicing-COMP Generalization (SCG). This is summarised in 5.1

Table 5.1 Focus Particles, Clausal Conjuncts and Sluicing

Language	Overt Foc	Coordinator	Follows SCG	Follows BG
Likpakpaanl	lé	lé	✓	×
Akan	nà	nà	✓	×
Dagbani	ká	ká	✓	×
Gungbe	wɛ'	bo/ kpo	×	✓
Nupe	no	to	×	✓
Yoruba	ni	si	×	✓

Interestingly, this classification cannot be attributed solely to language family since Twi and Gungbe, both Kwa languages, exhibit divergent patterns in the realisation of sluicing. We observe that languages like Likpakpaanl (a Mabia language) and Akan (a Kwa language), with morphological focus markers, delete their focus markers under sluicing, while others, like Yoruba, Gungbe, and Nupe, do not.

This cross-linguistic variation highlights the need for a principled account that captures the intricate interactions between focus marking, clausal coordination, and the conditions governing the retention of focus markers under sluicing across diverse language families. These observations require further research to determine the precise nature of the relationship between clausal coordination and sluicing in these languages.

5.9 Chapter summary

This chapter investigated sluicing constructions in Likpakpaanl and examined the theoretical implications of the data. An analysis of Likpakpaanl sluicing was proposed by adopting Merchant (2001)'s notion that all properties of the E-feature are hosted on a syntactic element. I also proposed the projection of a licensing phrase (LP) in line Aelbrecht (2010), but the difference is that the LP in Likpakpaanl does not only license ellipsis through Agree but bears the E-feature. The investigations showed that Likpakpaanl sluicing is derived through the A-bar wh-movement of the wh-phrase from its base position to Spec-FocP, followed by successive cyclic movement into Spec-LP, a higher functional projection hosting the ellipsis E-feature that licenses sluicing Merchant (2001); Aelbrecht (2010). For inverse copular sluicing, Hatakeyama (1997)'s analysis was adopted, arguing that the predicate complement (the wh-phrase) is initially merged as the complement of vP, undergoing successive cyclic movement through the specifier positions of FocP and LP to its landing site in Spec-PredP.The data demonstrated that the overt focus particle accompanying ex-situ wh-movement is obligatorily absent under sluicing, supporting Merchant (2001)'s Sluicing-COMP Generalisation, which dictates that only the wh-operator can be retained in sluicing, while non-operator material must be omitted.

The chapter also discussed various island diagnostics, including complex NP constraints, adjunct islands, and coordinate structure constraint violations, indicating that these island effects are ameliorated under sluicing. Additional empirical evidence also showed that wh-phrases restricted to occurring in-situ in Likpakpaanl cannot license sluicing in the language. Sluicing obviates syntactic islands because the PF-uninterpretable feature (*), which causes island violations in non-elliptical constructions, is eliminated from the PF object(Merchant, 2001, 2004). Thus, the absence of island effects motivates an analysis where Likpakpaanl sluicing is derived by overt focus movement of the wh-phrase from its base-generated position through Spec-FocP to the specifier of LP, followed by deletion targeting the FocP constituent containing the trace of the displaced wh-phrase. The Likpakpaanl data provide novel evidence that cross-linguistic sluicing does not uniformly equate to TP-ellipsis, as some languages may target larger domains like FocP, contributing to the cross-linguistic variation observed in the syntactic environments that license sluicing. In Likpakpaanl, sluicing instantiates FocP ellipsis, but in languages like Gungbe, the E-feature on Foc deletes its TP complement.

Cross-linguistic data revealed variation regarding the Sluicing-COMP Generalisation and Baltin (2010)'s Generalisation on sluicing. The data show that Kwa languages like Gungbe, Nupe, and Yoruba satisfy Baltin's prediction by having their overt focus head survive ellipsis, while Akan, also a Kwa language, violates the BG like Likpakpaanl. I hypothesised that a link between the choice of overt focus marker and clausal conjunctive particles and their effects on surviving sluicing. Consequently, languages using different particles for focus marking and clausal coordination will tend to obey BG, while those employing the same particle for both functions will violate BG but satisfy the SCG. This divergence highlights the need for further research into the precise relationship between clausal coordination and sluicing and developing a theoretical account to capture these intricate interactions. I assume that accounting for the nature of this relationship between the choice of the focus particle and the clausal conjunct is crucial for advancing our understanding of the conditions governing the presence of focus markers under sluicing across diverse language families.

Chapter 6

Likpakpaanl Fragment Answers

6.1 Introduction

The goal of this chapter is to investigate Likpakpaanl fragment answers and demonstrate that they are elliptical constructions which ential a movement of the fragment element to the specifier of a Licensing Phrase (LP) followed by FocP deletion. Evidence for proposing a movement approach to these fragments comes from connectivity, island and post-position stranding effects. The Chapter is divided into six sections. Section 6.2 provides an overview of fragment answers and the approaches used to analyse fragmentary answers. In Section 6.3, I present the properties of Likpakpaanl fragments in matrix and embedded clauses and investigate how Likpakpaanl fragment answers are derived in Section 6.5. I propose that fragment answers can be accounted for using Merchant (2001, 2004)'s movement plus ellipsis approach. Using island and connectivity effects to support the movement approach, I argue in Section 6.6 that grammatical dependencies exist between the fragment answers and their correlates in the non-elliptical antecedent clauses. Therefore, these dependencies indicate that fragment answers in Likpakpaanl are derived by moving the remnant fragment to a left-peripheral position followed by deleting the remaining clause, as assumed by Merchant (2001, 2004, 2019). The main ideas discussed in the chapter are summarised in Section 6.7.

6.2 Defining Fragment Answers

Fragment answers to short responses provided to various types of questions or statements. The form of such short answers varies depending on the context. For example, we can have fragment answers to wh-question (293-b), or they can also surface as replies to polar (yes/no) questions (294-b). Additionally, fragment answers may arise in contrastive fragments when a speaker seeks to correct or clarify a declarative statement made by an interlocutor (295-b).

(293) a. [Q:] Who did she see?
b. [B:] JOHN.
c. [B':] She saw JOHN.

(Merchant, 2004, 673)

(294) a. [Q:] Does Abby speak GREEK fluently?
b. [B:] No, ALBANIAN.
c. [B':] No, she speaks ALBANIAN fluently.

(Merchant, 2004, 688)

(295) a. [A:] Bement a tóba Peti.
PRT.GO.PAST.3SG the lake.Ill Peti
'Peti entered the lake.'
b. [B:] Nem, Misi.
no Misi
'No, Misi.' (i.e. 'Misi entered the lake.')

(Lipták, 2020, 211)

A common feature among these three forms of short answers is the assumption that they involve elliptical XPs, which do not have the same overt propositional content as their non-elliptical counterparts (293-c) but convey the same semantic interpretation as their non-elliptical counterparts (294-c).

In generative syntax, there has been a long-standing debate as to whether fragment answers and other elliptical constructions involve only the overtly expressed structure or they possess an underlying syntax that is pronounced at LF. Progovac (2013) provides a comprehensive review of this controversy surrounding the analysis of fragment answers and elliptical structures within the framework of generative grammar. He compares and contrasts the work of various scholars, such as Morgan (1989, 1973) and Barton (1990) regarding their approaches to Fragment structures. These varying opinions have led to the emergence of the Direct Interpretation (DI) approach and the Ellipsis Approach (EA) or Movement plus deletion approach in analysing fragment answers.

The Direct Interpretation (DI) approach, led by Barton and Progovac (2005), Culicover and Jackendoff (2005), and Stainton (2006) contends that fragment answers are bare structures without additional syntax beyond the surface fragment element. For instance, Culicover and Jackendoff (2006) support the DI approach using examples like 296. They argue that there are interpretive differences between the fragment answer in (296-b) and its non-elliptical equivalent in (296-c) because both are not semantically identical. Consequently, while (296-a) can be followed by the fragment DP ‘scotch’ in (296-b), its interpretation will mean that Ozzie is just *imagining* Harriet’s drinking of scotch expressed in (296-c) but not the fact that Harriet has in reality being drinking scotch, which is the meaning expressed in (296-d).

(296) a. Ozzie fantasises that Harriet’s been drinking.
b. Yeah, scotch
c. [A1:] Ozzie fantasises that Harriet’s been drinking scotch
d. [A2:] Harriet’s been drinking scotch

(Culicover and Jackendoff, 2006, 414)

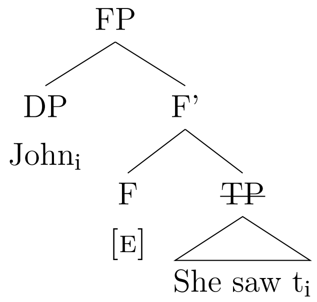
Culicover and Jackendoff (2006) therefore, contend that the different semantic interpretation associated with the fragment answer and its non-elliptical counterpart suggests that there is no underlying syntax but the bare answers that we see. There is also the Ellipsis Approach (EA), originating from Morgan (1973)’s seminal work, which proposed an analysis of fragment answers as containing more linguistic material than what is pronounced. This idea has been developed over the years by Merchant (2001, 2004, 2019), Lasnik (2001),van Craenenbroeck (2010), Weir (2014) and Bassong (2014) and others. The EA posits that fragment answers possess complete structures that undergo ellipsis if they satisfy the focus condition of ellipsis. Thus, in fragment answers to constituent questions, it is assumed that only the new information must be overtly expressed in the syntax, while the old information (e-GIVEN) is elided.

Using example (293), repeated here as 297, Merchant (2004) argues that the fragment DP *John* in (297-b) is derived by first undergoing A-bar movement from its base position as the object complement of *saw* to the specifier of a functional projection (FP), followed by the deletion of the remaining TP from which *John* was extracted. Merchant (2004) adopts Rizzi (1997) Cartographic framework and posits that the FP is a functional projection that projects a Focus Phrase (FocP) in the left periphery and attracts focused constituents to its Specifier. Within this EA framework, the functional head F accommodates Merchant (2001)’s E-feature, which licenses the fragment construction. According to Merchant, fragments are derived in the same way as sluicing; the only difference is that the ’remnants’ of fragments are DPs or adjuncts, while those of sluices are wh-interrogative phrases.

(297) a. Who did she see?
b. John.
c. She saw John

The representation in (298) demonstrates how fragments are derived using the movement plus deletion approach.

(298)



(Merchant, 2004, 675)

This study adopts the ellipsis or movement plus PF deletion approach by arguing that fragment answers involve two derivational steps: moving the fragment element to the Spec of a functional projection and deleting the remnant clause. Drawing on language-internal evidence, I argue that Likpakpaanl fragments are not bare structures but have an underlying syntactic representation that is not pronounced.

6.3 Properties of Likpakpaanl Fragment Answers

Wh-questions and fragment answers show an intrinsic relationship because the former determines the focus structure by marking what is questioned, while fragment answers provide the missing new information that is requested by the question. This question-answer congruence is achieved by eliding the already given information in line with constraints like e-GIVENness Merchant (2001); Schwarzschild (1999); Rooth (1992b). Therefore, before we examine the data on fragment answers, let us recall some basic facts about focus movement in Likpakpaanl.

Constituent questions in Likpakpaanl can be asked using an in-situ (300-b) or an ex-situ (300-a) approach. The responses to in-situ questions like (300-b) can be in-situ (299-b) or ex-situ (299-c) and both are accepted as an answer to the question 'Who did Binlu see?'.

(299) a. Binlu nàn kàn ŋmà? (In-situ wh-question)
B. PST see who
'Who did Binlu see?'

b. Binlu nàn kàn Francis *(là). (In-situ answer)
B. PST see F. FOC
'Binlu saw FRANCIS.'

c. Francis$_i$ *(lé) Binlu nàn kàn t$_i$. (Ex-situ non-elliptical answer)
F. FOC B. PST see
'Binlu saw FRANCIS.'

Ex-situ wh-questions, which are formed by moving the wh-phrase to Spec-FocP, optionally focus marked, can, however, be answered using only ex-situ answer or a fragment answer as illustrated in (300-b) and (300-c), respectively. Although the focus particle is optional in the wh-question, it is obligatory in the full answer but deleted in the fragment answer as (300-c) shows.

(300) a. Ŋmà$_i$ (lé) Binlu nàn kàn t$_i$? (Ex-situ wh-question)
who FOC B. PST see
'Who did Binlu see?'

b. Francis *(lé) Binlu nàn kàn t$_i$. (Ex-situ non-elliptical answer)
F. FOC B. PST see
'Binlu saw FRANCIS.'

c. Francis$_i$ ~~lé~~ ~~Binlu nàn kàn~~ t$_i$. (Fragment answer)
F. FOC B. PST see
'FRANCIS.'

The analysis proposed for Likpakpaanl focus constructions is discussed in detail in Section 3.11. I propose that the wh-phrase in the complement of the verb 'see' has a valued focus feature and, therefore, serves as a matching goal for the unvalued focus features on the probe FocP. Once an Agree relation is established between the wh-phrase and Foc and the unvalued features on Foc are valued, the EPP feature on Foc^0 extracts the wh-phrase to the left periphery in Spec-FocP. The rest of this section will investigate fragment answers in Likpakpaanl, examining their distribution in matrix and embedded clauses. The data will show that while the focus particle is elided in both root and embedded fragment answers, fragment answers in the latter require the overt realisation of the complementiser head, **kè**, for the embedded fragment to be grammatical.

6.3.1 Fragment Answers in Matrix Clauses

This subsection explores fragment answers to wh-questions and polar questions in Likpakpaanl.

6.3.1.1 Fragment Answers to Constituent Questions

Answers to Likpakpaanl constituent questions can be complete sentences, as in (301-b) or fragments, as in (301-c). The examples show that the direct object of the verb **bà** is extracted to the left periphery; however, the non-elliptical response requires the focus particle to be present for the sentence to be licit. The fragment answer, on the other hand, cannot be overtly focused marked (301-d). The observed asymmetry between fragment and non-fragmentary answers in Likpakpaanl already leads to whether there is a correlation between focus movement and fragment answers. This question will be answered in section 6.5.

(301) a. [Q:] Bà$_i$ (lé) Taanan nàn bì kìr t$_i$?
what FOC T. FOC IPFV pluck
'What was Taanan plucking?'

b. [A:] Gbàndɔ$_i$ *(lé) Taanan bì kìr t$_i$. (Sentential answer)
3.SG.pawpaw FOC T. IPFV pluck
'It is PAWPAW that Taanan is plucking./ Taanan is plucking PAWPAW.'

c. [A1:] Gbàndɔ$_i$ ~~(lé) Taanan bì kìr~~ t$_i$. (Fragment answer)
3.SG.pawpaw FOC T. IPFV pluck
'Pawpaw.'

d. [A2:] * Gbàndɔ$_i$ lé. (Intended Fragment answer)
3.SG.pawpaw FOC
Intended: 'Pawpaw.'

In di-transitive constructions, either the direct or the indirect object can be questioned. For example, in (302-a), the wh-phrase 'who', which is the indirect internal argument of the verb **tìì** 'give', is extracted to the left edge for an ex-situ question. In the fragment response, 'Mpopiin', the new information is A-bar moved to a clause-initial position and the old information with the focus head is elided.

(302) a. [Q:] Ŋmà$_i$ (lé) Taanan nàn tìì t$_i$ sìkìr?
who FOC T. PST give 3.SG.sugar
'Who did Taanan give sugar?'

b. [A:] Mpopiin$_i$ [$_{FocP}$ ~~(lé)~~ [TP ~~Taanan nàn tìì~~ t$_i$ ~~sìkìr~~.]]
M. FOC T. PST give 3.SG.sugar
'Mpopiin.'

c. [A1:] Mpopiin$_i$ *(lé) Taanan nàn tìì t$_i$ sìkìr.
M. FOC T. PST give 3.SG.sugar
'Taanan gave MPOPIIN sugar.'

Fragment elements are not always DPs but can also be adverbs, as in (303-b) or adjectives (304-b). The temporal adverb in the ex-situ wh-question in (303-b) is moved to Spec-FocP in the left periphery. As the previous examples show, the fragment answer occurs without the focus particle, although it is obligatory in the non-fragmentary answer (303-c).

(303) a. [Q:] [$_{\text{FocP}}$ Bà-dààl$_i$ (lé) [$_{\text{TP}}$ Tachin gà gbìì bànchì gbààn
what-day FOC T. FUT dig 3.SG.cassava DEF
t$_i$?]]

'When will Tachin dig the cassava?'

b. [A:] Dìn$_i$ [$_{\text{FocP}}$ ~~(*lé)~~ [TP ~~Tachin gà gbìì bànchì gbààn~~ t$_i$.]]
today FOC T. FUT dig 3.SG.cassava DEF
'Today.'

c. [A:] Dìn$_i$ *(lé) Tachin gà gbìì bànchì gbààn t$_i$.
today FOC T. FUT dig 3.SG.cassava DEF
'Tachin will dig the cassava TODAY.'

It is important to state that the adjectives used as fragment answers must first be nominalised; otherwise, the resulting fragment will be ungrammatical. Nominalised adjectives acquire the NC features of the head noun they modify and distinguish singular and plural[1]. The D-link wh-question in (304-a) taps into the common ground that 'Sanja wore a shirt' and the speaker now seeks specific information about the type of shirt 'Sanja wore'.

The fragment answer in (304-b) relies on the shared discourse context for interpretation, and the noun class prefix is used as an anaphoric reference to the 'shirt'. The fragment answer is also picking out 'black' out of the alternative set of colours of shirts that Sanja could wear, and it expresses the same meaning as the non-fragment answer in (304-c).

(304) a. Sanja nàn pèè **lì**-bùákù-l **lì**-là-lì?
S. PST wear 7.SG-shirt-7 7.SG-which-7
'Which shirt did Sanja wear?'

b. **Lì**-màmàn-l gbààn$_i$.
7.SG-red-7 DEF
'the red (one)'

c. **Lì**-bùákù-l **lì**-màmàn-l gbààn$_i$ lé Sanja nàn pèè t$_i$.
7.SG-shirt-7 7.SG.red-7 DEF FOC S. PST wear
'Sanja wore the red one (shirt).'

[1]Consider (i-a) below where the predicative adjective is not nominalised and consequently does not show any agreement morphology with the subject as we see in the fragment answer in (304-b). (i-b) is ungrammatical because the predicative adjective must be adjacent to the head noun.

(i) a. Lì-jà-l gbààn **pììn**.
7.SG-chair DEF white
'The chair is white.'

b. ***Lì**-jà-l gbààn **lì**-pììn.
7.SG-chair DEF 7.SG-white
Intended: 'The chair is white.'

A comparison of the fragment answer in (304-b) to the non-elliptical response in (304-c) demonstrates that head nouns in complex DPs modified by adjectives can be omitted in Likpakpaanl fragments, leaving only the adjective as a remnant. Thus, the fragment answer originates as part of the complex DP comprising the noun and adjective. The head noun, already e-GIVEN in the sense of Schwarzschild (1999); Merchant (2001), triggers morphological agreement with the adjective in its base position[2]. The fragment answer in (304-b) at the start of the derivation is part of the complex DP, where it is assigned a number feature by the head of NumP in the DP. After that, the DP, with its interpretable and valued features, checks the *uVal* features on FocP. Instead of the EPP feature on FocP attracting the entire DP complex, only the adjective is sub-extracted[3] to the left periphery to derive the fragment answer, stranding the head noun at the ellipsis site as (305-b) illustrates. The PF content of the stranded NP and TP are then deleted. It must, however, be noted that such sub-extraction from the DP in non-elliptical contexts, as **??** demonstrates, is ungrammatical.

(305) a. Lì-màmàn-l gbààn$_i$ ~~lé~~ ~~Sanja~~ ~~nàn~~ ~~pèè~~ [~~lì-bùákù-l~~ t$_i$.]
7.SG-red-7 DEF FOC S. PST wear 7.SG-shirt-7
'the red one'

b. *Lì-màmàn-l gbààn$_i$ lé Sanja nàn pèè lì-bùákù-l t$_i$.
7.SG-red-7 DEF FOC S. PST wear 7.SG-shirt-7
Intended: 'Sanja wore the red one'

The data has thus far focused on fragment answers in response to constituent

[2]An in-depth analysis of the agreement morphology between the noun and the d-link wh-phrase has been discussed in section 3.5.3

[3]I use the term in the sense of Corver (2017) to describe the phenomenon where a constituent is moved out of a larger phrase in which it is embedded. He notes that though such extractions violate Ross (1967) *Left Branch Constraint*, there is evidence in languages like Dutch (i) and French (ii) permit "Exceptional subextraction" by allowing a left branch constituent to be extracted from a nominal or adjectival construction.

(i) a. Wat$_i$ heeft Jan [t$_i$ voor boeken] verkocht?
wat has Jan for books sold
'What kind of books did John sell?'

b. *Welke$_i$ heeft Jan [t$_i$ boeken] verkocht?
which has Jan books sold
Intended 'Which books did Jan sell?'

(ii) a. Combien$_i$ a-t-il vendu [t$_i$ de livres]?
how-many has-he sold of books
'How many books did he sell?'

b. *Quels$_i$ a-t-il vendu [t$_i$ livres]?
which has-he sold books
Intended: "Which books did he sell?"

(Corver, 2017, 8-9)

interrogatives, such as wh-questions seeking more specific information about a previously mentioned entity. The discussion in the next section examines the nature of responses to polar interrogatives in Likpakpaanl.

6.3.1.2 Fragment Answers to Polar Questions

Polar questions (henceforth, PQ) are interrogative questions that elicit the equivalent of a 'Yes' or a 'No' answer or questions whose expected answer either affirms a proposition or denies it. Polar interrogatives represent a grammatical category whose fundamental illocutionary force is to solicit a response that affirms or denies the veracity of the proposition expressed by the interrogative radical (Krifka, 2011, 1747). Siemund (2001) posits that polar questions may possess a positive or negative polarity depending on whether they elicit a 'Yes' or a 'No' answer. He contends that a question that elicits a positive response is said to possess a positive polarity and vice versa. In other words, the felicitous responses to polar constructions are restricted to the affirmative and negative polarity values encoded, respectively.

Dryer (2005) argues that rising intonation is the most prevalent strategy for forming polar interrogatives cross-linguistically. However, Clements and Rialland (2008) contend that from their data collection from seventy (70) African languages, thirty-four (34) are a counterexample to Dryer (2005)'s claims since they use a clause-final falling rather than rising intonation in polar interrogatives. They note specifically that among the Moba sub-phylum of the Mabia (Gur) languages, to which Likpakpaanl belongs, the use of final lengthening is also associated with other interrogative markers, particularly falling intonation(Clements and Rialland, 2008, 77). Thus, using a rising pitch to signal polar questions is not a universally observed phenomenon cross-linguistically, as Dryer (2005) suggested. The data from Likpakpaanl corroborate the assumption by Clements and Rialland (2008) that most Mabia languages use falling intonation patterns in polar sentence constructions.

Likpakpaanl employs two principal strategies for forming PQs. The primary strategy is the use of sentence-final falling intonation coupled with the lengthening of the final vowel sound. (306-a) shows a declarative sentence, while example (306-b) represents a PQ counterpart. The PQ has an addition of vowel and falling intonation contours.

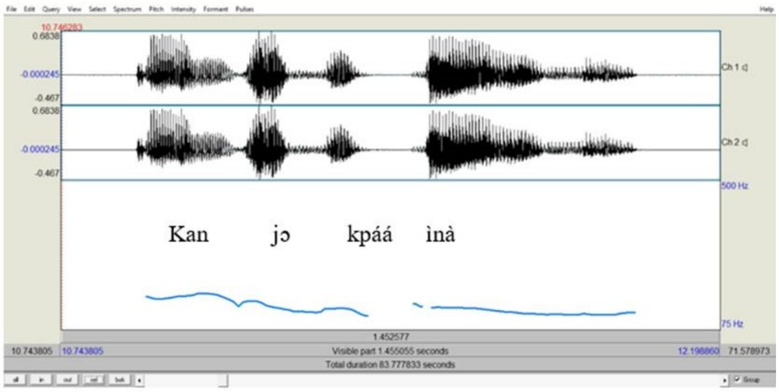
(306) a. Kpadin kpáá ì-nà.
K. herd 6.PL-cow
'Kpadin herds cattle.'

b. Kpadin kpáá ì-nà-à?

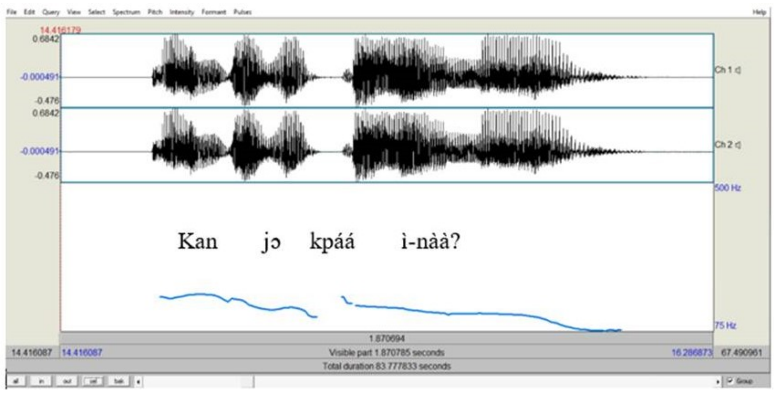
K. herd 6.PL-COW-INTER
'Does Kpadin herd cattle?'

The data shown in ((306-a)) and the accompanying spectrographic illustration in 307 serve as evidence for the co-occurrence of a clause-final falling pitch and vowel lengthening in Likpakpaanl polar interrogatives. The illustration in (307-a) also depicts the intonational patterns observed in a declarative utterance, while (306-b).

(307) a. The Spectrograph of a declarative sentence.



b. The spectrograph of a polar question.



An examination of the spectrographic representation in (307-a) shows that the pitch associated with the declarative has a low pitch. In contrast, the polar interrogative representation in (307-b) has a falling pitch contour, accompanied by an increase in the duration of the final vowel segment, as indicated by the extended wavelength[4]. Moreover, the pitch track duration of the polar interrogative sentence is longer than its declarative counterpart.

[4] A more detailed analysis of the pitch contour and formant frequency values need to be carried out to predict the exact differences in the pitch contours between declarative and polar questions. Dehé and Braun (2020) highlight the importance of considering not just boundary tones or edge tones, but also pitch accents and their distribution when modelling the interface between prosodic form and semantic interpretation. Their argument stems from the observation that differences in rising and falling intonation alone do not provide enough empirical evidence

According to Cahill (2012), in Konni, a Mabia language spoken in Yikpabongo in the North-East region of Ghana, polar questions are formed by lengthening the final vowel or nasal, accompanied by various falling final pitches as illustrated in (308-b).

(308) a. Ú ŋmìá gúúm bú.
3SG IPFV roll rope
'S/he is rolling the rope.'

b. ú ŋmìá gúúm bú-ú.
3SG IPFV roll rope-INTER
'Is she/he rolling the rope?'

(Cahill, 2012, 91)

Similar sentence-final falling intonational patterns have been proposed for polar interrogatives in Sisali (Dumah, 2017), Buli (Ferreira and Ko, 2003), and Dagbani Issah (2015), among others, thereby confirming Clements and Rialland (2008)'s assertion.

The second strategy employed in polar question formation involves using a clause-final interrogative particle **àà**. This interrogative particle occurs when the final syllable of the word preceding the question particle ends with a consonant sound. An illustration of this phenomenon is provided in the examples in 309.

(309) a. Bòndààn nàn lúú tì-bùákú-r.
B. PST weave 14.PL-basket-14
'Bòndààn wove baskets.'

b. Bòndààn nàn lúú tì-bùákú-r àà?
B. PST weave 14.PL-basket-14 INTER
'Did Bòndààn weave baskets?'

Answers to PQs can be a simple affirmative 'Yes' (310-b) or an elaborate affirmative confirming the question's proposition as shown in (310-c). The response can also be a corrective or contrastive fragment answer as in (310-d) with the main proposition following the response particle either elided or present as in the non-elliptical sentences (310-e). According to Lipták (2020), contrastive fragments frequently manifest as corrective statements targeting the focal correlate. Thus, in (310-d), the corrective fragment indicates that it is **Bondaan**, and not **Ponja** as presumed, who cooked the rice.

to determine the mapping to interpretation. More detailed prosodic analysis will be required to to distinguish between polar and declarative tones in Likpakpaanl.

.

(310) a. [Q:] Ponja lé nàn ŋààn ì-mùùl àà?
P. FOC PST cook 6.PL-rice INTER
'Is it Ponja who cooked rice?'

b. [A:] Één.
Yes
'Yes.'

c. [A:] Één, Ponja *(lé) nàn ŋààn ì-mùùl.
Yes, P. FOC PST cook 6.PL-rice
'Yes, Ponja cooked rice.'

d. [A:] Aàyì, Bondaan.
no, B.
'No, it's Bondaan (not Ponja).'

e. [A1:] Aàyì, Bondaan *(lé) nàn ŋààn ì-mùùl.
no, B. FOC PST cook 6.PL-rice
'No, Bondaan cooked rice (not Ponja).'

Comparing the non-elliptical responses in (310-c) and (310-e) to their elliptical counterparts in (310-b) and(310-d) shows, once again, the absence of the focus particle in fragmented answers. The discussion in this subsection centred around fragment answers to matrix wh-questions and polar questions. Nonetheless, it is worth mentioning that fragment answers can also be found as responses to embedded questions within more complex clause structures. The next section explores how fragment answers are derived from embedded questions.

6.3.2 Embedded Fragment Answers

There are two fundamental characteristics concerning embedded fragment answers (EFAs) in Likpakpaanl. Firstly, like root fragments, the overt focus particle in non-elliptical embedded fragments is omitted (311-b). Secondly, EFAs also require the obligatory presence of the overt complementiser **ké** for the sentence to be grammatical, as shown in (311-b). While (311-c) will be a grammatical response to a matrix question, it has a degraded meaning when used to answer a question like (311-a). This example illustrates that embedded fragments have the same propositional content as their non-fragmented counterpart, as depicted in (311-d).

(311) a. Sanjà lèn kè Pàndò gà màà kì-dìì-k gbáán bà-dààl?
S. say COMP P. FUT build 11.SG-room-11 DEF what-day
'When did Naajo say (that) Pando will build the room?'

b. *(Kè) fén (*lé).
COMP tomorrow FOC
'That tomorrow.'

c. ?Fén.
tomorrow

d. Kè fén lé Pàndò gà màà kì-dìì-k gbáán t_i.
COMP tomorrow FOC P. FUT build 11.SG-room-11 DEF
'That Pando will build the room tomorrow.'

A notable difference between embedded fragment answers and sluicing constructions in Likpakpaanl is their structural requirements. While sluicing has additional projections, CP and PredP, LP, above the FocP, embedded fragment answers necessitate only a single CP projection above FocP to host the C-head, as exemplified in (311-b). The presence of a copular verb leads to ungrammaticality in embedded fragment answers, as demonstrated by the ill-formedness of (312-b).

(312) a. $[_{TP}$ Sanjà lèn $[_{CP}$ kè $[_{FOCP}$ bà-dààl$_i$ (lé) $[_{TP}$ Pàndò gà màà
S. say COMP what-day FOC P. FUT build
kì-dìì-k t_i]]]]?
11.SG-room-11
'When did Sanjà say (that) Pando will build a room?'

b. *$[_{CP}$ Kè $[_{PredP}$ nì yè fén.]]
COMP it COP tomorrow
Intended: 'That tomorrow.'

Issah (2020) highlights that in the Dagbani, a Mabia language, embedded fragments also necessitate the overt morphological presence of the CP-head **nì**, as demonstrated by grammatical examples in (313-c). The ungrammaticality of (313-b), according to Issah (2020), is because of the absence of the C-head.

(313) a. [Q:] Ò yèlí-yá nì bò kà Baba dá t_i?
3SG.ANIM say.PFV-DJ that what FOC B. buy.PFV
'What did s/he say that Baba has bought?'

b. [A1:] *Lòòrí (*kà).
car FOC

c. [A2:] Nì lòòrí (*kà).
That lorry FOC
'That a lorry

(Issah, 2020, 183)

Similarly, Bassong (2019) argues that in Basaá, a Bantu language spoken in Cameroon, the complementiser **má** is obligatory in embedded fragments, as (314-b) shows. It can also be noticed that the focus particle in Basaá also undergoes ellipsis in embedded fragment answers.

(314) a. Kíí$_i$ í mudaaá$_i$ a-n-sɔmb t$_i$.
9.what 9.REL 1.woman 1.SM-PST-buy
'What is it that the woman has bought?'

b. **Makúɓé má** ~~mɔ-n~~ ~~pro~~$_i$ ~~ai-n-sɔmb~~ t$_i$
6.bananas 6.REL 6-FOC 1.SM-PST-buy
'Bananas.'

c. Makúɓé ma mɔ-n pro$_i$ ai-n-sɔmb t$_i$
6.bananas 6.REL 6-FOC 1.SM-PST-buy
'Bananas are what she will buy.'

(Bassong, 2019, 924)

Despite the requirement for the presence of the C-head in embedded fragments, Temmerman (2013) notes that the Dutch complementiser cannot co-occur with the embedded fragment answer, unlike the patterns observed in Likpakpaanl, Basaá, and Dagbani, among others. She further argues that English fragments cannot be embedded because native speakers perceive such embedded fragments to be semantically degraded if the subject is altered.

Examining fragmentary constructions in Likpakpaanl leads to the following descriptive generalisations:

i. The overt focus particle in non-elliptical fragments does not survive the ellipsis process in root and embedded fragment answers.

ii. In fragmentary answers, only the new information, the fragment element, remains phonologically visible, while shared (old) information is elided.

iii. Embedded fragments must be preceded by the overt complementiser head **kè** for the fragment answer to be accepted.

6.4 Constraints on Likpakpaanl Fragment Answers

In Likpakpaanl, fragments are permissible in ex-situ wh-questions, where the wh-phrase has undergone wh-movement, but not in in-situ wh-questions. Recall that in 5.5, it was shown that adjunct wh-phrases like *Reason* **bàŋà** and *Manner* **kìnyé**, which occur solely in-situ, could not participate in sluicing constructions. While the manner adverb **kìnyé** can felicitously occur in-situ, as exemplified in 315, it cannot undergo wh-movement to the left periphery of the clause, as illustrated by the ungrammaticality of (315).

(315) Q: Mbeebi gà tér ù-bò gbààn **kìnyé**?
M. FUT help 1.SG-child DEF how
'How will Mbeebi help the child?'

Q: *kìnyé$_i$ (lé) Mbeebi gà tér ù-bò gbààn t$_i$?
how FOC M. FUT help 1.SG-child DEF

Following the assumption that fragment answers are derived via A-bar movement of the fragment constituent to the left periphery followed by PF deletion of the remnant, it is predicted that **kìnyé** should be unable to occur as a fragment answer since it cannot be extracted from its base-generated position to a left periphery position. This prediction is born out since the question (316-a), the fragment answer (316-b), and the non-elliptical response (316-c) are all ungrammatical.

(316) a. [Q:] *kìnyé$_i$ (lé) Ndoombi nàn tóór tì-bɔr gbààn t$_i$.
how FOC N. PST repair 14.SG-case DEF
Intended: 'How did Ndoombi solve the problem?'

b. [A:] *Ì-chììn pónnì$_i$ (lé).
6.PL-wisdom inside FOC
Intended: 'With wisdom.'

c. [A1:] *Ì-chììn pónnì$_i$ lé Ndoombi nàn tóór tì-bɔr
6.PL-wisdom inside FOC N. PST repair 14.SG-case
gbààn t$_i$?
DEF
Intended: 'Ndoombi solved the problem with wisdom.'

Similarly, the *reason* wh-adjunct **bàŋà**[5] is banned in fragmentary answers, as evidenced by the ungrammaticality in (317-c). Thus, the fragment answer and their non-elliptical counterparts (317-d) are ungrammatical in Likpakpaanl.

(317) a. [Q:] **Bàŋà** Mbeebi gà tér ù-bò gbààn?
why M. FUT help 1.SG-child DEF
'Why will Mbeebi help the child?'

b. [Q1:] ***Bàŋà** (*lé) Mbeebi gà tér ù-bò gbààn?
why FOC M. FUT help 1.SG-child DEF
'Why will Mbeebi help the child?'

c. [A:] *Bà-pù ù-tè (lé) kpó.
what-on 3SG-father FOC die
Intended: 'Because his father died.'

d. [A1:] *Mbeebi gà tér ù-bò gbààn bà-pù ù-tè kpó.
M. FUT help 1.SG-child DEF what-on 3SG-father die
Intended: 'Mbeebi will help the child because his father died.'

[5]Note that **bàŋà** does not allow even optional focus fronting like the other adverbs in Likpakpaanl since it does undergo focus movement. It also always occurs in a preverbal position, see: (317-b).

Another constraint in Likpakpaanl fragment answers is that verb or verb phrase fragment answers are banned. This observation is unsurprising since Merchant (2004) has predicted that languages like Likpakpaanl, which disallow VP-fronting, should block VP fragment answers.

(318) a. Bà$_i$ (lé) Manasseh nàn ŋà lì-bùù-l gbààn t$_i$?
what FOC M. PST do 7.SG-pot-7 DEF
'What did Manasseh do to the pot?'

b. *Wìì (lì).
break 7.SG.it
Intended:'Broke it.'

c. Manasseh nàn wìì lì-bùù-l là.
M. PST do 7.SG-pot-7 DEF FOC
'Manasseh BROKE the pot?'

Unlike in Likpakpaanl, where even the entire VP cannot be fronted as a fragment answer, Merchant (2004) contends that in English, fragment predicate answers are allowed if the fragment consists of the entire VP, not just a verb as in (319-b). He posits that such fragment predicates require the phrasal movement of the entire verb phrase, the smallest constituent that can be used as a fragment, not just the head movement of the verb.

(319) a. What did she do with the spinach?

b. Washed (it).

(Merchant, 2004, 698)

The discussion in this subsection examined fragment answers to constituent and polar questions in root clauses and embedded contexts in Likpakpaanl. I claim that while root fragment answers consist solely of the A-bar moved fragment constituent, embedded fragments require the overt presence of the complementiser head **ké**. This asymmetry between root and embedded fragment answers in Likpakpaanl provides insights into the underlying syntactic derivation of these constructions. The next subsection provides a formal analysis of fragment answers in Likpakpaanl, adopting the PF-deletion approach proposed within the ellipsis literature

6.5 Towards an Analysis of Likpakpaanl Fragment Answers

This section systematically analyses the derivation of fragment answers from a Minimalist approach. I argue that just as in the case of sluicing, the Licensing Phrase (LP) with an EPP and E feature to license fragmentary answers as proposed by Merchant (2001, 2004). I also argue that the lack of fragment answers in in-situ questions suggests that overt focus movement feeds ellipsis and supports a movement approach. The first part of the analysis examines the syntax of fragment answers to matrix and embedded questions.

6.5.1 Syntax of Likpakpaanl Fragments

Using the fragment response in (320-b) where we have the fragment DP at the clause-initial position, I demonstrate that Likpakpaanl fragment answers are derived through three successive steps: Agreement, A-bar movement, and Phonological Form (PF) deletion.

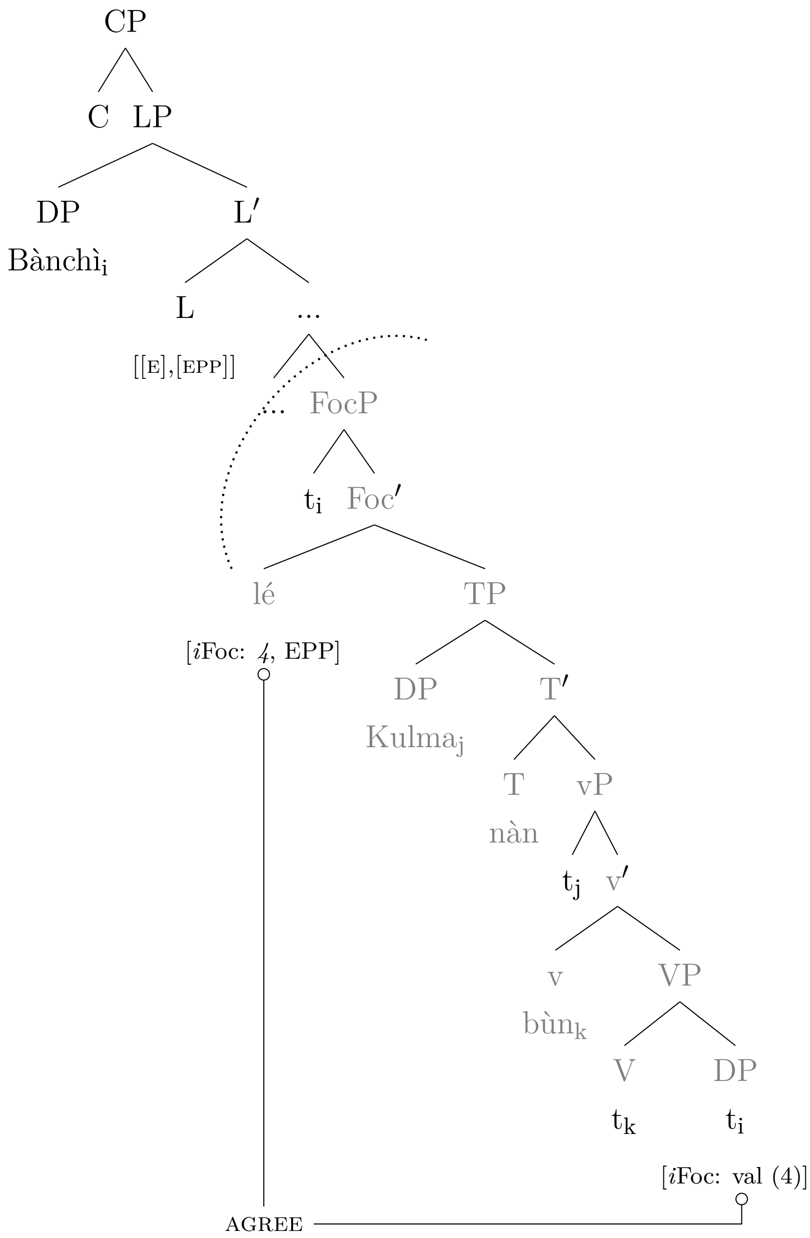
(320) a. [Q:] Bà$_i$ (lé) Kulma nàn bùn t$_i$?
what FOC K. PST plant
'What did Kulma cultivate?'

b. [A:] Bànchì$_i$.
3.SG.cassava
'Cassava.'

c. [A1:] Bànchì$_i$ *(lé) Kulma nàn bùn t$_i$.
3.SG.cassava FOC K. PST plant
'Kulma cultivated cassava.'

The structural representation in (321) is derived as follows: The fragment DP **Baanchi** bearing an interpretable focus feature [*i*Foc] is merged as the internal argument of VP. The verb **bùn** undergoes head movement from V^0 to v^0, forming the vP projection. TP is merged above vP, and the EPP feature on T attracts the external argument **Kulma** from Spec-vP to Spec-TP. Next, the unvalued focus feature [*u*Val] is valued by the valued feature on the focused element. The EPP on the Foc^0 extracts the fragment DP to Spec-Foc, resulting in focus movement. A further movement is required for the fragment answer, and this is because the E feature must be in a head complement relationship with the elements in the E-site (see Merchant, 2001, 2004). Thus, the licensing phrase is projected above FocP, with the ellipsis feature and an EPP feature on its head L. The focused DP **Baanchi** undergoes a successive cyclic movement from Spec-FocP to Spec-LP to satisfy the EPP on L. Once Spec-LP is filled, the E feature on L^0 licenses the ellipsis of the entire FocP constituent and the TP to give us the fragment answer. This is represented in the structure in (321).

(321)



Having examined fragment answers to matrix questions, I now focus on embedded fragment answers, which, as the data demonstrated, require the C^0 for the answer to be grammatical. Thus, embedded fragments give rise to an additional CP projection above the LP. Despite this extra CP layer, the E-feature still resides on the head of the LP, providing a unified account for both root and embedded fragment answers. The derivation of the embedded fragment in (322-b) proceeds through the successive operations of Agree, movement, and deletion of the underlying PF content in the remnant clause. In (322-a), the wh-phrase is questioned inside an embedded clause, and as indicated earlier, the fragment answer (322-b), as well as the non-elliptical answer in (322-c), must be preceded by the complementiser **ké**.

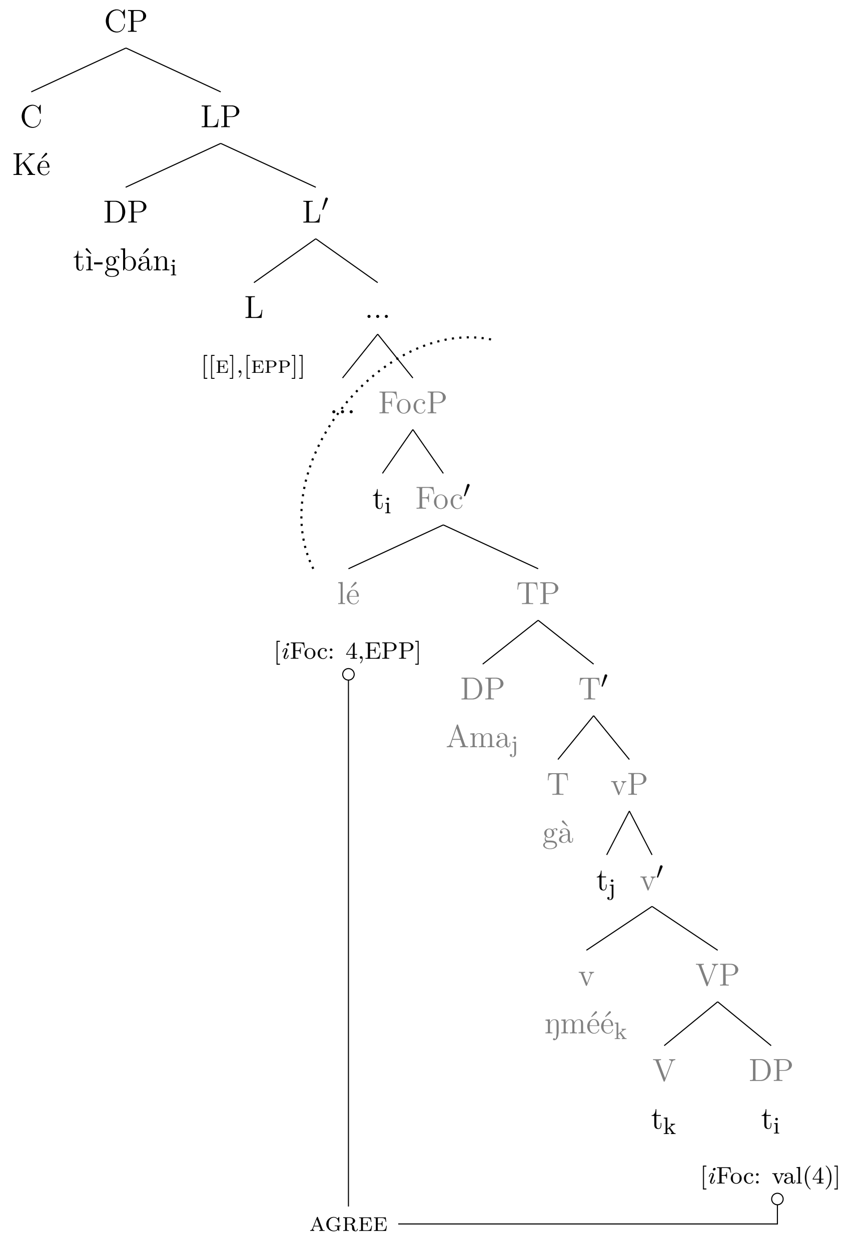
(322) a. [Q:] Afia lén ké Amà gà ŋméé bà?
A. say.PFV COMP A. FUT write what
'What did Afia say Ama will write?'

b. [A:] Ké tì-gbán.
COMP 14.PL-book
'that books.'

c. [A1:] Ké tì-gbán lé Amà gà ŋméé t_i.
COMP 14.PL-book FOC A. FUT write
'That Ama will write books.'

The derivation of the embedded fragment answer in (322-b) in Likpakpaanl systematically occurs through different syntactic operations. In the first place, the DP **tì-gbán** 'book' is internally merged as the complement of the VP 'write', form where it undergoes head movement from V to little v, projecting vP. I argue that the external argument to the verb 'write', **Ama** merged in Spec-vP moves to Spec-TP to satisfy the EPP feature on T. The focus head **lé** with [*i*Foc: *u*Val] feature projects a focus phrase above TP. The DP element **tì-gbán** in the complement of the verb with matching value features [*u*Foc: Val] features enters into Agree with FocP to value the [*u*Val] on the probe, Foc^0. After valuation, the DP **tì-gbán** moves from its base position to Spec-FocP to satisfy the EPP feature on Foc. Next, the left-peripheral LP is merged above FocP, and the EPP on L^0 attracts the fragment DP to its specifier. Once Spec-LP is filled, the E-feature on L^0 is activated, triggering FocP 'deletion.' Finally, a CP layer is projected above LP, with the overt complementiser "Ké" realised in C^0, yielding the embedded fragment answers shown on the surface structure in (322-b). The derivation is illustrated in (323).

(323)



The analysis presented so far has demonstrated that both matrix and embedded fragments in Likpakpaanl are derived via an identical set of operations: movement of the focused phrase to Spec-LP, followed by PF deletion of the FocP complement. I now turn to investigate the syntax of polar fragment answers.

6.5.2 Polar Response Particles and Sluicing

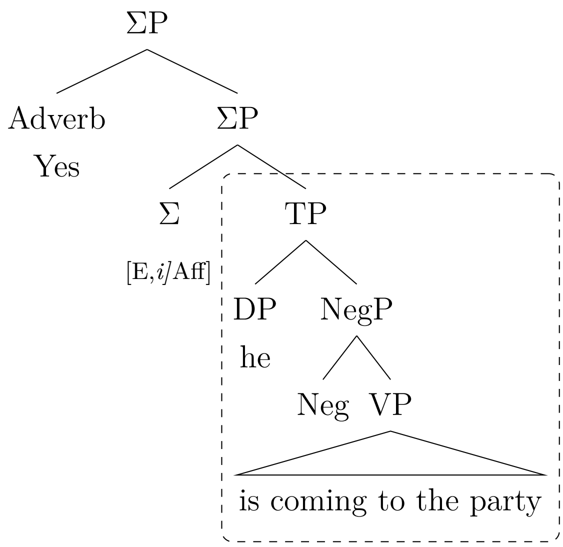
Kramer and Rawlins (2009) and Holmberg (2013), have argued that the simple responses ‘Yes’ and ‘No’ given to polar questions, like fragment answers to constituent questions, have an elaborate underlying syntax. They note that such short response particles (*ResPrts*) represent an entire proposition (the TP) that is not overtly realised because the PF information has been elided. In their analysis of polar response particles, Kramer and Rawlins (2009) proposed that solitary ‘Yes’ and ‘No’ response particles such as those in (324) involve elided structure.[6]

(324) a. [Q:] Is Alfonso coming to the party?
b. [A:] Yes, ~~he coming to the party~~.
c. [A1:] No, ~~he is not coming to the party~~.

(Kramer and Rawlins, 2009, 483)

Kramer and Rawlins (2009) argue that attempts at providing a formal syntactic analysis of polar questions within the Minimalist Programme. Their proposal posits the existence of a Polarity Phrase (ΣP) in such fragments, whose head bears both interpretable and uninterpretable polarity features, with the response particle (*ResPrt*) occupying the specifier position. Specifically, a *Yes* response, as exemplified in (324-b), is analysed as carrying an interpretable affirmative [*i*Aff] polarity feature. Conversely, a *No* answer like in (324-c) is taken to express an uninterpretable [*u*Aff] affirmative polarity feature. Kramer and Rawlins (2009) propose that the ellipsis-licensing E-feature is merged on this null Polarity head, as represented in (325) (cf. Kramer and Rawlins, 2009, 484).

(325)



Kramer and Rawlins (2009) assume that *ResPrts* are sentential adverbs, given that they exhibit parallel syntactic behaviour to adverbs such as ‘maybe’, ‘perhaps’, ‘certainly’, and ‘definitely’, which can also surface as fragment answers. This adverbial analysis for *ResPrts* forms a core component of their account of

[6]Holmberg (2013) also assumes that *ResPrts* possess a polarity variable that expresses two possible values, an affirmative ‘Yes’ or negative ‘No’, assigned a value by the focused polarity expression.

the syntax of polar fragment responses. I adopt the proposal by Kramer and Rawlins (2009) regarding the presence of a ΣP in polar fragment answers but deviate from their analysis in two ways. Firstly, while I assume the response particle is base-generated in Spec-ΣP, I do not support the proposal that Σ^0 bears Merchant (2001)'s E-feature responsible for licensing ellipsis, but instead the L^0 contra Kramer and Rawlins (2009)'s claims. Secondly, I propose that the ellipsis-licensing E-feature is instead merged below ΣP. Empirical motivation for this proposal comes from answers to corrective or contrastive polar fragments, as in (326-c), where the contrastive/corrective fragment element must follow the response particle.

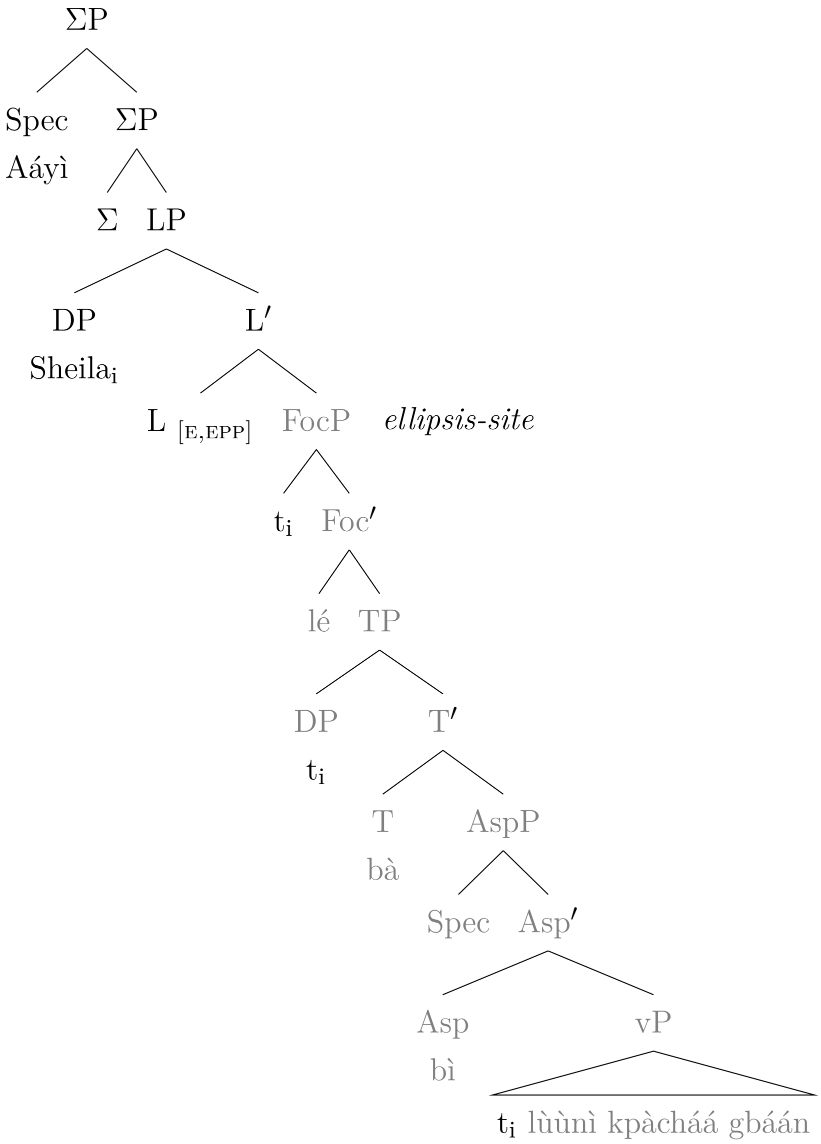
(326) a. [Q:] [TP Dipo bà [$_{AspP}$ bì [$_{vP}$ t_i lùùnì kpàcháá gbáán
D. PST IMPFV sharpen 3.SG.cutlass DEF
àà?]]]
INTER
'Was Dipo sharpening the cutlass?'

b. [A:] Aáyì, Sheila lé bà bì lùùnì kpàcháá gbáán.
no, S. FOC PST IMPFV sharpen 3.SG.cutlass DEF
'No, Sheila was sharpening the cutlass.'

c. [A1] Aáyì, Sheila ~~lé bà bì lùùnì kpàcháá gbáán~~.
no, S. FOC PST IMPFV sharpen 3.SG.cutlass DEF
'No, Sheila.'

In the fragment response in (326-c), the fragment DP **Sheila**, bearing new information focus, undergoes A-bar movement from Spec-TP through Spec-FocP to Spec-LP, paralleling fragment answers in root and embedded clauses. Crucially, the polar response particle **Aáyì** occurs in Spec-ΣP above LP, as illustrated in (327). The analysis indicates that the final landing site of **Sheila** is in Spec-LP since the response particle already fills Spec-ΣP. Thus, we have an analysis similar to fragment answers in embedded clauses where we have the CP head projecting above the LP even though L^0 licenses the ellipsis.

(327)



The ellipsis analysis proposed for Likpakpaanl polar response particles, 'Yes' and 'No', shows that they are merged in Spec-ΣP above the LP and the focus phrase, which can optionally undergo ellipsis under Merchant (2001)'s e-GIVENness and identity condition. This supports the view held by Holmberg (2013); Kramer and Rawlins (2009) that response particles and their full sentential counterparts have the same underlying structure.

This section presented a unified analysis of fragment types in Likpakpaanl. I showed that regardless of whether the ellipsis is an answer to a matrix, an embedded, or a polar question, the LP bears the ellipsis licensing feature. Moreover, the analysis indicates that ellipsis targets the FocP and the entire TP for deletion rather than solely the TP and demonstrates that sluicing and fragment answers in Likpakpaanl constitute FocP ellipsis, contrasting with the TP ellipsis found in

English as summarised in 328.

(328)
a. $[_{\text{LP}}$ E-Feature $\cancel{[_{\text{FocP}}\ [_{\text{TP}}\ [_{\text{vP}}\ [_{\text{VP}}\ ]]]]}]$ (Fragments to root questions)
b. $[_{\text{CP}}$ ké $[_{\text{LP}}$ E-Feature $\cancel{[_{\text{FocP}}\ [_{\text{TP}}\ [_{\text{vP}}\ [_{\text{VP}}]]]]}]]$ (Embedded fragments)
c. $[_{\Sigma\text{P}}$ $[_{\text{LP}}$ E-Feature $\cancel{[\cdots[_{\text{FocP}}\ [_{\text{TP}}\ [_{\text{vP}}\ [_{\text{VP}}]]]]]}]]$ (Polar response particles)

6.6 Evidence for Movement in Fragment Answers

In the previous section, I argued that Likpakpaanl fragment answers are derived through A-bar movement of the fragment element to the left peripheral position, Spec-LP, followed by 'deletion' of the FocP complement of L^0. The motivation for the assumption is based on the theoretical framework proposed by Merchant (2001, 2004, 2013a), proposing a move plus PF deletion for elliptical constructions like fragments. This section provides empirical support for opining that Likpakpaanl fragment answers have complete underlying syntactic representations similar to their non-elided counterparts. I adduce evidence from the binding and the sensitivity of fragments to island constraints and the distribution of strong pronouns in Likpakpaanl (cf. Merchant, 2001, 2004, 2013a).

6.6.1 Binding and Connectivity Effects

One of the diagnostic tests used to demonstrate that fragment answers arise from A-bar movement followed by PF deletion has come from binding and reconstruction effects. Chomsky (1981)'s Government and Binding Theory (GB) theory (stated in 17) outlines how anaphors (reflexives and reciprocals), pronouns, and referential (R)-expressions are distributed within a clause structure (see also: Reinhart and Reuland, 1993). Consequently, proponents of the movement approach contend that if fragment answers have the same underlying syntax as their non-fragmented counterparts, the same binding constraints are predicted to apply to both constructions.

I first examine how reconstruction effects provide evidence for the A-bar movement of fragment answers. Reconstruction occurs when an A-bar displaced constituent at the left periphery is interpreted in a lower structural position. According to Fox (1999); Aguero Bautista (2001), reconstruction can lead to scope ambiguities because the displaced wh-element is syntactically interpreted as if it were still its canonical position. For instance, Aguero Bautista (2001) contends that in (329), the wh-object receives a single, individual reading in Spanish without reconstruction. However, reconstruction yields the availability of a pair-list reading, indicating that 'each witness' refers to a different person as the direct object of *pegar* 'hit'.

(329) A **quién** dijo cada testigo que Maria le queria pegar?
to whom said each witness that M. himD wanted to.hit
'Whom did each witness say that María wanted to hit?'

(Aguero Bautista, 2001, 172)

Turning to Likpakpaanl fragment answer in (330-c), I argue that it exhibits reconstruction effects, resulting in scope ambiguity. The sentence can be interpreted to indicate that 'every chief' watched some 'pictures of himself'. This interpretation is only possible because the complex wh-phrase has undergone A-bar movement from its base-generated position as the direct object of the verb **gèè** 'like' where the antecedent QP c-commands it. The quantifier phrase **ù-bɔr mɔmɔk** 'every chief' binds the anaphor **ù-bà** 'himself' satisfying Principle A of the GB Theory.

(330) a. Fótó-tììb bì-là-bì$_i$ lé ù-bɔr mɔmɔk gèè t$_i$?
picture-4.PL 4.PL-which-4 FOC 1.SG-chief every like
'Which pictures does every chief like?'

b. Ù-bà àà-fótó-tììb$_i$ lé ù-bɔr mɔmɔk gèè t$_i$.
3.SG-self POSS-picture-4.PL FOC 1.SG-chief every like
'Every chief$_i$ likes pictures of himself$_i$.'

c. Ù-bà àà-fótó-tììb$_i$ (*lé).
3.SG-self POSS-picture-4,PL FOC
'Pictures of himself.'

Since Binding *Principle A* of the GB Theory requires anaphors to be bound in their governing category, it means that the reflexive emanated as the complement of the vP where is it properly governed and bound by the quantifier phrase **ù-bɔr mɔmɔk** 'every chief' before extraction to the left periphery. The fragment answer in (330-c) 'pictures of himself' in Likpakpaanl allows for two possible interpretations: The first is the pair-list reading where the fragment answer is reconstructed in its base position, represented by the trace. By this interpretation, it means that for *each chief X, X likes pictures of X* suggesting that each chief likes pictures depicting *themselves.* This pair list is possible because of the reconstruction of the fragment element in a position c-commanded by the quantifier phrase.

Further evidence for the movement analysis of fragment answers is that the binding conditions constraining reflexives in fragments and non-fragmentary sentences are the same as illustrated in (331-b) and (331-c). The reflexive pronoun **ù-bà** is bound by its antecedent **Mpopiin** with which it agrees with phi-feature. The fact that **bì-bà**, a plural reflexive, cannot be bound by a singular antecedent reveals that anaphoric dependencies must be appropriately established between the reflexives and their corresponding antecedents before extracting the fragment reflexive to the left periphery.

(331) a. [Q:] Mpopiin gà tèr ŋmà?
M. FUT help who
'Who will Mpopiin help?'

b. Mpopiin$_i$ gà tèr ù-bà$_i$/*bì-bà$_j$ là.
M. FUT help 1SG-self/2PL-self FOC
'Mpopiin will help herself/*themselves.'

c. Ù-bà$_i$.
1SG-self
'Herself.'

There are reciprocals in Likpakpaanl as shown in (332), regulated by binding *Principle A* . However, the reciprocal pronoun **tɔb** does not specify any difference between *each other* and *one another* as we find in English.

(332) B$_i$ fé mànn/tér tɔb fénnà.
3.PL HEST.PST visit/help RECP yesterday
'They visited/helped each other/one another yesterday.'

(Acheampong, 2015, 78)

Now, let us examine the distribution of the reciprocal as a fragment answer. In (333-c), the fragment answer 'each other' is grammatical because the reciprocal is bound by the compound subject **John and Mercy**. The same is observed in the non-fragmented counterpart in (333-b), with both obeying Principle A of the GB theory.

(333) a. John nì Mercy géé ŋmà?
J. and M. love who
'Who does John and Mercy love?'

b. John nì Mercy$_k$ géé tɔb$_k$ là.
J. and M. love RECP FOC
'John and Mercy said they loved one another.'

c. Tɔb$_k$.
RECP
'Each other/One another'

The evidence from binding and connectivity effects in Likpakpaanl strongly suggests that fragment answers involve movement of the fragment element to a left peripheral position prior to ellipsis rather than being base-generated. This movement-based analysis is supported by the fact that the fragment elements show the same binding requirements and connectivity effects that are observed in their non-elliptical counterparts. Furthermore, the pattern observed in fragment answers parallels what is found in sluicing constructions in the language, where the focus particle **lé** is notably absent in the elliptical answer despite being present in non-elliptical contexts.

6.6.2 Distribution of Strong and Weak Pronouns

While Principle A prescribes that anaphors be bound in their local domain, Principle B requires a pronoun to remain free (i.e., unbound) within its governing category. The effect of this constraint is exemplified in (334) where the emphatic pronoun **bìmà**, which is supposed to be free, is illicitly bound by the c-commanding DP antecedent **Ponpiir ni Gulpii** in (334-c) in violation of the principle. Thus, the fragment answer in (334-b) and its non-fragment counterpart in (334-c) are ungrammatical because the emphatic pronoun is bound to the antecedent, violating the binding principle B.

(334) a. Ponpiir ni Gulpii gà tér ŋmà-màm dìn?
P. and G. FUT help who-4.PL today
'Who will Ponpiir and Gulpii help today?'

b. *Bìmà$_i$ lé Ponpiir ni Gulpii$_i$ gà tér t$_i$ dìn?
them.EMPH FOC P. and G. FUT help today
Intended: 'Ponpiir and Gulpii$_i$ help them$_i$ today.'

c. *Bìmà$_i$
them.EMPH
Intended: 'They$_i$'

The data further indicates that the same constraint extends to weak pronouns like **ù**. In (335-b), the weak pronoun **ù** cannot be co-indexed with and bound by the DP antecedent **Bòndòbòl**. Example (335-c) is grammatical when the pronoun remains free as required, supporting a syntactic identity between the fragment answer and full sentential counterparts.

(335) a. Bindibil nàn séén sìmà tìì ŋmá?
B. PST boil 3.SGgroundnut give who
'Who did Bindibil boil groundnut for?'

b. Bòndòbòl$_i$ nàn séén sìmà tìì ù$_i$. là
B. PST boil 3.SGgroundnut give 3SG FOC
'Bindibil$_i$ boil groundnut for him$_{*i/j}$'

c. *U$_i$.
3SG
Intended: '*him$_i$'

6.6.3 Fragments and Island Constraints

Another test used in the literature to support the claim that focus or ex-situ movement feeds fragment answers comes from island effects. Thus, the assumption is that if fragments involve movement tot to a derived position, then they should be sensitive to locality constraints on movement. Merchant (2004, 2013a) argues that fragment constituents that originate from a syntactic island should be ungrammatical, and he shows, using examples like (336), that this is born out in English. The full sentence in (336-c) shows that the ungrammatical fragment DP, 'Charlie' in (336-b) is blocked from extraction since the escape hatch in spec-CP is occupied by the DP 'the same Balkan language'. Thus, he contends that if a phrase cannot undergo overt movement out of the island, as in (336-c), then it cannot equally be extracted out of the island to a higher position, accounting for the ungrammaticality of the fragment answer in(336-b).

(336) a. Does Abby speak the same Balkan language that **Ben** speaks?
b. No, **Charlie**.
c. No, she speaks the same Balkan language that **Charlie** speaks.

(Merchant, 2004, 688)

Merchant (2004) also asserts that when a constituent that substitutes for a wh-phrase in a polar question is located within a syntactic island, the resulting fragment will be illicit. Consequently, the ungrammaticality of (337-b) stems from the fact that the fragment element 'Beth' is situated within the adjunct island, as demonstrated by the full sentence in (337-c). The ungrammaticality of the fragment answer in (337-b) suggests that the overt movement of the remnant constituent to a derived position at the left periphery is blocked by the intervening adjunct island.

(337) a. Did Ben leave the party because Abby wouldn't dance with him?
b. No, Beth.
c. No, he left the party [$_{\text{Adjunct island}}$ because] Beth wouldn't dance with him.

(Merchant, 2004, 688)

Following the assumptions of Merchant (2004), I show that Likpakpaanl fragment answers are derived by movement, and they are, therefore, predicted to exhibit sensitivity to syntactic island constraints. The data support this assumption and provide further empirical support for a movement plus PF deletion. The example in 338 demonstrates when the answer to a wh-question is contained within a relative clause island, it cannot be used as a fragment answer in Likpakpaanl. The fragment answer (338-b) and the non-fragmented ex-situ counterpart (338-c) are both ungrammatical because the fragment DP is moved out of the island. The answer to the question is perfectly grammatical if the DP "rice" remains inside the island as shown (338-d).

(338) a. Néínà nàn kàn [$_{\text{Island NP}}$ ì-nà [$_{\text{CP}}$ ì ŋmɔn bànchì
N. PST see 6.PL-cow REL eat 3.SG.casssava
nà-à?]]
REL.DEF-INTER
'Did Neina see the cows that ate the cassava?

b. *Ààyí ì-mùùl.
no 6.PL-rice
Intended: 'No, the rice.'

c. *Ààyí ì-mùùl$_i$ lé Néínà nàn kàn [$_{\text{Island NP}}$ ì-nà [$_{\text{CP}}$ ì
no 6.PL-rice FOC N. PST see 6.PL-cow REL
ŋmɔn t$_i$ nà]]
eat REL.DEF
Intended:'No, Neina saw the cows that ate the rice.'

d. Ààyí Néínà nàn kàn ì-nà ì ŋmɔn ì-mùùl nà là.
no N. PST see 6.PL-cow REL eat 6.PL-rice REL.DEF FOC
'No, Neina saw the cows that ate the rice.'

Further evidence that fragment answers involve movement and are subject to locality constraints comes from the example in (339-b) illustrating that the extraction of fragment elements from adjunct islands is disallowed in Likpakpaanl. In (339-a), the speaker asks whether 'Kobina' went to the farm before 'Kojo' returned home. The intended fragment answer in (339-b) is 'No, Mosia', but this is ungrammatical because the fragment DP is contained within the adjunct island. Since adjunct islands are strong islands, extracting 'Mosia' from this island violates locality constraints on movement.

(339) a. [Q:] Kobina nàn bùèn kì-sàà-k [$_{\text{Adjunct island}}$ wààr lé
K. PST go 11.SG-farm-11 before CONJ
Kojo kùn àà]?
K. return.home INTER
'Did Kobina go to the farm before Kojo returned home?'

b. *Ààyí, Mosia.
no M.
Intended: 'No, Mosia'

c. *Ààyí, Mosia$_i$ lé Kobina nàn bùèn kì-sàà-k [$_{\text{Adjunct island}}$
no, M. FOC K. PST go 11.SG-farm-11
wààr lé t$_i$ kùn.
before CONJ return.home
Intended:'No, Kobina went to the farm before Kojo returned home'

d. Ààyí Kobina nàn bùèn kì-sàà-k [$_{\text{Adjunct island}}$ wààr lé
no, K. PST go 11.SG-farm-11 before CONJ
Mosia kùn.
M. return.home
'No, Kobina went to the farm before Kojo returned home'

This island violation is also evident in the ungrammaticality of the (339-c), where 'Mosia' has been overtly extracted from the adjunct island, leading to unacceptability. The grammatical version in (339-d) demonstrates that the sentence is well-formed when the adjunct island is not violated. It is worth noting that Griffiths and Lipták (2014) provides data on island sensitivity in fragments. This data points to a phenomenon similar to island insensitivity in fragment answers. This poses a theoretical challenge to Merchant (2004)'s analysis, where he predicts that all fragment answers involve overt movement of the remnant to the left periphery and should uniformly exhibit island effects. However, Griffiths and Lipták (2014) demonstrates that only contrastive fragments show sensitivity to islands. Non-contrastive fragments, such as (340-b) with an indefinite correlate, consistently show insensitivity to island violations.

(340) a. [A:] I imagine John wants a detailed list.
b. [B:] I'm afraid he does. Very detailed.

(Griffiths and Lipták, 2014, 193)

According to Barros et al. (2014) and Griffiths and Lipták (2014), the island insensitivity effect in non-contrastive fragments cannot be straightforwardly captured under Merchant (2004)'s movement and deletion approach, which predicts island sensitivity across the board. The asymmetry between contrastive and non-contrastive cases suggests that different derivations may underlie these fragment types. According to Griffiths and Lipták (2014), fragment answers, like sluicing, might involve repair mechanisms that bypass island violations under specific discourse conditions. This presents an empirical challenge to the uniformity proposed in Merchant (2004)'s original account.

6.7 Chapter Summary

This chapter outlined the basic properties of Likpakpaanl fragment answers. I showed that fragments can occur as answers to matrix, embedded, and polar questions. I argued and showed that fragments are derived via A-bar movement to a left peripheral position. However, the data revealed that constraints requiring only elements capable of undergoing A-bar movement to the left periphery could constitute a fragmented answer. I demonstrated that fragment answers to a matrix and embedded fragments are derived by the movement to Spec-LP even though embedded fragments require the obligatory complementiser **ké**. I argued, following Merchant (2001, 2004), that ellipsis is licensed by an E-feature that is hosted on the L^0 in the case of matrix and embedded fragment answers. I adopted Kramer and

Rawlins (2009)'s analysis, but contrary to their claim that *ResPrts* are sentential adverbs, I assume the *Yes/No* particles are base-generated in Spec-ΣP, whose head bears Merchant (2001)'s E-feature responsible for licensing ellipsis.

Consistent with the observation in the chapter on sluicing, this chapter showed that the obligatory focus particle present in non-elliptical answers to constituent questions is absent in fragment answers. I proposed that both fragments and sluicing are derived via the same syntactic process: an initial Agree relation between the [*i*Foc] feature on the fragment and the Focus head uninterpretable [*u*Foc] feature. After agreeing and checking the features on Foc-head, the fragment element undergoes successive-cyclic A-bar movement through FocP to Spec-LP. The E-feature on LP then licenses deletion of the FocP complement and TP. Consequently, I contend fragment answers and sluicing in Likpakpaanl constitute instances of FocP ellipsis. Adopting a movement plus PF-deletion approach, I adduced evidence from connectivity effects (binding principles) and island sensitivity to support the claim that fragment derivation involves overt A-bar movement of the remnant to Spec-LP from its base position before ellipsis applying.

Chapter 7

Summary and Conclusions

7.1 Introduction

This dissertation aims to contribute to the study of elliptical constructions with a focus on the syntax of Modal Complement Ellipsis (MCE), Sluicing, and Fragment answers in Likpakpaanl, a Mabia (Gur) language belonging to the Gurma sub-group of the Oti-Volta branch of the North Central Mabia (Gur) languages spoken in the Eastern Corridor of the Northern, Oti and North-East regions of Ghana (West Africa). In the preceding chapters, I discussed some aspects of the Grammar of Likpakpaanl, the structure of wh-questions in matrix and embedded clauses. I also extensively discussed the derivation of modal complement ellipsis, a type of VP ellipsis where the modal complement is elided, sluicing and fragment answers. The empirical evidence supported a movement account of Likpakpaanl elliptical constructions and their syntactic structure. I proposed a Licensing Phrase in the left periphery, above topic and focus phrases, which bears the Merchant (2001)'s ellipsis [E]-feature to license MCE, sluicing, and fragments. I also demonstrated that sluicing and fragment answers in Likpakpaanl target FocP and not TP for deletion. The dissertation is organised into seven chapters spanning literature on ellipsis, the grammar of Likpakpaanl, and ellipsis in Likpakpaanl (MCE, sluicing, and fragments) themes on wh-questions and their answers. The structure of this chapter is as follows: Section 7.2 provides an overall summary of the dissertation. In Section 7.3, I outline the scientific contributions that this dissertation makes to the syntax and theory of elliptical constructions. Finally, Section 7.4 presents open issues and recommendations for further research on this topic.

7.2 Summary of Findings of the Study

The core objective of this dissertation is to describe aspects of the grammar of Likpakpaanl, investigate and provide a theoretical account for the derivation of Modal Complement Ellipsis, Sluicing, and Fragment answers of Likpakpaanl. The dissertation presents an empirical investigation into the interaction and agreement morphology between the noun classes and the demonstrative, relative, and 'which' interrogative markers that have not been systematically researched. Chapter 3 of the dissertation further examines the formation of wh-questions and A-bar constructions in Likpakpaanl. I reviewed the previous studies on Likpakpaanl noun class (NC) classifications by Tait and Strevens (1954) and Winkelmann (2012) and offered a novel analysis that gives a minimalist account of Chomsky (1995, 2000) of the agreement morphology between noun classes and other elements like demonstrative, relative operators and interrogative marker. I also argued that the number feature on the NC is the head of a functional category, number Phrase (NumP), while gender is a feature realised on the syntactic head of the noun phrase. Thus, number and gender features are conflated on each NC. NumP has interpretable phi-features [iϕ] while the nouns have unvalued phi-probes [uϕ]. The motivation for that conclusion comes from the agreement patterns within the nominal phrase, where nouns agree with targets like demonstratives in number and gender features. I demonstrated that nominal morphology within the noun phrase is derived through Agree between unvalued noun probes and the interpretable specifications on Num.

Moreover, I examined the syntax of Likpakpaanl wh-questions. I showed that ex-situ wh-questions initial and Agree relation between the left peripheral focus head with an interpretable but unvalued focus feature and the wh-phrase with an uninterpretable but valued focus feature, the overt movement of wh-phrase from its base generated positions to the Spec-FocP and the optional presence of overt focus **lé**. Adopting a Probe Goad approach of the Minimalist programme, I argued that the overt syntactic movement of wh-phrases to the left periphery is triggered by an EPP feature on FocP that needs to be satisfied.

Chapter 4 was dedicated to the syntax of Likpakpaanl Modal Complement Ellipsis (a type of Verb Phrase ellipsis), where the complement to the modal auxiliary is targeted for deletion. I showed that although Likpakpaanl has different modals, there is a selectional restriction on those that can license MCE. These are the root modals expressing possibility **ŋmàà** and desire **bàn** that can license MCE. I adopt the standard assumption that Likpakpaanl modal complement ellipsis is licensed by (Merchant, 2001)'s an E-feature which is syntactically merged on modal head **ŋmàà**. The syntactic component of the E-feature C-selects the root modal **ŋmàà**, while the semantic category ensures that only elements with an antecedent (e-GIVEN) can undergo deletion while the phonological component of the E feature ensures that the PF content of ModP's complement, the vP is unpronounced at LF. While **ŋmàà** targets a vP complement for 'deletion,' the ellipsis site of the modal **bàn** is bigger since it elides a TP complement. The syntactic distribution of **bàn** is similar to Dagnac (2010)'s observation that the

French modal **pouvoir** 'can' in and the Italian **potere** 'can' license the ellipsis of a TP clausal complement rather than a VP. I also presented empirical data demonstrating the constraints on licensing modal complement ellipsis (MCE) in Likpakpaanl. Specifically, I claimed that modals expressing obligation like **yè kè** 'must', possibility, **nìbààkàn** 'maybe', mild obligation **lì** 'should', as well as tense, aspect and negation particles, are blocked from licensing MC ellipsis in the language.

Additionally, I proposed that when **ŋmàà** and **bàn** co-occur, it is the higher modal **ŋmàà** that licenses MCE, suggesting that there is only one position in the clause spine where the E-feature can be hosted. Consequently, **bàn** moves to the higher modal position occupied by **ŋmàà** when it is the only modal to license Likpakpaanl MCE.

A major observation from the data showed a morphological change of the modal auxiliary from **ŋmàà** to **ŋmàn** when it licenses ellipsis. I account for this observation by drawing on Halle and Marantz (1993)'s Distributed Morphology framework and propose that the modal head **ŋmàà** may undergo a morphological modification to **ŋmàn** after the ellipsis operation takes place, and the PF content of the ModP complements is null. Thus, I propose that the ellipsis operation at the syntactic level interacts with the morphological component, leading to observable changes in the realisation of the verbal form of the modal. Using evidence from Antecedent-Contained Deletion, focus and topic movement, island constraints, missing antecedents, and anaphoric binding, I demonstrated that the E-site in Likpakpaanl MCE has an underlying syntactic structure that becomes phonologically null when the E-feature was merged on the modal head.

In Chapter 5, I examined the syntactic derivation of sluicing constructions in Likpakpaanl and proposed an analysis that supports Merchant (2001)'s idea that the E-feature is hosted on a single syntactic head that licenses ellipsis and not spread across functional heads as Aelbrecht (2010) and Stigliano (2022) propose. I argued that the E-feature, with its semantic, syntactic, and phonological feature bundles, are merged on the licensing phrase (LP), which elides the FocP Complement. I further contended that Likpakpaanl sluicing is derived via the A-bar wh-movement of the wh-phrase from its base position to Spec-FocP, followed by successive cyclic movement to Spec-LP. Thus, sluicing in Likpakpaanl targets the FocP ellipsis rather than the TP ellipsis observed in English.

I adduced evidence from complex noun phrases, relative clauses, coordinate structure constraint islands, and internal language evidence, where in-situ wh-phrases like **kinye** 'how' cannot participate in sluicing to support a movement-based account to sluicing. Adopting Mendes and Kandybowicz (2023) *Salvation by Deletion* hypothesis and Merchant (2004), PF-theory of islands, I posited that the absence of island effects in sluicing is due to the fact the PF-uninterpretable feature causing the ungrammaticality is eliminated under ellipsis. The data showed that the optional focus particle accompanying ex-situ wh-movement is obligatorily absent under sluicing, satisfying the prediction of Merchant (2001)'s *Sluicing-COMP Generalisation*, which stipulates that all non-operators should be deleted under ellipsis (sluicing).

I compared the derivation of sluicing from languages with overt focus markers to present a cross-linguistic variation in the survival of overt focus markers under sluicing. Thus, while languages like Gungbe, Yoruba, and Nupe follow Baltin (2010)'s Generalisation by retaining the focus head, others like Likpakpaanl, Dagbani, and Akan violate it but uphold Merchant (2001)'s SCG. This divergence is hypothesised to stem from whether the same particle marks focus and clausal conjunction in a given language. These findings provide new evidence on the syntactic derivation of sluicing and its cross-linguistic variation, which motivates further research into the precise relationship between focus, coordination, and the licensing of sluicing across diverse language families.

The basic properties and derivation of fragment answers were outlined in Chapter 6. I showed that fragments can occur as answers to matrix, embedded, and polar questions. I showed that even though matrix and embedded fragments are derived by A-bar movement, embedded fragments require the obligatory complementiser **ké**. I accounted for the derivation of fragments, following Merchant (2001, 2004)'s Movement plus PF deletion approach where an E-feature licenses ellipsis. After checking and valuing the focus features on FocP in the left periphery, I contended that the fragment constituent is extracted to Spec-FocP to check FocP's EPP feature. When the licensing phrase (LP) bearing the E-feature and an EPP is merged, the fragment element in Spec-FocP is further moved to Spec-LP, with the E-feature deleting its FocP complement. Consequently, the focus particle, obligatory in non-fragmentary responses, does not survive in fragments in Likpakpaanl. I also discussed the syntax of polar response particles as fragment answers by adopting Kramer and Rawlins (2009)'s approach by assuming that *Yes/No* particles are base-generated in the specifier of a polarity phrase (Spec-ΣP), whose head bears Merchant (2001)'s E-feature responsible for licensing ellipsis. I demonstrated that the ellipsis-licensing E-feature is merged on the LP merged below -ΣP. The A-bar movement plus PF-deletion account was supported by empirical arguments such as binding and connectivity effects, the ban on weak pronouns from occurring in fragments, and the sensitivity of fragment answers to locality constraints.

7.3 Summary of the Contributions of the Study

This dissertation has made empirical and theoretical contributions to the study of Likpakpaanl, an understudied language, and the theory of ellipsis. While primarily investigating the syntax of ellipsis in Likpakpaanl, focusing on Verb Phrase ellipsis, sluicing, and fragment answers, the dissertation also explored aspects of the language's clause structure, tense, aspect, and noun class system, which have received limited linguistic investigation.

The dissertation contributes to understanding the syntax of Likpakpaanl and Mabia languages by providing a theoretical analysis of tense, aspect, modals, and noun class systems previously discussed from a descriptive perspective. It also advances the theory of ellipsis by supporting the typological variation of the ellipsis feature proposed by Stigliano (2022), Aelbrecht (2010) and Merchant (2001), indicating that the components of E-feature can either be distributed across different lexical elements or merged on a single functional head. The dissertation further demonstrates that, unlike English Verb Phrase Ellipsis (VPE), which deletes a VP complement, the E-feature in Modal Complement Ellipsis in Likpakpaanl can target a larger complement, such as an entire clause (TP) or a verb phrase (VP), supporting Dagnac (2010)'s view of modal ellipsis in French, Spanish, and Italian. Thus, the study shows that cross-linguistically, the different statuses of modals directly affect ellipsis licensing and the size of the E-site.

The dissertation has made significant and specific contributions to the study of Likpakpaanl. First, it has provided a novel account of Likpakpaanl noun class (NC) system and the agreement morphology, driven by features, with demonstrative pronouns, relative operators, and D-linked interrogative markers. It also provides a systematic analysis of the inventory of 15 NCs, showing that they conflate for number (singular and plural) and gender. Theoretically, the study demonstrated that in Likpakpaanl, number is the head of a functional category (NumP), while gender is a feature realised on the syntactic head of the noun phrase. Thus, even though Tait and Strevens (1954), Winkelmann (2012) and Bisilki and Akpanglo-Nartey (2017) have worked on the noun classes of Likpakpaanl, their works did not give any a detailed syntactic analysis as has been done in this dissertation.

Secondly, this work is crucial to the syntactic presentation of aspects of the structure of Likpakpaanl wh-questions and A-bar movements. The study accounts for the formation of in-situ and ex-situ questions and the asymmetry in the distribution of focus head **lé**, contributing to the theory of focus and focus marking in Likpakpaanl. Apart from providing empirical data showing the optional marking of focus on ex-situ wh-questions and the obligatory marking on the response to the same question, a theoretical analysis was offered, focusing on broader syntactic issues regarding the A-bar movement to the left-periphery

Furthermore, this dissertation comprehensively analyses three elliptical constructions in the language: modal complement ellipsis, sluicing and fragments. These elliptical phenomena have not been previously investigated in Likpakpaanl and have received little attention in African languages. The findings, therefore, provide a foundation upon which future research can be built. For instance, the derivation of modal complement ellipsis is significant in studying ellipsis licensing in Likpakpaanl and other Mabia languages. The theoretical motivation for projecting a Modal Phrase bearing the E-feature also contributes to a better understanding the interaction between modality and ellipsis licensing in the language.

Finally, the dissertation analyses the derivation of sluicing and fragment answers and provides a uniform account, showing that the predicate construction feeds sluicing and that the focus particle must be contained in the ellipsis site. The study presents interesting empirical facts showing that sluicing and fragment answers in Likpakpaanl are not TP, as in English, Gungbe or Nupe ellipsis, but FocP ellipsis. The research findings contribute to the ongoing debate in generative linguistics about the nature and size of the ellipsis site in ellipsis.

These empirical facts, coupled with the evidence supporting A-bar movement plus ellipsis account for all the elliptical phenomena examined in this dissertation, advance our understanding of a syntactic appreciation of Merchant (2001, 2004, 2019)'s PF-theory of ellipsis and the structuralist approach in the study of MCE, sluicing and fragment answers. The findings in this dissertation, in the distribution of focus particles and the adherence or violation of Merchant (2001)'s Sluicing-Comp Generalisation. The findings about the cross-linguistic variation in the presence/absence of the focus marker in ellipsis sets the tone for a broader investigation into the distribution of focus markers, clausal coordination and ellipsis.

7.4 Recommendations for Future Research

This dissertation on the syntax of ellipsis in Likpakpaanl has laid a strong foundation for future research in several areas. One fertile ground for further exploration is the noun class system and its interactions with ellipsis. While the research provided a detailed syntactic analysis of the noun class system, more work could be done to investigate the interactions between different noun classes, number, and gender features across various constructions and contexts.

Another promising avenue for future research is expanding the study of ellipsis to other Mabia languages. The dissertation's analysis of modal complement ellipsis, sluicing, and fragment answers in Likpakpaanl is a solid foundation for comparative research within this language family. Such cross-linguistic comparisons could yield valuable insights into the typological variation of ellipsis and shared properties across Mabia languages.

Furthermore, a comparative analysis of sluicing, verb phrase ellipsis, and fragment answers across these languages could reveal similarities and differences, contributing to our understanding of elliptical constructions within this language family. Furthermore, investigating other ellipsis phenomena, such as gapping and noun phrase ellipsis, which are not covered in the dissertation, could broaden our knowledge of ellipsis in Mabia languages.

The dissertation's findings on the distribution of focus particles and their adherence or violation of Merchant (2001)'s Sluicing-COMP Generalisation and Baltin (2010)'s Generalisation open up exciting avenues for future research. A cross-linguistic study that examines why languages like Gungbe, Nupe, and Yoruba, which use different particles for focus and clausal coordination, retain their focus particle under sluicing, and others like Likpakpaanl, Dagbani, and Akan, which use the same focus particle as a clausal coordinator, delete it under ellipsis would provide insights into the intricate relationships between focus marking, clausal coordination, and ellipsis.

Finally, exploring the interfaces between ellipsis and other linguistic domains, such as information structure, discourse pragmatics, and semantics, could provide valuable insights into how ellipsis is influenced by and interacts with these factors in Likpakpaanl.

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LIST OF PUBLICATIONS

1. Acheampong, S. O. (2024). Resumption and long-distance wh-movement in Likpakpaanl. In A. Himmelreich, D. Hole, J. Mursell (Eds.), *To the left, to the right, and much in between: A Festschrift for Katharina Hartmann.* (pp. 27–42). Frankfurt: Goethe University Frankfurt.

2. Abubakari, H., Issah, S. A., **Acheampong, S. O.**, Luri, M. D., Napari, J. N. (2023). Mabia languages and cultures expressed through personal names. *International Journal of Language and Culture*, 1–28.

3. Issah, S.A., **Acheampong, S. O.** (2021). Interrogative pronouns in Dagbani and Likpakpaanl. *Ghana Journal of Linguistics* 10 (2), 30-57.

CURRICULUM VITAE

SAMUEL OWOAHENE ACHEAMPONG

Research Interest

My research interest is descriptive and theoretical linguistics of the Mabia and Ghanaian languages and the pedagogy of language in education. I also work on the morpho-syntax of Likpakpaanl, particularly the syntax of elliptical phenomena, reflexivity, and information structure.

Education

2021-2024: Ph.D. in Linguistics - Goethe-Universität Frankfurt am Main.
Dissertation title: The Syntax of Elliptical Constructions in Likpakpaanl.
Supervisor: Prof Dr. Katharina Hartmann
Co-supervisor: Prof Dr. Enoch Abo

2013-2015: M.Phil in Applied Linguistics
University of Education, Winneba, Ghana
Thesis title: Reflexivisation in Likpakpaanl.

2006-2010: B.Ed. (Arts) English
University of Cape Coast, Ghana
Thesis title: Factors that account for the poor performance of Pupils in English in the Basic Education Certificate Examination (BECE) in the Saboba District.

2000-2003: Teachers' Cert 'A'
Tamale Training College, Ghana

1997-1999: WASSCE (General Arts)
St. Charles Minor Seminary/Senior Secondary School, Tamale, Ghana

Employment History

April 2016 - July 2020: Language and Linguistics Tutor and Acting Head of the Department of Languages at St. Vincent College of Education, Yendi.

March 2013 - July 2020: Part-time tutor on University of Education Winneba (UEW) Distance Education Programme (Winneba and Tamale Study Centres). *Courses taught include* Phonetics and Phonology of English, Introduction to General Linguistics, and the Language Situation in Ghana.

January 2016 - July 2020: Part-time tutor on University for Development Studies (UDS) Distance Education Programme (Yendi Study Centre).
Courses taught include Introduction to Linguistics, English Language, and Communication Skills.

April 2018 - March 2019: Part-Time Communication Skills Lecturer at the Yendi College of Health Sciences.

September 2010 - August 2013: English Language and Literature in English Teacher at St. Charles Seminary/Senior High School, Tamale.

September 2003 - July 2006: English Language and RME teacher at St. Brigid's Junior Secondary Senior, Chereponi.

Peer-Reviewed Journal Publications

Acheampong, S. O. (2024). Resumption and Long-Distance Wh-Movement in Likpakpaanl. In A. Himmelreich, D. Hole, and J. Mursell (Eds.), *To the Left, to the Right, and much in between: A Festschrift for Katharina Hartmann.* (pp. 27–42). Frankfurt: Goethe University Frankfurt.

Abubakari, H., Issah, S. A., **Acheampong, S. O.**, Luri, M. D., and Napari, J. N. (2023). Mabia languages and cultures expressed through personal names. *International Journal of Language and Culture*, 1–28.

Issah, A. S., and **Acheampong, O. S.** (2021). Interrogative pronouns in Dagbani and Likpakpaanl. *Ghana Journal of Linguistics*, 10(2), 30-57.

Acheampong, O. S., Atintono, A. S., and Issah, A. S. (2019). The Morphosyntactic Characterisation of Likpakpaanl reflexives. *Journal of West African Languages*, 46(1), 123-141.

Conferences and Seminar Presentations

Acheampong, S. O., and Aremu, D. (2024). *On the Sluicing - COMP Generalisation in Mabia and Kwa.* Presented at the 32nd Conference of the Student Organisation of Linguistics in Europe (ConSOLE31), Queen Mary University of London, 17th-19th January 2024.

Acheampong, S. O. (2023). *Sluicing and Fragment Answers in Likpakpaanl.* Presented at the Syntax Colloquium, Goethe University Frankfurt am Main, 6th November 2023.

Acheampong, S. O., and Aremu, D. (2023). *Mono-Clausal and Bi-clausal Topics in Likpakpaanl.* Presented at the International Conference on Topic Focus and Subject: Between grammatical necessity and structural-functional Load, University of Osnabrück, 19th-23rd September 2023.

Acheampong, O. S. (2023). *On the Syntax of Tense and Aspect in Likpakpaanl.* Presented at the Conference on the syntax at the vP edge in African languages (SASAL III), Goethe University Frankfurt am Main, 29th-30th June 2023.

Acheampong, O. S. (2023). T*he Derivation and Analysis of Fragment Answers in Likpakpaanl.* Presented at the 31st Conference of the Student Organisation of Linguistics in Europe (ConSOLE31), Bielefeld University, 22nd-24th February 2023.

Book Section

Acheampong, S. O. (2021). Inspirational Poems-Likpakpaanl. In Ladé Wosornu (Ed.), *A selection of seven Wosornu poems translated into fifteen local Ghanaian languages.* (pp. 29-42). Accra: Bureau of Ghana Languages.

Workshops, Seminars, and Summer Schools Attended

July 18-July 30, 2022: African Linguistics School (ALS), Porto Novo, Benin.

October 2021-Date: Weekly seminar on syntactic effects on information structure, Goethe-Universität Frankfurt am Main.

August-September 2021: Weekly seminar on the comparative syntax of the D.P., Goethe-Universität Frankfurt am Main.

May-August 2021: Weekly seminar on Minimalist Approach to Core Syntax, Goethe-Universität Frankfurt am Main.

May-August 2021: Seminar on syntactic typologies, Goethe-Universität Frankfurt am Main.

Awards and Distinctions

May 2024: Three-Month Deutscher Akademischer Austauschdienst (DAAD) Completion Scholarship, Goethe University Frankfurt am Main, Germany.

Jan 2021-May 2024: Three-year Katholischer Akademischer Ausländer-Dienst (KAAD) scholarship for PhD in Linguistics, Goethe University Frankfurt am Main, Germany.

August 2013-2015: KAAD Scholarship for M.Phil in Applied Linguistics, University of Education, Winneba.

May 2010: Graduated with First Class in English, University of Cape Coast, Ghana.

Community Service

May 2014 - July 2020: Community volunteer for The Literacy and Development Programme (LDP-Ghana) and a part-time volunteer teacher of Likpakpaanl at St. Joseph Primary School, Yendi.

Jan 2007 - July 2020: Part-time volunteer teacher of Likpakpaanl at The Institute of Languages, Tamale.

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Appendix A

Sample Elicitation Questions for Modal Complement Ellipsis

Sample Questions were used to collect Data on Modal Complement Ellipsis in Saboba in August and September 2022. Some of the questions were given to respondents, and they were asked to give the English equivalent in Likpakpaanl. Others had the Likpakpaanl versions read to them for felicity judgments and semantic interpretations.

1. Nimomi gà ŋmàà gbìì n-nyìnbùn, lé Bilijo mù gà ŋmàn.(MCE)[1]
   Nimomi can dig a well, and Bilijo can, too

2. Nimomi gà ŋmàà gbìì n-nyìnbùn, lé Bilijo mù gà ŋmàà gbìì n-nyìnbùn.(Full)
   Nimomi can dig a well, and Bilijo can dig a well, too

3. Naajo nàn ŋmàà kàn Manyi àmáá Chati áá nàn ŋmàn. (MCE)
   Naajo was able to see Manyi, but Chati wasn't.

4. Naajo nàn ŋmàà kàn Manyi àmáá Chati áá nàn ŋmàà kàn Manyi.(Full)
   Naajo could see Manyi, but Chati couldn't see Manyi.

5. Kpààà gbààn gà ŋmàà gàà búsùb gbààn béé kì àà ŋmàn/*ŋmàà. (MCE)
   Can the cutlass cut the tree, or it can't?

6. Mataani bàn (ké) ù tér Taala lé Bileŋan mú bàn. (MCE)
   Mataani wants to help Taala, and Bileŋan too wants

7. Mataani bàn (ké) ù tér Taala lé Bileŋan mú bàn (ké) ù tér Taala. (MCE)
   Mataani wants to help Taala, and Bileŋan too wants to help Taala.

8. Ntaadi gà lì bán ù kárn kìgbàŋ gbààn, lé Kunjaam mù gà lì bàn.(MCE)
   Ntaadi will want to read the book, and Kunjaam will want to.

[1] MCE represents the Modal complement constructions and Full is the non elliptical one.

9. Ntaadi gà lì bán ù kárn kìgbàŋ gbààn, lé Kunjaam mù gà lì bàn ù kárn kìgbàŋ gbààn (Full).
   Ntaadi will want to read the book, and Kunjaam will want to read the book.

Appendix B

Sample Elicitation Questions for Sluicing and Fragment Answers

A: **Sluicing in Likpakpaanl**

Q1 Pambiin bì yìn unì ubàà, lé m bì bàè ké á daàk ké nì yè ŋmà.(SL)[1]
Pambiin is calling someone, but I don't know who.

Q2 Pambiin bì yìn unì ubàà, lé m bì bàè ké á daàk ké nì yè ŋmà (lé) Pambiin bì yìn.(NSL)
Pambiin is calling someone, but I don't know who Pambiin is calling.

Q3 Nilimor lén ké Bondan nàn pèn lìbóól lé Nambe bàn ù bé ké nì nàn yè lá ché. (SL)
Nilimor said Bondan roofed a hut and Nambe wanks to know where.

Q4 Nilimor lén ké Bondan nàn pèn lìbóól lé Nambe bàn ù bé ké nì nàn yè lá ché (lé) Bondaan nàn pèn lìbóól.(NSL)
Nilimor said Bondan roofed a hut and Nambe wanks to know where Bondan roofed a hut.

B: **Fragment Answers**

Q1 Tadime nàn kàn ŋmà?
Who did Tadime see?

(a) Maabe. (Frg)

(b) Maabe (*lé/là)
Mabee

[1] SL and Frg represent sluicing and fragment answers while NSL and N-Frg are the non-elliptical counterparts, respectively.

(c) ii. Tadime nàn kàn Maabe là. iii. Maabe lé Tadime nàn kàn.(N-Frg)
Tadimee said Maabe.

Q2 Poliye len ke Ama dàà bà?
What did Poliye say Ama bought?

(a) Ké Sima. (Frg)

(b) Ké Sima (*lé/là) (Frg)

(c) *Sima
That groundnuts

(d) Poliye lén ké Ama dàà sima là.(N-Frg)

(e) Sima lé Poliye lén ké Ama dàà
Poliye said that Ama bought groundnuts.